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FINANCIAL ADVISORY SERVICES FOR DORJILUNG HYDROPOWER PROJECT

DRAFT PRELIMINARY INVESTMENT REPORT

Draft for Discussion

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ABBREVIATIONS

Acronym	Definition
ABR	Aggregate Billing Rate
ACS	Average Cost of Supply
ADB	Asian Development Bank
ADF	Asian Development Fund
ADFB	African Development Fund Board
AFC	Annual Fixed Charges
AIIB	Asian Infrastructure Investment Bank
AKFED	Aga Khan Fund for Economic Development
APCPDCL	Andhra Pradesh Central Power Distribution Company Limited
APDCL	Assam Power Distribution Company Limited
APEPDCL	Andhra Pradesh Eastern Power Distribution Company Limited
APR	Annual Performance Review
APSPDCL	Andhra Pradesh Southern Power Distribution Company Limited
ARR	Aggregate Revenue Requirement
ASA	Average System Availability
AVVNL	Ajmer Vidyut Vitran Nigam Limited
B2B	Business to Business
BAU	Business As Usual
BBIN	Bangladesh, Bhutan, India and Nepal
BBINS	Bangladesh, Bhutan, India, Nepal and Sri Lanka
BDT	Bangladeshi Taka
BEA	Bhutan Electricity Authority
BEC	Bulk Electric Consumer
BERC	Bihar Electricity Regulatory Commission
BESCOM	Bangalore Electricity Supply Company
BESS	Battery Energy Storage System
BGICC	Bharat Gas Infrastructure Corporation
BHP	Bangladesh Energy Regulatory Commission
BIMSTEC	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
BLTPS	Bharat Aluminium Company Limited Thermal Power Station
BMP	Best Management Practices
BOOT	Build Own Operate Transfer
BOT	Build Operate Transfer
BPC	Bhutan Power Corporation
BPDB	Bangladesh Power Development Board
BRPL	BSES Rajdhani Power Limited
BRSR	Business Responsibility and Sustainability Report

Acronym	Definition
BSES	Bombay Suburban Electric Supply
BSPHCL	Bihar State Power Holding Company Limited
BYPL	BSES Yamuna Power Limited
CAGR	Compound Annual Growth Rate
CBET	Cross Border Electricity Trade
CCPP	Combined Cycle Power Plant
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CESCO	Central Electricity Supply Company
CHESCOM	Chamundeshwari Electricity Supply Corporation Limited
CHMP	Capital Health Management Plan
CHP	Combined Heat and Power
CO2	Carbon Dioxide
COD	Commercial Operation Date
COP26	26th Conference of the Parties
CSPDCL	Chhattisgarh State Power Distribution Company Limited
CSS	Central Sector Scheme
DAM	Day Ahead market
DBFOT	Design Build Finance Operate Transfer
DECC	Department of Energy and Climate Change
DERC	Delhi Electricity Regulatory Commission
DGPC	Druk Green Power Corporation
DGVCL	Dakshin Gujarat Vij Company Limited
DHBVN	Dakshin Haryana Bijli Vitran Nigam
DHEP	Doyang Hydro Electric Plant
DHI	Druk Holdings & Investments
DHP	Dagachhu Hydropower Plant
DHPP	Dorjilung Hydropower Project
DoE	Department of Energy
DISCOM	Distribution Company
DPR	Detailed Project Report
DSCR	Debt Service Coverage Ratio
DSRA	Debt Service Reserve Account
DTTILLP	Deloitte Touche Tohmatsu India LLP
DTTL	Deloitte Touche Tohmatsu Limited
DVC	Damodar Valley Corporation
DVVNL	Dakshinanchal Vidyut Vitran Nigam Limited
EFMP	Environmental Flow Management Plan
EGAT	Electricity Generating Authority of Thailand
EGCO	Electricity Generating Public Company Limited
EIB	European Investment Bank
EPC	Engineering, Procurement, and Construction

Acronym	Definition
EPS	Electric Power Survey
ERA	Energy Regulatory Authority
ESIA	Environmental and Social Impact Assessment
EUR	EURO
EURIBOR	Euro Interbank Offered Rate
FDI	Foreign Direct Investment
FI	Financial Institution
FSV	Final Scheduled volume
FY	Financial Year
G2G	Government-to-Government
GDAM	Green Day Ahead Market
GDP	Gross Domestic Product
GoA	Government of Austria
GESCOM	Gulbarga Electricity Supply Company
GHG	Greenhouse Gas
GMR	Geometrical Mean Radius
GOI	Government of India
GRIDCO	Grid Corporation of Odisha Limited
GSECL	Gujarat State Electricity Corporation Limited
GT	Gas Turbine
GTPS	Gas Turbine Power Station
GUVNL	Gujarat Urja Vikas Nigam Limited
GW	Gigawatt
HDSC	Hydro Debt Service Coverage
HEP	Hydro Electric Power
HESCOM	Hubli Electricity Supply Company
HFO	Heavy Fuel Oil
HPGCL	Haryana Power Generation Corporation Limited
HPO	Hydropower Purchase Obligations
HPP	Hydropower projects
HPPCL	Himachal Pradesh Power Corporation Limited
HPX	Hindustan Power Exchange Limited
HSPL	Himachal Sorang Power Limited
HV	High Voltage
HVAC	High Voltage Alternating Current
HVDC	High Voltage Direct Current
IBRD	International Bank for Reconstruction and Development
ICR	Interest Coverage Ratio
ICRA	Investment Information and Credit Rating Agency
IDA	International Development Association
IDC	Interest During Construction
IEA	International Energy Agency

Acronym	Definition
IEDC	Interest Expenses During Construction
IEPMP	Integrated Energy and Power Master Plan
IEX	Indian Energy Exchange
IFC	International Finance Corporation
IFIs	International Financial Institutions
INR	Indian Rupee
InVIT	Infrastructure Investment Trust
IPS	Industrial Promotion Services
IRR	Internal Rate of Return
ISGS	Inter-State Generating Station
JBIC	Japan Bank for International Cooperation
JBVNL	Jharkhand Bijli Vitran Nigam Limited
JICA	Japan International Cooperation Agency
JV	Joint Venture
KESCO	Kanpur Electricity Supply Company
KFAED	Kuwait Fund for Arab Economic Development
KHP	Kurichhu Hydropower plant
KHPL	Khorlochhu Hydropower
KPIC	Kshema Power and Infrastructure Company Private Limited
KW	Kilowatt
LALRP	Land Acquisition and Livelihood Restoration Plan
LC	Letter of Credit
LGBR	Load Generation Balance Report
LHPP	Large Hydro Power Plant
LLCR	Loan Life Coverage Ratio
LLP	Limited Liability Partnership
LS	Lower Stage
MBED	Market-Based Economic Dispatch
MCLR	Marginal Cost of Funds-based Lending Rate
MCP	Market Clearing Price
MDB	Multilateral Development Bank
MGVCL	Madhya Gujarat Vij Company Limited
MIPA	Moveable and Immovable Property Act
MLA	Multilateral Agencies
Mn	Million
MoENR	Ministry of Energy & Natural Resources
MoF	Ministry of Finance
MoU	Memorandum of Understanding
MNRE	Ministry of New and Renewable Energy
MPMKVVCL	Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Limited

Acronym	Definition
MPPKVVCL	Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited
MPPMCL	Madhya Pradesh Power Management Company Limited
MSEDCL	Maharashtra State Electricity Distribution Company Limited
MSTC	Metal Scrap Trade Corporation
MT	Million Tonnes
MU	Million Units
MV	Medium Voltage
MVVNL	Madhyanchal Vidyut Vitaran Nigam Limited
MW	Megawatt
MYT	Multi Year Tariff
NBPDCL	North Bihar Power Distribution Company Ltd
NEA	Nepal Electricity Authority
NEEPCO	North Eastern Electric Power Corporation Limited
NESCO	North Eastern Electricity Supply Company of Odisha Limited
NHP	Nikachhu Hydropower Plant
NHPC	National Hydroelectric Power Corporation
NLC	National Land Commission
NOC	No Objection Clearances
NPL	Nabha Power Limited
NPPF	National Pension and Provident Fund of Bhutan
NTGMP	National Transmission Grid Master Plan
Nu	Ngultrum
NTPC	National Thermal Power Corporation
NVVN	NTPC Vidyut Vyapar Nigam Ltd
OCF	Operational Cash Flow
OCR	Ordinary Capital Resources
OPEC	Organization of the Petroleum Exporting Countries
PAF	Plant Availability Factor
PAT	Profit After Tax
PFC	Power Finance Corporation
PGVCL	Paschim Gujarat Vij Company Limited
PLCR	Project Life Coverage Ratio
PLF	Plant Load Factor
PLN	Perusahaan Listrik Negara
PMF	Probable Maximum Flood
PPA	Power Purchase Agreement
PSA	Power Sales Agreement
PSP	Pumped Storage Plant
PTC	Power Trading Corporation
PVVNL	Paschimanchal Vidyut Vitran Nigam Limited

Acronym	Definition
RE	Renewable Energy
REL	Ruzizi III Energy Limited
RGOB	Royal Government of Bhutan
RMA	Royal Monetary Authority
RPO	Renewable Purchase Obligation
RSPCL	Rajasthan State Power Corporation Limited
RTM	Real-Time Market
RTS	Roof-Top Solar
RZB	Raiffeisen Bank International
SAARC	South Asian Association for Regional Cooperation
SAIL	Steel Authority of India Limited
SASEC	South Asia Subregional Economic Cooperation
SBI	State bank of India
SBPDCL	South Bihar Power Distribution Company Ltd
SECI	Solar Energy Corporation of India
SERC	State Electricity Regulatory Commission
SHDP	Sustainable Hydropower Development Project
SHPP	Small hydropower projects
SJVNL	Satluj Jal Vidyut Nigam Limited
SLDC	State Load Dispatch Centre
SNP	SN Power AS
SO	System Operator
SOFR	Secured Overnight Financing Rate
SPV	Special Purpose Vehicle
TANGEDCO	Tamil Nadu Generation and Distribution Corporation
THDC	Tehri Hydro Development Corporation Limited
THPP	Tala Hydropower Plant
TNEB	Tamil Nadu Electricity Board
TNW	Tangible Net Worth
TOL	Total Outside Liabilities
TPCL	Tata Power Company Limited
TPDDL	Tata Power Delhi Distribution Limited
TPS	Thermal Power Station
TPTCL	Tata Power Trading Company Limited
TSERC	Telangana State Electricity Regulatory Commission
UDAY	Ujwal DISCOM Assurance Yojana
UHBVN	Uttar Haryana Bijli Vitran Nigam
UPPCL	Uttar Pradesh Power Corporation Limited
USD	United States Dollar
US	Upper Stage
VAT	Value Added Tax
VRE	Variable Renewable Energy

Acronym	Definition
WACC	Weighted Average Cost of Capital
WBSEDCL	West Bengal State Electricity Distribution Company Limited
WESCO	Western Electricity Supply Company of Orissa
WR	Western Region

1. Executive Summary

About DHPP	The total generation at the P90 level is 4,453 MW, with approximately 85% occurring between May and November, identifying this period as the optimal export window. India and Bangladesh are recognized as key offtakers due to their proximity to the project. Based on this generation profile, determining the optimal offtake will depend on domestic demand at the time of project commissioning. Any surplus beyond domestic needs will then be evaluated for export markets.
Bhutan Power Market Scenario	Bhutan's peak electricity demand rose from 374 MW in 2020 to 1,026 MW in 2024 and is projected to reach 1,662 MW by 2026. Domestic electricity demand grew at a CAGR of 23%, outpacing supply growth at 7.4%, leading to reduced export capacity. However, Bhutan has an ambitious capacity expansion plan dominated by hydropower, complemented by solar projects, aiming to reach 78,150 GWh in supply by 2040. Bhutan Power Corporation Limited (BPCL) plays a critical role in managing distribution, infrastructure, and cross-border trade to support the country's growing energy needs and export ambitions.
Indian Power Market Scenario	The offtake assessment begins with an analysis of the Indian power market, where demand–supply is projected to grow steadily at approximately 4% annually until FY 2042. This is followed by a deeper regional-level demand–supply profiling to understand energy and peak deficits across different regions. Such insights are critical before evaluating various offtake options, including through State DISCOMs, sale through merchant markets or C&I consumers.
Indian Power Market- Offtake Assessment	A performance analysis of Indian DISCOMs highlights their long-term reliability amid evolving market dynamics. Several states—including Maharashtra, Gujarat, Bihar, Odisha, Uttar Pradesh, Rajasthan, Haryana, and Andhra Pradesh—are projected to require over 1 GW of hydropower capacity each within the next decade to ensure resource adequacy. Meanwhile, states such as Tamil Nadu, Telangana, Karnataka, and BRPL (Delhi) face high average power procurement costs (above INR 5 per kWh), presenting opportunities to replace expensive fuel-based power with more affordable hydropower. DHPP can be a viable supplier to these discoms provided the tariff aligns with the power procurement costs. Short-term trading through power exchanges are increasingly emerging as flexible alternatives for Discoms; however, prices in both DAM and RTM have trended downward over the past three years due to the growing addition of solar and storage projects limiting the potential to allocate offtake for short term power market for DHPP. The short and medium term trading through bilateral route is an attractive segment with short term contracts offering high volumes and prices. The commercial and industrial (C&I) segment demonstrates strong growth potential, supported by favorable consumption trends. The benchmark study reveals that government hydro projects predominantly secure long-term PPAs with State DISCOMs. However, a few government projects, such as Himachal Pradesh's hydroelectric plants, participate in short-term power markets due to the absence of long-term offtake agreements. Typically, government projects allocate up to 30% of their power to short-term markets owing to a lower risk appetite, whereas private projects may allocate up to 100%, reflecting greater flexibility and market exposure.

Bangladesh Power Sector Landscape	Natural gas has historically accounted for over 50% of Bangladesh's power generation, but declining reserves now present a significant energy security challenge. To address this, the government is increasingly turning to electricity imports from neighboring countries as a strategic measure to diversify and stabilize its energy supply. Bangladesh is rapidly expanding its clean energy capacity, targeting 40% of its total supply by 2041 and 66% by 2050. With approximately 20% of this clean energy expected to come from imports, Bhutanese hydropower is well-positioned to play a crucial role in helping Bangladesh achieve its ambitious sustainability goals.
Regulatory Landscape in South Asia	The determination of tariffs for hydropower generation plants in India is governed by a well-established regulatory framework administered by the Central Electricity Regulatory Commission (CERC) and the respective State Electricity Regulatory Commissions (SERCs), based on a cost-plus methodology. Recent tariffs for projects commissioned over the past 4–5 years have largely ranged between INR 4 to 5 per unit. Common reasons for tariff disallowances in some hydro projects include cost overruns due to project delays and under-approval of annual fixed charges. While Bangladesh has a regulatory framework in place, including institutions such as the Bangladesh Energy Regulatory Commission (BERC), its regulatory maturity and market mechanisms are still evolving and are not yet comparable to the more advanced regimes found in developed power markets.
Export Risk Analysis	Cross border electricity trade entails several key risks that need careful management. Curtailment risk arises from transmission constraints and demand fluctuations, potentially limiting the ability to fully evacuate power. Currency risk may also be significant due to revenue and cost mismatches when dealing with foreign exchange fluctuations in cross-border transactions. Additionally, participation in the merchant power market exposes projects to price volatility and market liquidity challenges.
Transmission Infrastructure in South Asia	Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. Currently, the power system of Bhutan is connected to the Indian grid, through 13 interconnecting lines with eight feeders at 400kV and three feeders at 220kV voltage levels from western Bhutan, and two S/C 132kV lines from eastern Bhutan and around 1,000 MW evacuation capacity is additionally planned between India and Bangaldesh.
Hydropower and Bhutan's Macroeconomic Structure	An analysis of principal repayments, interest outflows, and revenue collections, both domestic and export, indicates that the debt is self-liquidating. Total revenue generated from the project is sufficient to meet debt servicing obligations from the very first year. However, the second condition outlined in the Bhutan Debt Management Policy, which requires that debt servicing remain within 50% of total export revenue, is met only in the later years, around 2045. While this condition is not fulfilled in the initial years, it is important to recognise that the DHPP project's contribution goes well beyond export earnings. It is expected to act as a catalyst for broader economic transformation by enabling the growth of energy-intensive industries, improving energy security, and supporting infrastructure development. The availability of reliable and surplus electricity can attract manufacturing and service-sector investments, create jobs, and stimulate downstream economic activity. In this way, the project serves as an enabler of long-term economic diversification, reducing Bhutan's dependence on hydropower exports alone and creating the foundations for a more resilient and self-sustaining domestic economy.

Evalution of hydropower financing trends, risks and potential structure models for DHPP	To determine an optimal financial structure for DHPP, a study of Bhutan's hydropower financing evolution was conducted. Historically, Bhutan relied on bilateral funding from India for major projects like Chukha, Kurichhu, Tala, and Mangdechhu. Post-2010, DGPC began diversifying funding sources to reduce dependence on sovereign support, engaging multilateral institutions, private investors, and export credit agencies. The Nikachhu project marked Bhutan's first fully public-funded hydropower initiative. International case studies such as Ruzizi III (Africa) and Nam Ngiep 1 (Laos) were also analyzed. Lenders typically assess technical, environmental, geological, and commercial aspects before financing. Legally, DHPP can borrow from foreign lenders with prior government approval under Bhutan's MIPA Act, which also governs the enforceability of security interests and step-in rights. The financing process is expected to take 5–7 months post-PPA finalization. Key risks—such as land acquisition, geological surprises, hydrology shocks, and cost overruns—have been identified, with mitigation strategies in place based on lessons from similar projects.

2. Progress So Far

2.1.Key insights from Inception Report

The Inception Report offered a comprehensive overview of Bhutan’s macroeconomic environment and the power sector, detailing its institutional framework and key policies. It highlighted the importance and unique features of the Dorjilung Hydropower Project (DHPP), while also analyzing the key dynamics of South Asian regional power market. The report also covered Bhutan’s evolving financial strategy for large hydropower projects, noting a shift from traditional bilateral support to a more diversified approach involving joint ventures and partnerships with international financial institution.

Bhutan's economy is significantly driven by its hydropower sector, which contributes substantially to GDP, export revenues, and overall fiscal stability. The country possesses a vast hydropower potential, estimated at 37,000 MW[1], with approximately 2,453 MW currently installed. The development of this hydropower potential has demonstrably influenced key macroeconomic indicators. Historical data over the past two decades indicate a consistent positive correlation between the commissioning of major hydropower projects and subsequent increases in GDP, typically ranging from 2% to 4%[2]. A notable instance is the Tala Hydropower Plant, the commissioning of which precipitated a 10% rise in GDP. Such projects not only augment domestic electricity supply and enhance export revenues but also catalyze broader economic activity, thereby contributing to an improved trade balance.This pronounced reliance on hydropower development, while economically beneficial, also presents distinct financial considerations. The nation's debt-to-GDP ratio is elevated, standing at approximately 109.8%, with investments in hydropower projects constituting a significant component of the external debt portfolio (64%). However, the operational profitability of these hydropower assets has proven sufficient to service the associated debt obligations, as evidenced by a Hydro Debt Service Coverage (HDSC) ratio of 1.9 in Fiscal Year 2023, which is projected to improve further. Furthermore, hydropower generation contributes directly to governmental revenues through the provision of royalty energy, representing 21.1% of total tax revenue in 2023, thereby supporting public fiscal capacity

To effectively govern and strategically expand this pivotal sector, Bhutan has established a comprehensive institutional and regulatory architecture. A significant restructuring of the power sector occurred in 2002, which unbundled the previously vertically integrated system, clearly demarcating responsibilities for policy formulation, regulatory oversight, and operational management. The Ministry of Energy & Natural Resources, through its Department of Energy, assumes the central role in policy development and strategic planning. Regulatory functions are vested in the Energy Regulatory Authority (formerly the Bhutan Electricity Authority), while state-owned enterprises, notably Druk Green Power Corporation Limited (DGPC) for generation and Bhutan Power Corporation Limited (BPC) for transmission and distribution, manage operational aspects. This framework is further reinforced by a suite of policies aimed at promoting sustainable hydropower development, encouraging the diversification of renewable energy sources, and ensuring the establishment of equitable and transparent tariff mechanisms.

The efficacy of these institutional arrangements is continually tested by evolving sectoral dynamics, particularly the rapid escalation in domestic electricity demand. Over the last five years, domestic consumption has increased at an exceptional rate of approximately 38%[3], with threefold increase in peak demand to 1,026 MW[3] in FY 2024. This surge is primarily attributable to the expansion of energy-intensive industries and the development of new industrial parks. While Bhutan's overall hydropower generation has remained relatively stable, the absence of significant new capacity additions in recent years, juxtaposed with rising domestic requirements, has necessitated an increase in electricity imports from India, particularly during the lean winter months when hydrological conditions reduce domestic generation capacity.

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[1] Bhutan Energy Directory 2022
[2] World Bank, GDP Growth, 2005-2011 and 2012-2024
[3] DGPC Additional Data Workbook

In response to these evolving demand-supply dynamics and to harness its untapped potential, Bhutan has formulated ambitious strategic initiatives for capacity augmentation. The national energy plan envisages the addition of approximately 23,500 MW of hydropower capacity and 5,000 MW of solar capacity by the Fiscal Year 2040[4]. Central to this expansion strategy are projects such as the Dorjilung Hydropower Project (DHPP). With an expected installed capacity of 1,125 MW, DHPP is designed not only to address domestic energy deficits during periods of low supply but also to generate a substantial surplus for export. The project's strategic location on the Kurichhu river, coupled with a meticulously planned transmission evacuation system, is intended to ensure the reliable conveyance of generated power to both domestic consumers and regional export markets.

The ultimate viability and economic contribution of such large-scale projects are intrinsically linked to the ability to secure stable and remunerative offtake arrangements within the broader South Asian regional power market. Bhutan's hydropower exports are strategically positioned to cater to the escalating energy demands of neighboring countries. India, for instance, is pursuing an aggressive expansion of its renewable energy capacity and has instituted Renewable Purchase Obligations, including specific mandates for hydropower, thereby creating a receptive market for Bhutanese electricity. Similarly, Bangladesh, which currently relies heavily on expensive imported fossil fuels, presents a significant potential market for cleaner and more cost-effective hydropower imports. The emergence of Nepal as another regional hydropower exporter introduces a competitive element that underscores the importance of timely project execution and market engagement for Bhutan.

The substantial capital investment required for the development of these large-scale hydropower facilities necessitates financing strategies. The capacity of Bhutan's domestic banking sector is insufficient to independently finance projects of this magnitude. Consequently, Bhutan has progressively diversified its financing modalities, moving beyond traditional bilateral arrangements to embrace joint ventures, public-private partnerships (PPPs), and concessional funding from international financial institutions such as the World Bank and the Asian Development Bank, as well as commercial lending from Indian banks. This multifaceted financing approach, which is also envisaged for the Dorjilung Hydropower Project, aims to achieve a balance between ensuring financial sustainability and facilitating the accelerated development of the nation's hydropower resources.

DGPC, the sole hydropower operator in Bhutan, is a wholly owned subsidiary of Druk Holdings & Investments – a Royal Government of Bhutan (RGoB) company incorporated for holding all government owned enterprises in Bhutan across multiple sectors. India based rating agency ICRA, subsidiary of Moody’s Ratings, has assigned DGPC with an issuer rating of BBB+ with a stable outlook on March 13, 2024.

DGPC currently operates at an installed capacity of 2,453 MW with its projects housed under holding company i.e. DGPC (2,209 MW) and standalone subsidiaries (224 MW- Dahgachhu Hydropower Corporation Limited and Tangsbji Hydro Energy Ltd.). Druk Hydro Energy Limited (100% subsidiary) was incorporated in December 2021 to house small and medium scale hydropower projects. To support the hydropower projects, the company has incorporated Bhutan Automation Engineering Limited and Bhutan Hydro Power Services Ltd. to offer a portfolio of critical and specialized services.

DGPC has developed comprehensive business plan for multiple hydropower and solar projects to expand its footprint in the renewable energy sector and increase its capacity over the next 5 years. The financial structuring of these projects is based on an optimal debt-to-equity mix, ensuring financial sustainability while facilitating efficient capital mobilization.

The company currently has a total operating capacity of 2,453 MW hydropower projects housed either directly under DGPC or DGPC controlled subsidiaries/joint ventures. The existing operational capacity is projected to increase to 3,547 MW, that is almost ~1.5x times of the existing capacity over the period

[4] DGPC Energy Data Excel Workbook

of 5 years i.e., till 2030. This increase in operational capacity constitutes capacity addition of 494 MW and 600 MW in SHPP and LHPP (Khorluchhu HPP) respectively.

While hydropower remains the backbone of Bhutan's energy sector, DGPC is actively diversifying into solar energy to create a more resilient and sustainable energy mix. Recognizing the potential of solar power as a complimentary source of hydropower, DGPC has initiated the development of solar projects across locations. This diversification enhances energy security by ensuring stable supply during dry seasons when hydropower generation may fluctuate. DGPC has projected to develop a total of 965 MW of the operational solar capacity by 2030. The estimated project cost of these 16 solar projects is c. Nu 59,138 Nu with debt equity ratio of 70:30.

Under its five-year plan, DGPC envisages to undertake a capex of ~ Nu. 498,538 million encompassing development of 8.9 GW of under-construction/to be operational SHPPs, LHPPs and Solar power plants. On a cumulative basis this capex shall be funded on a Debt:Equity Ratio of 70:30, translating into a debt and equity infusion requirement of Nu. 348,883 million and Nu. 140,656 million respectively. Further, part of the equity requirement i.e. Nu. 50,658 million is expected to be funded by JV partners in certain projects, balance Nu. 89,998 million shall have to be sourced by DGPC from its resources.

DGPC has established strong financial relationships with multiple banking institutions, including domestic and foreign banks, Indian banks, and multilateral agencies. These strategic partnerships play a crucial role in securing debt and equity financing for the overall expansion of DGPC's projects. Predominantly DGPC's hydropower projects have been funded with support from Sovereign Partners. Over the years, DGPC has been actively diversifying its financing structure to reduce reliance on bilateral sovereign funding. Subsequently, DGPC has utilized a mix of funding from Commercial Banks, Multilateral Organizations and Strategic Partners (PPP mode) to finance its operational/under-construction projects.

Various routes have also been suggested to DGPC for arranging funds for pipeline projects like green Bonds, InVITs, concession-based model, PPP model and refinancing of existing projects. These routes have been suggested keeping in view the financial suitability and objectives of DGPC that are, lowered cost of capital, broadened investor base, reduced reliance on RGoB for fund raise, maintaining health GDP-Debt ratio and long term financing aligned to project's life cycle.

Dorjilung Hydropower plant is expected to be developed on a Public-Private Partnership model with World bank as the anchor. Tata Power Corporation Limited along with its trading arm is envisaged to be the off-taker for power exports and strategic partner with non-controlling stake in the project SPV. Envisaged financing structure is similar to Dagachhu Hydropower plant and financing structures being adopted in international market.

DHPP is expected to be developed at an estimated total cost of $1.7 billion. Funding mix for the project is yet to be determined. It has been observed that in the past and internationally, financing mix of 70:30 debt equity ratio is considered reasonable for projects of such nature, depending on tariff and financing viability. World Bank group, through its entities would be participating in part-financing of the project. Part of the funding for the project would be contributed to by Strategic partner and commercial debt financiers.

3. Our Approach to the Study

3.1. Our Approach towards the study:

This Preliminary Investment Report focuses on identifying potential off-take options and financial analysis for Bhutan's Dorjilung Hydropower Plant. Our approach segments the study under six key steps outlined below –

Table 1: Study Focus Areas and Objectives

KEY FOCUS AREA	OBJECTIVES
DHPP Generation Profile and Strategic Importance	Provide detailed technical overview of DHPP's generation capabilities and articulate the strategic importance of the project for Bhutan's energy security, export earnings, and regional energy cooperation
Indian Power Market Analysis - Demand, Supply, Hydropower's Role, Power Sector Reforms and Evolving Tariff Trends	Evaluate India as the primary export market for DHPP, understanding its electricity landscape, demand-supply dynamics, and the evolving role of hydropower, key policy inititations to reform power sector and evolving tariff landscape based on upcmong trends
Offtake Options and Discom Level Assessment in India	Identify and evaluate potential offtakers in the Indian power market
Bangladesh Power Market Assessment	Evaluate Bangladesh as a potential secondary export market for DHPP, considering its energy needs, import strategies, and the evolving framework for regional power trade
Regulatory, Risk and Tariff Framework in Export Markets	Provide clear understanding of the tariff determination regulations and prevailing regulatory challenges in CBET including India and Bangladesh, which will directly impact the pricing and bankability of power exported from DHPP. Additionally, study of export risk consideration for DHPP.
Transmission Infrastructure and Regional Energy Cooperation	Assess the adequacy of existing and planned transmission infrastructure for facilitating power exports from DHPP and to highlight the broader context of regional energy cooperation in South Asia

In accordance to these key steps' objectves, our approach has been structured into two key sections:

Offtake Analysis	Financial Analysis

3.2. Detailed Approach & Methodology

Table 2: Detailed Approach & Methodology

OFFTAKE ANALYSIS	**Our Understanding on DHPP**	• Analysis of generation profile under P-50 and P-90 scenarios. • Significance of the study
	Indian Power Market Analysis	• Comprehensive analysis of the Indian power market has been undertaken to evaluate both national and regional demand-supply dynamics • Critical insights into the evolving energy landscape • Regional assessment to identify potential offtake opportunities for surplus generation with examination of monthly demand and supply patterns across key Indian states.

		Profiling of Indian Offtake Options • On basis of Gross Input Energy, the largest States from each region have been selected for detailed DISCOM-level analysis. From each selected state, DISCOMs are evaluated on Installed Capacity, Average Cost of Supply (ACS), Average Revenue realized (ARR), Gap between ACS-ARR, Payable days, profitability, Uncontracted HPO Obligations and Merit Order Analysis of Power Purchase Cost
		• Power trading through both exchange and bilateral routes • Analyze major corporate houses, identifying their net-zero targets, and evaluating their future sustainability plans in alignment with national Renewable Purchase Obligation (RPO) and Hydro Purchase Obligation (HPO) mandates
		• Key Sector Reforms in Indian Power Markets • Evolving Trends affecting Pricing In India
	Bangladesh Power Sector Overview	• Reviewed Bangladesh's power sector overview: peak demand growth installed capacity, and fuel mix • Analyzed the power import trends of Bangaldesh • Assessed role of the Bangladesh Power Development Board as single buyer and its generation/purchase cost structures. • Reviewed Bangladesh's Integrated Energy and Power Master Plan (IEPMP) 2023 for future demand and supply and need for hydropower import
	Regulation, Risk and Transmission	• Examine regulatory framework for tariff determination of hydropower generation in South Asia, with focus on CBET regulations and regulatory landscape in India and Bangladesh • Case studies and tariff comparisons of various hydropower plants • Identify challenges encountered during tariff determination • Recommend areas of optimization • Study of export risk in context of DHPP.

FINANCIAL ANALYSIS	**Macroeconomic structure**	• An analysis of principal repayments, interest outflows, and both domestic and export revenue collections indicates that the debt is self-liquidating, with total revenue sufficient to meet debt servicing obligations from the first year. • Although the Bhutan Debt Management Policy requires debt servicing to remain within 50% of total export revenue, this condition is met only in the later years, around 2045. • Despite not meeting this condition initially, the DHPP project is expected to drive broader economic transformation by supporting energy-intensive industries, enhancing energy security, and attracting investments, thereby contributing to long-term economic diversification and resilience.
	Optimal financing structure	• Study of financing structure prevailing in Bhutan for HPPs and its evolution - two phases namely Bilateral mode of funding (1988-2010) and diversification in financing structures to reduce reliance on bilateral sovereign funding (2010 and onwards). • Significant transformation in Bhutan’s hydro journey • Funding structures of other similar international hydropower projects in Asia and Africa continents

	Considerations for India/international lenders	• Broadly divided into three sub-sections namely, key appraisal requirements by lenders for financing a hydropower project, security enforcement plan and financing timeline • First sub-section - discusses as to how banks / financiers typically require a thorough and comprehensive appraisal of under-construction hydropower projects to assess their financial viability and risk profile before extending loans or investments. DHPP's preparedness towards various appraisal aspects and requirements. • Next sub-section - security enforcement by Indian and international lenders. Details on DGPC obtaining a legal opinion for Tangsibji Hydro Energy Limited (THyE) Project from Bhutan Law Services regarding enforceability and validity of the proposed security package in Bhutan.
	Project risks-benchmarking and mitigation	• Range of material risks, including Land acquisition issues, geological surprises, construction delays leading to time and cost overruns, and regulatory challenges. • Devising mitigation strategies

4. Profile of Dorjilung Hydropower Plant

Purpose of the section:

As part of this study, the generation profile of the Dorjilung Hydropower Plant will be analyzed to assess its seasonal energy contribution to Bhutan's overall generation portfolio across various probability levels. By comparing this with Bhutan's projected electricity demand through 2040, the study aims to identify potential energy surpluses and deficits. This will support strategic planning for offtake opportunities, optimize domestic demand, exports and enhance overall demand-supply management.

Key Conclusion and takeaways

The total generation at the P90 level is 4,453 MW, with approximately 85% occurring between May and November, identifying this period as the optimal export window. India and Bangladesh are recognized as key offtakers due to their proximity to the project. Based on this generation profile, determining the optimal offtake will depend on domestic demand at the time of project commissioning. Any surplus beyond domestic needs will then be evaluated for export markets.

Bhutan's electricity sector is a cornerstone of its socio-economic development, with hydropower being the dominant source. The country has an installed hydropower capacity of approximately 2,453 MW[5], the sector not only ensures reliable and affordable electricity for domestic use but also generates significant revenue through exports, primarily to India. Bhutan's hydropower potential is estimated at 37,000 MW, with 33,000 MW[6] considered technologically and economically feasible.

Bhutan's electricity demand has surged at a compound annual growth rate (CAGR) of approximately 31% from FY2021 to FY2024, driven by rapid electrification, economic growth, and the government's push to transform the country into an industrial and service powerhouse. The sales mix includes both domestic demand and electricity exports to India. Domestic demand has grown at an exceptional rate of around 38% over the past five years, with peak demand tripling from 374 MW in FY2020 to 1,026 MW in FY2024.[7] This increase is primarily due to the rising high voltage (HV) and medium voltage (MV) demand from energy-intensive industries and the expansion of the Jigmeling and Motanga industrial

[5] DGPC Website
[6]Bhutan Energy Directory, Department of Energy Ministry of Energy and Natural Resources,2022
[7] DGPC Additional Databook Excel

parks along the India-Bhutan border. Bhutan's energy demand is projected to reach 20,968 GWh, requiring approximately 7,141 MW of capacity addition by FY2030.[8]

4.1. Generation profile of Dorjilung Hydropower Plant

Bhutan is set to harness its substantial yet untapped hydropower resources with the Dorjilung Hydropower Plant, which will have an anticipated capacity of 1,125 MW, making it one of the nation's largest hydropower projects. According to the Detailed Project Report (DPR), the expected commissioning of the project will be in FY2031, with an annual design energy output of 4,504 GWh. The project is expected to enhance Bhutan's domestic power supply and boost its export potential.

At P90 probability, Total anticipated generation capacity as per DPR is 4,504 GWh out of which 85% is available for export whereas 15% would be held for domestic consumption.

Generational Profile of DHPP[9]

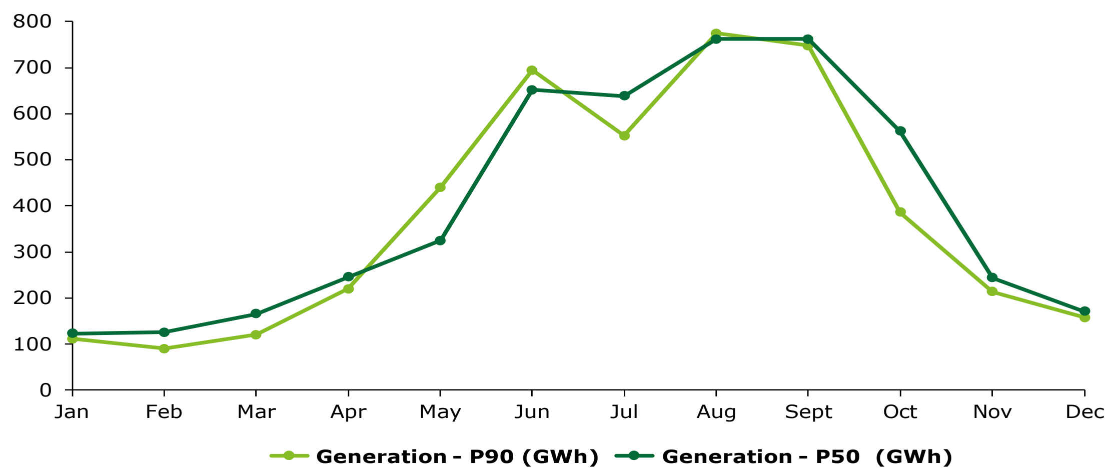

Figure 1: Generational profile of DHPP at P50 and P90 probabilities

Table 3: Month-wise Generational Profile of DHPP at P-90[10]

Generation - P90					
Month	Generation (GWh)	Aux Loss (1.12%)	Energy net (GWh)	Domestic Allocation (GWh)	Energy Available for Export (GWh)
Jan	111	1.24	110	110	-
Feb	90	1.00	89	89	-
Mar	120	1.35	119	119	-
Apr	219	2.46	217	217	-
May	440	4.93	435	-	435
Jun	695	7.78	687	-	687
Jul	552	6.18	545	-	545
Aug	774	8.67	765	-	765
Sept	747	88.37	739	-	739
Oct	386	4.32	381	-	381
Nov	214	2.39	211	-	211
Dec	158	1.76	156	156	-
Total	4504	50.45	4,453	690	3,764

[8] DGPC Energy Databook Excel
[9] DHPP Generation Data P50 and P90 Excel
[10] Source: Detailed Project Report for DHPP

At P50 probability, Total anticipated generation capacity as per DPR is 4,772 GWh out of which 83% is available for export whereas 17% would be held for domestic consumption.

Table 4: Month-wise Generational Profile of DHPP at P-50[11]

Generation – P50					
Month	Generation (GWh)	Aux Loss (1.12%)	Energy net (GWh)	Domestic Allocation (GWh)	Energy Available for Export (GWh)
Jan	123	1.38	122	122	-
Feb	125	1.40	124	124	-
Mar	165	1.85	163	163	-
Apr	245	2.75	243	243	-
May	325	3.64	321	-	321
Jun	651	7.29	644	-	644
Jul	638	7.15	631	-	631
Aug	761	8.53	753	-	753
Sept	762	8.53	753	-	753
Oct	562	6.29	556	-	556
Nov	244	2.73	241	-	241
Dec	171	1.91	169	169	-
Total	4,772	53.45	4,719	820	3,899

During the non-monsoon period (January to April), power generation capacity is expected to experience a notable decline, necessitating its allocation primarily for domestic consumption. Following April, projections indicate increase in generation capacity, which will be directed towards meeting export demands. Detailed Project Report has majorly focused on the quantum part which is available for export.

4.2. Need to Identify Offtake Markets for DHPP

Apart from DHPP, Bhutan is also expected to add significant installed capacity for electricity generation. The total generation of electricity is projected to rise from 27,371 GWh to 76,042 GWh from FY 2031 to FY 2040, growing at CAGR of 13.6%. Concurrently, the anticipated load is expected to increase from 22,061 GWh to 34,757 GWh from FY 2031 to FY 2040, growing at CAGR of 5.8%.

[11] Source: Information shared by Client (DGPC)

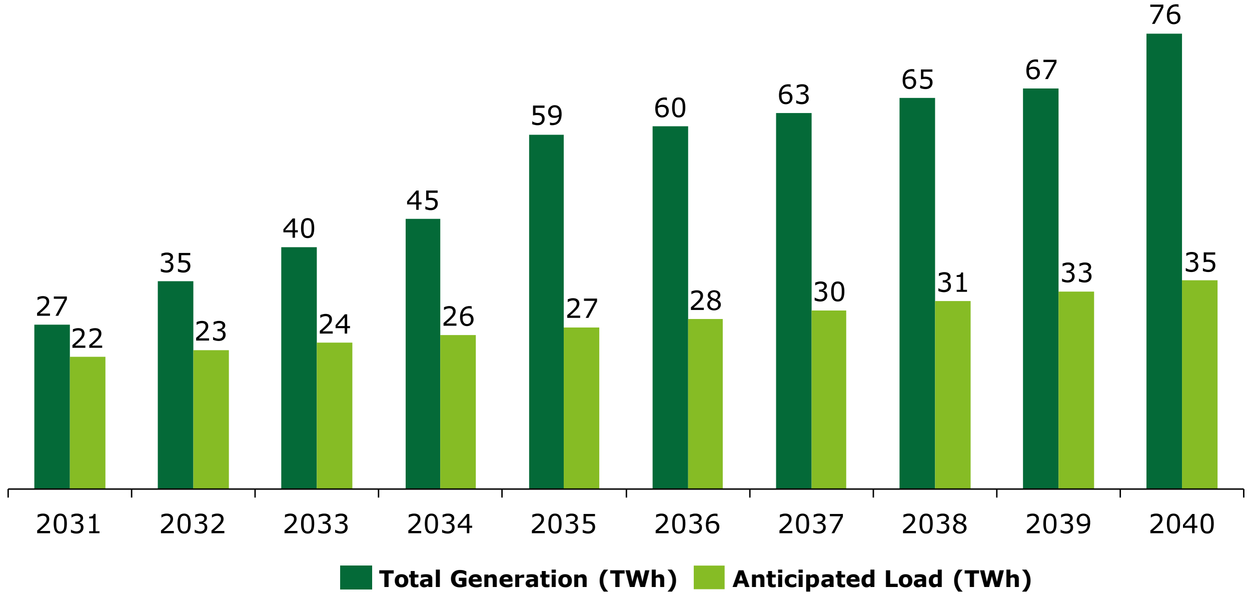

Figure 2: Projected Demand and Supply in Bhutan Power Market after commissioning of DHPP[12]

The projected surplus in generation capacity offers a valuable opportunity for regional energy trade. This excess electricity can be strategically utilized through offtake arrangements with neighboring countries, fostering regional cooperation, and contributing to Bhutan's economic development.

South Asian power sector is undergoing significant transformation due to rising energy demand. In the past two decades, energy demand in South Asia has surged by more than 50%[13], primarily driven by population growth and industrial expansion. This region heavily relies on energy imports, with approximately two-thirds of its primary energy being imported, making it vulnerable to global energy price fluctuations. To address these challenges, there is a strong push towards integrating renewable energy sources, such as hydropower, solar, and wind, to reduce dependence on fossil fuels and mitigate greenhouse gas emissions.

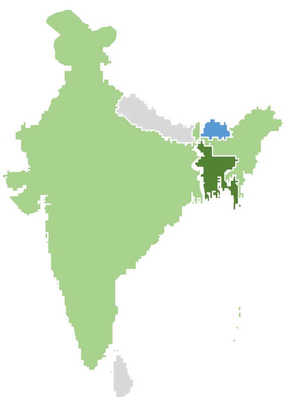

Figure 3: Map of BBINS (not to scale)

The BBIN region within South Asia comprising of Bangladesh, Bhutan, India, Nepal are already partnering within on cross border electricity trade. Most of the agreements and trade have been bilateral in nature with a few trilateral agreements emerging in recent years. Sri Lanka too may be connected to the Indian grid in a few years taking the trade potential to BBINS (with Sri Lanka). Both India and Bangladesh present opportunities for offtake in the region due to issues such as reliance on thermal fuel and shift towards variable renewable energy in India, liquid and imported fuels in Bangladesh. These are discussed in detail in subsequent sections.

India and Bhutan have a strong partnership in the hydropower sector, significantly benefiting Bhutan's economy and India's energy security. Bhutan exports 70% of its generated electricity to India, with India providing financial and technical assistance for various projects. In 2006, both countries signed a framework agreement to develop and trade hydropower, committing India to import at least 5,000 MW by 2020, later revised to 10,000 MW in 2009. India, as the world's third-largest power consumer, benefits significantly from Bhutan's surplus electricity. A joint initiative between Government of India

[12] DGPC Energy Data
[13] An integrated electricity market in South Asia is key to energy security, World Bank Blogs

and State's "Power for All"[14] has accelerated capacity expansion, with Bhutan playing a crucial role as both a buyer and seller of power. Evaluating the DISCOMs in the Indian power market is essential for assessing offtake options for DHPP in Indian power market.

Similar to India, Bangladesh is also emerging as a key market for cross border power trading in the region. It's installed power capacity stands at 27,424 MW[15], with only 2.78% from renewables, mainly off-grid solar. The country's heavy reliance on depleting natural gas reserves and rising energy demand underscore the need for diversification. Bangladesh is already engaged in cross-border electricity trade, importing ~2.5 GW[16] (primarily from India and marginal volume from Nepal). Given its rising electricity demand and commitment to clean energy, Bangladesh presents a viable offtake market for Bhutanese hydropower.

The consistent growth in both electricity generation and domestic load in Bhutan highlights the country's strong capacity to efficiently meet its projected energy demands. This dependable supply, combined with a significant surplus in generation capacity, presents a strategic opportunity to expand electricity exports. Leveraging this surplus through well-structured offtake arrangements with neighboring countries can maximize Bhutan's export potential while contributing meaningfully to regional energy security.

[14] Power for All, Ministry of Power
[15] Bangladesh Power Development Board
[16] Annual Report, Bangladesh Power Development Board

5. Bhutan Power Market Overview

Purpose of the Section
This section presents an understanding of key trends in Bhutan power market, focusing on long-term demand and supply projections, capacity addition plans and overview of performance of Bhutan Power Corporation Ltd (BPC). This section evaluates how the generational profile of the Dorjuling Hydropower Project (DHPP) will contribute to meeting Bhutan's projected electricity demand, it is essential to undertake a comprehensive analysis of the country's current and forecasted demand and supply dynamics. This assessment provides the necessary framework to determine the extent to which DHPP's generation capacity can be absorbed into the national grid and effectively support future energy requirements.
Key Conclusion and takeaways
Bhutan's peak electricity demand rose from 374 MW in 2020 to 1,026 MW in 2024 and is projected to reach 1,662 MW by 2026. Domestic electricity demand grew at a CAGR of 23%, outpacing supply growth at 7.4%, leading to reduced export capacity. However, Bhutan has an ambitious capacity expansion plan dominated by hydropower, complemented by solar projects, aiming to reach 78,150 GWh in supply by 2040. Bhutan Power Corporation Limited (BPCL) plays a critical role in managing distribution, infrastructure, and cross-border trade to support the country's growing energy needs and export ambitions.

5.1. Electricity Landscape in Bhutan

Bhutan is emerging as a key regional energy player leveraging its annual power surplus and strategic hydropower assets. It has a longstanding history of electricity trade with India, supported by robust infrastructure and enabling policies. The country's peak electricity demand has surged from 374 MW in 2020 to 1,026 MW in 2024[17] and is expected to reach 1,662 MW by 2026[18].

From FY2016 to FY 2025, electricity domestic demand in Bhutan has grown at a CAGR of 23%, i.e., from 2,009 GWh to 10,589 GWh. In the same period electricity supply in Bhutan grew at a CAGR of 7.4%, i.e., from ~7,490 GW to ~13,510 GWh. It is notable that export supply is currently in a declining phase, primarily driven by a surge in domestic energy demand. However the country has significant expansion plan which is expected to make the country a major export player in the region.

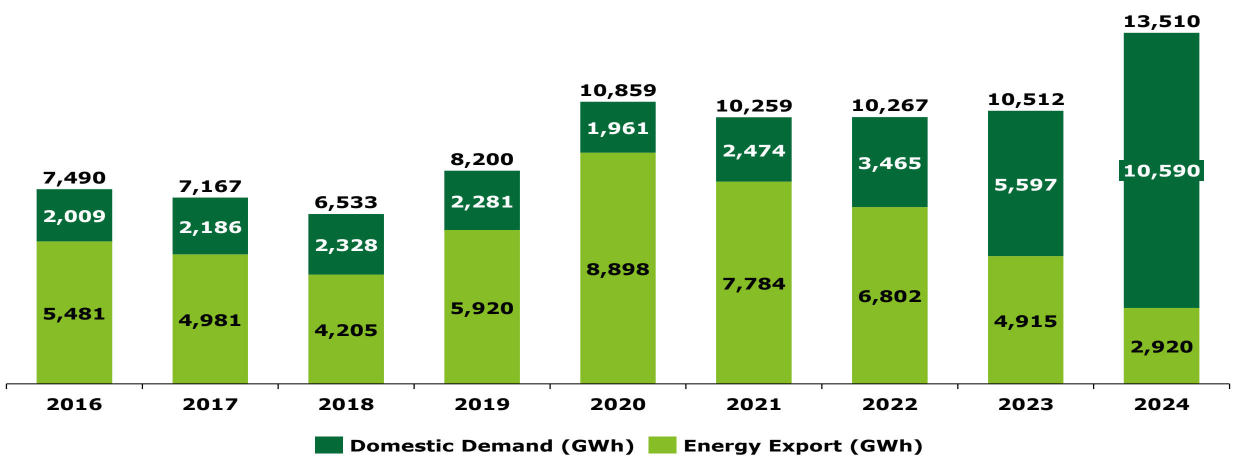

Figure 4: Allocation of Bhuatn's Power Generation

[17] DGPC Additional Data Workbook
[18] Distributed Solar for Public Infrastructure Project, ADB

5.2.Electricity Outlook in Bhutan

5.2.1 Demand Projections

The electricity demand in Bhutan is projected to grow at a lower pace than last few years as the demand matures. Under the base demand scenario, electricity demand is expected to increase from 12,341 GWh in 2025 to approximately 14,836 GWh by 2040. This represents a CAGR of around 1.23%, indicating moderate but consistent growth [19].

The forecast incorporates two sensitivity scenarios to reflect potential variations in Bhutan's future domestic electricity demand, drawing from historical growth trends. Scenario 1 assumes a conservative compound annual growth rate (CAGR) of 2.5%, while Scenario 2 reflects a more aggressive growth outlook with a 4.5% CAGR. These scenarios help account for sensitivities related to economic development, industrial expansion, population growth, and electrification efforts. Based on these projections, Bhutan's domestic electricity demand is expected to range between 14,836 GWh and 23,884 GWh by the forecast horizon. This wide range highlights the critical importance of adopting a flexible and scalable approach to energy planning, ensuring the system is resilient and responsive to varying growth trajectories and potential shifts in consumption patterns.[20].

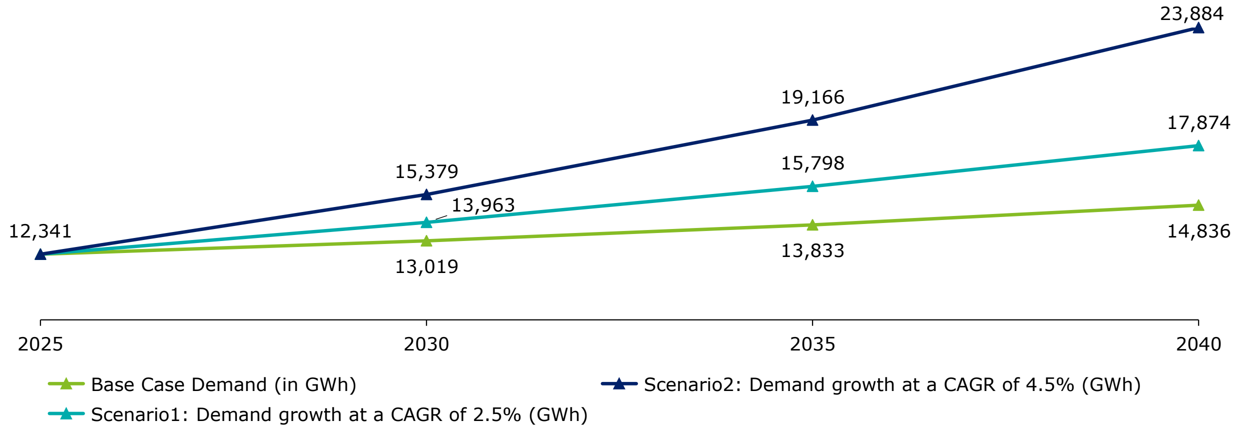

Figure 5: Demand ForecastP

5.2.2 Planned Capacity Addition

Bhutan's electricity capacity addition plan from FY 25 to FY 34 reflects an ambitious and well-structured strategy to meet its rapidly growing energy demand and establish itself as a major power exporter. This plan is dominated by hydropower projects, with a few solar initiatives integrated to diversify the energy mix. Starting from FY 25, major projects like Puna-II (1,020 MW), Kholongchhu (600 MW) in FY 29 and DHPP (1,125 MW) are expected to significantly boost generation capacity. Solar capacity is planned to be gradually introduced through rooftop and larger grid connected solar plants. By FY 34, large-scale storage-based hydropower projects such as Wangchhu and Gongri further enhance reliability and seasonal stability. This phased and geographically distributed rollout ensures Bhutan not only meets domestic needs but also strengthens its position as a sustainable energy exporter in the region.

[19] DGPC- Additional energy databook
[20] DGPC- Additional energy databook

Table 5: Plant Capacity Addition Plans[21]

SN	Projects	Plant Type	Capacity (MW)	Firm Power (MW)[22]	Energy (MU)	Expected COD
1	Punatsangchhu-II	Hydro	1,020	164	4,575	2024
2	Suchhu	Hydro	18	3	77	2024
3	Burgangchhu	Hydro	54	14	261	2025
4	Yungichhu	Hydro	32	8	158	2025
5	Druk Bindu-I&II	Hydro	26	4	110	2026
6	Begana	Hydro	25	2	104	2027
7	Gamri-I	Hydro	54	13	228	2027
8	Jomori	Hydro	90	15	367	2027
9	Punatsangchhu-I	Hydro	1,200	199	5,429	2028
10	Kholongchhu	Hydro	600	114	2,569	2029
11	Gamri-II	Hydro	85	14	303	2030
12	Yurmochhu	Hydro	24	6	103	2030
13	Parochhu	Hydro	33	6	124	2030
14	Dorjilung	Hydro	1,125	168	4,558	2031
15	Nyera Amari-I&II	Hydro	404	94	1,598	2031
16	Sherichhu	Hydro	53	13	230	2031
17	Jigmechhu	Hydro	64	16	261	2031
18	Chamkharchhu-I	Hydro	770	137	3,344	2032
19	Bunakha	Hydro	180	59	719	2032
20	Dangchhu	Hydro	170	28	685	2032
21	Khomachhu	Hydro	363	65	1,415	2033
22	Jongthang	Hydro	237	53	994	2033
23	Wangchhu	Hydro	900	150	2,304	2034
24	Gongri	Hydro	740	130	2,721	2034
25	Gongri-Jerichhu PSP	Hydro	1,800	450	3,942	2034
26	Sunkosh	Hydro	4,060	684	7,314	2034
27	Dorokha	Hydro	550	93	1,917	2035
28	Chamkharchhu-II	Hydro	590	95	2,448	2037
29	Chamkharchhu-IV	Hydro	364	59	1,501	2038
30	Kuri-Gongri	Hydro	2,800	504	9,012	2039
31	Sephu solar	Solar	23	6	NA	2025
32	RTS Sub Project–I,II&III	Solar	50	13	NA	2024-27
33	Jamjee	Solar	120	30	NA	2024-25
34	Wobthang	Solar	108	27	NA	2024-25
35	Apai Amai Pang	Solar	150	38	NA	2026-27
36	Gogana	Solar	75	19	NA	2026-27

[21] DGPC- Supply Demand Forecast
[22] Firm power capacity refers to the guaranteed availability of power generation or power-producing capacity at all times, even under adverse conditions

SN	Projects	Plant Type	Capacity (MW)	Firm Power (MW)[22]	Energy (MU)	Expected COD
37	Solar Phase-II	Solar	500	125	NA	2027-29
38	Solar Phase-III	Solar	1,500	375	NA	2030-34
39	Solar Phase-IV	Solar	2,500	625	NA	2035-40

5.2.3. Supply Forecast

Bhutan's electricity supply is expected to more than quintuple, reaching approximately 78,150 GWh by 2040. This represents an average annual growth rate of around 11%, indicating a robust expansion trajectory. Notable increases are observed in 2029 and 2030, coinciding with the anticipated commissioning of major hydropower projects.

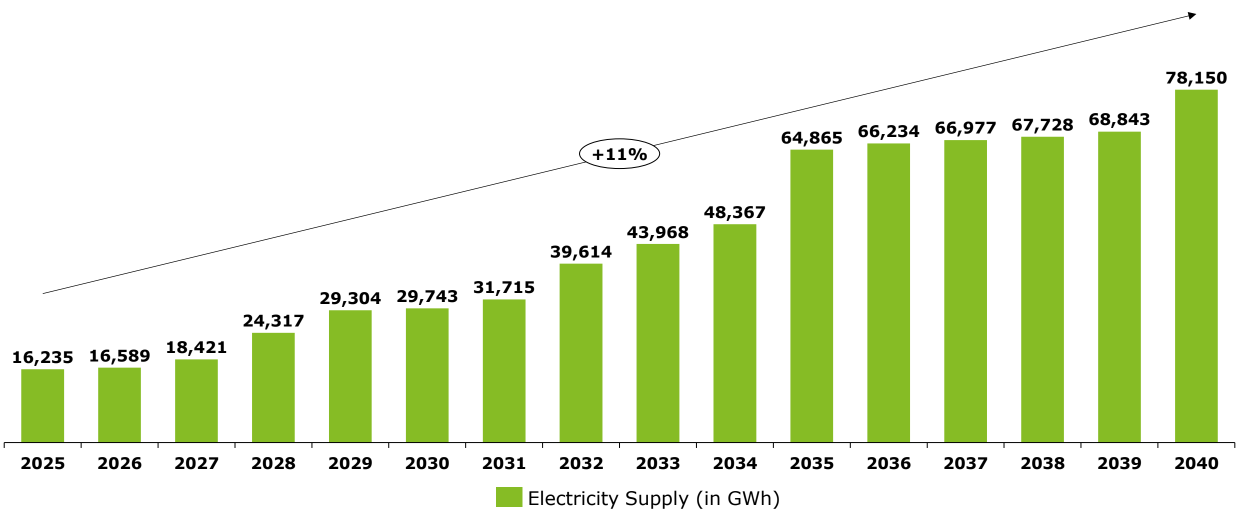

Figure 6: Electricity Supply Forecast (in GWh)

5.3. Electricity Demand Supply Outlook of Bhutan

5.3.1 Base Case Scenario

In the base case scenario, energy demand in Bhutan is projected to grow moderately at an annual rate of 1% to 1.5%. In contrast, planned capacity additions are much more ambitious, with a compound annual growth rate (CAGR) of approximately 11%. This significant difference between demand growth and capacity expansion will result in a substantial power surplus. Specifically, Bhutan is expected to have around 8,929 MW of excess generation capacity after fully meeting its domestic electricity needs.

This surplus presents a valuable opportunity for Bhutan to supply power to export markets, potentially enhancing its energy trade prospects and contributing to regional energy security.

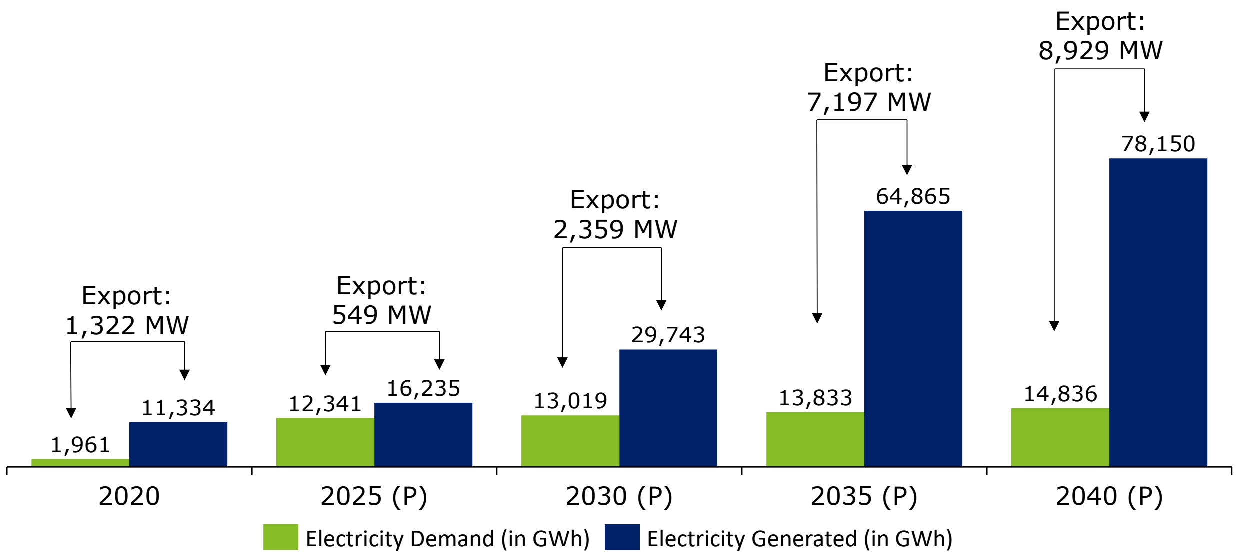

Figure 7:Demand Supply Outlook of Bhutan under Base Case Scenario[23]

5.3.2 Scenario 1: Demand Growth by 2.5%

Under Scenario 1, Bhutan's energy demand is projected to grow moderately at an annual rate of 2.5%, reflecting the significantly high historical growth observed over the past five years. In contrast, planned capacity additions remain consistent with the base case scenario. Despite the higher demand growth, Bhutan's supply is still expected to comfortably meet domestic needs, leaving an estimated surplus of approximately 8,501 MW available for export.

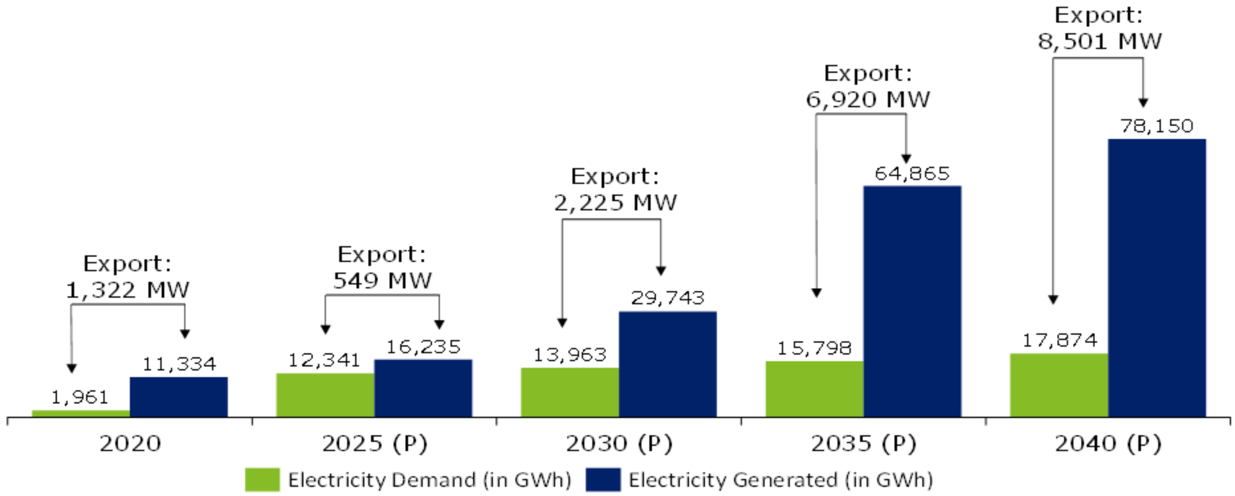

Figure 8: Demand Supply Outlook of Bhutan under Scenario 1

5.3.2 Scenario 2: Demand Growth by 4.5%

[23] Export was provided in MU, we have converted it into MW by taking PLF of 81% from past data of DGPC Demand and Supply Forecaast excel

Under Scenario 2, Bhutan's energy demand is projected to grow moderately at an annual rate of 4.5%, reflecting the significantly high historical growth observed over the past five years. In contrast, planned capacity additions remain consistent with the base case scenario. Despite the higher demand growth, Bhutan's supply is still expected to comfortably meet domestic needs, leaving an estimated surplus of approximately 6,336 MW available for export.

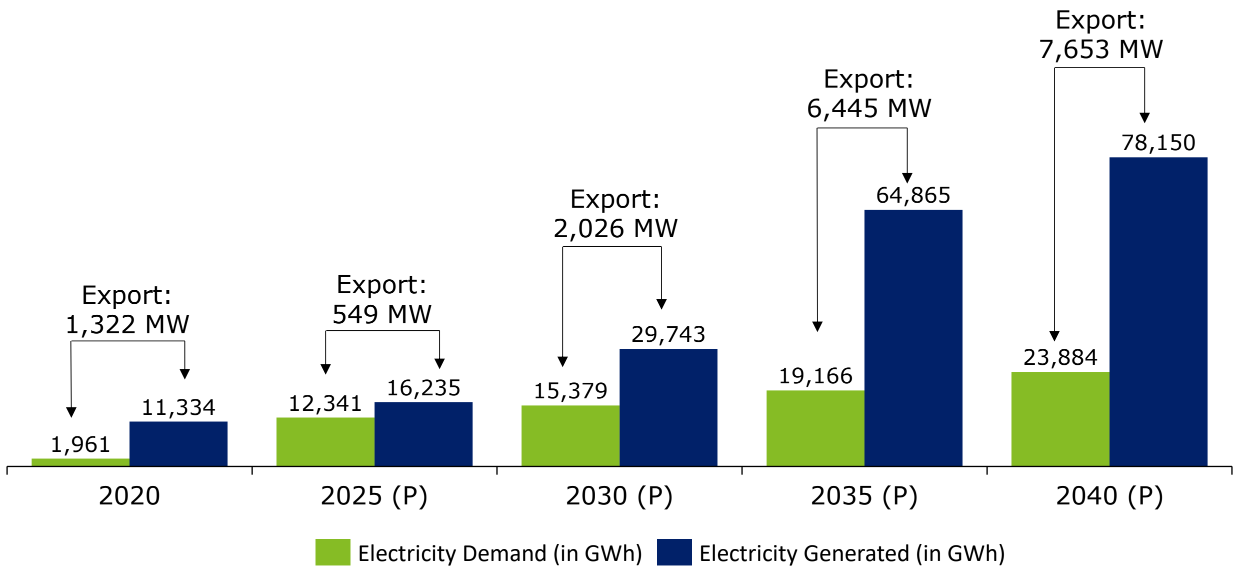

Figure 9: Demand Supply Outlook of Bhutan under Scenario 2

The commissioning of major hydropower projects over the next decade will significantly enhance generation capacity, but the success of these developments depends on Butwal Power Corporation Limited's (BPCL) ability to strengthen transmission and distribution infrastructure, manage seasonal variability, and support cross-border electricity trade. For overview of BPCL, refer Annexure A attached to report.

6. Indian Power Market Overview

Purpose of the Section
This section assesses the offtake market potential for surplus generation beyond BPCL's requirements. It evaluates India's electricity outlook through FY 2040, examining key demand and supply drivers. The analysis also explores regional variations in energy and peak deficits to capture local characteristics, enabling a more targeted assessment of offtake opportunities. Additionally, recent power sector reforms and evolving fuel-wise tariff trends provide a comprehensive perspective on the Indian power sector.
Key Conclusion and takeaways
The Indian power market is projected to grow steadily at around 4% annually through FY 2040, driven by urbanization, economic growth, and electrification. Electricity generation reached 475 GW in FY 2024-25, with approximately 9.9 GW of hydroelectric and pumped storage projects currently under construction. Hydro capacity is expected to increase from 47 GW in FY 2024-25 to between 54 and 62 GW by 2029-30. Regionally, the Northern region faces the highest energy deficits, while the Western region experiences the largest peak demand shortfalls. Consequently, cross-border power trade is anticipated to play a crucial role in reducing fossil fuel dependency, enhancing grid flexibility, and meeting Renewable Purchase Obligation (RPO) and Hydro Purchase Obligation (HPO) targets. Additionally, India has implemented several power sector reforms, including policies to promote renewables such as strengthened RPO obligations and RA adequacy framework. Measures to reduce dues, such as the PRAAPTI portal and Late Payment Surcharge (LPS) schemes, have also been introduced. Considering these reforms and the overall energy demand scenario, tariff projections have been made based on upcoming trends, including additions of thermal projects, delays in hydro capacity installations, and the integration of storage projects. These reforms are also helping to strengthen the cross border power market (including for Bhutanese hydropower) by opening up market aveneus and improving transactional frameworks for stakeholders. Tariff trends indicate that conventional fuel-based power prices have grown at a CAGR of 4–5% annually, while solar power prices have declined, with wind tariffs remaining relatively stable. Hydro tariffs have increased by about 3%, with new projects priced between INR 4 and 5 per unit. Energy storage costs have fallen significantly, with recent bids revealing tariffs as low as INR 3.33 per unit. These declining storage prices are expected to influence the short-term power market, particularly by affecting evening peak prices as surplus renewable energy generated during solar hours is redirected. The Indian power market presents an attractive opportunity for Bhutanese hydropower, provided it aligns with these emerging tariff trends.

6.1. Electricity Landscape in India

India is the largest electricity market in South Asia, and a potential market for power export from the Dorjilung hydropower project. The country has a deep history of cross border electricity trade with Bhutan, strong existing cross border transmission interconnections, and favourable policy and regulatory frameworks enabling such power trade.

In this context, this study has been conducted to understand the existing and future electricity demand, including seasonal trends in India under a range of scenarios, and identification of the likely mix of generation (capacity and energy) required to meet the forecasted electricity demand in the country.

One of the major drivers of growth in electricity demand is economic growth; and in this regard, India's GDP has grown by 8.2% in FY 2023-24[24] while same was 7.2% in FY 2022-23. The Government of India (GoI) has also set a target of becoming a $5 trillion economy by 2024-25.[25]

[24] Source: Press Information Bureau, GoI
[25] Source: Press Information Bureau, GoI

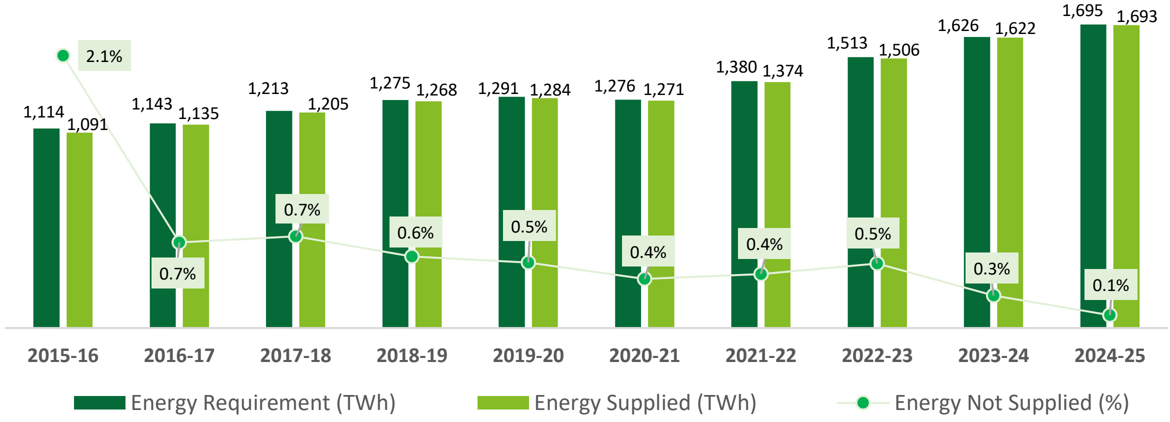

Figure 10: India Electricity Demand and Supply

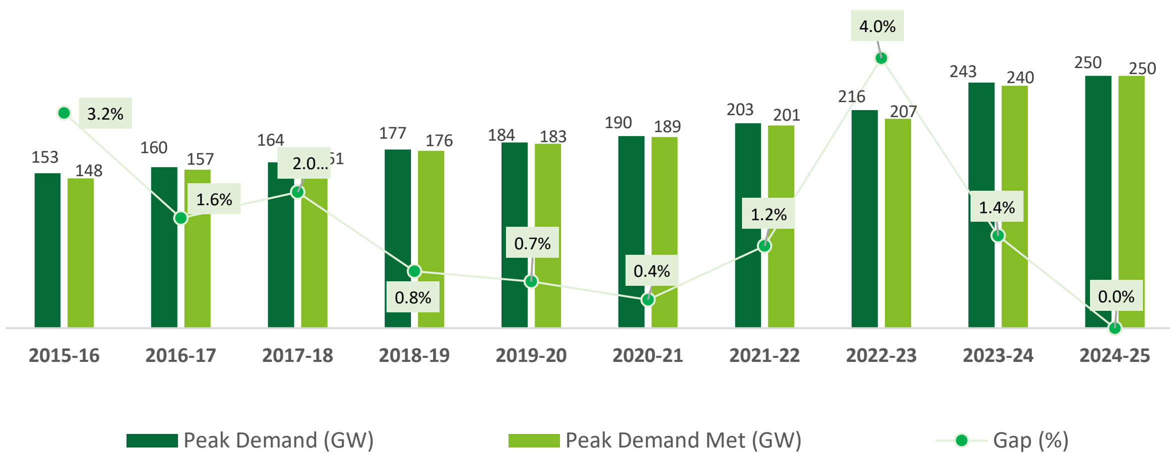

Figure 11: India Electricity Peak Demand and Peak Supply

In the past nine years, from FY2016 to FY 2025, electricity demand in India has grown at a compound average growth rate of 4.77%, i.e., from 1,114 TWh to 1,695 TWh, while at the same time, peak electricity demand in India grew at a compound average growth rate of 5.7%, i.e., from ~148 GW in FY 2015-16 to ~249 GW in FY 2024-25[26].

6.1.1. Electricity Demand Forecast

The Central Electricity Authority (CEA) publishes periodic forecasts for demand in the country. The energy and peak demand are expected to increase at a compound rate of 4%-5% per annum through to 2042 from the current level of around 1,695 TWh and 249 GW as on 31st March 2025.

Table 6: India: Electricity demand forecast*

As per CEA 20th EPS Vol 1	Scenarios	2021-22 (Base)	2023-24 (P)	2026-27 (P)	2031-32 (P)	2036-37 (P)	2041-42 (P)
	Moderate Scenario	1,382	1,552	1,849	2,297	2,796	3,343

[26] Source: Monthly Power Supply Reports, CEA

As per CEA 20th EPS Vol 1	Scenarios	2021-22 (Base)	2023-24 (P)	2026-27 (P)	2031-32 (P)	2036-37 (P)	2041-42 (P)
Energy Requirement (TWh)	Optimistic Scenario		1,572	1,908	2,474	3,095	3,776
Peak Demand (GW)	Moderate Scenario	203	226	264	329	405	488
	Optimistic Scenario		230	277	366	466	575
Base Corrected Scenario	**Scenarios**	**2021-22 (A)**	**2024-25(A)****	**2026-27 (P)**	**2031-32 (P)**	**2036-37 (P)**	**2041-42 (P)**
Energy Requirement (TWh)	Moderate Scenario	1,382	1,695	1,905	2,362	2,874	3,430
	Optimistic Scenario			1,930	2,498	3,128	3,824
Peak Demand (GW)	Moderate Scenario	203	249	277	343	423	510
	Optimistic Scenario			282	372	472	583

*Demand is Ex-Bus; Base corrected (P): CAGR is as per considerations in CEA 20th EPS report;**As per LGBR 2025-26 Report

The key demand drivers which are envisaged to influence the overall electricity demand in India are mentioned below:

Figure 12: Future Electricity Demand Drivers in India

6.1.2. Electricity Supply Forecast

The overall capacity addition in India stands at 475 GW, out of which thermal energy continues to play a significant role in India's electricity generation, having accounted for 51%[27] of the total electricity generation in FY 2024-25 (refer figure below). The Government of India (GoI) has set an ambitious plan of 500 GW renewable energy installed capacity by 2030 and in this regard has already achieved ~220 GW as on May 2025[28] (~44% of 2030 target).

[27] Source: National Power Portal, May 2025
[28] Source: National Power Portal, May 2025

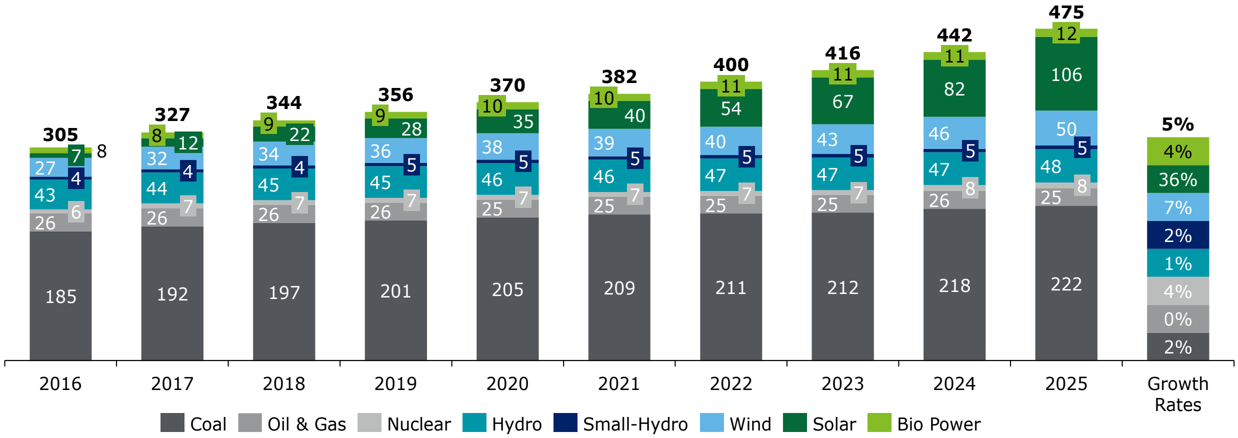

Figure 13: Source wise Installed Capacity (in GW) in India[29]

Electricity supply forecast by CEA and some other industry reports were reviewed. Key observations are as follows[30,31]:

India will experience steady growth in the electricity demand in the years up to 2050.

The share of coal based thermal power generation capacity in the year 2030 would be around 32-34% in the energy mix as against the present share of ~51%.

The share of coal-based power generation capacity is further expected to reduce to ~13% of total installed capacity by 2050 under Economic Transition Scenario (ETS) wherein cost-based technology changes taken in consideration.

RE penetration (particularly solar and wind) will accelerate because of aggressive policies and targets, coupled with lower capital costs and easier access to finance. The capacity share of renewables will grow from the current level of 44% to ~51-53% by 2030 while wind and solar alone would account for ~57% in 2050 under ETS scenario.

Hydropower development has slowed down in the last decade due to combination of technical and safeguard challenges. This is also reflected in CEA projections, as only ~7-15 GW of Hydro capacity addition is projected by 2029-30 from current Hydro installed capacity of ~47 GW as of FY 2024-25.

Cross-border power trade will become increasingly important to meet hydropower capacity targets, to underpin decarbonisation efforts and to provide balancing power requirements for integrating RE into the grid.

Figure 14: India Electricity Sector Forecast Observations

The evolving energy landscape in India highlights the significant shift towards renewable energy sources, driven by ambitious targets and supportive policies. This transition not only promises a more sustainable future but also underscores the importance of strategic planning and international cooperation in achieving energy security and environmental goals.

6.1.3. Role of Hydropower

India's hydropower sector has seen significant decline in recent decades: the share of hydropower in the total capacity has reduced from 50% in the 1960s to around 10% as of May 2025. This decline can be attributed largely to the relatively high development risk associated with hydropower, coupled with the proliferation of domestic coal resources and more recent shift towards cheaper solar power.

The continued push for adding renewable sources in the energy mix is creating infrastructural challenges for maintaining the national grid as the variable nature of renewable energy sources are inherently

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[29] Source: India Climate & Energy Dashboard
[30] Source: National Power Portal, May 2025
[31] Source: New Energy Outlook, Bloomberg, 2023

accompanied by intermittency in generation. This necessitates the development of storage solutions, which can rapidly respond to changes in variable renewable energy (VRE) output, thereby stabilizing the grid. Storage hydro plants are expected to play a pivotal role in grid balancing requirements, and also in providing ancillary services such as black start, voltage support, over-frequency reserve, instantaneous reserve and frequency keeping.

Lastly, hydropower is expected to resume its role as base load plant due to the obligated reduction in coal plants to meet GoI's net zero targets. Hydroelectric power projects and Pumped Storage Projects with an aggregate capacity of ~9.9 GW are under construction in the country as on April 2025[32]. The hydro capacity is projected to increase from 47 GW in FY 2024-25 to ~54-62 GW by 2029-30. Currently, PSPs with aggregate capacity of ~4 GW are under construction in the county while CEA in its revised report on optimal generation mix in 2030, has projected installed capacity of around ~19-25 GW of PSP by 2029-30 under different scenarios.

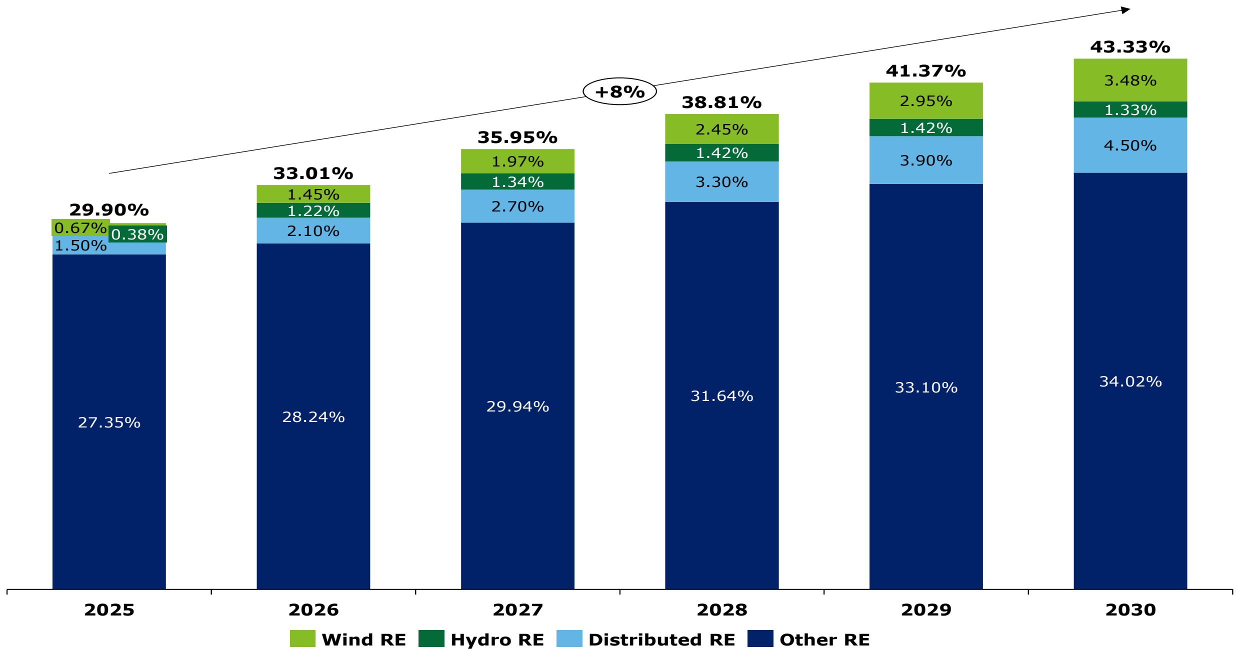


Figure 15: India's RPO trajectory till 2030

The Government of India, through recent policy updates and the revised Renewable Purchase Obligation (RPO) trajectory notified by the Ministry of Power, has broadened the definition of renewable energy to include large hydropower projects (LHPs) commissioned after March 8, 2019. As part of the new structure, the Hydro Purchase Obligation (HPO) has been created as a dedicated sub-category within the overall RPO targets, enabling States to meet their renewable energy obligations through procurement of hydropower. As a result of this policy guidance, distribution utilities are mandated to purchase a certain proportion of their electricity requirements from hydropower plants. The RPO trajectory[33] as notified by GoI is provided in the figure below:-

Several States have been assigned significant RPO targets for the year 2031–32, reflecting their contribution towards India's broader clean energy goals:

[32] Source: Hydroelectric Projects under Construction, CEA, April 2025
[33] Source: Ministry of Power, GoI, 2023

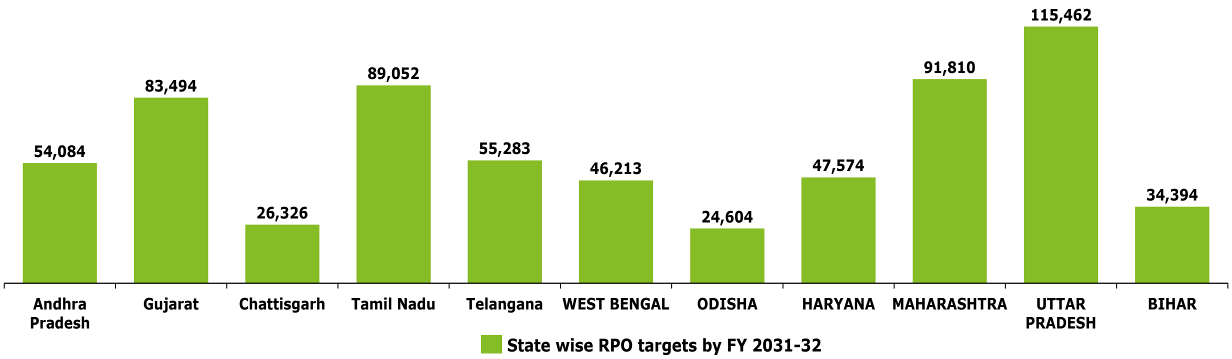


Figure 16: State wise RPO targets (in GWh) by FY 2031-32[34]

Cross-border power trade is therefore expected to play a key role in reducing the dependency on fossil fuels, providing flexibility to the grid and meeting RPO/HPO obligation. India will likely increasingly leverage the benefit of hydropower (and particularly storage hydro) from neighbouring nations such as Bhutan and Nepal for peaking and balancing services.

In this context, it is important to further deep dive and study the offtake options to further align the generation profile of DHPP with electricity landscape of different regions in India.

6.2. Region wise Electricity Demand and Supply

As India is predominantly a sub-tropical country, cooling loads during the period May-October dominate the heating loads during the period November – February. Demand profiles also vary regionally. The region-wise electricity demand projections for the period 2016-17 to 2031–32 is presented in the table below:

Table 7: Region wise projection for Demand and Supply[35]

Region	Energy Requirement (in TWh)				Peak Demand (in GW)			
	2016-17 (A)	2021-22 (A)	2026-27 (P)	2031-32 (P)	2016-17 (A)	2021-22 (A)	2026-27 (P)	2031-32 (P)
Northern Region	356	418	592	773	55	73	97	127
Western Region	352	428	596	763	50	65	89	114
Southern Region	307	351	460	596	44	61	80	107
Eastern Region	128	164	232	308	20	26	37	50
North-Eastern Region	15	18	24	32	2	3	4	6
All India	**1160**	**1382**	**1907**	**2473**	**161**	**203**	**277**	**366**

[34] Resource RPO trajectory based on forecasted demand as per Resource Adequacy Study of different States. Some states are yet to adopt the RPO set by Ministry of Power, India but are expected to align with the national trajectory in the long run
[35] CAGR is as per considerations in CEA 20th EPS report

6.2.1.1. Northern Region Electricity Demand and Supply

In the Northern Region, Chandigarh, Delhi, Haryana and Punjab had closely met the Energy Requirement in full. Himachal Pradesh, UT of J&K and Ladakh, Rajasthan, Uttar Pradesh and Uttarakhand experienced marginal gap between Energy Requirement and Energy Supplied in the range of 0.2% to 0.4%. However, the gap between the Peak Demand and Peak Met is 0.9%. The month wise regional profile is provided below:-

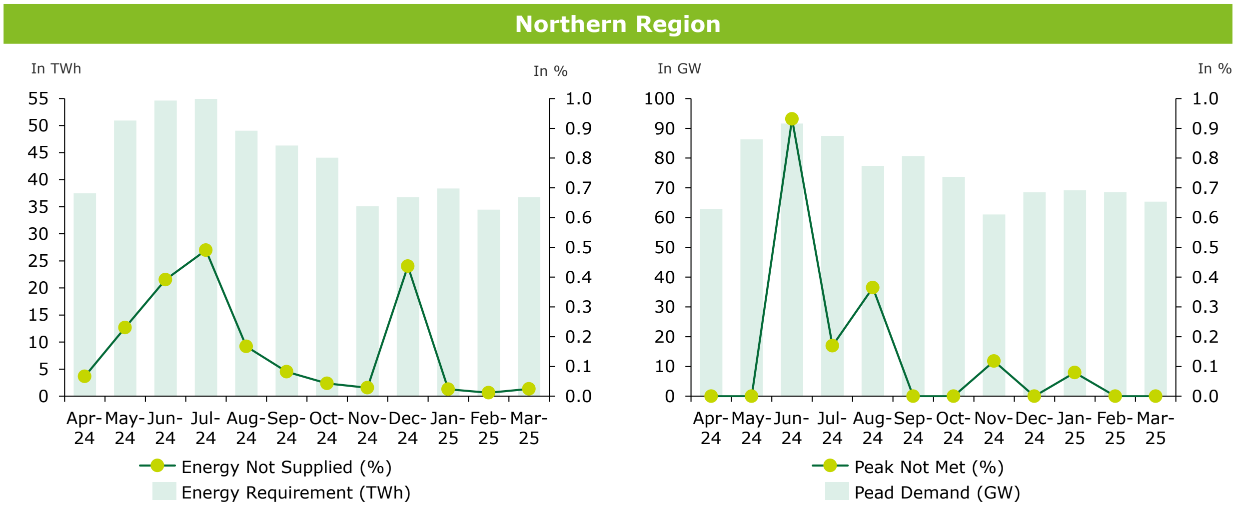


Figure 17: Northern Region monthly energy profile of FY 2024-25

- In the Northern Region, of the total 952 GWh energy deficit, approximately 75% occurs between May and September majorly due to hot summer months.
- The peak deficit in FY 2024 occurred between June and August, with the highest deficit of 845 MW recorded in June. This period aligns with the maximum generation phase for hydro projects, making it an ideal time to leverage hydroelectric power.

6.2.1.2. Southern Region Electricity Demand and Supply

In Southern Region, all the States and UTs i.e. Andhra Pradesh, Karnataka Kerala, Tamil Nadu, Puducherry, and Telangana closely met the Energy Requirement. However, Southern region has not faced a gap between the Peak Demand and Peak Met. The month wise regional profile is provided below:-

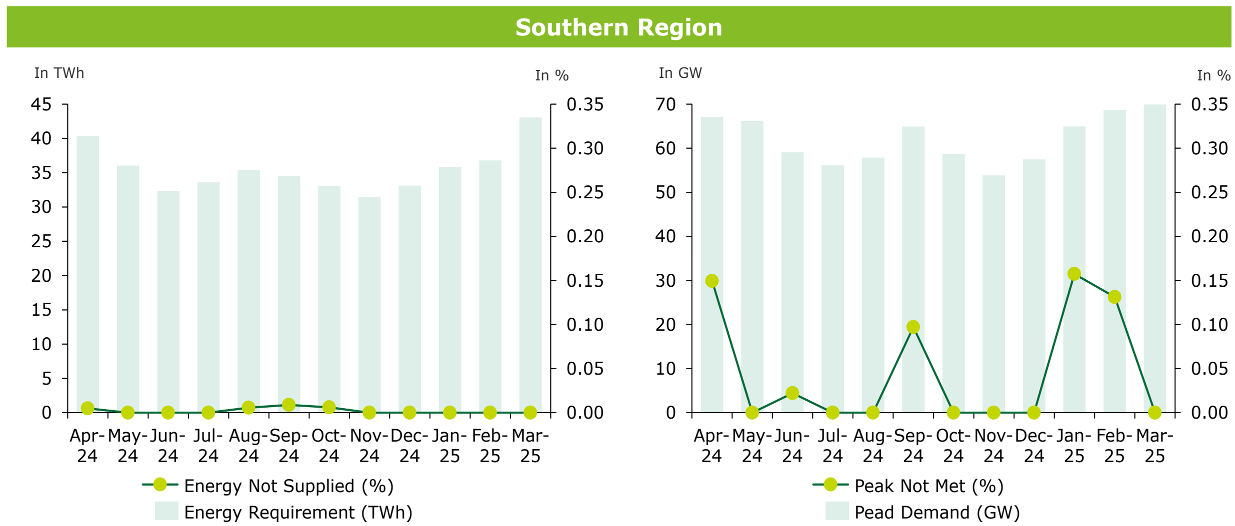


Figure 18: Southern Region monthly energy profile of FY 2024-25

- In FY 2024, the Southern Region experienced negligible energy deficits across most months. In FY 2024, the Southern Region experienced negligible energy deficits primarily due to a strong focus on renewable energy and a well-managed power supply system across most months. However, peak deficits of up to 100 MW were observed in January, February, and April.

6.2.1.3. Western Region Electricity Demand and Supply

In the Western Region, Gujarat, Maharashtra, DDD

NH and Goa had met the Energy Requirement in full while Madhya Pradesh and Chhattisgarh had experienced a marginal gap of 0.1% between Energy Requirement and Energy Supplied. However, Western had faced a gap of 0.4% between the Peak Demand and Peak Met.

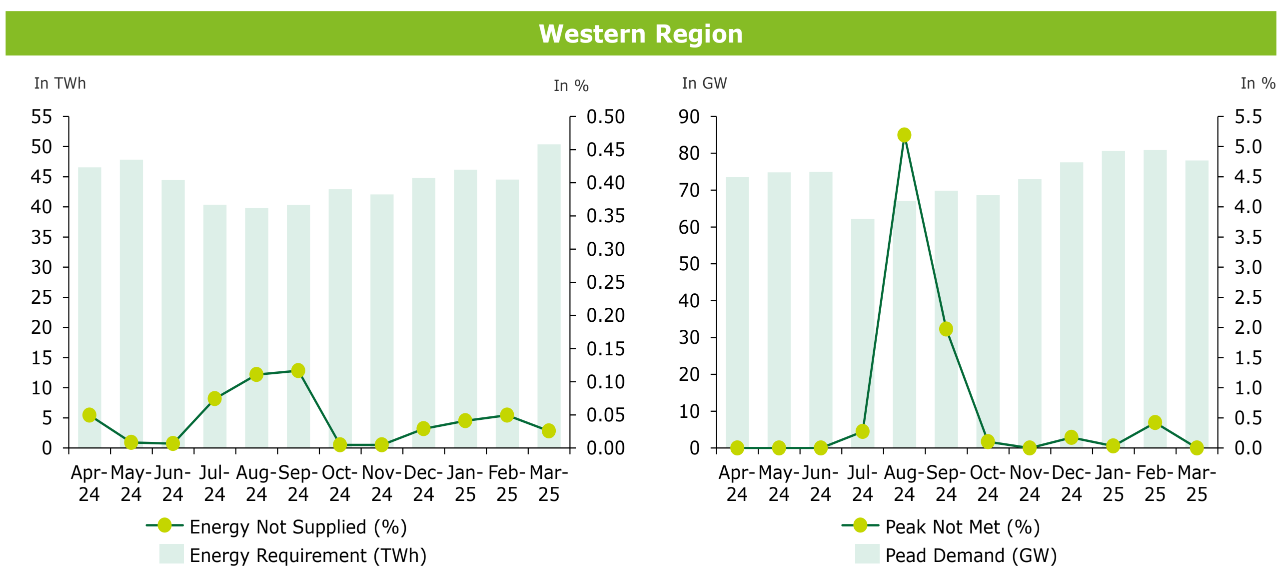


Figure 19: Western Region monthly energy profile of FY 2024-25

- In the Western Region, of the total 223 GWh energy deficit, approximately 55% occurs between July and September.
- The peak deficit is the highest among all other regions, standing at 3,306 MW in August 2024 and 1,350 MW in September. This period aligns with the maximum generation phase for hydro projects, making it an ideal time to leverage hydroelectric power.

6.2.1.4. Eastern Region wise Electricity Peak Demand and Peak Supply

In the Eastern Region, Sikkim and DVC met the Energy Requirement almost in full. West Bengal, Bihar, Jharkhand and Odisha experienced marginal Energy deficits in the range of 0.1% to 0.5%. However, Eastern had faced a gap of 0.3% between the Peak Demand and Peak Met.

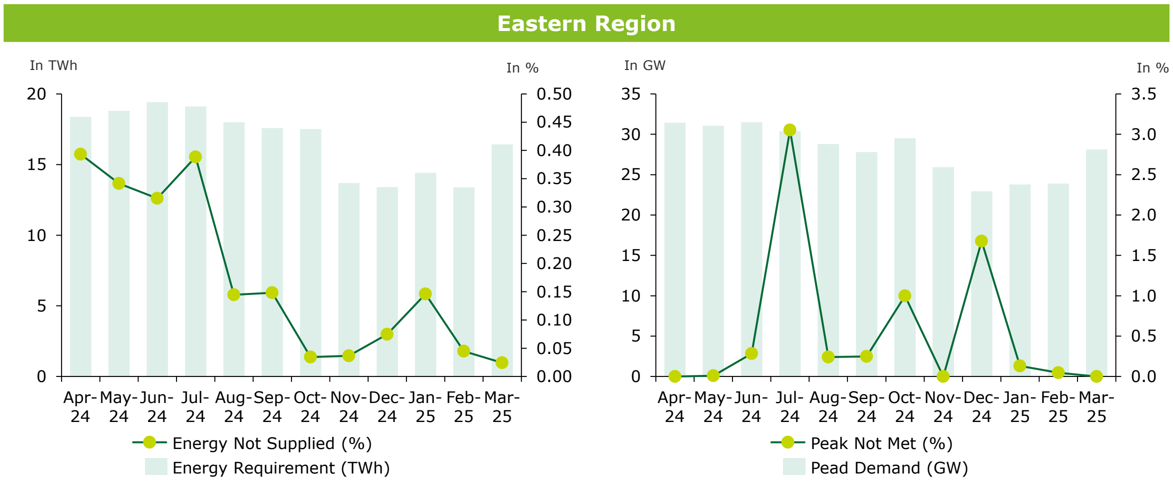


Figure 20: Eastern Region monthly energy profile of FY 2024-25

- In FY 2024, the Eastern Region experienced minimal energy deficits throughout most months. However, peak demand typically occurs between April and July due to hot, humid summer months, with the deficit reaching a peak of 900 MW in July. Consequently, this aligns with the generation profile of hydroelectric plants during these months.

6.2.1.5. North-Eastern Region wise Electricity Peak Demand and Peak Supply

In the North-Eastern Region, Arunachal Pradesh, Assam, Meghalaya, Mizoram, Nagaland, and Tripura completely met the Energy Requirement. However, Manipur experienced the energy deficit of 1.0%. North-Eastern Regions had faced a gap of 1.1% respectively between the Peak Demand and Peak Met.

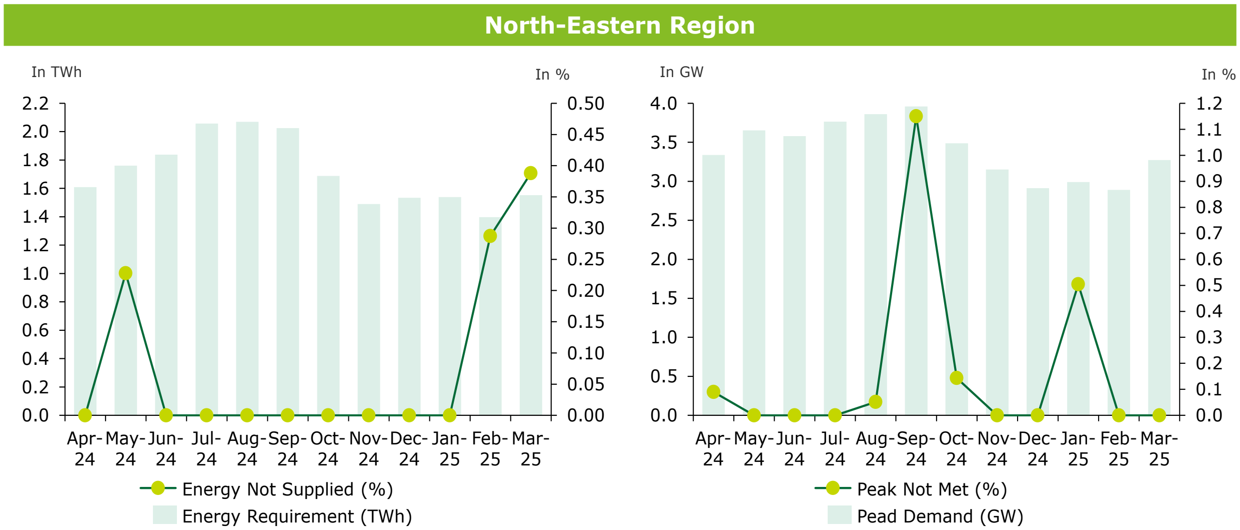


Figure 21: North-Eastern Region monthly energy profile of FY 2024-25

- In FY 2024, the North-Eastern Region experienced minimal energy deficits throughout most months. In FY 2024, the North-Eastern Region experienced minimal energy deficits throughout most months. The Peak Demand was ~3,914 MW with deficit of 45 MW in the month of September.

In summary, during the export period of DHPP, from May to November, the Northern region experiences the highest energy deficit, while the Western region faces the greatest peak demand deficit. The details are summarized in the table below:

Table 8: Summary of Region wise Energy and Peak Deficits in india

Region	Energy Deficits between May to November 2025	Peak Demand Deficits between May to November 2025
Northern	**745 GWh**	1,346 MW
Southern	17 GWh	76 MW
Western	133 GWh	**4,898 MW**
Eastern	261 GWh	1,422 MW
North-Eastern	6 GWh	52 MW

In response to the evolving dynamics of the regional power market in India, we plan to conduct a thorough assessment of the potential offtake opportunities for DHPP. This analysis will focus on major DISCOMs, evaluating their Gross Input Energy, Demand and Generation profiles, merit order, and retirement plans for existing assets. Additionally, we will assess the opportunity for DHPP to participate in power exchanges, open access arrangements, and potential integration with C&I consumers. The objective is to identify the most viable and strategic offtake pathways that align with the project's operational characteristics and market conditions, ensuring the optimal integration of DHPP.

6.3.Indian Power Sector Reforms

Over the last five to six years, the Government of India has undertaken a series of structural reforms to enhance transparency, market access, and financial discipline in the power sector. Key initiatives include the introduction of National Renewable Purchase Obligations (RPO), CBET trading through power exchanges, introduction of Green Day Ahead Market (GDAM) and Green Term Ahead Market (GTAM) market segments in power exchanges to promote renewable energy trading. Additionally, platforms like PRAPTI (Payment Ratification and Analysis in Power Procurement for Bringing Transparency in Invoicing of Generators), implementation of Letter of Credit (LoC) mechanisms and schemes related to raise funding for late payment surcharge (LPS) from Regional Energy Corporation (REC) /Power Finance Corporation (PFC) have significantly improved payment enforcement and reduced outstanding dues.

The key reforms have been enumerated below:

1. The Ministry of Power (MoP) issued the **Renewable Purchase Obligation (RPO)** and Energy Storage Obligation Trajectory for 2022–2030 in July 2022, which was later updated in September 2022 and again in October 2023[36]. As per Note 3 of the latest notification, hydro renewable energy component may also be met from Hydro Power Projects located outside India as approved by the Central Government on a case to-case basis. This has opened the hydropower market for Bhutan in India.

2. The Ministry of Power (MoP), in consultation with the Central Electricity Authority (CEA), has issued guidelines for the **Resource Adequacy (RA)** Planning Framework for India. This study provides an overview of state-wise generation capacities available round-the-clock to reliably meet demand under various scenarios. The RA framework defines the optimal capacity mix needed to satisfy projected demand at the lowest cost. It also assesses the timely addition of new generation capacity, energy storage, and other flexible resources required to reliably support future demand growth at an optimal system cost. This analysis offers valuable insights for Bhutanese power generators by outlining state-specific hydroelectric requirements, aiding in targeted DISCOM assessments. A detailed evaluation is presented in the Indian Power Market Offtake Assessment section

3. The Cross-Border Trade of Electricity in the **Day Ahead Market (DAM)** at the Indian Energy Exchange (IEX) began in 2021-22, with Bhutan joining on January 1, 2022. In 2023, the Government of India amended its procedures to allow Bhutanese power producers to participate in the **Real Time Market (RTM)** segment as well of Indian power exchanges. Both these reforms opened another avenue for offtake of Bhutanese hydropower in the Indian markets.

4. The **Green Day Ahead Market (GDAM)** and **Green Term Ahead Market (GTAM)** were launched in October 2021 and August 2020, respectively, to facilitate renewable energy trading within the Indian Power Exchange framework. The GTAM market encompasses renewable energy sources such as solar, non-solar, and hydro[37]. Trading in GDAM and GTAM markets enable the state DISCOMs to effectively fulfill their RPO targets. GDAM and GTAM promote transparency in the green power market by enabling open, competitive, and regulated trading of renewable energy. At present, cross-border electricity trade is not permitted in the exchange. However, with the ongoing reforms and strong policy advocacy in India's power sector, it is anticipated that Bhutan may be able to participate in GDAM.

5. Launched in May 2018, the PRAAPTI (Payment Ratification and Analysis in Power Procurement for Bringing Transparency in Invoicing of Generators) portal was introduced to improve transparency and accountability in power procurement payments across India. It tracks payment status from distribution companies (DISCOMs) to power generators, including cross-border suppliers. Effective from August 2019, the **Letter of Credit** (LoC) mechanism was mandated

[36] Noticfication on RPO Obligation, The Gazette of India, October 2023
[37] Ministry of Power

by the Ministry of Power to strengthen payment security for electricity generators. Under this reform, Indian DISCOMs are required to maintain LoCs with power producers, ensuring timely payments for scheduled power and reducing financial risk. As per the PRAAPTI portal, outstanding dues of DISCOMs increased from INR 422.13 billion[38] in March 2023 to INR 778.57 billion[39] in March 2024,further declining in March 2025 to INR 528.6 billion[40], indicating fluctuating financial stress and partial recovery in the power distribution sector.

For Bhutan, which exports hydroelectric power to India, PRAAPTI ensures visibility into payment timelines and outstanding dues from Indian DISCOMs. This transparency reduces financial risk, builds trust, and encourages sustained participation in India's power markets—especially through structured segments like CBET, GTAM, and GDAM.

6. The **Late Payment Surcharge (LPS)** framework in India, governed by the Electricity (Late Payment Surcharge and Related Matters) Rules, was introduced in 2022 and updated in 2025 to strengthen payment discipline in the power sector. Under the 2025 amendment, DISCOMs that default on payments may face curtailment of access to the national grid, which can be restored upon settlement of dues. Generators are required to offer surplus power on the exchange; failure to do so results in forfeiture of fixed charge claims. Additionally, surplus power pricing is capped at 120% of the energy charge plus transmission costs. These measures aim to improve liquidity, ensure timely payments, and optimize generation capacity. Since its implementation, the LPS framework has significantly reduced outstanding dues from INR 1.4 lakh crore in June 2022 to INR 91,051 crore[41] in March 2023 to further 49,452 crore[42] by February 2024.

The above reforms have helped create a more transparent, secure, and market-driven power ecosystem in India. These reforms are also helping to strengthen the cross border power market (including for Bhutanese hydropower) by opening up market aveneus and improving transactional frameworks for stakeholders.

[38] PRAAPTI Dues March 2023
[39] PRAAPTI Dues March 2024
[40] PRAAPTI Dues, March 2025
[41] Ministry of Power, 2023
[42] Ministry of Power, 2024

6.4. Evolving Tariff Trends in India

In this section we have assessed the historical tariff trends in India as well as the major developments in the sector governing future pricing outlooks. Historically from FY 2016-24, the tariffs from thermal energy have grown at a steady growth rate of 5% for both coal and oil & gas this is majorly due to rising coal prices and domestic coal shortages. In the same period due to focus on Renewable energy, the tariffs of solar based power plants have decreased by ~7% as the equipment cost has fallen significantly. The average tariff for hydro-based power plants is approximately INR 2.5 per unit; however, this rate primarily reflects costs associated with older and depreciated assets. The details of the growth trend of source wise electricity supply are provided in the figure below:-

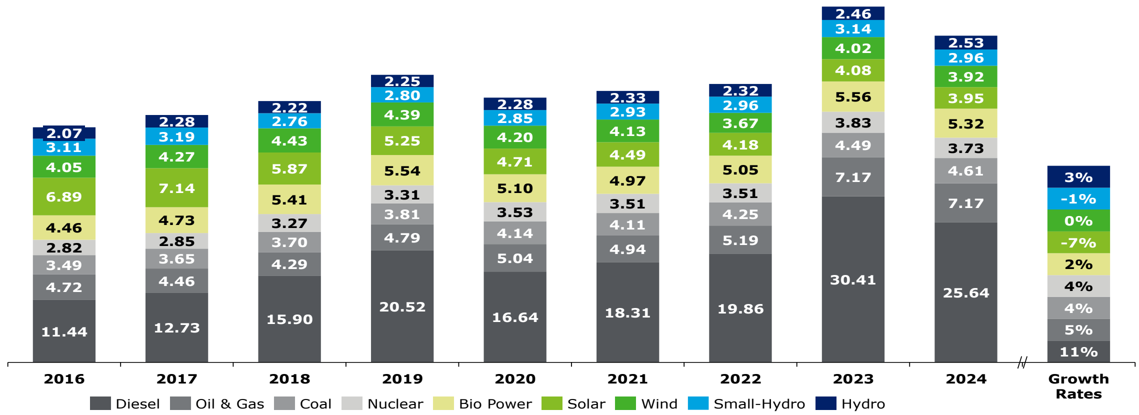


Figure 22: Source wise Electricity tariff (INR/kWh) in last 8 years[43]

The electricity market is evolving rapidly and dynamically, driven by technological advancements, changing regulatory frameworks, and shifting consumer demands. Several key trends are emerging that are expected to shape the future landscape of the market and resultant tariffs. These include increasing integration of renewable energy sources, greater adoption of energy storage solutions, the rise of decentralized generation, and the implementation of more sophisticated pricing and trading mechanisms. Understanding these trends is crucial for stakeholders to navigate and capitalize on the opportunities in the evolving electricity sector.

Table 9: Evolving Trends affecting Pricing In India

Addition of Thermal Power Plants	To meet the estimated electricity demand by the year 2031-32, generation planning studies have been carried out by Central Electricity Authority (CEA). As per the study results, it is envisaged that to meet the base load requirement of the country in 2032, the required coal & lignite based installed capacity would be 283 GW against the present installed capacity of 240 GW as on May 2025[44]. Based on the recent bids, the discovered tariff of upcoming thermal plant is expected to be ~INR 5.4 per unit[45]. These are for the coal plants coming online in 2030-31.
Delayed Hydro Project Installation	At present, ~9.9 GW are under construction in the country as on April 2025[46]. The hydro capacity is projected to increase from 47 GW in FY 2024-25 to ~54-62 GW by 2029-30 (under different scenarios), marking possibility of substantial increase in capacity. Currently, PSPs with aggregate capacity of ~4 GW are under construction in the county while CEA in its revised report on optimal generation mix in 2030, has projected installed capacity of around ~19-25 GW of PSP by 2029-30 under different scenarios. Historically, the average tariff for hydro-

[43] India's Climate and Energy Dashboard
[44] As per MoP, GoI
[45] Source Reuters News Report
[46] Source: Hydroelectric Projects under Construction, CEA, April 2025

	based power plants is approximately INR 2.5 per unit; however, this rate primarily reflects costs associated with older and depreciated assets. **However, the tariffs of projects commissioned in the last decade ranges between INR 4-5.5 per unit.**
Declining Solar Tariffs	The Government of India (GoI) has set an ambitious target of achieving 500 GW of renewable energy capacity by 2030, with approximately 220 GW already installed—of which nearly 50% is contributed by solar energy. The project cost of solar plant has also decreased from INR 4.5 crore to INR 4.1 crore which has resulted in reduction of tariff by 10-12%. Additionally, as discussed in Chapter 5, an analysis of the Indian Energy Exchange (IEX) over the past three years reveals that Market Clearing Prices (MCP) consistently peak and reaches even up to ~INR 8 per unit during the evening hours after 8 p.m. and early morning hours between 1 a.m. and 3 a.m. in both the Day-Ahead Market (DAM) and Real-Time Market (RTM) segments. However, in solar hours, the MCP is in the range of INR 2.5 to 3.5 per unit. Recently, MCPs have fallen to near zero during afternoon hours, primarily due to a surplus of solar generation coinciding with low electricity demand during the pre-monsoon season.
Energy Storage Solutions	In response to the falling tariffs during solar hours, the market is beginning to reward flexibility and time-shifting capabilities. Energy storage systems, particularly battery storage and PSPs, are emerging as critical assets. Projects integrated with storage can store excess solar energy generated during the day and dispatch it during peak periods, allowing developers to capture higher tariffs during peak hours. This arbitrage potential is altering the revenue model of solar projects—pushing the sector toward hybrid and round-the-clock (RTC) configurations. In energy storage systems, 13,050 MW/78,300 MWh Pumped Storage Projects are under construction/concurred and 14,970 MW/54,803 MWh Battery Energy Storage System are currently under various stages of construction/bidding[47]. Out of this, 6,853 MW/ 36,592 MWh of energy storage system (3,180 MW/19,080 MWh Pumped Storage Projects and 3,673 MW/ 17,512 MWh of Battery Energy Storage System) is likely to be added by 2025-26.[48] **As per the recent price bids, the discovered tariffs from The recent bids has fallen to INR 3.33 per unit[49].**

- Historically, tariffs for conventional fuel-based power have increased by 4–5% annually[50]
- Solar power price have declined by approximately 7%, with wind and hydro tariff remaining stable
- While the average tariff for hydro projects in India stands at INR 2.3 per unit, newer projects commissioned in the last five years have tariffs ranging from INR 4- 5.5 per unit.
- Additionally, thermal power prices are projected to reach a landed tariff of INR 5.3 per unit by 2031-32 as evidenced in the recent competitive bids
- Energy storage costs have also been trending downward, with recent bids discovering tariffs as low as INR 3.33 per unit. These prices from energy storage is expected to impact the short term power market, most notably the evening peak prices as surplus RE power during solar hours is being redirected.

The Indian power market can be an attractive market for Bhutanese hydropower provided it aligns with the emerging tariff trends in India

47 PIB, March 2025
48 PIB, March 2025
49 As per Tariff Discovery
50 Except Diesel which has risen at 11% CAGR

7. Indian Power Market – Offtake Assessment

Purpose of the Section

This section evaluates the Indian power market with a focus on identifying viable offtake channels, including long-term Power Purchase Agreements (PPAs) with State DISCOMs, offtake via power exchanges, and opportunities within the commercial and industrial (C&I) segment. It assesses the performance of State DISCOMs based on factors such as capacity addition plans, merit order dispatch, PRAAPTI portal data, ACS versus ARR, and payable days. Furthermore, the study explores alternative market mechanisms by analyzing short- and medium-term power markets, examining key price and volume trends in DAM, RTM, and bilateral transactions. Additionally, a benchmark study of offtake structures for both public and private hydroelectric projects in India has been conducted to evaluate risk-based offtake strategies.

Key Conclusion and takeaways

A performance analysis of Indian DISCOMs highlights their long-term reliability amid evolving market dynamics. Several states—including Maharashtra, Gujarat, Bihar, Odisha, Uttar Pradesh, Rajasthan, Haryana, and Andhra Pradesh—are projected to require over 1 GW of hydropower capacity each within the next decade to ensure resource adequacy. Meanwhile, states such as Tamil Nadu, Telangana, Karnataka, and BRPL (Delhi) face high average power procurement costs (above INR 5 per kWh), presenting opportunities to replace expensive fuel-based power with more affordable hydropower. DHPP can be a viable supplier to these discoms provided the tariff aligns with the power procurement costs.

Short-term trading through power exchanges are increasingly emerging as flexible alternatives for Discoms; however, prices in both DAM and RTM have trended downward over the past three years due to the growing addition of solar and storage projects limiting the potential to allocate offtake for short term power market for DHPP. The short and medium term trading through bilateral route is an attractive segment with short term contracts offering high volumes and prices.

The commercial and industrial (C&I) segment demonstrates strong growth potential, supported by favorable consumption trends. The benchmark study reveals that government hydro projects predominantly secure long-term PPAs with State DISCOMs. However, a few government projects, such as Himachal Pradesh's hydroelectric plants, participate in short-term power markets due to the absence of long-term offtake agreements. Typically, government projects allocate up to 30% of their power to short-term markets owing to a lower risk appetite, whereas private projects may allocate up to 100%, reflecting greater flexibility and market exposure.

This report explores three primary methods for selling Bhutanese hydropower in India:

1. Long-term Power Purchase Agreements (PPAs) with Distribution Companies (DISCOMs)
2. Short Term Power Market Analysis through
   - Indian Power Market Exchange
   - Bilateral Trades
3. Targeting large corporate houses like Commercial and Industrial (C&I) segments

7.1. Offtake Analysis at Indian DISCOM level

India's power sector is highly diversified, offering numerous opportunities to bridge the gap between electricity demand and supply. The country hosts around 70 Distribution Companies (DISCOMs) with significant variation in energy input, operational scale, and market reach.

For the purpose of offtake analysis, a filtering approach has been adopted to focus on State's with high Gross Input Energy. This method ensures that the analysis targets larger markets where the potential for offtake arrangements is more substantial. Prioritizing high-demand regions and major State's as well as DISCOMs allows for a more strategic assessment of export opportunities and enhances the relevance of the findings in the context of regional energy planning and cross-border trade.

Further for analyzing each DISCOMs, certain parameters has been selected. The rationale for selecting parameters has been stated below:

Table 10: Parameters considered for Analyzing DISCOMs

Parameters	Rationale for Parameter Selection
Average Cost of Supply	Average Cost of Supply has been considered as one of the parameters as it reflects the cost incurred by DISCOMs to procure and supply electricity.
Average Revenue Realized	Average Revenue Realized (ARR) reflects the average amount of revenue collected per unit of electricity supplied, on a subsidy-received basis. It is one of the crucial metrics for evaluating the financial performance of a DISCOM. A higher ARR, compared to the ACS, suggests better financial health and profitability.
Gap between Average Cost of Supply vs Average Revenue Realized	Average Cost of Supply (ACS) and Average Revenue Realized (ARR) is a key indicator of DISCOM's of financial health and profitability. A smaller gap suggests that the DISCOM is recovering its costs effectively, which is vital for maintaining operations without incurring losses.
Resource Adequacy Report	Resource adequacy covers future as well projected demand of different states and utilities in India. This metric gives overview of requirement of quantum of power required from different sources like wind, hydro and solar which needs to be furnished.
Profit After Tax (PAT)	Positive PAT indicates financial health and efficient operations, making DISCOMs attractive to investors and ensuring they can meet contractual obligations. This stability reduces risk and enhances consumer confidence in the reliability of electricity supply.
Payable Days	High Payable days indicates that DISCOMs take longer to pay their bills, which can signal cash flow issues and potential payment delays.

Region-wise detailed analysis has been performed, refer below sections:

7.1.1. Western Region

This region is one of the most industrialized and power-intensive zones in the country and contributes a large share of India's thermal power, especially coal-based generation. DISCOMs in Gujarat and Maharashtra have a history of engaging in long-term power purchase agreements, including with Bhutanese projects like Tala and Chhukha. List of States of Western Region have been stated below[51]:

Table 11 : State-wise Gross Input Energy[52]

State	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Maharashtra	172,082	39%
Gujarat	125,896	28%
Madhya Pradesh	96,499	22%
Chhattisgarh	45,453	10%
Goa	5,517	1%
Dadra & Nagar Haveli, Daman & Diu	-	-

[51] Central Electricity Authority
[52] Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

Among the above-mentioned States, Maharashtra, Gujarat, Madhya Pradesh, and Chhattisgarh has been selected further for detailed level Analysis as it covers approximately 99% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 12: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Maharashtra	Maharashtra State Electricity Distribution Company Limited (MSEDCL)	1,66,970	91%
	Adani Electricity Mumbai Limited	10,779	6%
	Brihan Mumbai Electric Supply & Transport Undertaking	5,112	3%
	Tata Power Company Limited	-	0%
	Total of Maharashtra	**1,82,861**	
Madhya Pradesh	Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Ltd	34,017	35%
	Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Ltd	33,522	35%
	Madhya Pradesh Poorv Kshetra Vidyut Vitaran Company Ltd	28,961	30%
	Total of Madhya Pradesh	**96,500**	
Gujarat	Paschim Gujarat Vij Company Limited	46,544	37%
	Uttar Gujarat Vij Company Limited	33,057	26%
	Dakshin Gujarat Vij Company Limited	31,515	25%
	Madhya Gujarat Vij Company Limited	14,781	12%
	Torrent Power Limited	-	0%
	Total of Gujarat	**1,25,897**	
Chhattisgarh	Chhattisgarh State Power Distribution Company Limited	45,453	100%
	Total of Chhattisgarh	**45,453**	

From the above-mentioned list, DISCOM has been selected from each State based on its contribution of atleast 90% of the State's total Gross Input Energy is most to that State, refer below mentioned table for the analysis of DISCOMs:

1. Region: West

State: Maharashtra

Maharashtra has historically made substantial investments in rural electrification, renewable energy development, and capacity expansion. In FY 2025, the state recorded a peak electricity demand of approximately 30.7 GW[53]. The state has an installed generation capacity of 53.3 GW[54], comprising 55% from fossil fuel-based projects, 43% from renewable energy sources (RES), including hydro, and remaining 2% from nuclear projects. Maharashtra is among the leading states in renewable energy adoption, hosting more than 10.4% of India's renewable energy capacity. The state's energy requirement and peak demand from FY 2021-22 to FY 2031-32 has been detailed below-

[53] Load Generation Balance Report, 2025
[54] Government of India, Central Electricity Authority, 2024

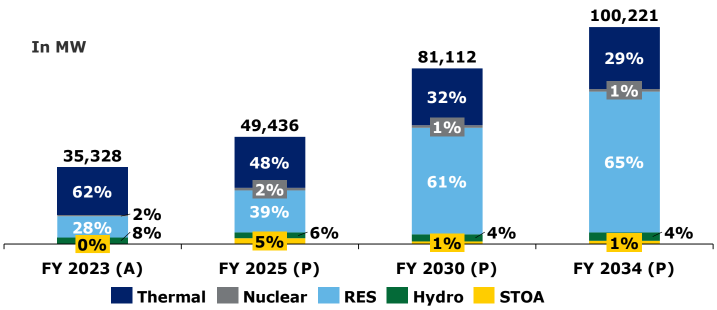
Table 13: Energy Demand of Maharashtra

Particulars	Units	2021-22 (Actual)[55]	2024-25 (Actual)[56]	2026-27 (Projected)[57]	2031-32 (Projected)
Energy Requirement	TWh	170	203	231	265
Peak Demand	GW	26	31	35	45

Fuel Mix of Maharashtra:[58]

Maharashtra's contracted power capacity reached 35 GW by the end of March 2023, of this capacity, 62% is derived from fossil fuel-based sources, primarily coal and natural gas and the rest is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, small hydro, wind and other RE. This composition reflects the state's continued reliance on conventional energy sources while also indicating a growing commitment to renewable energy integration.

The projected contracted capacity for the year FY 2033-34 is 100 GW, out of which 72% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at CAGR of 11% over the period of next 10 years Capacity wise generation mix from FY 2023-34 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Small Hydro
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Storage
RES	Distributed Renewable Energy (DRE)
Nuclear	Nuclear
Hydro	Hydro

Figure 23: Fuel wise capacity mix of Maharashtra[59]

Financial Health of Maharashtra's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 26th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 40% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 60%, are those that are still within their respective due dates but have not yet been fully or partially paid :[60]

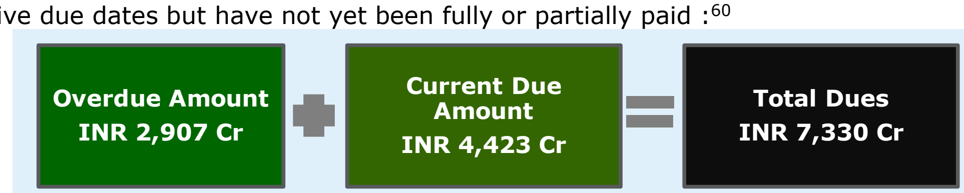


Figure 24: Snapshot of PRAAPTI – Maharashtra

Utility Name : Maharashtra State Electricity Distribution Co. Ltd. (MSEDCL):

[55] Load Generation Balance Report, 2022
[56] Load Generation Balance Report, 2025
[57] National Electricity Plan 2022-32
[58] Resource Adequacy Report, Central Electricity Report, GOI, 2024

[59] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[60] Dashboard for Maharashtra, PRAAPTI, as accessed on 26th May 2025

Maharashtra State Electricity Distribution Company Limited (MSEDCL) is the one of the largest power distribution company both in the country and in Asia, serving over 27.43 million[61] consumers. Established in 2005, MSEDCL operates an extensive network to ensure reliable power supply, contributing significantly to the state's energy infrastructure. MSEDCL serves a diverse consumer base, with the industrial sector accounting for approximately 42% of total electricity consumption. This is followed by the agricultural sector at 22%, domestic consumers at 20%, and the commercial segment at 9%, with the remaining share attributed to other sectors. [62]

Performance History of MSEDCL:

In FY 2023–24, MSEDCL reported consistent increase in gross input energy over the past four years at CAGR of 8% similarly, gross energy sold has also demonstrated steady growth of 10%. The Average Cost of Supply (ACS) declined marginally to INR 8.96 per kWh while Average Revenue realized (ARR) has increased to INR 8.67 per kWh in FY 2023-24, narrowing the gap between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past four years, a decline was observed over in FY 2023-24. Payable Days, while still high, they have steadily decreased over the past three years and improved compared to the national average (excluding private DISCOMs) of 130 days in FY 2023-24.[63]

Table **14**: Performance Report Snapshot

Key Parameters	Units	FY 2020-21[64]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	105,484	116,663	126,741	139,470
Gross Input Energy	GWh	132,485	144,235	155,096	166,970
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.91	7.83	9.32	8.96
Average Revenue Realized (ARR)	INR /kWh	7.37	7.60	7.60	8.67
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.54)	(0.23)	(1.72)	(0.29)
PAT (Profit/(Loss))	INR in Crore	(1,322)	280	(5,199)	(4,517)
Payable Days	Days	178	177	113	109

[61] India Climate & Energy Dashboard
[62] India Climate & Energy Dashboard- Maharashtra
[63] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[64] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of Maharashtra (MSEDCL):

MSEDCL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2023 to 29% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. As per Resource Adequacy report, MSEDCL has plans to increase its capacity mix by 1,063 MW through hydropower till 2034. The state's planned capacity addition for next 10 years has been stated below:

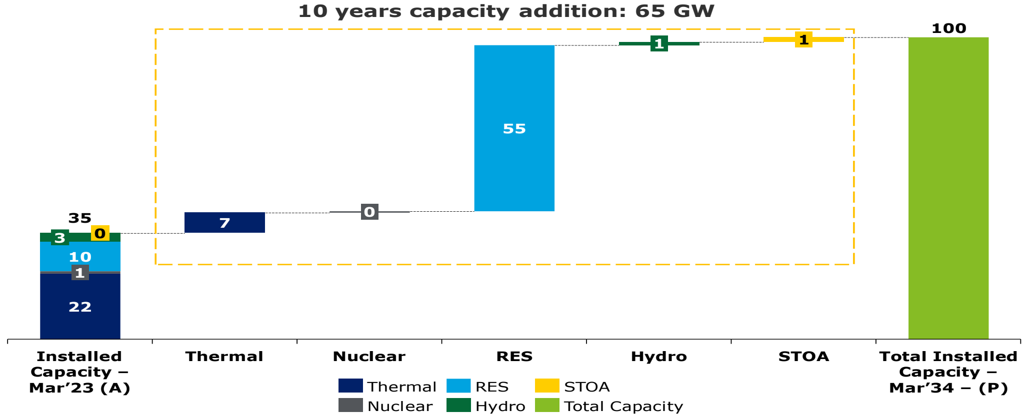
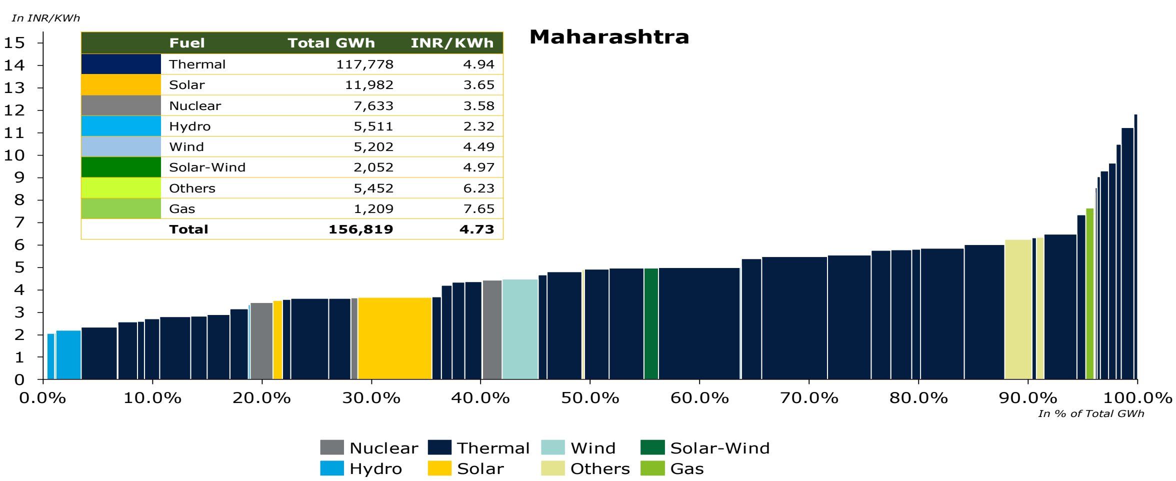


Figure **25**: Capacity Addition Plan for next 10 years of Maharashtra

Merit Order Analysis [65]:-

Maharashtra's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 156,819 GWh. Of this, thermal power accounts for approximately 75% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 7.97 per kWh. Notably, some of the older thermal assets, such as the Parli- Unit6&7 thermal Power Station operated by MAHAGENCO, report a generation cost of INR 9.64 per kWh and Parli Replacement U8 reports generation cost of INR 11.85 per kWh. Similarly, the gas-based GTPS Uran plant had tariff of at INR 7.65 per kWh, while some cogeneration and biomass projects average around INR 6.98 per kWh.



Fuel	Total GWh	INR/KWh
Thermal	117,778	4.94
Solar	11,982	3.65
Nuclear	7,633	3.58
Hydro	5,511	2.32
Wind	5,202	4.49
Solar-Wind	2,052	4.97
Others	5,452	6.23
Gas	1,209	7.65
Total	**156,819**	**4.73**

Figure 26: Merit Order Analysis of Maharashtra

[65] MYT Order, Maharashtra State Electricity Commission, 2024

1. Region: West

State: Gujarat

Gujarat is one of the India's most industrialized state, has made significant investments in renewable energy, resulting in a substantial share of solar and wind power in its overall energy mix, with peak electricity demand of approximately 25.6 GW[66] in FY 2025. The state has an installed generation capacity of approximately 57 GW[67], with about 42% from fossil fuel-based sources, 56% from renewable energy sources (RES), including hydro, and the remaining 2% from nuclear power.

Table **15**: Energy Demand of Gujarat

Particulars	Units	2021-22 (Actual) [68]	2024-25 (Actual) [69]	2026-27 (Projected) [70]	2031-32 (Projected) [71]
Energy Requirement	TWh	124	152	183	243
Peak Demand	GW	19	26	38	36

Fuel Mix of Gujarat[72]

Gujarat's contracted power capacity reached 33 GW by the end of March 2024, of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas and the rest is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, small hydro, wind and other RE.

The projected contracted capacity for the year 2034-35 is 118 GW, out of which 83% through non-fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at CAGR of 10% over the next 10 years. Capacity wise generation mix from 2023-35 has been illustrated below:

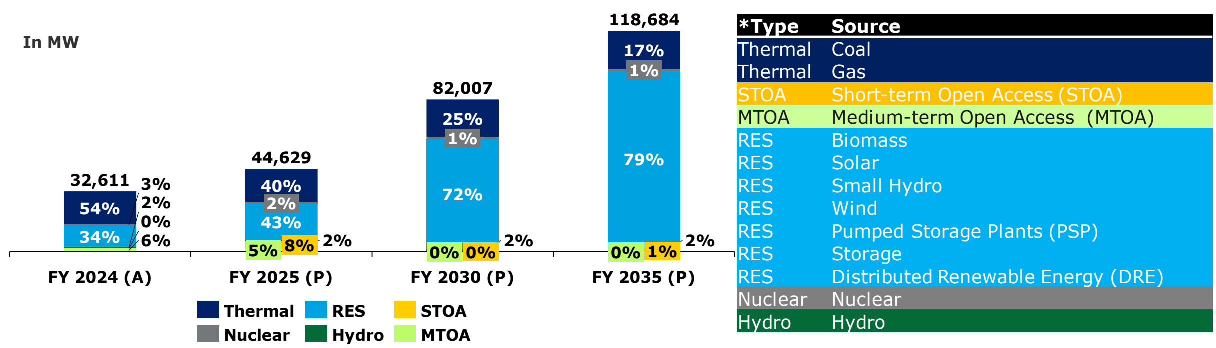


Figure 27: Fuel wise capacity mix of Gujarat[73]

Financial Health of Gujarat's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 50% of the total, refer to payments that have surpassed their due dates and remain either partially or fully

[66] Load Generation Balance Report, 2025
[67] Government of India, Central Electricity Authority, 2024
[68] Load Generation Balance Report, 2022
[69] Load Generation Balance Report, 2025
[70] National Electricity Plan 2022-32
[71] National Electricity Plan 2022-32
[72] Resource Adequacy Report of Gujarat

[73] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

unpaid. In contrast, Current dues, accounting for the remaining 50%, are those that are still within their respective due dates but have not yet been fully or partially paid : [74]

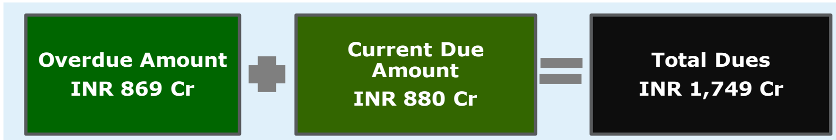


Figure 28: Snapshot of PRAAPTI - Gujarat

Utility Name 1: Gujarat Urja Vikas Nigam Limited (GUVNL)

Gujarat's power distribution is managed primarily by four State-owned distribution companies (DISCOMs) under the umbrella of Gujarat Urja Vikas Nigam Limited (GUVNL). Gujarat Urja Vikas Nigam Limited (GUVNL), established in 2004, is a State-owned holding company responsible for managing Gujarat's power sector following the unbundling of the Gujarat Electricity Board.:

- Dakshin Gujarat Vij Company Limited (DGVCL)
- Madhya Gujarat Vij Company Limited (MGVCL)
- Paschim Gujarat Vij Company Limited (PGVCL)
- Uttar Gujarat Vij Company Limited (UGVCL)

GUVNL serving over 17.47 million[75] consumers, across a diverse customer base. The industrial sector accounts for approximately 65% of total electricity consumption, followed by the agricultural sector at 15%, domestic consumers at 12%, the commercial segment at 5%, and the remaining share attributed to other sectors. [76]

Parameter History of GUVNL:

In FY 2023-24, GUVNL reported consistent increase in gross input energy over the past four years at CAGR of 10% similarly, gross energy sold has also demonstrated steady growth of 11%. The Average Revenue Realized (ARR) consistently remained slightly above the Average Cost of Supply (ACS), resulting in a positive gap that significantly widened in FY 2023-24. This improvement contributed to a sharp rise in profitability, which reached ₹4,339 crore in FY 2023-24. Additionally, the reduction in payable days to just 3 suggests enhanced financial discipline and timely payments. [77]

Table 16:Performance Report Snapshot

Key Parameters	Units	FY 2020-21[78]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	79,680	95,327	104,078	109,205
Gross Input Energy	GWh	94,585	110,622	121,872	125,896
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	6.12	6.46	7.23	7.12
Average Revenue Realized (ARR)	INR /kWh	6.20	6.52	7.25	7.78
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	0.08	0.06	0.02	0.67
PAT (Profit /(Loss))	INR in Crore	429	371	147	4,339
Payable Days	Days	-	-	-	3

[74] Dashboard for Gujarat, PRAAPTI, as accessed on 27th May 2025
[75] India Climate & Energy Dashboard
[76] India Climate & Energy Dashboard- Gujarat
[77] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[78] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of GUVNL: [79]

Gujarat's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 54% in FY 2024 to 17% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. As per Resource Adequacy report, GUVNL has plans to increase its capacity mix by 1,098 MW through hydropower till 2035. The state's planned capacity addition for next 10 years has been stated below:

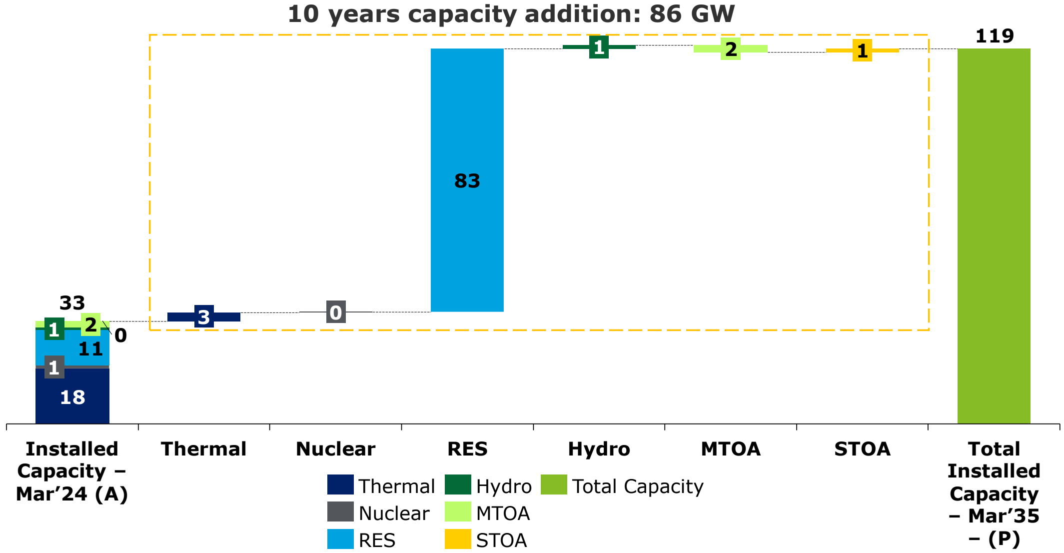


Figure 29: Capacity Addition Plan for next 10 years of Gujarat

Merit Order Analysis:- [80]

Gujarat's energy generation landscape is characterized by a diverse mix of sources, with a total generation of 149,290 GWh. Thermal power remains the leading contributor, generating 59,993 GWh. According to the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal, lignite, and competitively bid sources, with an average cost of INR 11.02 per kWh. Older thermal assets such as NTPC Khargone and NTPC Solapur report significantly high generation costs of INR 30.25 and INR 25.83 per kWh, respectively, mainly due to elevated fixed charges. Similarly, the gas-based plant operated by Gujarat State Energy Generation reports a high generation cost of INR 20.60 per kWh. Lignite-based plants, including GSECL Kutch Lignite and GSECL BLTPS, average around INR 7.57 per kWh. Additionally, competitively bid power from Adani Power Limited is priced at INR 13.42 per kWh, further contributing to the high cost profile of the upper quartile.

[79] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[80] Truing up for FY 2022-23 and Determination of Tariff for FY 2024-25, Gujarat Electricity Regulatory Commission, 2024

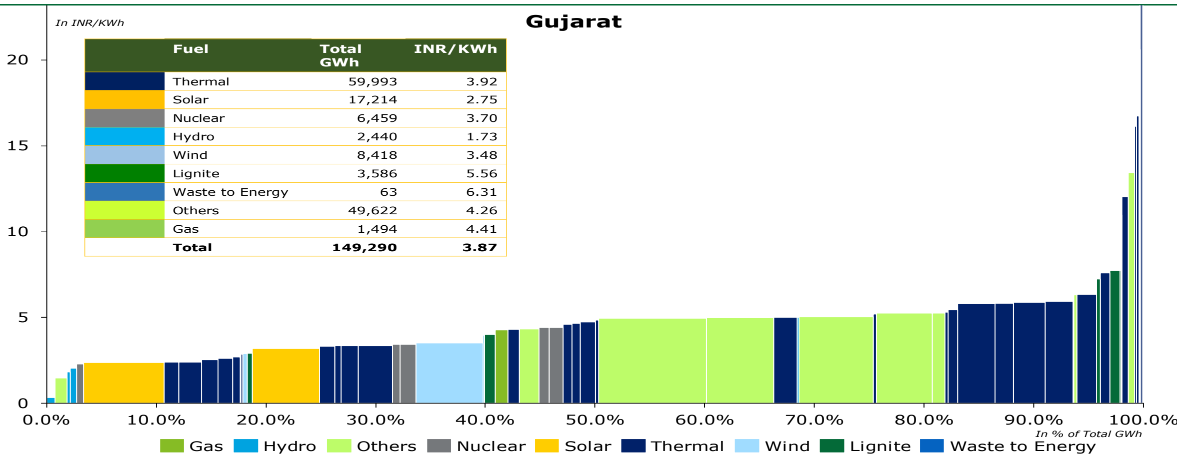


Figure 30: Merit Order Analysis of Gujarat

1. Region: West

State: Madhya Pradesh

In FY 2025, Madhya Pradesh recorded a peak electricity demand of approximately 19.37 GW[81], of which 19.18 GW was met, resulting in a peak deficit of around 1%. The state has an installed generation capacity of about 28.6 GW[82], comprising approximately 57% from fossil fuel-based sources, 41.3% from renewable energy sources (RES), including hydro, and the remaining 1.7%[83] from nuclear power. Madhya Pradesh is steadily advancing in renewable energy adoption and significantly contributing to India's overall renewable energy capacity.

Table **17**: Energy Demand of Madhya Pradesh

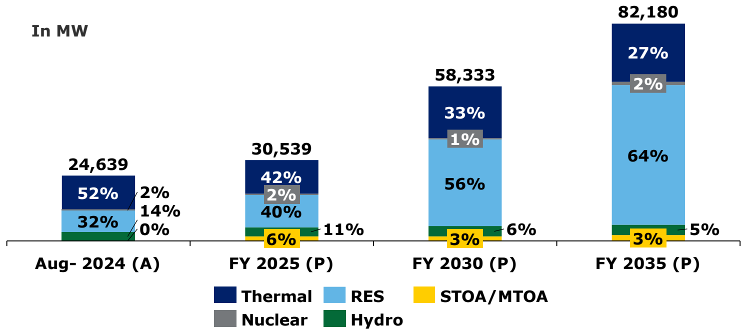
Particulars	Units	2021-22 (Actual) [84]	2024-25 (Actual)	2026-27 (Projected) [85]	2031-32 (Projected)
Energy Requirement	TWh	85	104	129	169
Peak Demand	GW	16	19	22	27

Fuel Mix of Madhya Pradesh[86]

Madhya Pradesh's contracted power capacity stood at 25 GW with 52% derived from fossil fuel-based generation, primarily coal and natural gas and the rest is derived from non-fossil fuel-based sources, including nuclear, bioenergy, hydro, solar, wind and other RE.

The projected contracted capacity for the year 2034-35 is 82 GW, out of which 73% is anticipated to come from non-fossil fuel sources. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:

81 Load Generation Balance Report, 2025
82 All India Installed Capacity, Central Electricity Authority, 2024
83 Resource Adequacy Report of Madhya Pradesh
84 Load Generation Balance Report, 2022
85 National Electricity Plan 2022-32
86 Resource Adequacy Report of Madhya Pradesh



*Type	Source
Thermal	Coal
Thermal	Gas
STOA/ MTOA	Short-term Open Access (STOA) Medium-term Open Access (MTOA)
RES	Bioenergy
RES	Solar
RES	Wind
RES	Banking
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 31: Fuel-wise capacity mix of Madhya Pradesh[87]

Financial Health of Madhya Pradesh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 37% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 63%, are those that are still within their respective due dates but have not yet been fully or partially paid :[88]

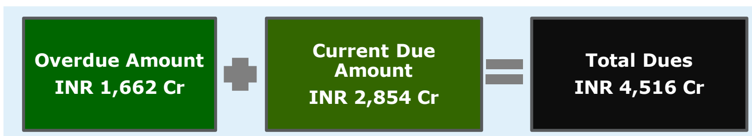


Figure 32: Snapshot of PRAAPTI- Madhya Pradesh

Utility Name 1: M.P. Power Management Company Limited (MPPMCL)

MP Power Management Company has been designated as the holding company for all electricity distribution companies (DISCOMs) in Madhya Pradesh. The state's electricity distribution is managed by three regional DISCOMs:

- Madhya Pradesh Poorv Kshetra Vidyut Vitaran Company Limited (MPPoKVVCL) – East
- Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Limited (MPMKVVCL) – Central
- Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited (MPPaKVVCL) – West

These DISCOMs serve over 18.06 million[89] consumers across the eastern, central, and western regions of the state. In FY 2022–23, the agriculture sector accounted for the largest share of electricity consumption at 37%, followed by the industrial sector at 25%, and domestic consumption at 24% and remaining attributed to commercial and other sectors. [90]

Performance History of MPPMCL:

Over the past four years, gross energy sold has shown consistent growth at CAGR of 7% similarly, gross input energy has increased steadily at 8%. The average cost of supply (ACS) has fluctuated slightly but increased to INR 8.22 per kWh in FY 2023–24, while the average revenue realized (ARR) has plateaued at INR 7.54 per kWh, persistent gap between ACS and ARR has been noted. PAT has remained negative over past four years. Additionally, payable days have remained high, hovering above 200 days, which is higher than the national average (excluding private DISCOMs) payable days of 130[91].

[87] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[88] Dashboard for Madhya Pradesh, PRAAPTI- as accessed on 27th May 2025.
[89] India Climate & Energy Dashboard
[90] India Climate & Energy Dashboard- Madhya Pradesh
[91] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

Table **18**: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[92]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	58,828	63,058	67,794	71,365
Gross Input Energy	GWh	80,523	83,100	89,958	96,499
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	8.01	7.94	7.92	8.22
Average Revenue Realized (ARR)	Rs/kWh	7.30	7.41	7.54	7.54
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(0.71)	(0.53)	(0.38)	(0.68)
PAT (Profit/ (Loss))	Rs in Crore	(4,152)	(3,315)	(2,585)	(4,886)
Payable Days	Days	230	217	210	211

Capacity Addition Plans of MPPMCL:

Madhya Pradesh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 42% in FY 2025 to 27% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. Additionally, Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited (MPPKVVCL) has planned to increase its capacity by 339 MW through hydropower till 2035.

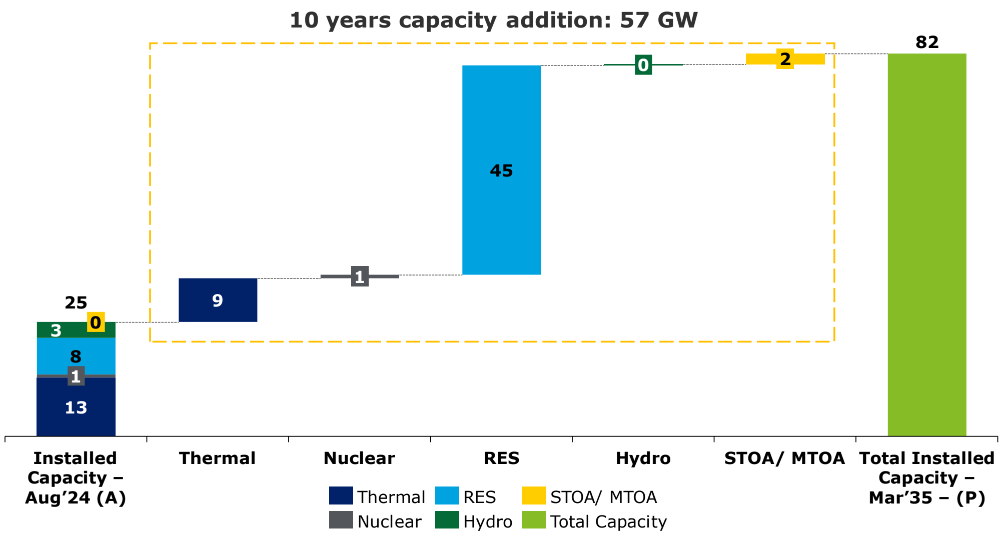


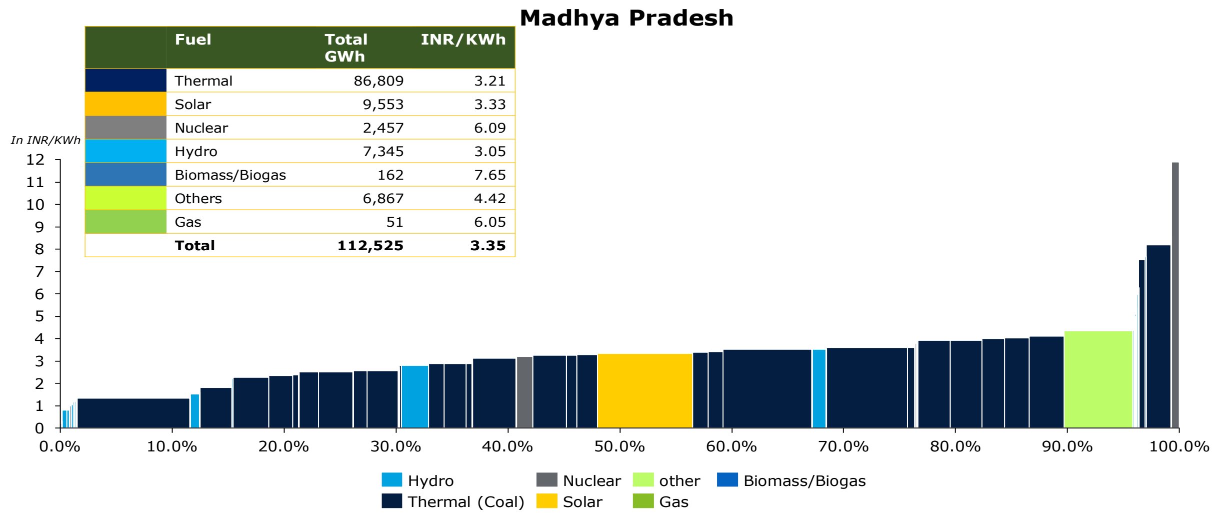
Figure 33: Capacity Addition plan for next 10 years of Madhya Pradesh

Merit Order Analysis [93]:-

Madhya Pradesh's energy generation landscape is characterized by a diverse mix of sources, with thermal power continuing to dominate due to its relatively low average cost of INR 3.21 per kWh. Solar energy stands out as a key renewable contributor, offering both substantial output and cost efficiency at INR 3.33 per kWh. According to the approved ARR tariff for FY 2025–26, the fourth quartile (75%–100%) of power purchase cost is primarily composed of thermal power plants, with an average generation cost of INR 4.49 per kWh. However, some older thermal assets, such as NTPC Khargone and NTPC Lara, report significantly higher generation costs of INR 8.18 and INR 7.53 per kWh, respectively. The nuclear plant at KAPP Kakrapar is the most expensive, with a generation cost

[92] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[93] MYT Order, Madhya Pradesh Electricity Regulatory Commission

of INR 11.90 per kWh. Others include non-solar renewable sources contribute at an average tariff of INR 4.36 per kWh, reflecting a moderate cost profile within the state's energy mix.



Fuel	Total GWh	INR/KWh
Thermal	86,809	3.21
Solar	9,553	3.33
Nuclear	2,457	6.09
Hydro	7,345	3.05
Biomass/Biogas	162	7.65
Others	6,867	4.42
Gas	51	6.05
Total	**112,525**	**3.35**

Figure 34: Merit Order Analysis of Madhya Pradesh

1. Region: West

State: Chhattisgarh

In FY 2025, Chhattisgarh recorded a peak electricity demand of approximately 6.5 GW[94], which has met without any surplus or deficit. The state has an installed generation capacity of around 14.33 GW, with about 85.3% derived from fossil fuel-based sources, 13.8% from renewable energy sources (RES), including hydro, and the remaining 0.9% from nuclear power[95].The state has made notable progress in renewable energy adoption, with over 3,000 MW[96] of installed renewable capacity, contributing around 2.5% to India's total renewable energy portfolio.

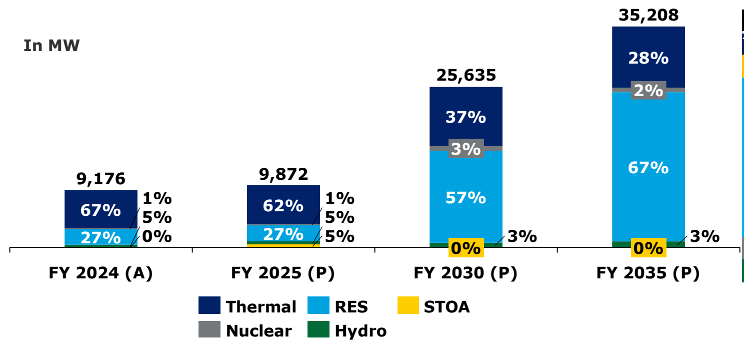
Table **19**: Energy Demand of Chhattisgarh

Particulars	Units	2021-22 (Actual) [97]	2024-25 (Actual)	2026-27 (Projected) [98]	2031-32 (Projected) [99]
Energy Requirement	TWh	32	43	47	64
Peak Demand	GW	5	7	7	10

Fuel Mix of Chhattisgarh[100]

As of March 2024, Chhatisgarh's contracted power capacity stood at 9 GW with 67% derived from fossil fuel-based sources primarily coal and the rest from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, hybrid, wind power and other RE technologies.

The projected contracted capacity for the year 2034-35 is 35 GW, with 72% expected from non-fossil fuel and 28% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 11% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:



*Type	Source
Thermal	Coal
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Hybrid
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Storage
RES	Distributed Renewable Energy (DRE)
Nuclear	Nuclear
Hydro	Hydro

Figure 35: Fuel-wise capacity mix of Chhattisgarh[101]

Financial Health of Chhattisgarh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 12% of the total, refer to payments that have surpassed their due dates and remain either partially or fully

[94] Load Generation Balance Report, 2025
[95] Government of India, Central Electricity Authority, 2024
[96] The Role of Renewable Energy in Electricity Production in Chhattisgarh, SSR Journal of Engineering and Technology (SSRJET), 2024
[97] Load Generation Balance Report, 2022
[98] National Electricity Plan 2022-32
[99] National Electricity Plan 2022-32
[100] Resource Adequacy report of Chhattisgarh
[101] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

unpaid. In contrast, Current dues, accounting for the remaining 88%, are those that are still within their respective due dates but have not yet been fully or partially paid :[102]

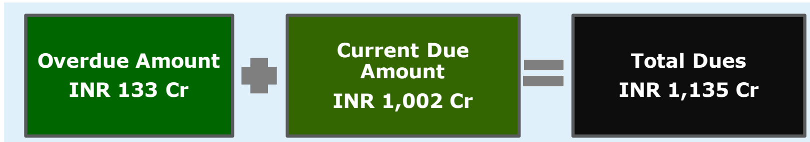


Figure 36: Snapshot of PRAAPTI -Chhattisgarh

Utility Name 1: Chhattisgarh State Power Distribution Company Limited (CSPDCL)

Chhattisgarh's electricity distribution is primarily managed by the Chhattisgarh State Power Distribution Company Limited (CSPDCL), a State-owned utility responsible for supplying power across urban and rural areas, serving over 6.31 million[103] consumers. CSPDCL serves a diverse consumer base, with the industrial sector accounting for approximately 63% of total electricity consumption. This is followed by the agricultural sector at 15%, domestic consumers at 14%, and the commercial segment at 4%, with the remaining share attributed to other sectors.[104]

Performance History of CSPDCL:

Chhattisgarh has demonstratedshown consistent growth over the past four years, whereas gross energy sold increasing at CAGR of 10%, while10% similarly gross input energy has grownalso risen steadily at 9%. The average cost of supply (ACS) peaked at INR 6.61 per /kWh in FY 2022–23 but declined to INR 6.13 per /kWh in FY 2023–24. Notably, the gap between ACS and Meanwhile, the average revenue realized (ARR) turned positive for improved to INR 6.14/kWh, resulting in a marginal surplus of INR 0.01/kWh—the first time in four years, reaching INR 0.01 per kWh.positive gap in four years. Profit After Tax (PAT), which had remained negative in previous years, showed a turnaround, turned positive in FY 2023–24 with a profit of INR 41 crore in FY 2023–24., indicating a potential financial recovery. Additionally, payable days reduced significantly from 202 to 108, which is notably better than the national average (excluding private DISCOMs) of 130 days, indicating improved financial discipline and liquidity.[105]

Table **20**: Performance Report Snapshot

Key Parameters	Units	FY 2020-21[106]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	29,244	31,124	32,483	38,515
Gross Input Energy	GWh	35,302	37,951	39,325	45,453
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	5.70	6.11	6.61	6.13
Average Revenue Realized (ARR)	INR /kWh	5.56	5.72	6.26	6.14
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.14)	(0.39)	(0.35)	0.01
PAT (Profit/(Loss))	INR in Crore	(420)	(1,214)	(1,133)	41
Payable Days	Days	202	181	114	108

[102] Dashboard for Chhattisgarh, PRAAPTI- as accessed on 27th May 2025
[103] India Climate & Energy Dashboard
[104] India Climate & Energy Dashboard- Chhattisgarh
[105] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[106] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of CSPDCL:

Chhattisgarh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2025 to 28% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. Additionally, CSPDCL had planned to purchase 568 MW through hydropower till 2035.

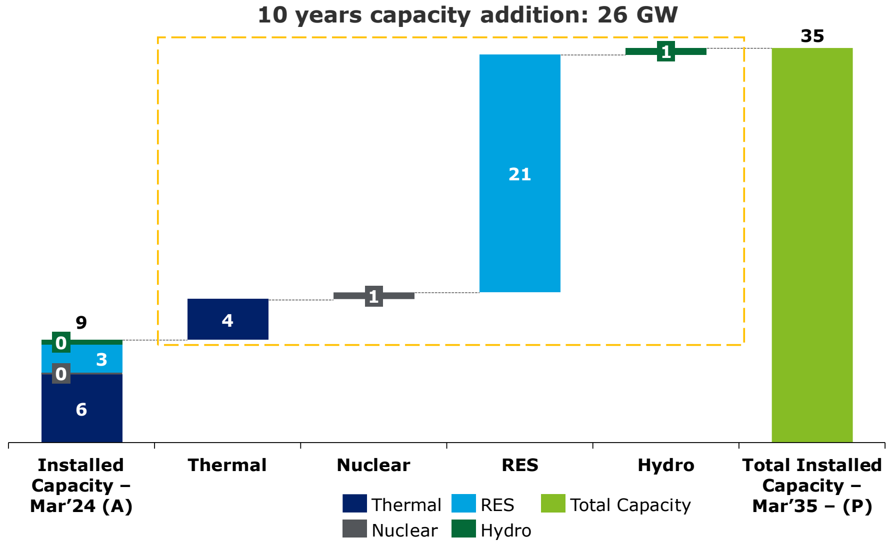
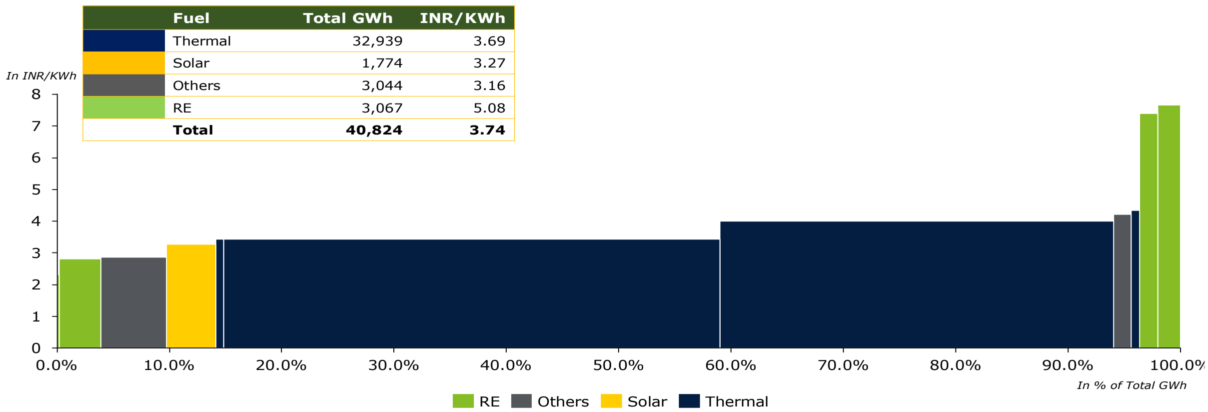


Figure 37: Capacity Addition Plan for next 10 years of Chhattisgarh

Merit Order Analysis[107]:

Chhattisgarh's power generation is dominated by thermal energy, contributing over 80% of the total at an average cost of INR 3.69 per kWh. Solar and other renewables contribute smaller shares but are generally more cost-effective, while non-solar renewables are the most expensive at INR 5.08 per kWh. As per the FY 2025–26 tariff order, the costliest quartile (75%–100%) is mainly thermal, averaging at INR 4.18 per kWh, with older assets like NTPC SAIL reaching INR 4.34 per kWh. Additionally, Hydropower and biomass are the priciest, averaging at INR 7.53 per kWh.

Chhattisgarh



Fuel	Total GWh	INR/KWh
Thermal	32,939	3.69
Solar	1,774	3.27
Others	3,044	3.16
RE	3,067	5.08
Total	**40,824**	**3.74**

Figure 38: Merit Order Analysis of Chhattisgarh

107 Tariff Order for FY 2024-25, Chhattisgarh State Electricity Regulatory Commission

7.1.1. Eastern Region

The Eastern region is a key part of India's energy landscape, with abundant coal reserves and major thermal power plants, especially in Jharkhand and Odisha. West Bengal and Odisha have relatively efficient DISCOMs with a track record of long-term power purchase agreements, including with Bhutanese hydropower projects, and are actively integrating renewables. Bihar and Jharkhand face higher losses and financial stress but are undergoing reforms under RDSS and UDAY. The region's proximity to Bhutan and established transmission links makes it well-positioned for hydropower imports to support grid balancing and renewable integration. List of All States of Eastern have been Stated below[108]:

Table 21: State-wise Gross Input Energy[109]

State	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
West Bengal	53,974	35%
Bihar	47,296	31%
Odisha	35,444	23%
Jharkhand	15,785	10%
Sikkim	1,076	1%
Andaman & Nicobar Islands	378	0%

Among the above-mentioned States, West Bengal, Bihar, Odisha and Jharkhand has been selected further for detailed level Analysis as it covers approximately 99% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 22: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
West Bengal	West Bengal State Electricity Distribution Company Limited	53,974	98%
	India Power Corporation Limited	943	2%
	Total of West Bengal	**54,917**	
Odisha	Tata Power Western Odisha Distribution Limited (erstwhile WESCO)	12,752	36%
	Tata Power Central Odisha Distribution Limited (erstwhile CESCO)	11,299	32%
	Tata Power Northern Odisha Distribution Limited (erstwhile NESCO)	7,047	20%
	Tata Power Southern Odisha Distribution Limited (erstwhile SouthCO)	4,345	12%
	Total of Odisha	**35,443**	
Jharkhand	Jharkhand Bijli Vitran Nigam Limited	15,785	100%
	Total of Jharkhand	**15,785**	
Bihar	South Bihar Power Distribution Company Limited	26,259	56%
	North Bihar Power Distribution Company Limited	21,037	44%
	Total of Bihar	**47,296**	

[108] Central Electricity Authority
[109] Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

2. Region: East

State: West Bengal

West Bengal recorded peak electricity demand of 12.65 GW in FY 2025. The state has an installed generation capacity of 11 GW[110], with 80% from fossil fuel-based sources and the rest from renewable energy sources (RES), including hydro. West Bengal is a key player in India's electricity sector, with WBSEDCL providing power to over 96% of the state. Though the state is 10th biggest electricity user in India, it used only about of 2112 MW renewable energy potential, showing big chances for growths [111].

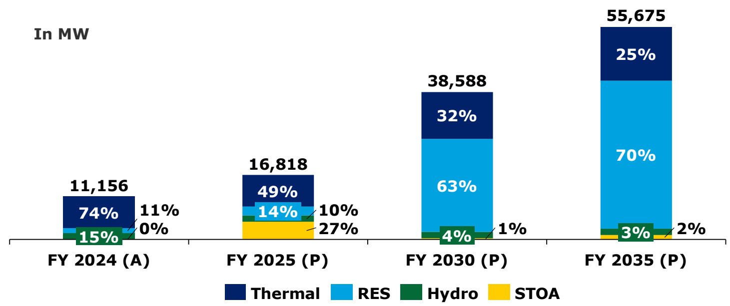
Table **23**: Energy Demand of West Bengal

Particulars	Units	2021-22 (Actual) [112]	2024-25 (Actual) [113]	2026-27 (Projected) [114]	2031-32 (Projected) [115]
Energy Requirement	TWh	54	71	76	99
Peak Demand	GW	9	13	13	17

Fuel Mix of West Bengal [116]

West Bengal's contracted power capacity reached 11 GW as of March 2024 with 74% is derived from fossil fuel-based sources, primarily coal and the remaining from non-fossil fuel-based energy, including hydro, solar, battery, wind, PSP and Other RE.

The projected contracted capacity for the year 2034-35 is 56 GW, with 75% derived from non-fossil fuel sources and 25% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 14% over the next 10 years. Capacity wise generation mix from FY 2024-35 has been illustrated below:



*Type	Source
Thermal	Coal
STOA	Short-term Open Access (STOA)
RES	Solar
RES	Hybrid
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Battery
RES	Distributed Renewable Energy (DRE)
Hydro	Hydro

Figure 39: Fuel-wise capacity mix of West Bengal[117]

Financial Health of West Bengal's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 42% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 58%, are those that are still within their respective due dates but have not yet been fully or partially paid :[118]

[110] All India Installed Capacity, Central Electricity Authority of India, 2025
[111] Indian Climate & Energy Dashboard
[112] Load Generation Balance Report, 2022
[113] Load Generation Balance Report, 2025
[114] National Electricity Plan 2022-32
[115] National Electricity Plan 2022-32
[116] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[117] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[118] Dashboard for West Bengal, PRAAPTI- as accessed on 27th May 2025

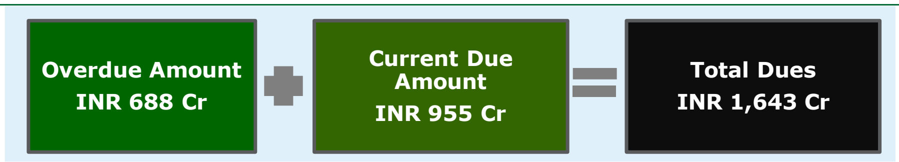


Figure 40: Snapshot of PRAAPTI - West Bengal

Utility Name : West Bengal State Electricity Distribution Company Limited (WBSEDCL)

WBSEDCL is responsible for supplying electricity to 23.10 million consumers across West Bengal. The company is focused on ensuring reliable power distribution and continuous infrastructure development to deliver consistent and efficient services. WBSEDCL caters to a diverse range of consumer categories, with the industrial sector accounting for around 41% of total electricity consumption. This is followed by the domestic consumers at 32%, commercial segment at 12%, agricultural sector at 3%, and the the remaining share attributed to other sectors. [119]

Performance History of WBSEDCL:

In FY 2023–24, WBSEDCL reported consistent increase in gross input energy over the past four years at CAGR of 7% similarly, gross energy sold has also demonstrated steady growth of 8%. The average cost of supply (ACS) has shown fluctuations during the period, peaking in FY 2023–24. Concurrently the average revenue realized (ARR) has also increased, helping to narrow the gap between cost and revenue. Although the gap between ARR vs ACS remained negative in most years, the deficit has reduced significantly. Profitability has shown a positive turnaround, moving from losses in FY 2020–21 and FY 2021–22 to profits in FY 2022–23 and FY 2023–24. Payable days, though declining over the past three years, rose again in FY 2023–24 and remain above the national average (excluding private DISCOMs) of 130 days. [120]

Table **24** : Performance Report Snapshot

Key Parameters	Units	FY 2020-21[121]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	34,522	43,054	41,888	43,636
Gross Input Energy	GWh	44,221	51,253	51,958	53,974
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	7.84	6.21	7.53	7.86
Average Revenue Realized (ARR)	Rs/kWh	6.61	6.45	7.14	7.65
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(1.23)	0.24	(0.40)	(0.21)
PAT (Profit/(Loss))	Rs in Crore	(190)	(205)	21	109
Payable Days	Days	175	147	143	162

Capacity Addition Plans of WBSEDCL:

WBSEDCL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 74% in FY 2024 to 25% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise.

[119] India Climate & Energy Dashboard
[120] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[121] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

As per Resource Adequacy report, WBSEDCL has plans to increase its capacity mix by 90 MW through hydropower till 2035. The state's planned capacity addition for next 10 years has been stated below:[122]

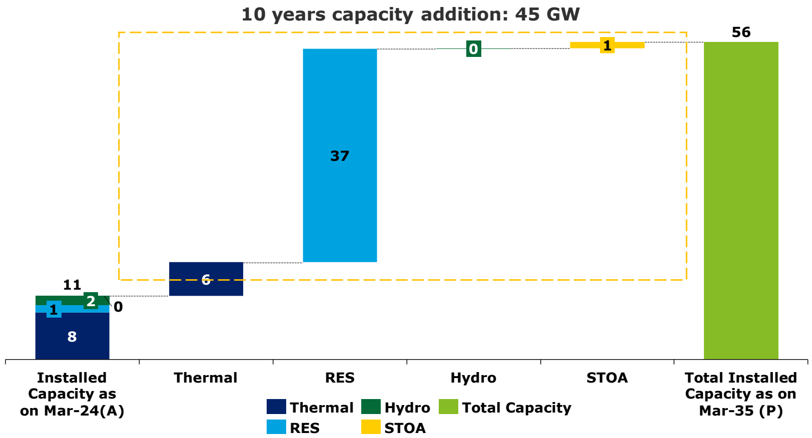
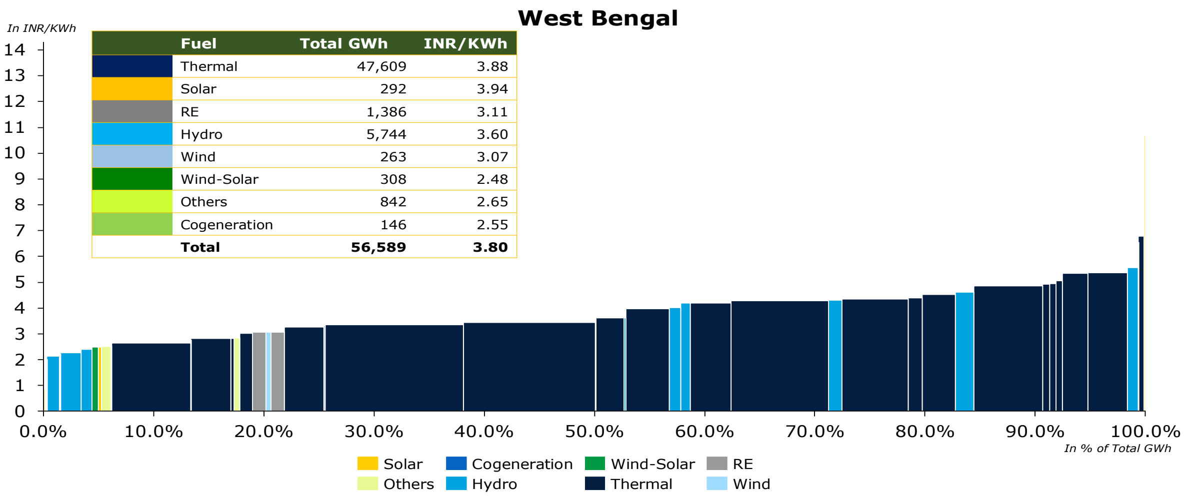


Figure 41: Capacity addition plans for next 10 years of West Bengal

Merit Order Analysis[123]:

West Bengal's energy generation landscape is characterized by a diverse mix of sources, with a total generation of 56,589 GWh. Thermal power remains the leading contributor, generating 47,609 GWh. According to the approved tariff order for FY 2023–24, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal and hydropower plant , with an average cost of INR 5.06 per kWh. Older thermal assets such as NVVN (Bundled Power) and Hiranmaye Energy Limited report significantly high generation costs of INR 6.78 and INR 5.37 per kWh, respectively. Similarly, the hydropower plant of NHPC's Teesta Low Dam-III (TLDP-III) reports a high generation cost of INR 5.57 per kWh.



Fuel	Total GWh	INR/KWh
Thermal	47,609	3.88
Solar	292	3.94
RE	1,386	3.11
Hydro	5,744	3.60
Wind	263	3.07
Wind-Solar	308	2.48
Others	842	2.65
Cogeneration	146	2.55
Total	**56,589**	**3.80**

Figure 42: Merit Order Analysis of West Bengal

[122] Report on Resource Adequacy of West Bengal upto 2034-35, Central Electricity Authority, GoI
[123] MYT 2023-24, West Bengal Electricity Regulatory Commission, 2023-24

2. Region: East

State: Jharkhand

Jharkhand's peak electricity demand for FY 2025 stood at 2.3 GW, which was fully met, resulting in minimal peak deficit of 0.1%[35]. The state has an installed generation capacity of 2.37 GW[124], with 87% sourced from fossil fuel and remaining 13% from renewable energy sources (RES), including hydro.
Jharkhand's electricity profile is heavily reliant on coal-based thermal power, which contributes over 85% of its utility scale generation.

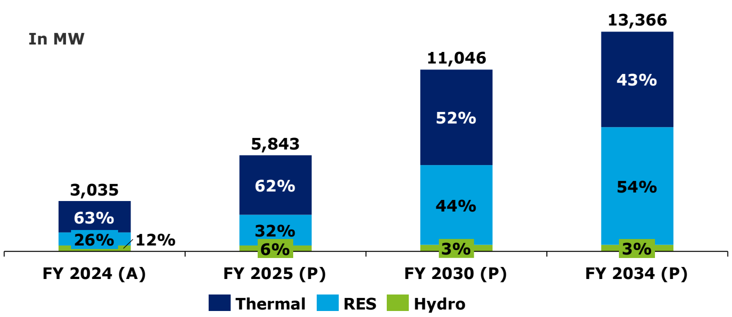
Table **25**: Energy Demand of Jharkhand

Particulars	Units	2021-22 (Actual) [125]	2024-25 (Actual) [126]	2026-27 (Projected) [127]	2031-32 (Projected) [128]
Energy Requirement	TWh	10.9	15.2	25.5	33.8
Peak Demand	GW	1.9	2.3	3.8	5.00

Fuel Mix of Jharkhand [129]

Jharkhand's contracted power capacity reached 3 GW by the end of March 2023, of this 63% is derived from fossil fuel-based sources, primarily coal while remaining 37% from non-fossil fuel-based energy sources, including hydro, solar, storage, wind and Distributed RE.

The projected contracted capacity for the year 2033-34 is 13 GW, with 57% from non-fossil fuel and remaining 43% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 10% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:



*Type	Source
Thermal	Coal
RES	Solar
RES	Wind
RES	Storage
RES	Distributed Renewable Energy (DRE)
Hydro	Hydro

Figure 43: Fuel-wise capacity mix of Jharkhand[130]

Financial Health of Jharkhand's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 3% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 97%, are those that are still within their respective due dates but have not yet been fully or partially paid:[131]

[124] All India Installed Capacity, Central Electricity Authority of India, 2025
[125] Load Generation Balance Report, 2022
[126] Load Generation Balance Report, 2025
[127] National Electricity Plan 2022-32
[128] National Electricity Plan 2022-32
[129] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[130] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[131] Dashboard for Jharkhand, PRAAPTI- as accessed on 27th May 2025

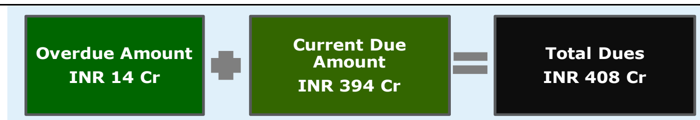


Figure 44: Snapshot of PRAAPTI - Jharkhand

Utility Name : Jharkhand Bijli Vitran Nigam Limited (JBVNL)

Jharkhand Bijli Vitran Nigam Limited (JBVNL) is the largest electricity distribution company in Jharkhand, formed after the unbundling of the Jharkhand State Electricity Board in 2013. JBVNL serves over 5.84 million consumers, serving majorly industrial consumers followed by domestic, commercial, agriculture and others. [132]

Performance History of Jharkhand:

Jharkhand's power sector exhibited a steady increase in both gross energy sold and input energy at CAGR of 11% and 6% respectively. The average cost of supply (ACS) remained consistently high, peaking at INR 11.27 per kWh in FY 2023–24, whereas average revenue realized (ARR) also improved during this period. The gap between ARR and ACS remained negative, while slightly narrowing in FY 2023–24.[133]

Table 26:Performance Report Snapshot

Key Parameters	Units	FY 2020-21[134]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	8,026	9,491	9,820	10,928
Gross Input Energy	GWh	13,290	14,369	14,468	15,785
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	10.09	9.06	10.93	11.27
Average Revenue Realized (ARR)	INR /kWh	7.35	6.91	7.14	8.90
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(2.74)	(2.15)	(3.79)	(2.37)
PAT (Profit/(Loss))	INR in Crore	(2,200)	(2,038)	(3,720)	(2,587)
Payable Days	Days	555	536	433	420

[132] India Climate & Energy Dashboard
[133] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[134] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of JBVNL:

Jharkhand 's energy capacity is expected to grow significantly, with a notable shift toward more diversified energy mix. The share of coal expected to decline from 63% in FY 2023 to 43% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise, reflecting the state's commitment to cleaner and more sustainable energy solutions. JBVNL plans to procure additional capacity of 20 MW hydropower till 2034. Capacity Addition requirement has been illustrated below[135]:

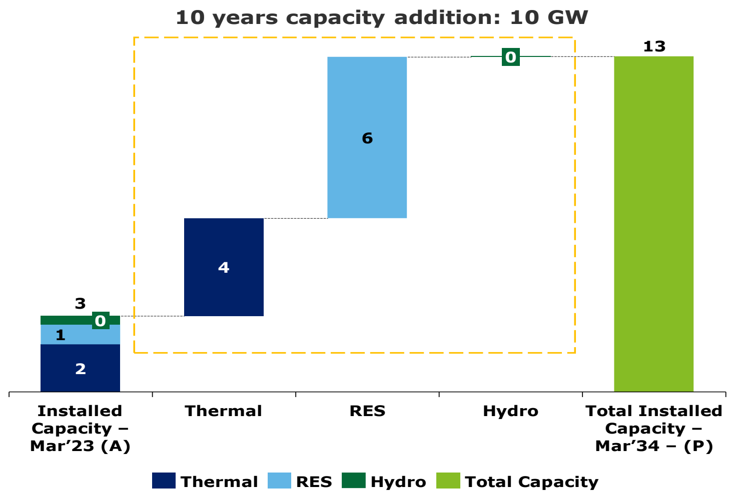
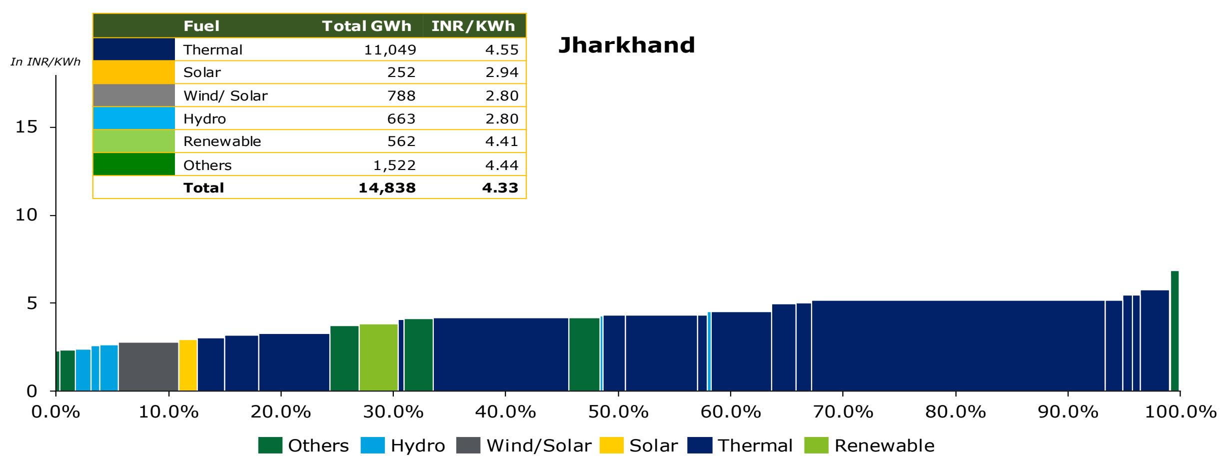


Figure 45: Capacity addition plan for next 10 years of Jharkhand

Merit Order Analysis[136]:

Jharkhand's energy generation landscape features a diverse mix of sources, with total generation reaching 14,838 GWh. Thermal power remains the dominant contributor, accounting for 11,049 GWh. As per the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal and solar sources, with an average cost of ₹7.28 per kWh. Notably, the state reports significantly high generation costs from State IPPs (MNRE-I) and SECI (MNRE-II), at ₹17.95 per kWh and ₹6.49 per kWh respectively. These elevated costs are largely attributed to high fixed charges and annual tariff escalations applicable in FY 2024–25.



Fuel	Total GWh	INR/KWh
Thermal	11,049	4.55
Solar	252	2.94
Wind/ Solar	788	2.80
Hydro	663	2.80
Renewable	562	4.41
Others	1,522	4.44
Total	**14,838**	**4.33**

Figure 46: Merit Order Analysis of Jharkhand

[135] Report on Resource Adequacy of Jharkhand upto 2033-34, Central Electricity Authority, GoI
[136] Tariff 2024-25, Jharkhand Electricity Regulatory Commission, 2025

2. Region: East

State: Bihar

Bihar's peak electricity demand for FY 2025 was 8.08 GW of with 7.85 GW was met, resulting in a peak deficit of 2.8%[35] . The state has an installed generation capacity of 7.90 GW[137], with 92% from fossil fuel-based sources and remaining from renewable energy sources (RES), including hydro. Despite historical challenges in power infrastructure, Bihar is steadily expanding its renewable energy footprint and currently contributes around 2.1% of India's total renewable energy capacity.

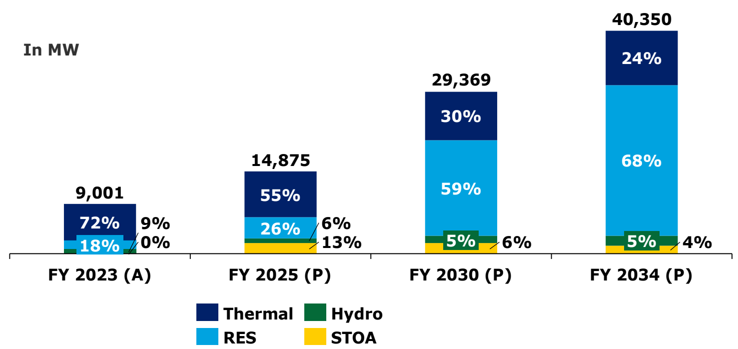
Table 27: Energy Demand of Bihar

Particulars	Units	2021-22 (Actual) [138]	2024-25 (Actual) [139]	2026-27 (Projected) [140]	2031-32 (Projected)
Energy Requirement	TWh	36	44	58	83
Peak Demand	GW	7	8	11	15

Fuel Mix of Bihar [141]

Bihar's contracted power capacity reached 9 GW by the end of March 2023. Of this, 72% is derived from fossil fuel-based sources, primarily coal and remaining 28% from non-fossil fuel-based energy, including hydro, solar, battery, wind, PSP and other RE.

The projected contracted capacity for the year 2033-34 is 40 GW, with 76% from non-fossil fuel and 24% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:



*Type	Source
Thermal	Coal
STOA	Short-term Open Access (STOA)
RES	Solar
RES	Biomass
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Battery
RES	Distributed Renewable Energy (DRE)
Hydro	Hydro

Figure 47: Fuel-wise capacity mix of Bihar[142]

Financial Health of Bihar's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 6% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 94%, are those that are still within their respective due dates but have not yet been fully or partially paid:[143]

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[137] All India Installed Capacity, Central Electricity Authority of India, 2025
[138] Load Generation Balance Report, 2022
[139] Load Generation Balance Report, 2025
[140] National Electricity Plan 2022-32
[141] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[142] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[143] Dashboard for Bihar, PRAAPTI- as accessed on 27th May 2025

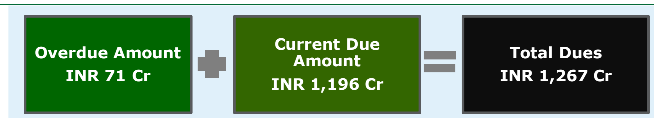


Figure 48: Snapshot of PRAAPTI -Bihar

Utility Name : Bihar State Power Holding Company Limited (BSPHCL)

Bihar's electricity distribution is managed by two State-owned DISCOMs under Bihar State Power Holding Company Limited (BSPHCL):

- North Bihar Power Distribution Company Ltd. (NBPDCL)
- South Bihar Power Distribution Company Ltd. (SBPDCL)

Together, they serve over 19.08 million[144] consumers, with the domestic sector accounting for the largest share, followed by industrial, agricultural and commercial sectors.

Performance History of BSPHCL:

Bihar's power sector demonstrated steady increase in gross energy sold at 18% and gross input energy at 11% . The average cost of supply (ACS) showed a gradual decline from ₹9.06/kWh in FY 2020–21 to INR 8.42 per kWh in FY 2023–24 meanwhile, the average revenue realized (ARR) steadily increased, reaching INR 8.79 per kWh in FY 2023–24. This led to a positive gap between ARR and ACS in the latest year, marking a financial turnaround after years of deficits. PAT followed a similar trend, shifting from significant losses in FY 2020–22 to a modest profit in FY 2022–23 and a substantial profit of INR 1,411 crore in FY 2023–24. Payable days, though still relatively high, remained stable, ranging over the four-year period.[145]

Table 28: Performance Report Snapshot

Key Parameters	Units	FY 2020-21[146]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	23,475	31,311	37,685	38,333
Gross Input Energy	GWh	34,205	40,799	47,731	47,296
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	9.06	8.31	8.52	8.42
Average Revenue Realized (ARR)	INR /kWh	7.99	7.63	8.52	8.79
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(1.07)	(0.68)	-	0.37
Profitability (Surplus/(Deficit))	INR in Crore	(2,523)	(2,128)	7	1,411
Payable Days	Days	104	148	104	109

Capacity Addition Plans of BSPHCL:

Bihar's energy capacity is projected to grow significantly, with a notable shift toward more diversified energy mix. The share of coal expected to decline from 72% in FY 2023 to 24% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. NBPDCL plans to procure additional capacity of 1,000 MW through hydropower till 2034. Capacity Addition requirement has been illustrated below[147]:

[144] India Climate & Energy Dashboard
[145] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[146] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[147] Report on Resource Adequacy of Bihar, Central Electricity Authority, GoI

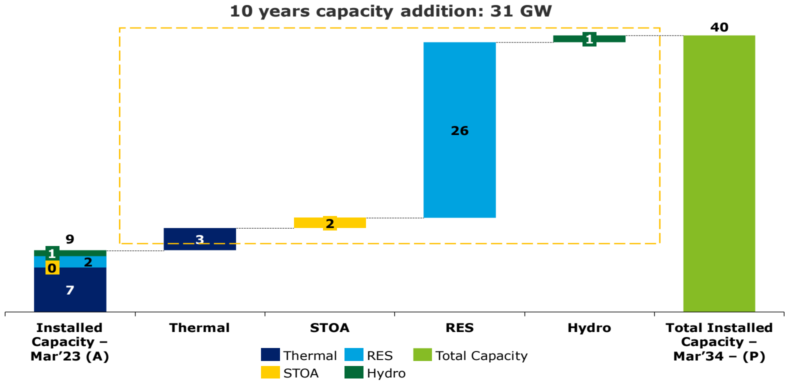
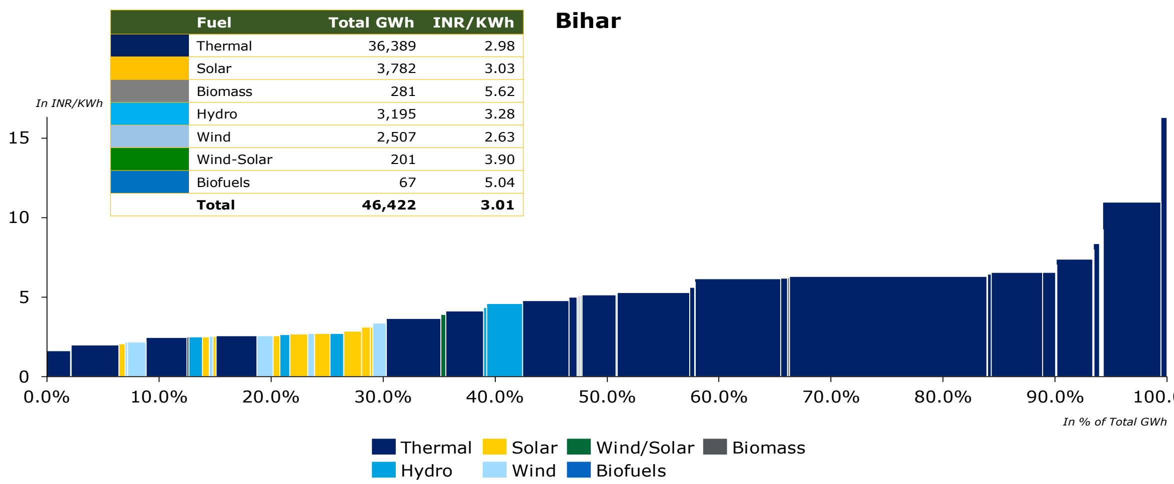


Figure 49: Capacity addition plan for next 10 years of Bihar

Merit Order Analysis[148]:

Bihar's energy generation landscape features a diverse mix of sources, with a total generation of 46,422 GWh. Thermal power remains the dominant contributor, accounting for 33,404 GWh. According to the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal power plants, along with some solar and biomass facilities, at an average cost of INR 8.14 per kWh. Certain thermal assets, such as Barh Stage I Unit III and Barh Stage II, report significantly high generation costs of INR 16.29 and INR 10.95 per kWh, respectively, largely due to elevated fixed costs. Similarly, solar-based projects like the Acme Cleantech and Welspun Renewables report high generation costs of INR 8.74 and INR 8.60 per kWh, respectively. Additionally, the Siddhashram Rice Mill Cluster Pvt. Ltd., a biomass-based plant, records generation cost among renewables at approximately INR 9.22 per kWh.



	Fuel	Total GWh	INR/KWh
	Thermal	36,389	2.98
	Solar	3,782	3.03
	Biomass	281	5.62
	Hydro	3,195	3.28
	Wind	2,507	2.63
	Wind-Solar	201	3.90
	Biofuels	67	5.04
	Total	**46,422**	**3.01**

Figure 50: Merit Order Analysis of Bihar

[148] True-Up for FY 2022-23, Annual Performance Review for FY 2023-24, Annual Revenue Requirement(ARR) and Determination of Retail Tariff forFY2024-25, Bihar Electricity Regulatory Commission

2. Region: East

State: Odisha

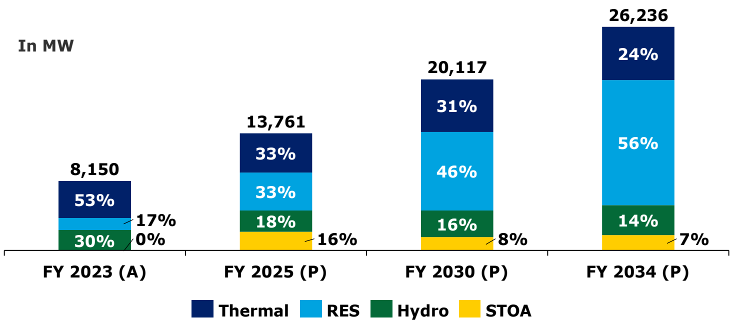
Odisha's peak electricity demand for FY 2025 was 6.9 GW[35]. The state has an installed generation capacity of 8.33 GW[149], with 64% sourced from fossil fuel-based generation and remaining 36% from renewable energy sources (RES), including hydro. Despite its current dependence on coal, Odisha is actively working to diversify its energy mix by expanding its renewable energy portfolio. The state plans to shift its power mix from the current 45:55 ratio of of renewable to conventional energy to 55:45 over the next decade [150].

Table 29: Energy Demand of Odisha

Particulars	Units	2021-22 (Actual) [151]	2024-25 (Actual) [152]	2026-27 (Projected) [153]	2031-32 (Projected)
Energy Requirement	TWh	38	43	49	59
Peak Demand	GW	5.6	7	8	10

Fuel Mix of Odisha[154]

Odisha's contracted power capacity reached 8 GW by the end of March 2023, with 53% sourced from fossil fuel-based generation, primarily coal and the remaining 47% from non-fossil fuel-based sources, including hydro, solar, battery, wind, PSP and other RE. The state's projected contracted capacity for FY 2033-34 is expected to rise significantly to 26 GW. Of this, 76% is anticipated from non-fossil fuel sources while 24% from fossil fuels. This shift highlights Odisha's strategic focus on sustainability and energy diversification. The state's contracted capacity is projected to grow at CAGR of 6% over the ten-year period. Capacity wise generation mix from FY 2023-34 has been illustrated below:



*Type	Source
Thermal	Coal
STOA	Short-term Open Access (STOA)
RES	Solar
RES	Biomass
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Battery
RES	Distributed Renewable Energy (DRE)
Hydro	Hydro

Figure 51: Fuel-wise capacity mix of Odisha[155]

Financial Health of Odisha's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 16% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 84%, are those that are still within their respective due dates but have not yet been fully or partially paid :

[149] All India Installed Capacity, Central Electricity Authority of India, 2025
[150] Green Initiative of Government of Odisha
[151] Load Generation Balance Report, 2022
[152] Load Generation Balance Report, 2025
[153] National Electricity Plan 2022-32
[154] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[155155] As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

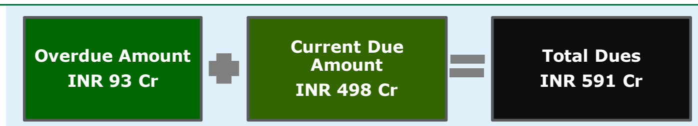


Figure 52: Snapshot of PRAAPTI - Odisha[156]

Utility Name 1: Grid Corporation of Odisha Limited (GRIDCO)

GRIDCO plays a central role in ensuring power availability and financial stability in Odisha's electricity sector. It supplies power to the four Tata Power-led DISCOMs (TPCODL, TPSODL, TPNODL, TPWODL) and manages power purchase agreements, renewable energy obligations, and grid coordination. Odisha's power distribution sector is managed by four regional Tata Power-led distribution companies, operating as joint ventures between Tata Power and the Government of Odisha:

- TP Central Odisha Distribution Ltd. (TPCODL)
- TP Southern Odisha Distribution Ltd. (TPSODL)
- TP Northern Odisha Distribution Ltd. (TPNODL)
- TP Western Odisha Distribution Ltd. (TPWODL)

These DISCOMs serve both urban and rural consumers across the State, covering around 9.89 million consumers, with the industrial sector forming the largest share, followed by domestic, commercial, agricultural and others.

Performance History of GRIDCO:

Odisha's power sector has demonstrated steady growth, with gross energy sold increasing at a CAGR of 15% and gross input energy at 13% over the past four years. The average cost of supply (ACS) rose consistently from INR 5.81 per kWh in FY 2020–21 to INR 6.47 per kWh in FY 2023–24, reflecting rising operational costs. Meanwhile, the average revenue realized (ARR) also improved, contributing to better financial performance. Profit after tax (PAT) remained positive for the last three consecutive years, indicating sustained profitability. Additionally, payable days declined significantly from 106 in FY 2020–21 to just 54 in FY 2023–24, highlighting improved financial discipline and timely settlement of dues. [157]

Table 30: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[158]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	18,717	21,500	26,851	28,710
Gross Input Energy	GWh	24,532	27,443	33,573	35,444
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	5.81	5.88	6.13	6.47
Average Revenue Realized (ARR)	INR /kWh	5.35	6.35	6.44	6.34
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.46)	0.47	0.31	(0.12)
PAT (Profit/(Loss))	INR in Crore	(909)	236	253	307
Payable Days	Days	106	56	56	54

Capacity Addition Plans of Odisha:

156 Dashboard for Odisha, PRAAPTI- as accessed on 27th May 2025
157 Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
158 Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Odisha's energy capacity is projected to grow significantly with diversified energy mix. The share of coal expected to decline from 54% in FY 2023 to 24% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. TPCODL plans to procure additional capacity of 1,101 MW through hydropower till 2034. Capacity Addition requirement has been illustrated below[159]:

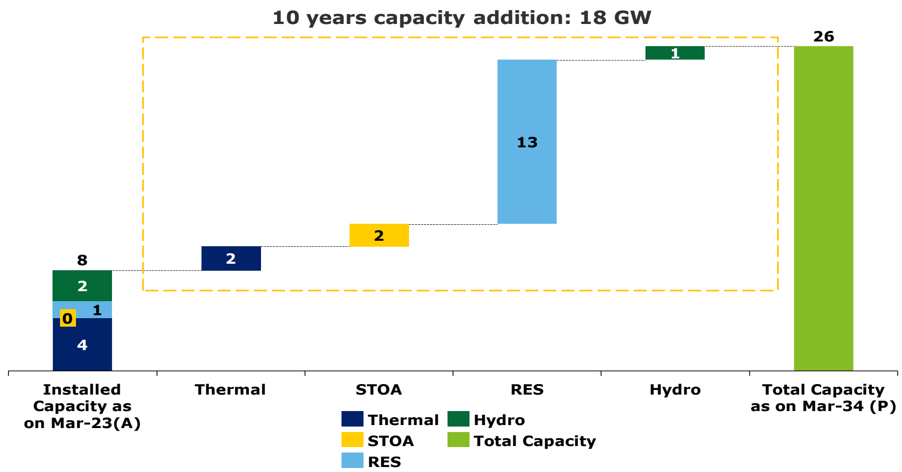
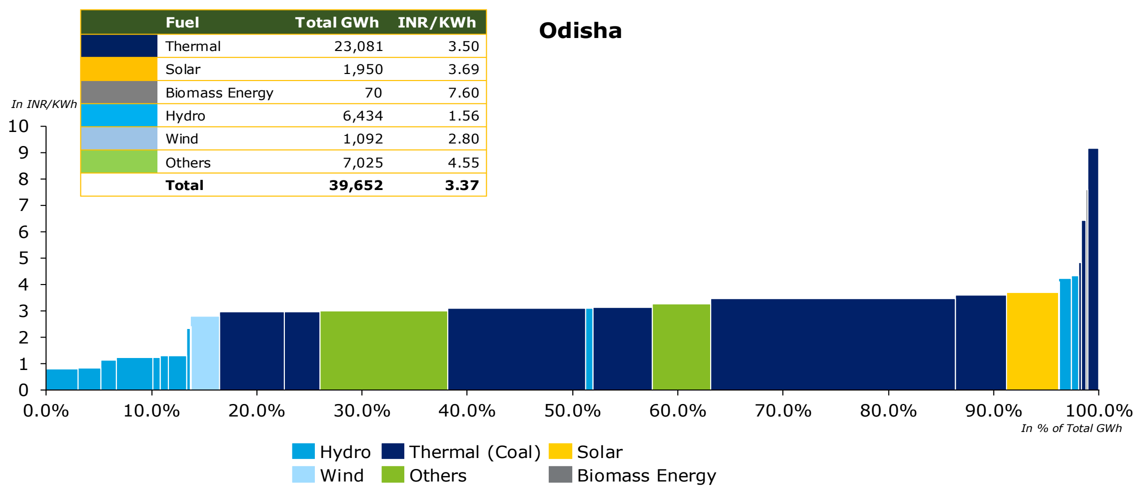


Figure 53: Capacity addition plan for next 10 years of Odisha

Merit Order Analysis[160]:-

Odisha's energy generation landscape features a diverse mix of sources, with total generation reaching 39,652 GWh. Thermal power remains the dominant contributor, accounting for 22,081 GWh. As per the approved tariff order for FY 2023–24, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal and solar and hydro sources, with an average cost of ₹5.15 per kWh. Notably, the state reports significantly high generation costs from Barh I, at ₹9.17 per kWh. These elevated costs are largely attributed to high fixed charges.



Fuel	Total GWh	INR/KWh
Thermal	23,081	3.50
Solar	1,950	3.69
Biomass Energy	70	7.60
Hydro	6,434	1.56
Wind	1,092	2.80
Others	7,025	4.55
Total	**39,652**	**3.37**

Figure 54: Merit Order Analysis of Odisha

[159] Report on Resource Adequacy of Odisha, Central Electricity Authority, GoI
[160] True-Up for FY 2023-24, Odisha Electricity Regulatory Commission, 2023-24

7.1.2. North-Eastern Region

The North-East region plays a strategic role in India's energy sector due to its rich hydropower potential and border proximity. Assam and Tripura have relatively better-developed infrastructure, while other states face challenges like limited grid access and high transmission losses. The region is gradually adopting solar and small hydro projects to enhance energy access. Its integration with the national grid and proximity to Bhutan and Bangladesh also position it well for cross-border electricity trade.

List of All States of Eastern & North-Eastern Region have been Stated below[161]:

Table 31: State-wise Gross Input Energy[162]

State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Assam	13,344	56%
Tripura	3,174	13%
Meghalaya	2,633	11%
Arunachal Pradesh	1,577	7%
Manipur	1,174	5%
Nagaland	1,036	4%
Mizoram	886	4%

Among the above-mentioned States, Assam has been selected from North-Eastern Region for further for detailed level Analysis. Refer below mentioned table:

3. Region: North-East

State: Assam

Assam's peak electricity demand for FY 2025 was 2.81 GW, with 2.68 GW met, resulting in a peak deficit of 4.4%.[35] Assam has an installed generation capacity of 2.37 GW[163], with 68% from fossil fuel-based projects, 32% from renewable energy sources (RES), including hydro. Assam is committed to increasing its renewable energy adoption, aiming to achieve an additional 11,700 MW[164] of renewable power projects by 2030.

Assam is the largest State of North-Eastern part of India, which has approximately 6.4 million[165] electricity consumers. As of March 2023, Assam's total contracted capacity stands at 2,430 MW[166], with 31% derived from hydro and 9% from solar Energy.

Table 32: Energy Demand of Assam

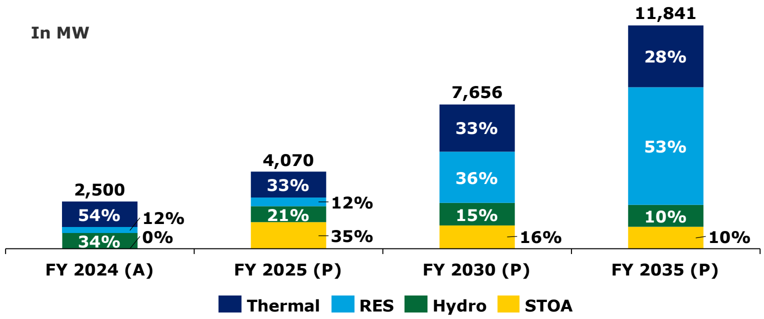
Particulars	Units	2021-22 (Actual) [167]	2024-25 (Actual) [168]	2026-27 (Projected) [169]	2031-32 (Projected)
Energy Requirement	TWh	11	13	15	20
Peak Demand	GW	2	3	3	4

[161] Central Electricity Authority
[162] Report on Performance Utilities, Power Finance Corporation Limited, 2023-24
[163] All India Installed Capacity, Central Electricity Authority of India, 2025
[164] Assam: Policy targets 11,700-MW renewable energy in State, 2025
[165] APDCL
[166] Resource Adequacy Report of Assam, Central Electricity Authority, 2024
[167] Load Generation Balance Report, 2022
[168] Load Generation Balance Report, 2025
[169] National Electricity Plan 2022-32

Fuel Mix of Assam [170]

Assam's contracted power capacity reached 25 GW by the end of March 2024. Of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2034-35 is 118 GW, out of which 72% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 17% over the period of next 10 years. Capacity wise generation mix from FY 2024-35 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Solar
RES	Storage
RES	Wind
RES	Distributed Renewable Energy (DRE)
Hydro	Hydro

Figure 55: Fue-wise capacity mix of Assam*

Financial Health of Assam's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 1% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 99%, are payments still within their respective due dates but not yet fully or partially paid:[171]

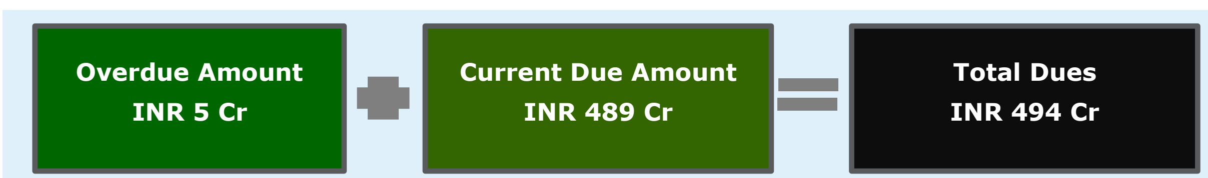


Figure 56: Snapshot of PRAAPTI - Assam

Utility Name : Assam Power Distribution Company Limited (APDCL)

Assam Power Distribution Company Limited (APDCL) is a pivotal entity in the state's electricity sector, catering to approximately 6.7 million consumers. APDCL's advancement of solar energy projects has enhanced energy security and socio-economic growth, positioning the company as a key player in Assam's sustainable energy development. Electricity consumers are distributed unevenly across sectors, with domestic usage at 42.99%, followed by industrial (26.3%), commercial (13.62%), and the remaining in other categories.

[170] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[171] Dashboard for Assam, PRAAPTI- as accessed on 27th May 2025
***** As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type

Performance History:

In FY 2023–24, APDCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold also demonstrated steady growth of 8%. The Average Cost of Supply (ACS) declined marginally to INR 9.51 per kWh, while the Average Revenue Realized (ARR) increased to INR 9.77 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite losses having quadrupled over the past few years, an increase was again observed in FY 2023–24. Payable days have steadily decreased over the past four years and remain favorable compared to the national average (excluding private DISCOMs) of 130 days in FY 2023–24. [172]

Table 33:Performance Report Snapshot

Key Parameters	Units	FY 2020-21[173]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	8,786	9,535s	10,290	10,955
Gross Input Energy	GWh	11,004	11,953	12,813	13,344
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.48	7.98	10.03	9.51
Average Revenue Realized (ARR)	INR/kWh	8.15	8.33	9.25	9.77
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.33)	0.35	(0.78)	0.26
Profitability (Surplus/(Deficit))	INR in Crore	(292)	336	(800)	384
Payable Days	Days	65	37	59	41

Capacity Addition Plans of APDCL

As stated above, Assam's energy capacity is projected to grow significantly and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 54% in FY 2024 to 28% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. APDCL plans to purchase additional capacity through short term open access (STOA) whereas no such plan for Hydropower has been noted till 2035. Capacity Addition requirement has been illustrated below[174]:

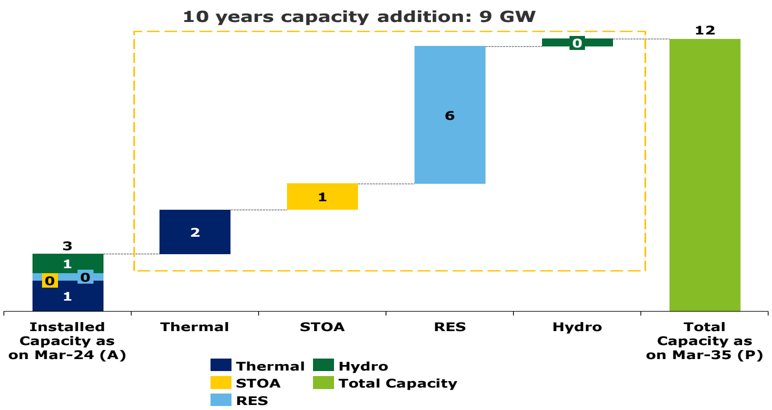
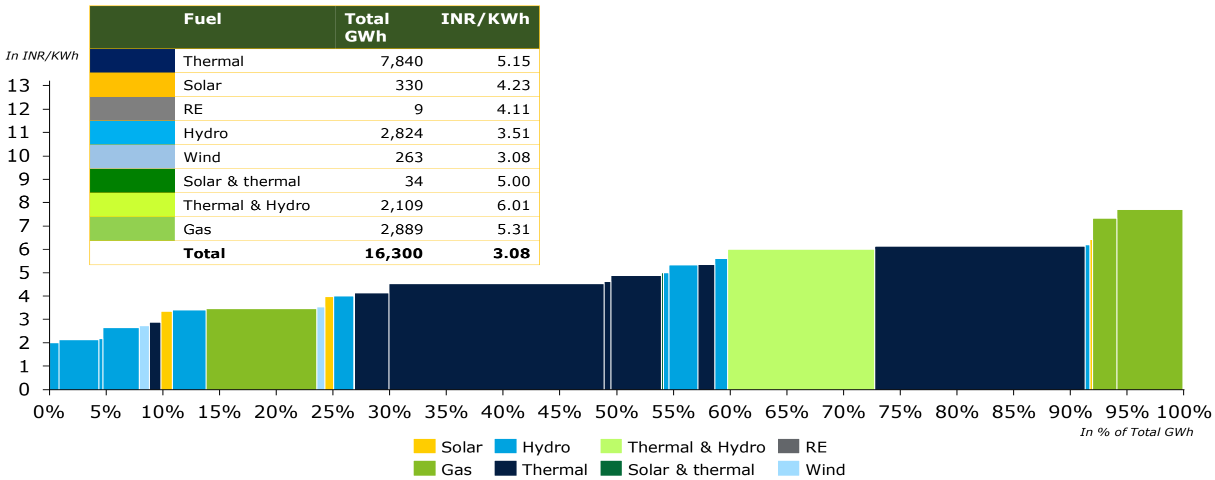


Figure 57: Capacity Addition Plan for next 10 years of Assam

[172] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[173] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[174] Report on Resource Adequacy of Assam upto 2034-35, Central Electricity Authority, GoI

Merit Order Analysis[175]:-

Assam's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 16,300 GWh. Of this, thermal power accounts for approximately 48% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 6.15 per kWh. Additionally, the hydropower plant at DHEP has a generation cost of INR 6.19 per kWh.



Fuel	Total GWh	INR/KWh
Thermal	7,840	5.15
Solar	330	4.23
RE	9	4.11
Hydro	2,824	3.51
Wind	263	3.08
Solar & thermal	34	5.00
Thermal & Hydro	2,109	6.01
Gas	2,889	5.31
Total	**16,300**	**3.08**

Figure 58: Merit Order Analysis of Assam

175 True-Up for FY 2023-24, Assam Electricity Regulatory Commission, 2023-24

7.1.3. Northern Region

The Northern region of India presents strong potential for offtake from Bhutan's Dorjilung Hydropower Project. The region has high and seasonal electricity demand, particularly during summer peaks, making it well-suited for reliable peaking power. Additionally, the Hydro Purchase Obligation (HPO) mandates clean energy procurement, aligning with Dorjilung's profile. Financially stable DISCOMs in Delhi and Haryana further enhance the region's viability for long-term power offtake agreements. List of All States of Northern Region have been Stated below[176]:

Table 34: State-wise Gross Input Energy[177]

State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Uttar Pradesh	1,34,269	29%
Rajasthan	1,12,663	25%
Punjab	71,876	16%
Haryana	66,375	15%
Delhi	37,767	8%
Uttarakhand	17,233	4%
Himachal Pradesh	15,551	3%
Ladakh	315	0%
Chandigarh	-	0%
Jammu & Kashmir	-	0%

Among the above-mentioned States, Uttar Pradesh, Rajasthan, Punjab, Haryana, and Delhi has been selected further for detailed level Analysis as it covers approximately 93% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 35: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Uttar Pradesh	Pashchimanchal Vidyut Vitran Nigam Ltd	39,108	28%
	Purvanchal Vidhyut Vitran Nigam Ltd	32,930	24%
	Dakshinanchal Vidyut Vitran Nigam Ltd	29,628	21%
	Madhyanchal Vidyut Vitran Nigam Ltd	28,314	21%
	Kanpur Electricity Supply Company	4,289	3%
	Noida Power Company Limited	3,615	3%
	Total of Uttar Pradesh	**1,37,884**	
Rajasthan	Jaipur Vidyut Vitran Nigam Limited	42,907	38%
	Jodhpur Vidyut Vitran Nigam Limited	38,762	34%
	Ajmer Vidyut Vitran Nigam Limited	30,994	28%
	Total of Rajasthan	**1,12,663**	

[176] Central Electricity Authority

[177] Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Punjab	Punjab State Power Corporation Limited	71,876	100%
	Total of Punjab	**71,876**	
Haryana	Dakshin Haryana Bijli Vitran Nigam	39,880	60%
	Uttar Haryana Bijli Vitran Nigam	26,495	40%
	Total of Haryana	**66,375**	
Delhi	BSES Rajdhani Power Limited	15,835	40%
	Tata Power Delhi Distribution Limited	12,686	32%
	BSES Yamuna Power Limited	9,246	24%
	New Delhi Municipal Council	1,568	4%
	Total of Delhi	**39,335**	

4. Region: North

State: Uttar Pradesh

Uttar Pradesh's peak electricity demand for FY 2025 was about 30.62 GW. The state has an installed generation capacity of 33.82 GW, with 71.8% from fossil fuel–based projects, 27.4% from renewable energy sources (RES), including hydro, and the remaining 0.8% from nuclear projects. The state is actively investing in renewable energy through various schemes like the Kusum Yojana and the Village Electrification Program, aiming to enhance its capacity and reduce reliance on fossil fuels.

Table 36: Energy Demand of Uttar Pradesh

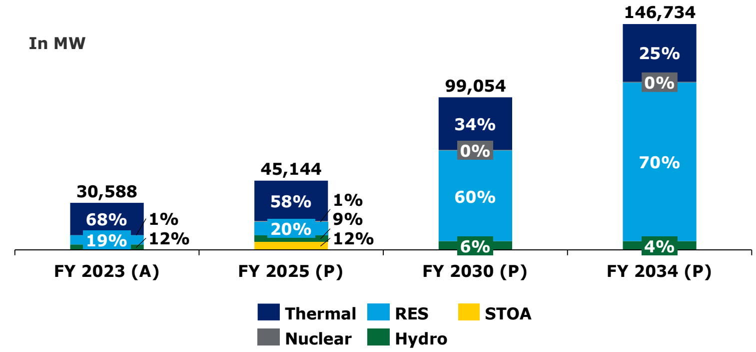
Particulars	Units	2021-22 (Actual) [178]	2024-25 (Actual) [179]	2026-27 (Projected) [180]	2031-32 (Projected) [181]
Energy Requirement	TWh	129	165	191	254
Peak Demand	GW	25	31	33	44

Fuel Mix of Uttar Pradesh [182]

Uttar Pradesh’s contracted power capacity reached 31 GW by the end of March 2023. Of this capacity, 68% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 147 GW, out of which 75% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 11% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:

[178] Load Generation Balance Report, 2022
[179] Load Generation Balance Report, 2025
[180] National Electricity Plan 2022-32
[181] National Electricity Plan 2022-32
[182] Resource Adequacy Report, Central Electricity Report, GOI, 2024
* As per the Government of India’s classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Wind
RES	Storage
RES	Distributed Renewable Energy (DRE)
Nuclear	Nuclear
Hydro	Hydro

Figure 59: Fuel-wise capacity mix of Uttar Pradesh*

Financial Health of Uttar Pradesh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 48% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 52%, are payments still within their respective due dates but not yet fully or partially paid: :[183]

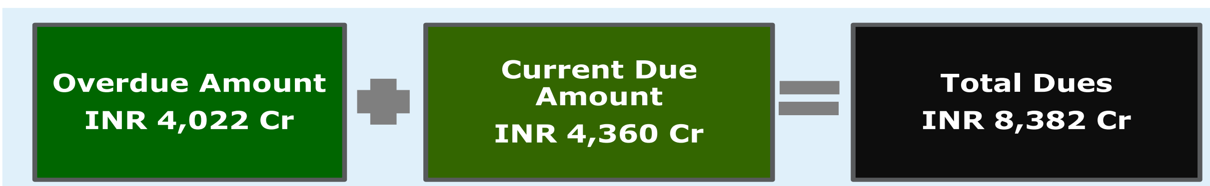


Figure 60: Snapshot of PRAAPTI - Uttar Pradesh

Utility Name: Uttar Pradesh Power Corporation Limited (UPPCL)

Uttar Pradesh Power Corporation Limited (UPPCL) is a state government-owned utility responsible for electricity transmission and distribution across Uttar Pradesh. Established in 2000, UPPCL ensures reliable power supply to millions of consumers through its extensive network and advanced technologies. For efficient operation & management, UPPCL is divided further divisions:

- Dakshinanchal Vidyut Vitaran Nigam Limited (DVVNL)
- Madhyanchal Vidyut Vitaran Nigam Limited (MVVNL)
- Pashchimanchal Vidyut Vitaran Nigam Limited (PVVNL)
- Purvanchal Vidyut Vitaran Nigam Limited (PUVVNL)
- Kanpur Electricity Supply Company (KESCO)

UPPCL serves 30.88 million consumers with a diverse consumer base, where the domestic sector accounts for approximately 41% of total electricity consumption. This is followed by the agricultural sector at 16%, industrial consumers at 13%, and the commercial segment at 7%, with the remaining share attributed to other sectors

Performance History:

In FY 2023–24, UPPCL reported consistent increase in gross input energy over the past four years at CAGR of 6% similarly, gross energy sold has also demonstrated steady growth of 8%. The Average Cost of Supply (ACS) declined marginally to INR 9.20 per kWh while Average Revenue realized (ARR) has increased to INR 8.57 per kWh in FY 2023-24, narrowing the gap between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past four years, a decline was

[183] Dashboard for Uttar Pradesh, PRAAPTI- as accessed on 27th May 2025

observed over in FY 2023-24. Payable Days, while still high, they have steadily decreased over the past three years and unfavourable compared to the national average (excluding private DISCOMs) of 130 days in FY 2023-24. [184]

Table 37: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[185]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	90,372	93,745	108,410	113,477
Gross Input Energy	GWh	113,859	116,885	130,059	134,269
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	8.64	9.25	9.74	9.20
Average Revenue Realized (ARR)	Rs/kWh	7.46	8.55	8.14	8.57
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(1.18)	(0.70)	(1.60)	(0.63)
PAT (Profit/(loss))	Rs in Crore	(10,660)	(6,498)	(15,512)	(7,058)
Payable Days	Days	208	203	172	157

Capacity Addition Plans of UPPCL:-

Uttar Pradesh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 68% in FY 2023 to 25% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. UPPCL has planned to procure 1,808 MW of hydropower till 2034. The state's planned capacity addition for next 10 years has been stated below[186]:

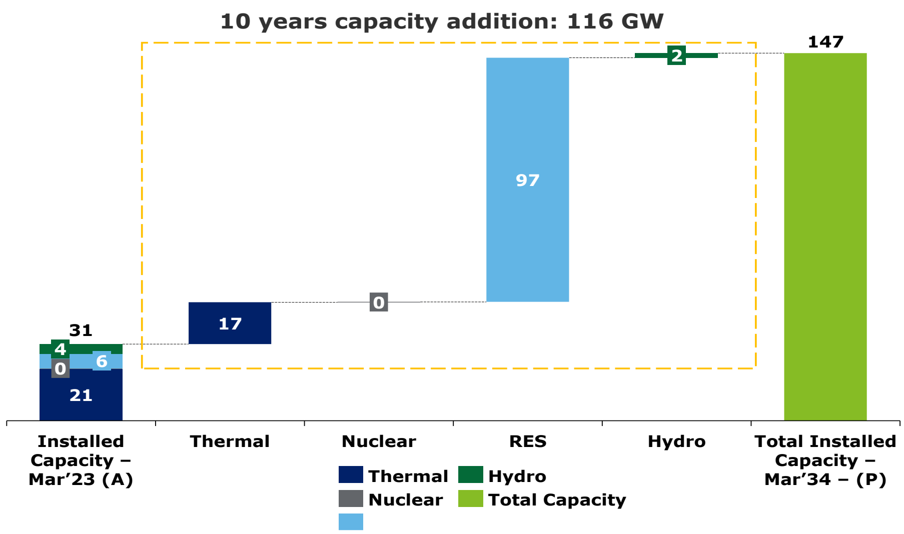


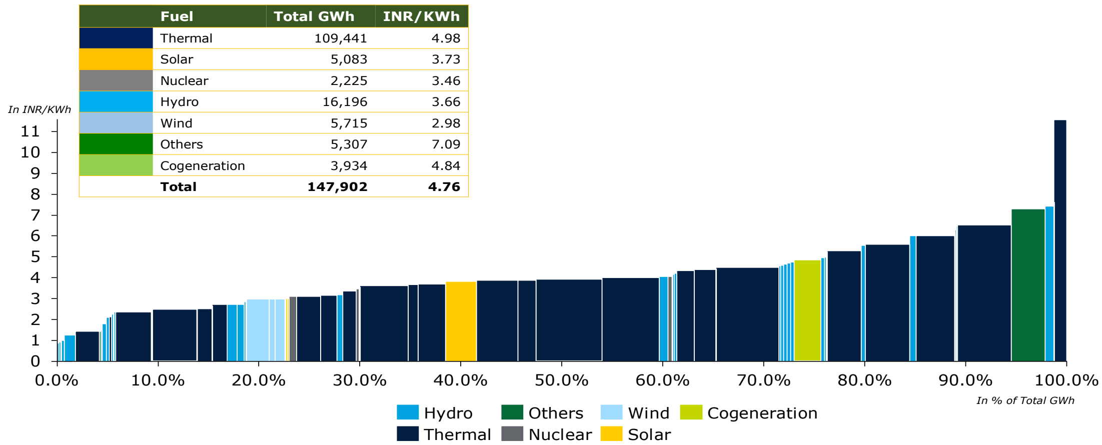
Figure 61: Capacity Addition Plan for next 10 years of Uttar Pradesh

Merit Order Analysis[187]:-

Uttar Pradesh's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 147,902 GWh. Of this, thermal power accounts for approximately 74% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission,

[184] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[185] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[186] Report on Resource Adequacy of Uttar Pradesh, Central Electricity Authority, GoI
[187] True-Up for FY 2022-23, Uttar Pradesh Electricity Regulatory Commission, 2024

the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 7 per kWh. However, some older thermal assets, such as KSK Mahanandi and Ghatampur, report significantly higher generation costs of INR 11.58 and INR 6.53 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 7.07 per kWh.



Fuel	Total GWh	INR/KWh
Thermal	109,441	4.98
Solar	5,083	3.73
Nuclear	2,225	3.46
Hydro	16,196	3.66
Wind	5,715	2.98
Others	5,307	7.09
Cogeneration	3,934	4.84
Total	**147,902**	**4.76**

Figure 62: Merit Order Analysis of Uttar Pradesh

4. Region: North

State: Rajasthan

Rajasthan's peak electricity demand for FY 2025 was approximately 19.17 GW. The state has an installed generation capacity of approximately 51.29 GW, with about 29.2% from fossil fuel-based sources, 69.7% from renewable energy sources (RES), including hydro, and the remaining 1.1% from nuclear power. Rajasthan is a front-runner in renewable energy adoption, contributing over 11.5% to India's total renewable energy capacity, with a strong focus on solar and wind power.

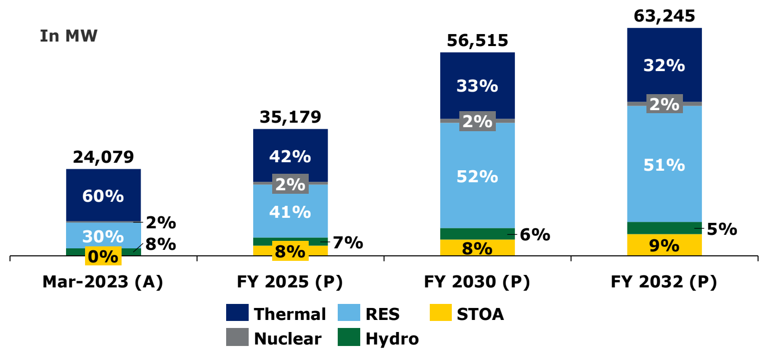
Table 38: Energy Demand of Rajasthan

Particulars	Units	2021-22 (Actual) [188]	2024-25 (Actual) [189]	2026-27 (Projected) [190]	2031-32 (Projected) [191]
Energy Requirement	TWh	90	114	134	172
Peak Demand	GW	16	19	21	27

Fuel Mix of Rajasthan[192]

Rajasthan's contracted power capacity reached 24 GW by the end of March 2023. Of this capacity, 60% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted Capacity for the year 2031-32 is 63 GW, out of which 68% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 8 years. Capacity wise generation mix from 2023 to 2032 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 63: Capacity Mix of Rajasthan*

Financial Health of Rajasthan's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 33% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 67%, are payments still within their respective due dates but not yet fully or partially paid:[193]

[188] Load Generation Balance Report, 2022
[189] Load Generation Balance Report, 2025
[190] National Electricity Plan 2022-32
[191] National Electricity Plan 2022-32
[192] Resource Adequacy Report of Rajasthan
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[193] Dashboard for Rajasthan, PRAAPTI- as accessed on 27th May 2025

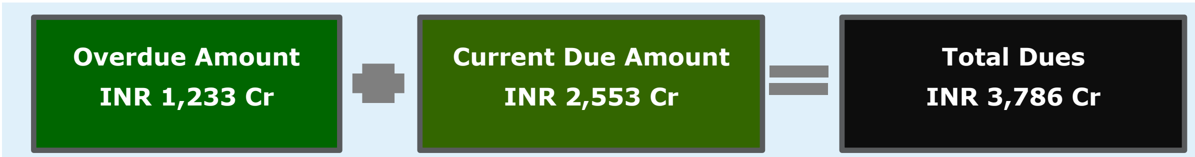


Figure 64: Snapshot of PRAAPTI - Rajasthan

Utility Name : Rajasthan State Power Corporation Limited (RSPCL)

Rajasthan State Power Corporation Limited (RSPCL) is a state government-owned utility responsible for electricity transmission and distribution across Rajasthan. Established in 1999, RSPCL ensures reliable power supply to millions of consumers through its extensive network and advanced technologies. For efficient operation & management, RSPCL is divided further divisions

Rajasthan's power distribution is managed primarily by three State-owned DISCOMs:

- Jaipur Vidyut Vitran Nigam Ltd (JVVNL)
- Ajmer Vidyut Vitran Nigam Ltd (AVVNL)
- Jodhpur Vidyut Vitran Nigam Ltd (JdVVNL)

These utilities collectively serve 16.32 million[194] consumers in the state vast and diverse consumer base majorly to Agriculture followed by Industrial, Domestic, Commercial and Others.

Performance History:

In FY 2023–24, RSPCL reported consistent increase in gross input energy over the past four years at CAGR of 9% similarly, gross energy sold has also demonstrated steady growth of 9%. The Average Cost of Supply (ACS) has increased to INR 8.81 per kWh and Average Revenue realized (ARR) has increased to INR 8.86 per kWh in FY 2023-24, improving the margin between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past three years, a turnaround was observed in FY 2023–24, with the company returning to profitability. Payable Days, while still high, they have steadily decreased over the past four years and improve compared to the national average (excluding private DISCOMs) of 130 days in FY 2023-24[195].

Table 39: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[196]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	66,553	73,189	80,951	87,023
Gross Input Energy	GWh	87,083	94,302	1,03,662	112,663
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.73	8.20	8.44	8.81
Average Revenue Realized (ARR)	INR/kWh	8.39	8.14	8.13	8.86
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.34)	(0.06)	(0.31)	0.05
Profitability (Surplus/(Deficit))	INR in Crore	(2,217)	(4720	(2,514)	505
Payable Days	Days	256	203	88	63

Capacity Addition Plans of RSPCL:-

[194] India Climate & Energy Dashboard
[195] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[196] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Rajasthan's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 60% in FY 2023 to 32% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. RSPCL plans to procure 1,342 MW of hydropower till 2032. The state's planned capacity addition for next 10 years has been stated below.[197]

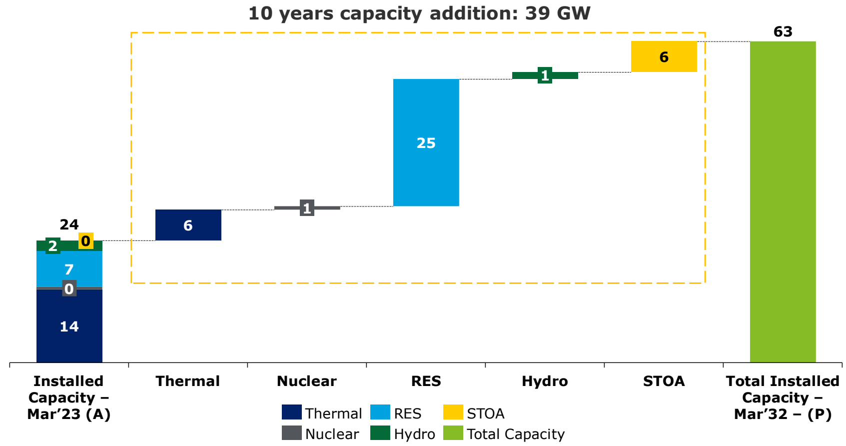


Figure 65: Capacity addition plan for next 10 years of Rajasthan

Merit Order Analysis [198]:-

Rajasthan's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 106,169 GWh. Of this, thermal power accounts for approximately 71% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 5.03 per kWh. However, Coastal Gujarat report significantly higher generation costs of INR 5.21 per kWh. Additionally, the Hydropower plant at PTC(Sikkim Urja Ltd) has a generation cost of INR 4.89 per kWh.

Fuel	Total GWh	INR/KWh
Thermal	75,862	4.48
Solar	833	2.92
Nuclear	3,209	3.65
Hydro	3,159	3.68
Hybrid	23,106	3.56
Total	**106,169**	**4.22**

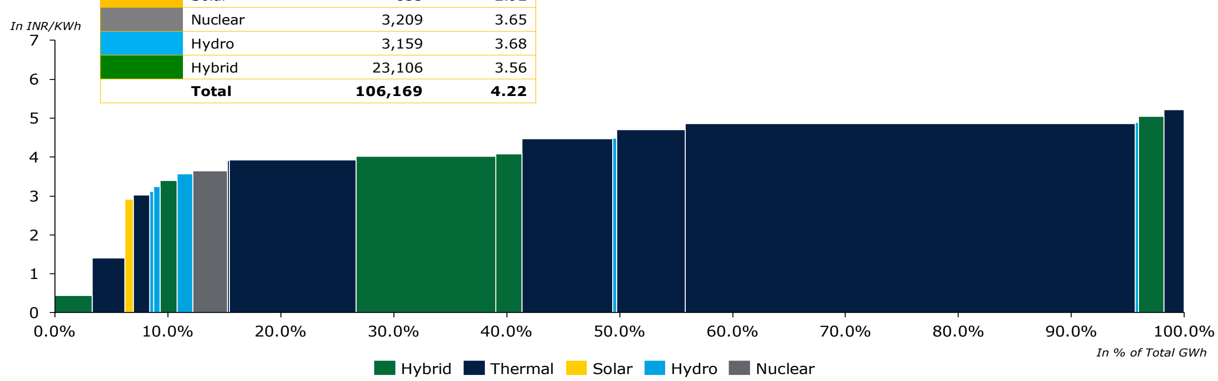


Figure 66: Merit Order Analysis of Rajasthan

[197] Resource Adequacy Report, Central Electricity Report, GOI, 2024
[198] MYT Order, Rajasthan Electricity Regulatory Commission, Jaipur

4. Region: North

State: Punjab

Punjab's peak electricity demand for FY 2025 was 16.06 GW. The state has an installed generation capacity of approximately 14.56 GW, with around 57.4% from fossil fuel-based sources, 41.2% from renewable energy sources (RES), including hydro, and the remaining 1.4% from nuclear power. Punjab is steadily increasing its renewable energy footprint and currently accounts for about 2.5% of India's total renewable energy capacity, with a strong focus on solar and wind power.

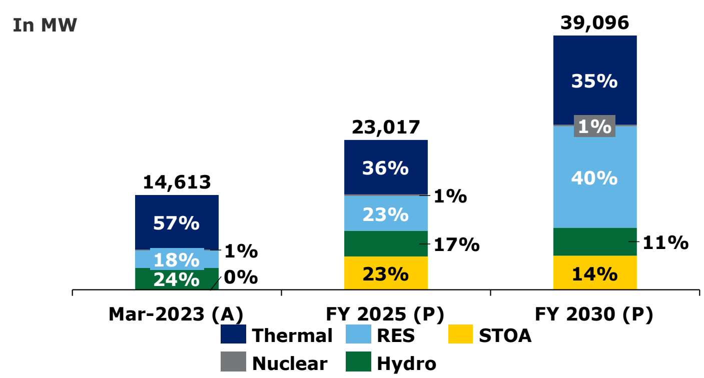
Table 40: Energy Demand of Punjab

Particulars	Units	2021-22 (Actual)	2024-25 (Actual)	2026-27 (Projected)	2031-32 (Projected)
Energy Requirement	TWh	63	77	82	105
Peak Demand	GW	14	16	17	21

Fuel Mix of Punjab [199]

Punjab's contracted power capacity reached 15 GW by the end of March 2023. Of this capacity, 57% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted Capacity for the year 2029-30 is 39 GW, out of which 65% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 17% over the next 6 years. Capacity wise generation mix from 2023 to 2030 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass/ Bioenergy
RES	Solar
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 67: Fuel-wise capacity mix of Punjab*

Financial Health of Punjab's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 25% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 75%, are payments still within their respective due dates but not yet fully or partially paid:[200]

[199] Resource Adequacy Report of Punjab
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[200] Dashboard for Punjab, PRAAPTI- as accessed on 27th May 2025

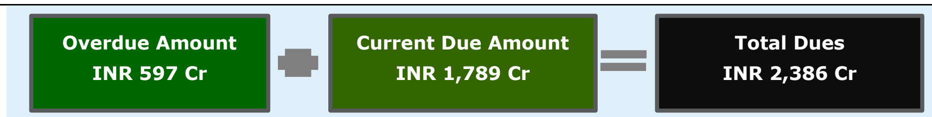


Figure 68: Snapshot of PRAAPTI - Punjab

Utility Name : Punjab State Power Corporation Limited (PSPCL)

Punjab State Power Corporation Limited (PSPCL) is a government-owned entity responsible for electricity generation and distribution in Punjab. Established in 2010, PSPCL operates several thermal and hydropower plants, ensuring a reliable energy supply. It plays a crucial role in maintaining a stable and efficient electricity network, supporting the state's economic growth. The major consumer base in Punjab comprises industrial consumers, followed by domestic, agricultural, commercial, and other categories

Performance History:

In FY 2023–24, PSPCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 4%. Similarly, gross energy sold also demonstrated steady growth of 4%. The Average Cost of Supply (ACS) decreased to INR 7.06 per kWh, while the Average Revenue Realized (ARR) increased to INR 7.19 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite incurring losses in FY 2022–23, a turnaround was observed in FY 2023–24, with the company returning to profitability. Payable days increased in FY 2022–23 but decreased again in FY 2023–24, which is an improvement compared to the national average (excluding private DISCOMs) of 130 days in the same year. [201]

Table 41: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[202]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	49,729	53,166	59,089	61,136
Gross Input Energy	GWh	58,137	62,154	69,269	71,876
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	6.60	6.59	7.46	7.06
Average Revenue Realized (ARR)	Rs/kWh	6.89	6.79	6.65	7.19
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	0.29	0.20	(0.81)	0.13
Profitability (Surplus/(Deficit))	Rs in Crore	1,446	1,069	(4,776)	800
Payable Days	Days	65	45	55	40

Capacity Addition Plans of PSPCL:-

Punjab's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 57% in FY 2023 to 35% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below[203]:

[201] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[202] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[203] Report on Resource Adequacy of Punjab, Central Electricity Authority, GoI

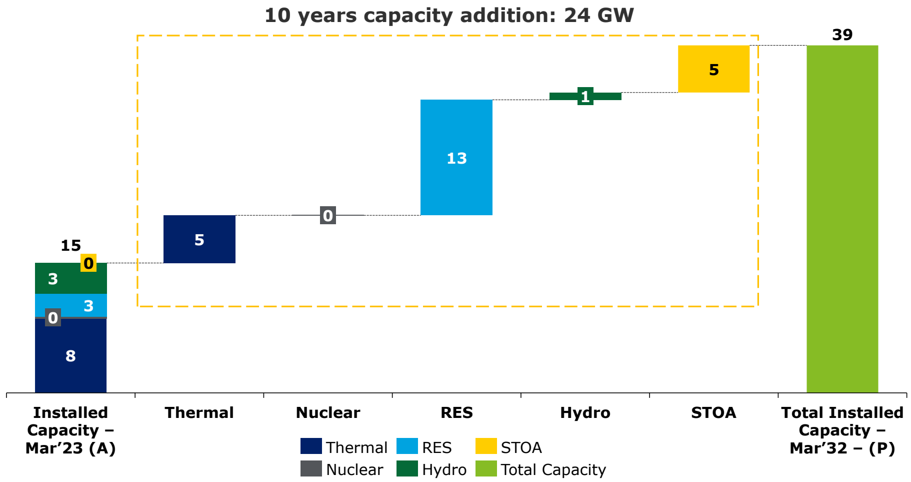
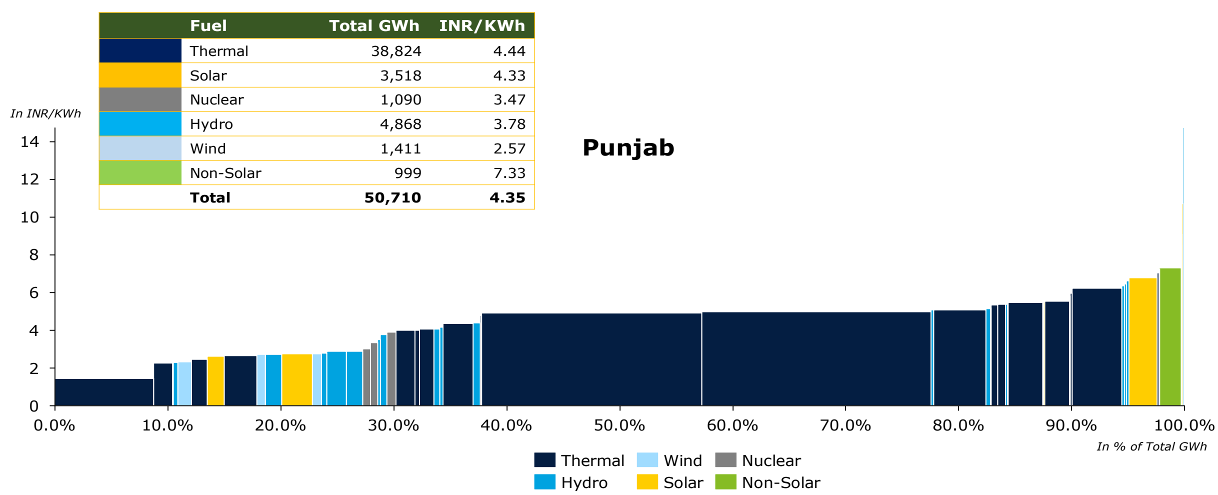


Figure 69: Capacity Addition Plan for next 10 years of Punjab

Merit Order Analysis[204]:-

Punjab's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 50,710 GWh. Of this, thermal power accounts for approximately 77% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 5.96 per kWh. However, some older thermal assets, such as NPL UI and Unchahar-1, report significantly higher generation costs of INR 8.53 and INR 7.24 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 7.33 per kWh, reflecting a moderate cost profile within the state's energy mix.



Fuel	Total GWh	INR/KWh
Thermal	38,824	4.44
Solar	3,518	4.33
Nuclear	1,090	3.47
Hydro	4,868	3.78
Wind	1,411	2.57
Non-Solar	999	7.33
Total	**50,710**	**4.35**

Figure 70: Merit Order Analysis of Punjab

[204] True-Up for FY 2023-24 and ARR for 3rd MYT Control Period, Punjab Electricity Regulatory Commission, 2024

4. Region: North

State: Haryana

Haryana's peak electricity demand for FY 2025 was approximately 14.66 GW. The state has an installed generation capacity of 14.15 GW, with about 65.2% from fossil fuel-based sources, 34.2% from renewable energy sources (including hydro), and the remaining 0.7% from nuclear power. The state is steadily expanding its renewable energy footprint and currently hosts around 2.3% of India's total renewable energy capacity.

Table **42**: Energy Demand of Haryana

Particulars	Units	2021-22 (Actual) [205]	2024-25 (Actual) [206]	2026-27 (Projected) [207]	2031-32 (Projected) [208]
Energy Requirement	TWh	55	70	83	115
Peak Demand	GW	12	15	16	22

Fuel Mix of Haryana [209]

Haryana's contracted power capacity reached 14 GW by the end of March 2024. Of this capacity, 62% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The total projected contracted Capacity for the year 2034-35 is 49 GW, out of which 71% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:

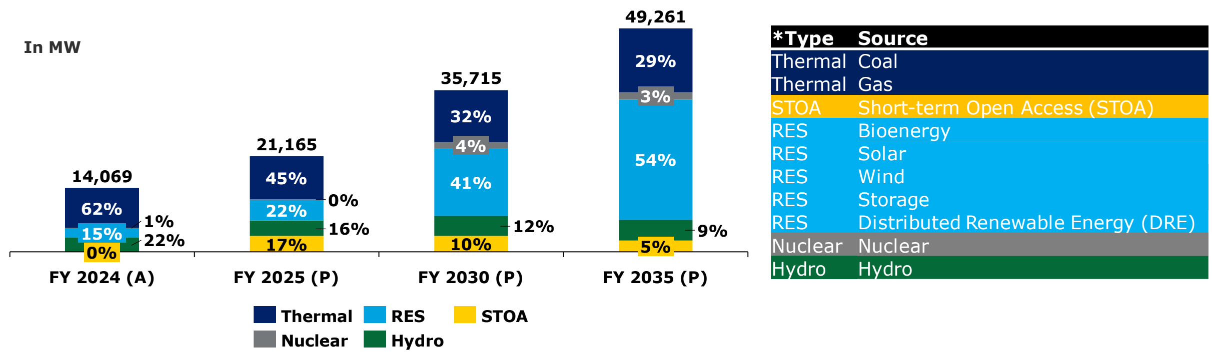


Figure 71: Fuel-wise capacity mix of Haryana*

Financial Health of Haryana's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 5% of the total, refer to payments that have surpassed their due dates and remain either partially or fully

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[205] Load Generation Balance Report, 2022
[206] Load Generation Balance Report, 2025
[207] National Electricity Plan 2022-32
[208] National Electricity Plan 2022-32
[209] Resource Adequacy Report of Haryana
***** As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

unpaid. In contrast, current dues, accounting for the remaining 95%, are payments still within their respective due dates but not yet fully or partially paid:[210]

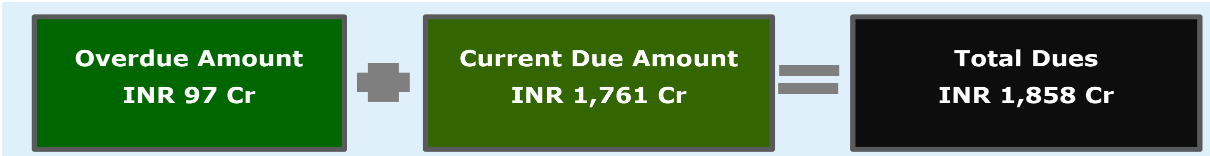


Figure 72: Snapshot of PRAAPTI - Haryana

Utility Name : Haryana Power Generation Corporation Limited (HPGCL)

Haryana Power Generation Corporation Limited (HPGCL) is a government-owned entity responsible for electricity generation in Haryana. Established in 1997, HPGCL operates several thermal and hydropower plants, ensuring a reliable energy supply. Haryana's electricity sector is managed by two main distribution companies: Uttar Haryana Bijli Vitran Nigam (UHBVN) and Dakshin Haryana Bijli Vitran Nigam (DHBVN). Both the Utilities serve 7.94 million consumers[211] majorly to Industrial consumers, followed by Domestic, Agriculture, Commercial and Others.

Performance History:

In FY 2023–24, HPGCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold demonstrated steady growth of 10%. The Average Cost of Supply (ACS) decreased to INR 7.16 per kWh, and the Average Revenue Realized (ARR) also decreased to INR 7.21 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite increasing profits in FY 2022–23, a turnaround was observed in FY 2023–24, with the company experiencing reduced profitability. Payable days increased in FY 2022–23 but decreased again in FY 2023–24, improving compared to the national average (excluding private DISCOMs) of 130 days in the same year. [212]

Table 43: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[213]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	43,165	47,501	54,445	57,433
Gross Input Energy	GWh	53,762	56,856	63,286	66,375
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	6.50	6.72	7.88	7.16
Average Revenue Realized (ARR)	Rs/kWh	6.64	6.89	8.06	7.21
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	0.15	0.18	0.18	0.05
Profitability (Surplus/(Deficit))	Rs in Crore	637	849	975	276
Payable Days	Days	55	45	62	38

Capacity Addition Plans of HPGCL:-

Haryana's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2025 to

[210] Dashboard for Haryana, PRAAPTI- as accessed on 27th May 2025
[211] India Climate & Energy Dashboard
[212] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[213] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

29% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below[214]:

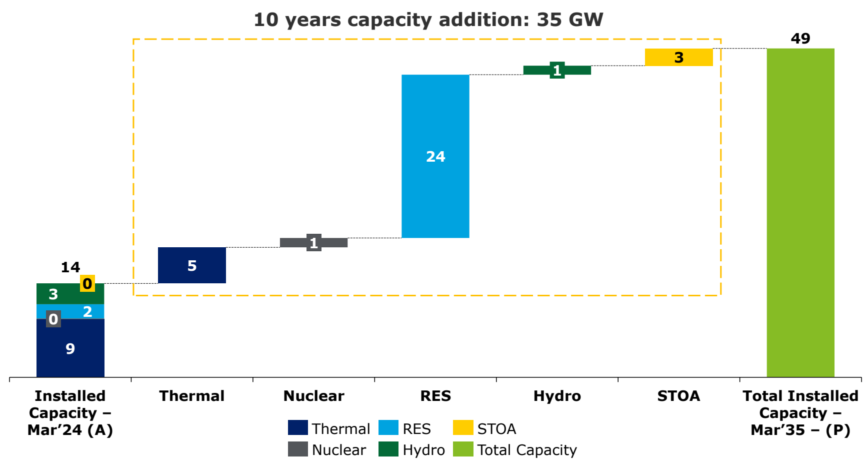
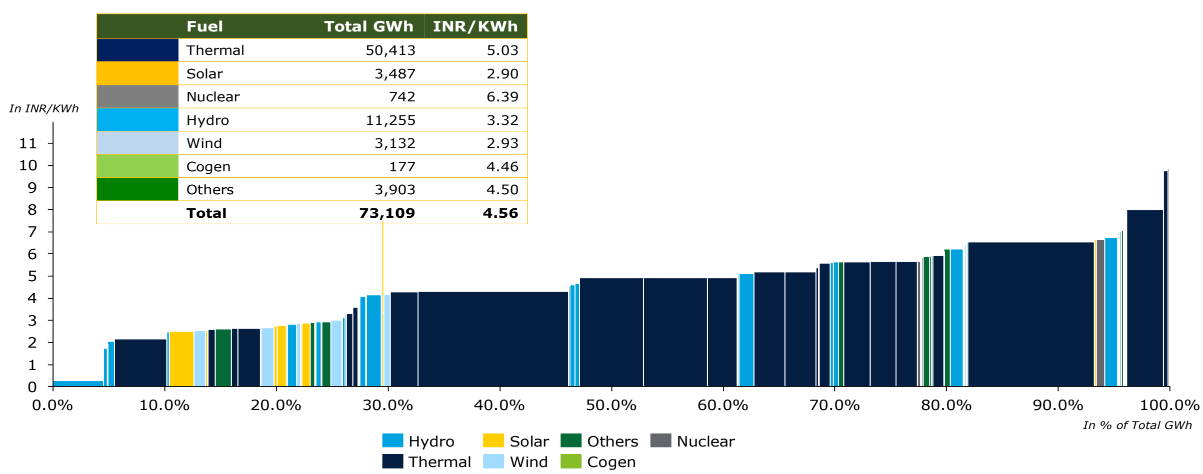


Figure 73: Capacity Addition Plan for next 10 years of Haryana

Merit Order Analysis[215]:-

Haryana's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 73,109 GWh. Of this, thermal power accounts for approximately 69% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of thermal power plants, with an average generation cost of INR 8.05 per kWh. However, some older thermal assets, such as Feroz Gandhi Unchahar-3 TPS and Feroz Gandhi Unchahar-1 TPS, report significantly higher generation costs of INR 11.95 and INR 11.27 per kWh, respectively. Additionally, the nuclear plant has an average generation cost of INR 6.16 per kWh. Other non-solar renewable sources contribute at an average tariff of INR 6.84 per kWh, reflecting a moderate cost profile within the state's energy mix.



Fuel	Total GWh	INR/KWh
Thermal	50,413	5.03
Solar	3,487	2.90
Nuclear	742	6.39
Hydro	11,255	3.32
Wind	3,132	2.93
Cogen	177	4.46
Others	3,903	4.50
Total	**73,109**	**4.56**

Figure 74: Merit Order Analysis of Haryana

[214] Report on Resource Adequacy of Haryana, Central Electricity Authority, GoI
[215] True-Up for FY 2022-23, Haryana Electricity Regulatory Commission, 2024

4. Region: North

State: Delhi

Delhi's peak electricity demand for FY 2025 was 8.66 GW. The city has an installed generation capacity of about 7.1 GW, with approximately 82.2% sourced from fossil fuel-based projects, 16.4% from renewable energy sources (RES), including solar and waste-to-energy, and the remaining 1.5% from nuclear power. The city contributes around 2.5% of India's total renewable energy capacity, supported by a strong policy push toward clean energy integration, electric mobility, and demand-side management.

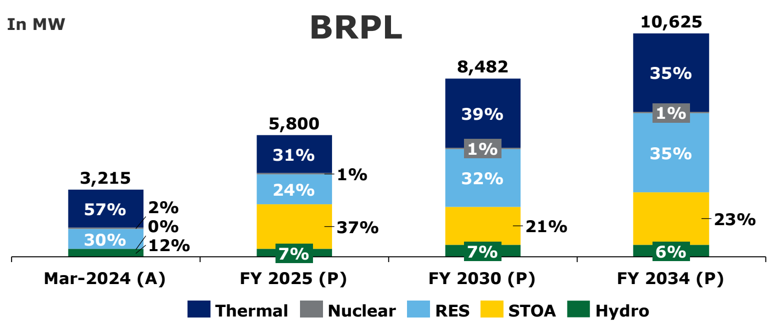
Table **44**: Energy Demand of Delhi

Particulars	Units	2021-22 (Actual) [216]	2024-25 (Actual) [217]	2026-27 (Projected) [218]	2031-32 (Projected) [21]
Energy Requirement	TWh	31	38	42	53
Peak Demand	GW	7	9	9	12

Fuel Mix of Delhi

BRPL's contracted power capacity reached 32 GW by the end of March 2024. Of this capacity, 57% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 106 GW, out of which 65% through non-fossil fuel. BRPL is projected to grow substantially, with contracted capacity rising at a CAGR of 14% over the next 9 years Capacity wise generation mix from 2024-34 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Hybrid
RES	Wind
RES	Distributed Renewable Energy (DRE)
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 75: Capacity Mix of BRPL – Delhi[219]*

BYPL's contracted power capacity reached 20 GW by the end of March 2023. Of this capacity, 70% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 41 GW, out of which 59% through non-fossil fuel. BYPL is projected to grow substantially, with contracted capacity rising at a CAGR of 8% over the next 9 years Capacity wise generation mix from 2024 to 2034 has been illustrated below:

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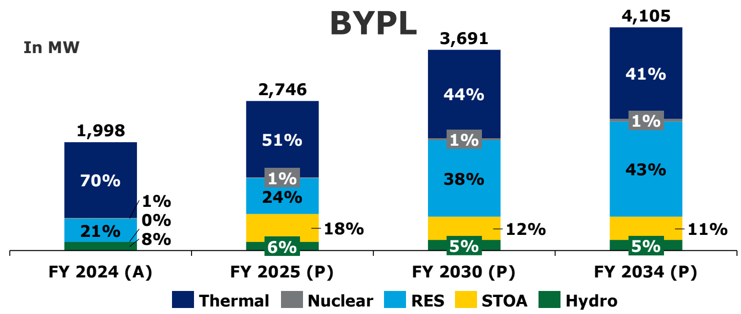
[216] Load Generation Balance Report, 2022
[217] Load Generation Balance Report, 2025
[218] National Electricity Plan 2022-32
[219] Resource Adequacy Report of BRPL-Delhi, FY 2025-34
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

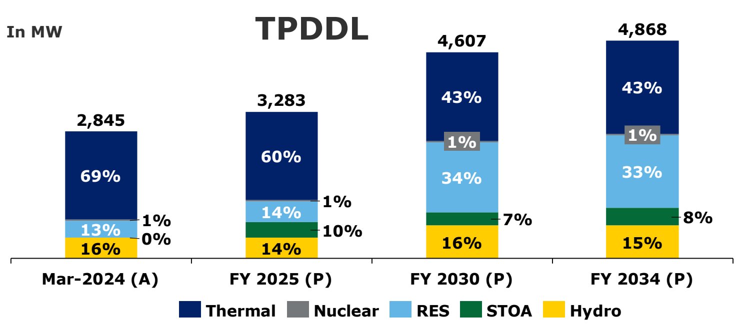


*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Hybrid
RES	Wind
RES	Distributed Renewable Energy (DRE)
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 76: Capacity Mix of BYPL- Delhi[220]*

TPDDL's contracted power capacity reached 28 GW by the end of March 2024. Of this capacity, 69% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 49 GW, out of which 57% through non-fossil fuel. TPDDL is projected to grow substantially, with contracted capacity rising at a CAGR of 6% over the next 9 years. Capacity wise generation mix from 2024 to 2034 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Hybrid
RES	Wind
RES	Battery
RES	Storage
RES	Distributed Renewable Energy (DRE)
RES	FDRE
Nuclear	Nuclear
Hydro	Hydro

Figure 77: Capacity Mix of TPDDL - Delhi[221]*

Financial Health of Delhi's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 60% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 40%, are payments still within their respective due dates but not yet fully or partially paid:[222]

[220] Resource Adequacy Report of BYPL- Delhi, FY 2025-34
[221] Resource Adequacy Report of TPDDL-Delhi, FY 2025-34
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[222] Dashboard for Delhi, PRAAPTI- as accessed on 27th May 2025

Overdue Amount
INR 269 Cr

Current Due Amount
INR 179 Cr

Total Dues
INR 448 Cr

Figure 78: Snapshot of PRAAPTI - Delhi

Utility Name 1: BSES Rajdhani Power Limited (BRPL)
Utility Name 2: BSES Yamuna Power Limited (BYPL)
Utility Name 3: Tata Power Delhi Distribution Limited (TPDDL)

Delhi's electricity distribution is managed by three main companies: BSES Rajdhani Power Limited (BRPL), BSES Yamuna Power Limited (BYPL), and Tata Power Delhi Distribution Limited (TPDDL). BRPL and BYPL serve the southern and eastern parts of Delhi, respectively, while TPDDL caters to the northern and north-western regions. They are regulated by the Delhi Electricity Regulatory Commission (DERC), which ensures compliance with standards and addresses consumer grievances. These Utilities together serve approx. 7.16 million[223] consumers, majorly catering to Domestic category at 53%, followed by Commercial, Industrial, Agriculture and Others.

Performance History:

In FY 2023–24, Delhi reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold demonstrated steady growth of 8%. The Average Cost of Supply (ACS) decreased to INR 8.33 per kWh, while the Average Revenue Realized (ARR) increased to INR 8.99 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite increasing profits in FY 2022–23, a turnaround was observed in FY 2023–24, with the company experiencing reduced profitability. Payable days, while still high, steadily decreased until FY 2022–23 but increased again in FY 2023–24, remaining higher than the national average (excluding private DISCOMs) of 130 days in FY 2023–24. [224]

Table **45**: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[225]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	27,410	29,728	32,748	34,082
Gross Input Energy	GWh	30,512	33,259	36,602	37,767
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.07	7.70	8.50	8.33
Average Revenue Realized (ARR)	INR/kWh	8.86	7.96	8.49	8.99
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	0.79	0.26	(0.01)	0.58
Profitability (Surplus/(Deficit))	INR in Crore	4,858	1,047	2,096	1,681
Payable Days	Days	343	335	271	277

Capacity Addition Plans:-

BRPL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 57% in FY 2024 to 35% by FY

[223] India Climate & Energy Dashboard
[224] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[225] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.[226]

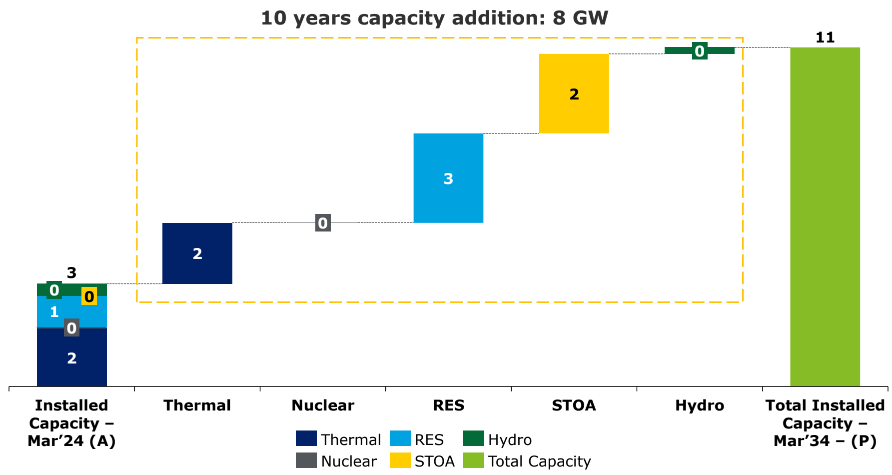


Figure 79: Capacity Addition Plan for next 10 years of BRPL

BYPL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 70% in FY 2024 to 41% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.[227]

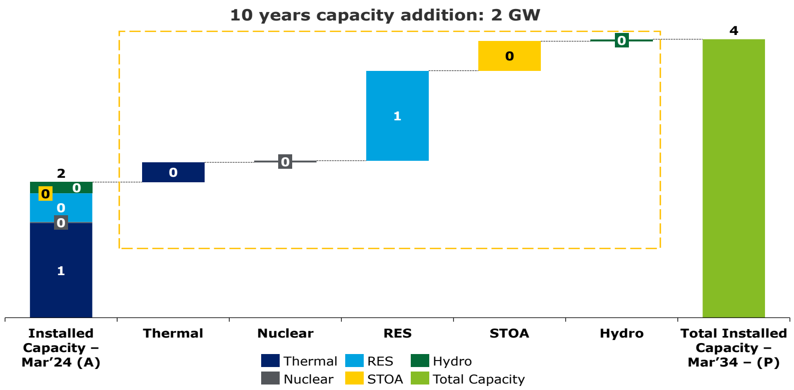


Figure 80: Capacity Addition Plan for next 10 years of BYPL

TPDDL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 69% in FY 2024 to 43% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.[228]

[226] Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI
[227] Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI
[228] Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI

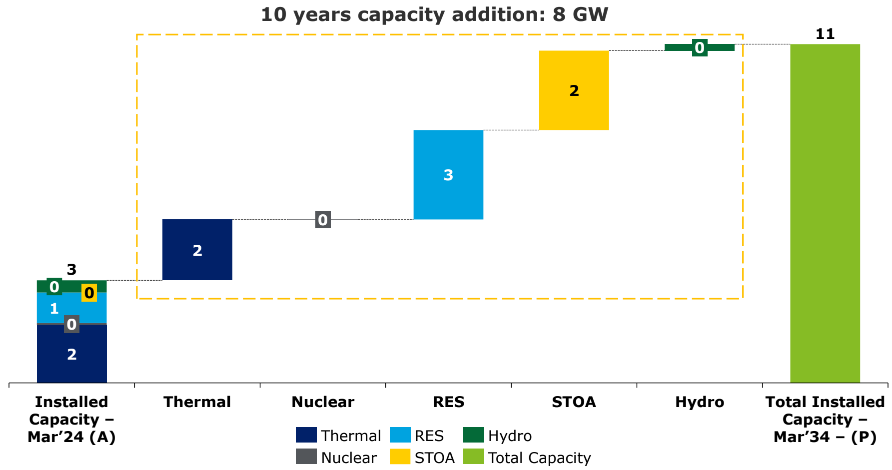
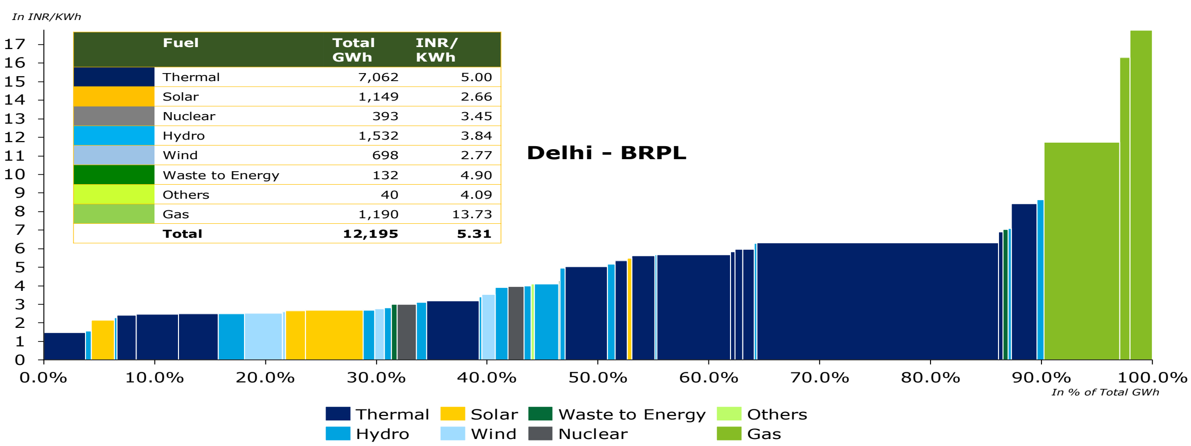


Figure 81: Capacity Addition Plan for next 10 years of TPDDL

Merit Order Analysis[229]:-

BRPL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 12,195 GWh. Of this, thermal power accounts for approximately 58% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of Solar power plants, with an average generation cost of INR 3.31 per kWh. However, some solar assets, such as Acme and Thyagraj solar, report significantly higher generation costs of INR 5.50 and INR 3.58 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 4.09 per kWh, reflecting a moderate cost profile within the state's energy mix.



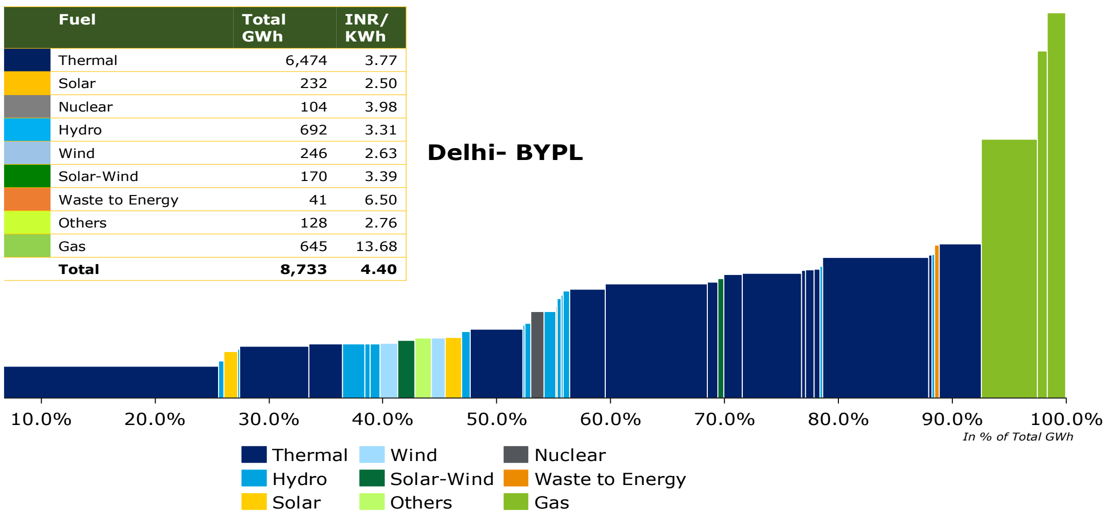
Fuel	Total GWh	INR/KWh
Thermal	7,062	5.00
Solar	1,149	2.66
Nuclear	393	3.45
Hydro	1,532	3.84
Wind	698	2.77
Waste to Energy	132	4.90
Others	40	4.09
Gas	1,190	13.73
Total	**12,195**	**5.31**

Figure 82: Merit Order Analysis – BRPL

BYPL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 8,733 GWh. Of this, thermal power accounts for approximately 74% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of thermal power plants, with an

[229] Aggregate Approval of True Up of Expenses upto FY 2022-23 & Aggregate Revenue Requirement (ARR) fo FY 2024-25, Delhi Electricity Regulatory Commission, 2024

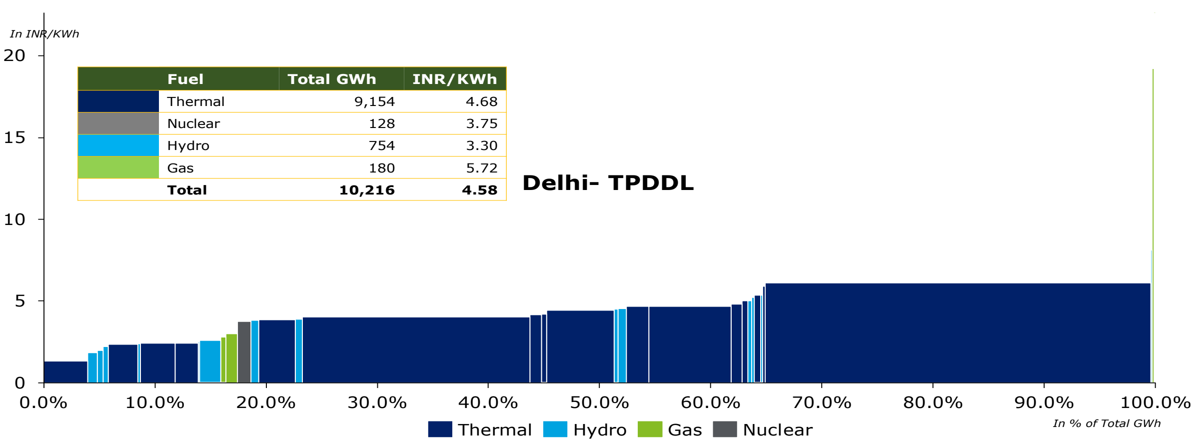
average generation cost of INR 1.47 per kWh. Other non-solar renewable sources contribute at an average tariff of INR 2.76 per kWh, reflecting a moderate cost profile within the state's energy mix.



Fuel	Total GWh	INR/ KWh
Thermal	6,474	3.77
Solar	232	2.50
Nuclear	104	3.98
Hydro	692	3.31
Wind	246	2.63
Solar-Wind	170	3.39
Waste to Energy	41	6.50
Others	128	2.76
Gas	645	13.68
Total	**8,733**	**4.40**

Figure 83: Merit Order Analysis – BYPL

TPDDL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 10,216 GWh. Of this, thermal power accounts for approximately 90% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of thermal power plants, with an average generation cost of INR 6.09 per kWh. Hydropower contributes at an average tariff of INR 8.13 per kWh, reflecting a moderate cost profile within the state's energy mix.



Fuel	Total GWh	INR/KWh
Thermal	9,154	4.68
Nuclear	128	3.75
Hydro	754	3.30
Gas	180	5.72
Total	**10,216**	**4.58**

Figure 84: Merit Order Analysis - TPDDL

7.1.4. Southern Region

The Southern region of India showcases a well-diversified electricity profile, marked by a balanced mix of energy sources. It relies significantly on thermal power for consistent base load supply, while hydropower supports peak demand and enhances sustainability. The region has also made substantial progress in integrating renewable energy, particularly solar and wind, reflecting its commitment to clean energy. Nuclear power, though limited in scale, contributes to energy security and grid stability. Overall, the Southern region demonstrates a forward-looking approach to energy management, emphasizing efficiency, reliability, and environmental responsibility.

List of All States of Southern Region have been Stated below[230]:

Table 46: State-wise Gross Input Energy[231]

Region	State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
South	Tamil Nadu	1,10,695	27%
South	Andhra Pradesh	80,664	20%
South	Telangana	86,163	21%
South	Karnataka	91,324	22%
South	Kerala	32,868	8%
South	Puducherry	4,499	1%
South	Lakshadweep	-	0%

Among the above-mentioned States, Tamil Nadu, Karnataka, Andhra Pradesh and Telengana has been selected further for detailed level Analysis as it covers approximately 91% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 47: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Andhra Pradesh	Andhra Pradesh Eastern Power Distribution Company Limited	30,038	37%
	Andhra Pradesh Southern Power Distribution Company Limited	32,781	41%
	Andhra Pradesh Central Power Distribution Company Limited	17,845	22%
	Total of Andhra Pradesh	**80,664**	
Karnataka	Bangalore Electricity Supply Company Limited	42,587	47%
	Chamundeshwari Electricity Supply Corporation Limited	10,262	11%
	Gulbarga Electricity Supply Company Limited	11,273	12%
	Hubli Electricity Supply Company Limited	19,146	21%
	Mangalore Electricity Supply Company Limited	8,056	9%
	Total of Karnataka	**91,324**	

[230] Central Electricity Authority
[231] Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

Telengana	Telangana State Northern Power Distribution Company Limited	24,908	29%
	Telangana State Southern Power Distribution Company Limited	61,255	71%
	Total of Telengana	**86,163**	
Tamil Nadu	Tamil Nadu Generation and Distribution Corporation Limited	110,695	100%
	Total of Tamil Nadu	**110,695**	

5. Region: South

State: Tamil Nadu

In FY 2025, Tamil Nadu's peak electricity demand reached 20.78 GW, with no reported surplus or deficit. The state boasts an installed generation capacity of approximately 42.77 GW[232], comprising around 37.5% from fossil fuel-based sources, 59.1% from renewable energy sources (RES) such as solar and waste-to-energy, and 3.4% from nuclear power. A national leader in renewable energy adoption—particularly in wind and solar—Tamil Nadu contributes over 14%[233] of India's total renewable capacity. The state has also made significant strides in supporting its clean energy transition through investments in grid-scale energy storage, green energy corridors, and smart grid technologies.

Table **48**: Energy Demand of Tamil Nadu

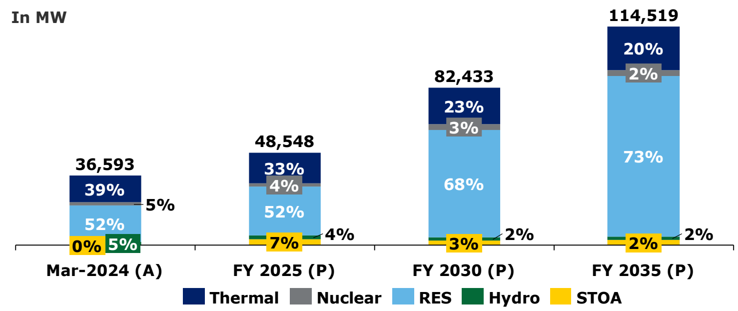
Particulars	Units	2021-22 (Actual) [234]	2024-25 (Actual) [235]	2026-27 (Projected) [236]	2031-32 (Projected) [237]
Energy Requirement	TWh	110	130	144	186
Peak Demand	GW	17	21	22	28

Fuel Mix of Tamil Nadu [238]

As of March 2024, Tamil Nadu's contracted power capacity reached approximately 37 GW, establishing the state as a key contributor to India's power infrastructure. Of this, 39% is sourced from fossil fuels—primarily coal and natural gas—while the remaining 61% comes from non-fossil fuel sources, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE) technologies. Looking ahead, the projected contracted capacity for FY 2034–35 is expected to reach around 114 GW, with 80% derived from non-fossil fuel sources and 20% from fossil fuels. The state's energy

[232] All India Installed Capacity, Central Electricity Authority of India, 2025
[233] Press Information Bureau, Government of India, 2025
[234] Load Generation Balance Report, 2022
[235] Load Generation Balance Report, 2025
[236] National Electricity Plan 2022-32
[237] National Electricity Plan 2022-32
[238] Resource Adequacy Report, Central Electricity Report, GOI, 2024
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

capacity is projected to grow at CAGR of approximately 9%, rising from 49 GW in FY 2025 to 115 GW by FY 2035. The evolution of the generation mix from FY 2024–35 is illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Biomass
RES	Solar
RES	Hydro DRE
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Storage
RES	Solar DRE Rooftop
Nuclear	Nuclear
Hydro	Hydro

Figure 85: Fuel-wise capacity mix of Tamil Nadu*

Financial Health of Tamil Nadu's Distribution Companies:

The analysis of payable dues is a vital indicator of the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—comprising 41% of the total—represent payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, which account for the remaining 59%, are still within their respective due dates but have not yet been settled:[239]

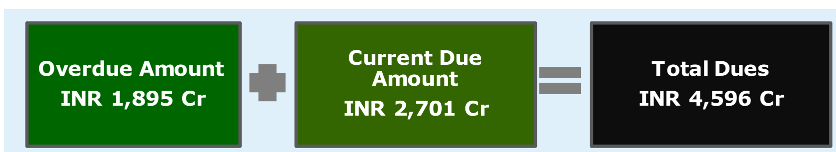


Figure 86: Snapshot of PRAAPTI Portal - Tamil Nadu

Utility Name 1: Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO)

Tamil Nadu's electricity distribution is managed by the Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO), a subsidiary of Tamil Nadu Electricity Board (TNEB). TANGEDCO oversee both power generation and distribution across the state. TANGEDCO serves over 3 crore consumers, including domestic, agricultural, commercial, and industrial categories.

Performance History of TANGEDCO:

From FY21 to FY24, energy input has increased at a compound annual growth rate (CAGR) of approximately 8%, reflecting the state's growing power demand. During the same period, the ACS-ARR gap has reduced by nearly 93%, indicating stronger revenue realization and improved billing and collection efficiencies. A similar positive trajectory is observed in Profit After Tax (PAT) due to decrease in ACS, increasing trend of ARR, high collection and billing efficiency, reduction in A&T losses among others. Although payable days remain relatively high, they have shown a consistent decline over the past three years. However, they still exceed the national average (excluding private DISCOMs) of 130 days recorded in FY 2023-24 [240]

Table 49: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[241]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	73,622	82,076	87,917	92,735

[239] Dashboard for Tamil Nadu, PRAAPTI, as accessed on 27th May 2025
[240] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[241] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Gross Input Energy	GWh	88,099	98,037	103,245	110,695	
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	9.14	8.76	10.42	10.79	
Average Revenue Realized (ARR)	INR/kWh	7.37	7.65	9.37	10.66	
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(1.77)	(1.11)	(1.05)	(0.13)	
PAT (Profit/(Loss))	INR in Crore	(13,066)	(9,130)	(9,192)	(1,196)	
Payable Days	Days	257	223	170	184	

Capacity Addition Plans of TANGEDCO:

TANGEDCO's installed capacity is projected to grow substantially in the coming years, with a progressively diversified energy mix. The share of renewable energy sources is expected to rise from 52% in FY 2024 to 73% by FY 2035, signaling a strategic shift toward cleaner and more sustainable power generation. According to the Resource Adequacy Report, hydropower capacity is set to increase by only 20 MW, indicating a relatively modest expansion in this segment compared to other renewables. The required capacity additions over this period are illustrated below [242]

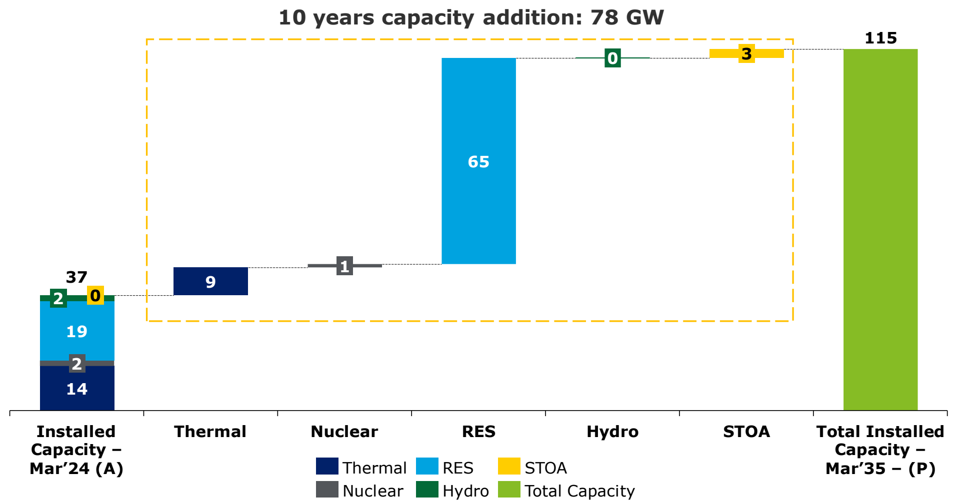


Figure 87: Capacity Addition for the next 10 years of Tamil Nadu

Merit Order Analysis : [243]

As per the true-up order for FY 2023–24 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio is primarily composed of solar, gas, and thermal power plants, with an average generation cost of ₹7.56 per kWh during FY23-24. Notably, cogeneration from sugar mills

[242] Report on Resource Adequacy of Tamil Nadu, Central Electricity Authority, GoI
[243] Approval of True Up of TNPDCL (formerly TANGEDCO) for FY 2023-24, Tamil Nadu Electricity Regulatory Commission,2024

and short-term power transactions report significantly higher power purchase costs of ₹9.69/kWh and ₹9.56/kWh, respectively.

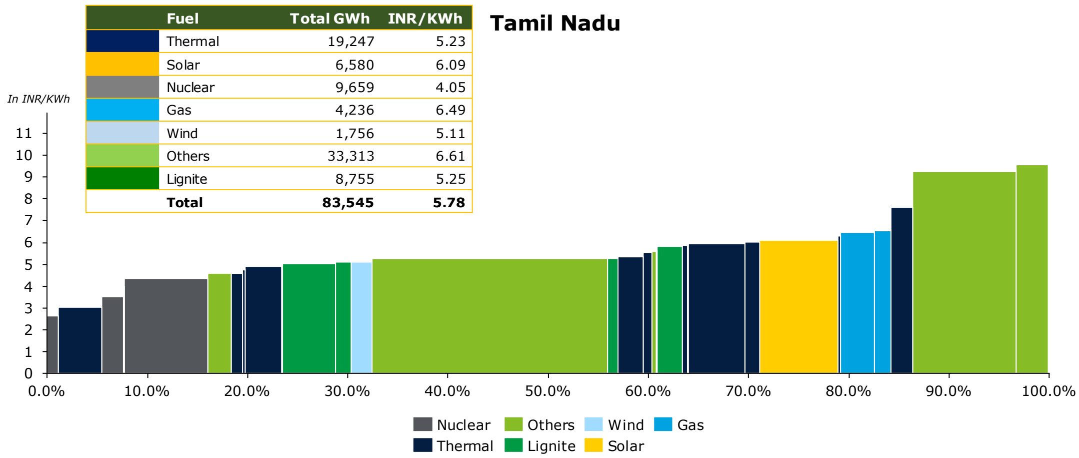


Figure 88: Merit Order Analysis of Tamil Nadu

5. Region: South

State: Telangana

In FY 2025, Telangana's peak electricity demand was approximately 17 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 20 GW[244], with roughly 61% from fossil fuel-based sources, 38% from renewable energy sources such as solar and waste-to-energy, and around 1% from nuclear power.

Table **50** : Energy Demand of Telengana

Particulars	Units	2021-22 (Actual) [245]	2024-25 (Actual) [246]	2026-27 (Projected) [247]	2031-32 (Projected) [248]
Energy Requirement	TWh	71	88	93	121
Peak Demand	GW	15	17	20	27

Fuel Mix of Telangana [249]

Telengana's contracted power capacity reached ~ 19 GW by the end of March 2024, positioning it as a key contributor to India's power infrastructure. Of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas and 46% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2034-35 is ~ 73 GW, out of which 77% through non-fossil fuel and 23% from fossil fuel based. The state's energy capacity is projected to grow at CAGR of approximately 9%, rising from 19 GW in FY 2025 to 73 GW by FY 2035. Capacity wise generation mix from FY 2024-35 has been illustrated below:

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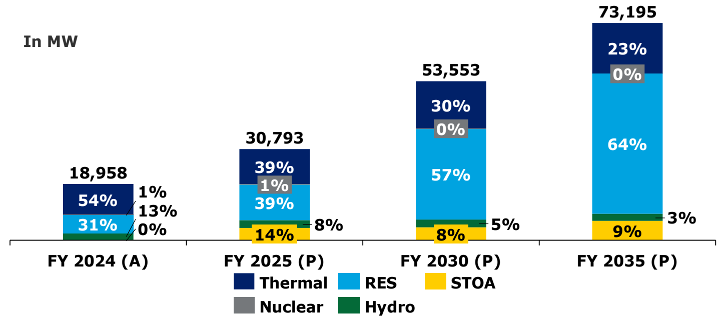
[244] All India Installed Capacity, Central Electricity Authority of India, 2025
[245] Load Generation Balance Report, 2022
[246] Load Generation Balance Report, 2025
[247] National Electricity Plan 2022-32
[248] National Electricity Plan 2022-32
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.
[249] Resource Adequacy Report, Central Electricity Report, GOI, 2024



*Type	Source
Thermal	Coal
Thermal	Gas
STOA	Short-term Open Access (STOA)
RES	Non-Conventional Energy Source (NCES)
RES	Solar
RES	Wind
RES	Storage
RES	Distributed Renewable Energy (DRE)
Nuclear	Nuclear
Hydro	Hydro

Figure 89: Fuel-wise capacity mix of Telangana*

Financial Health of Telangana's Distribution Companies:

The analysis of payable dues is a key metric for assessing the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—constituting 50% of the total—refer to payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, which make up the remaining 50%, are still within their respective due dates but have not yet been settled:[250]

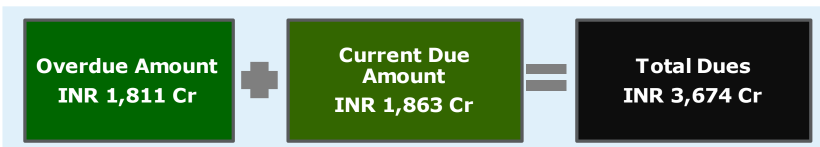


Figure 90: Snapshot of PRAAPTI Portal - Telangana

Utility Name 1: The Southern Power Distribution Company of Telangana Limited (TSSPDCL)
Utility Name 2: The Northern Power Distribution Company of Telangana Limited (TSNPDCL)

Telangana is managed by two State-owned DISCOMs. These utilities operate under the regulatory oversight of the Telangana State Electricity Regulatory Commission (TSERC) and serve both urban and rural regions across the State. Together, these utilities serve 17.20 million[251] consumers serving majorly Agriculture sector, followed by Industrial, Domestic, Commercial and Others.

Performance History:

From FY21 to FY24, energy input has increased at a CAGR of ~ 9%, reflecting the state's growing power demand. During the same period, the ACS-ARR gap has reduced by nearly 20%, indicating enhanced revenue realization, improved billing and collection efficiencies. Although payable days remained relatively high in FY22, they have remained steady over the past two years. However, they still significantly exceed the national average (excluding private DISCOMs) of 130 days recorded in FY 2023-24 [252]

Table 51: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[253]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	59,451	65,900	69,136	75,179

[250] Dashboard for Telangana, PRAAPTI, as accessed on 27th May 2025

[251] India Climate & Energy Dashboard

[252] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[253] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Gross Input Energy	GWh	66,468	74,651	78,117	86,163
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.22	7.48	8.83	8.74
Average Revenue Realized (ARR)	INR/kWh	6.03	7.39	7.25	7.87
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(1.19)	(0.09)	(1.58)	(0.86)
PAT (Profit/(Loss))	INR in Crore	(6,686)	(831)	(11,103)	(6,351)
Payable Days	Days	297	380	310	310

Capacity Addition Plans of Telangana:

As outlined above, Telangana's energy capacity is set for substantial growth, accompanied by a major transformation in its energy mix. The share of coal and gas is projected to decline significantly—from 54% in FY 2023 to just 23% by FY 2035. In contrast, the contribution of non-fossil and renewable energy (RE) sources is expected to more than double, rising from 31% to 64%. This shift is driven by increasing energy demand and the accelerated adoption of renewable technologies across the state. The transition aligns closely with India's national clean energy targets for 2030. The required capacity additions to support this transformation are illustrated below [254]:

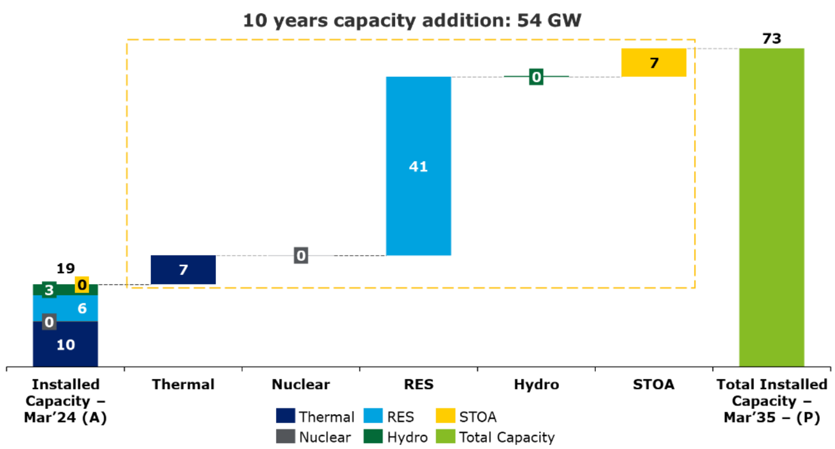


Figure 91: Capacity Addition for next 10 years of Telangana

Merit Order Analysis[255]:-

Telangana's power generation is predominantly reliant on thermal and solar energy, which remains one of the most expensive generation sources for the state. According to the revised Aggregate Revenue Requirement (ARR) order for FY 2025–26 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio primarily comprises thermal, solar, and biomass power plants, with an average generation cost of ₹7.63 per kWh during FY 2023–24. Notably, solar generation under JNNSM Phase I incurs significantly higher power purchase costs—₹10.4 per kWh—

[254] Report on Resource Adequacy of Telangana, Central Electricity Authority, GoI
[255] Revised Aggregate Revenue Requirement, Telangana Electricity Regulatory Commission, 2024-25

due to legacy feed-in tariffs fixed by the Central Electricity Regulatory Commission (CERC) for a 25-year period.

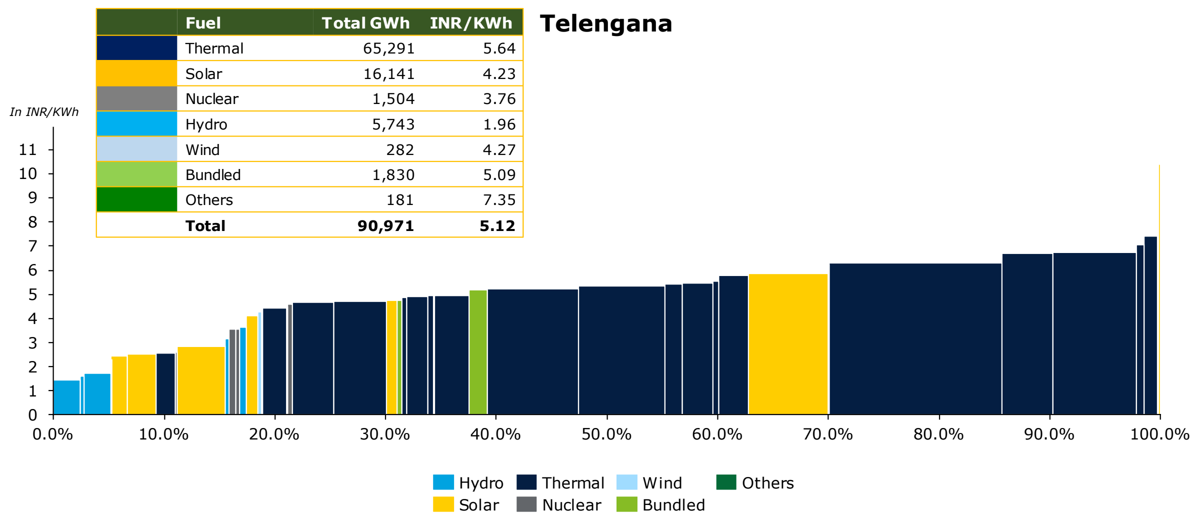


Figure 92: Merit Order Analysis of Telengana

5. Region: South

State: Andhra Pradesh

In FY 2025, Andhra Pradesh's peak electricity demand was approximately 14 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 28 GW [256], with around 57% sourced from fossil fuel-based projects, 43% from renewable energy sources (RES) such as solar and waste-to-energy, and a small fraction—about 0.5%—from nuclear power. Andhra Pradesh has made notable progress in renewable energy adoption, contributing roughly 8.5% of India's total renewable capacity. The state is actively expanding its solar and wind infrastructure, with strategic plans to increase the share of renewables to 74% by 2035[228]. This positions Andhra Pradesh as a frontrunner in India's clean energy transition, aligning with national goals for sustainability and energy security.

Table **52**: Energy Demand of Andhra Pradesh

Particulars	Units	2021-22 (Actual) [257]	2024-25 (Actual) [258]	2026-27 (Projected) [259]	2031-32 (Projected)
Energy Requirement	TWh	69	79	98	137
Peak Demand	GW	12	14	18	24

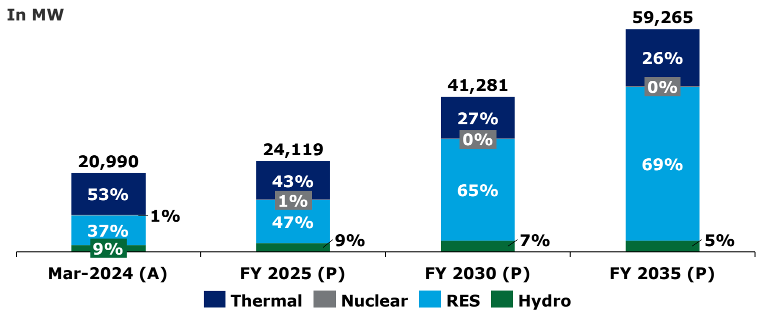
Fuel Mix of Andhra Pradesh [260]

Andhra Pradesh's contracted power capacity reached ~ 21 GW by the end of March 2024, positioning it as a key contributor to India's power infrastructure. Of this capacity, 53% is derived from fossil

[256] All India Installed Capacity, Central Electricity Authority of India, 2025
[257] Load Generation Balance Report, 2022
[258] Load Generation Balance Report, 2025
[259] National Electricity Plan 2022-32
[260] Resource Adequacy Report, Central Electricity Report, GOI, 2024
* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

fuel-based sources, primarily coal and natural gas and 47% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2034-35 is ~59 GW, out of which 74% through non-fossil fuel and 26% from fossil fuel based. The's energy capacity is projected to grow at CAGR of approximately 9%, rising from 24 GW in FY 2025 to 59 GW by FY 2035. Capacity wise generation mix from FY 2024-35 has been illustrated below:



*Type	Source
Thermal	Coal
Thermal	Gas
RES	Biomass
RES	Solar
RES	Wind
RES	Pumped Storage Plants (PSP)
RES	Distributed Renewable Energy (DRE)
Nuclear	Nuclear
Hydro	Hydro

Figure 93: Fuel-wise capacity mix of Andhra Pradesh*

Financial Health of Andra Pradesh's Distribution Companies:

The analysis of payable dues is a critical tool for assessing the financial health, payment behavior, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—comprising 49% of the total—represent payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 51%, are still within their respective due dates but have not yet been settled. [261]

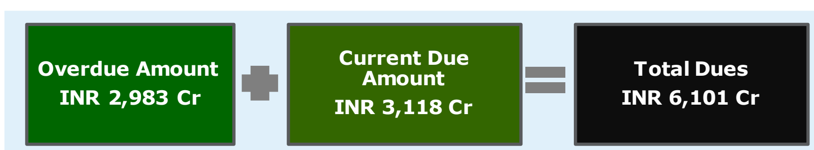


Figure 94: Snapshot of PRAAPTI portal – Andhra Pradesh

Utility Name1: Southern Power Distribution Company of Andhra Pradesh Limited (APSPDCL)
Utility Name2: Eastern Power Distribution Company of Andhra Pradesh Limited (APEPDCL)
Utility Name3: Central Power Distribution Company of Andhra Pradesh Limited (APCPDCL)

Electricity distribution in Andhra Pradesh is managed by three state-owned DISCOMs, each responsible for serving distinct geographical regions and consumer segments across the state. These DISCOMs play a vital role in ensuring reliable power supply, managing demand, and implementing both state and central energy policies. Collectively, they serve approximately 15.4 million consumers[262], with the industrial sector accounting for the largest share at 43%, followed by domestic, agricultural, commercial, and other categories

Performance History:

Between FY21 and FY24, energy input grew at a compound annual growth rate (CAGR) of approximately 6%, reflecting the state's rising power demand. Over the same period, the ACS-ARR gap narrowed by only 8%, indicating modest improvements in revenue realization and billing and

[261] Dashboard for Andra Pradesh, PRAAPTI, as accessed on 27th May 2025
[262] India Climate & Energy Dashboard

collection efficiency. However, the Average Cost of Supply (ACS) has continued to rise since FY21, pointing to increasing power purchase costs. On a positive note, payable days have steadily declined over the past three years and now remain below the national average (excluding private DISCOMs) of 130 days recorded in FY 2023–24 [263]

Table 53: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[264]	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	62,493	69,745	71,463	72,628
Gross Input Energy	GWh	68,564	76,122	78,424	80,664
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.14	6.89	7.87	9.24
Average Revenue Realized (ARR)	INR/kWh	7.14	6.54	7.17	8.86
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	-	(0.35)	(0.70)	(0.38)
PAT (Profit/(Loss))	INR in Crore	44	(2,458)	1,736	169
Payable Days	Days	217	166	91	105

Capacity Addition Plans of Andra Pradesh:

Andhra Pradesh's energy capacity is projected to grow significantly, accompanied by a marked diversification of its energy mix. The share of coal and gas is expected to decline from 53% in FY 2024 to 26% by FY 2035. In contrast, the contribution of non-fossil and renewable energy sources is set to rise from 46% to 74% over the same period, driven by increasing demand and the rapid adoption of renewable technologies across the state. This transition aligns closely with India's national clean energy targets for 2030. The required capacity additions to support this transformation are illustrated below. [265]

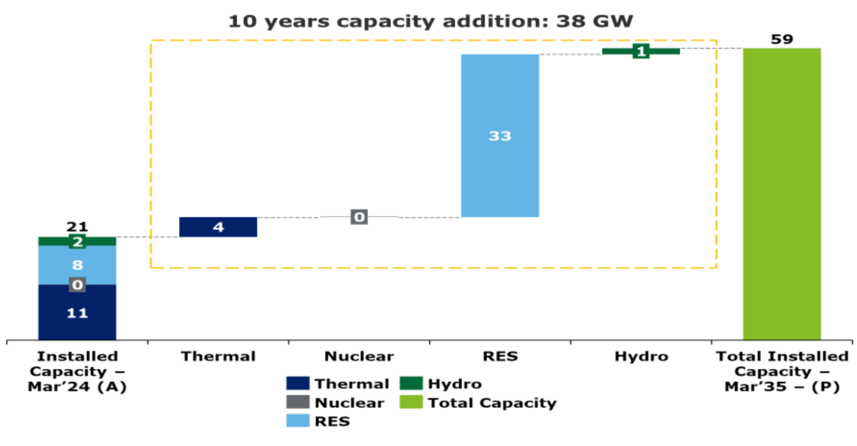
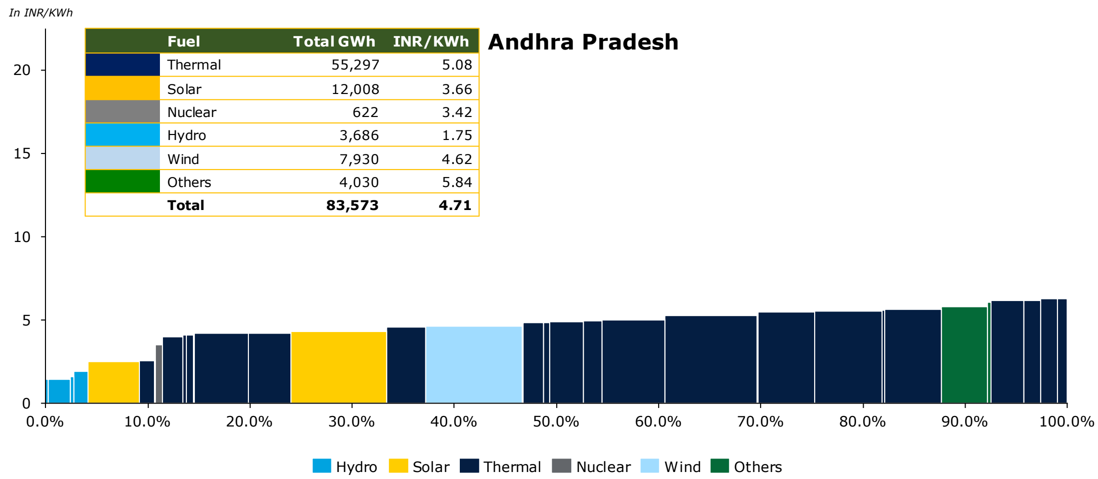


Figure 95: Capacity Addition Plan for next 10 years of Andra Pradesh

Merit Order Analysis[266]:-

[263] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[264] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23
[265] Report on Resource Adequacy of Andhra Pradesh, Central Electricity Authority, GoI
[266] RST Order, Andhra Pradesh Electricity Regulatory Commission, 2025-26

Andhra Pradesh’s electricity generation is dominated by thermal power, followed by solar and wind. While thermal energy has the highest share, it is also among the costlier sources. As per the revised Retail Supply Tariffs order for FY 2025–26 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio is primarily composed of thermal and hydropower plants, with an average generation cost of ₹7.82 per kWh during FY25-26. Notably, generation from PABRHES report significantly higher power purchase costs of ₹20.5/kWh, due to the high fixed cost of the hydropower plant



Fuel	Total GWh	INR/KWh
Thermal	55,297	5.08
Solar	12,008	3.66
Nuclear	622	3.42
Hydro	3,686	1.75
Wind	7,930	4.62
Others	4,030	5.84
Total	**83,573**	**4.71**

Figure 96: Merit Order Analysis of Andhra Pradesh

5. Region: South

State: Karnataka

In FY 2025, Karnataka's peak electricity demand was approximately 18 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 35 GW[267], with around 31% from fossil fuel-based sources, 67% from renewable energy sources such as solar and waste-to-energy, and about 2% from nuclear power.
Karnataka has been a frontrunner in solar energy, home to some of India's largest solar parks, including the Pavagada Solar Park. The state contributes over 12%[268] of the country's total renewable energy capacity, ranking among the top three states in renewable energy deployment.

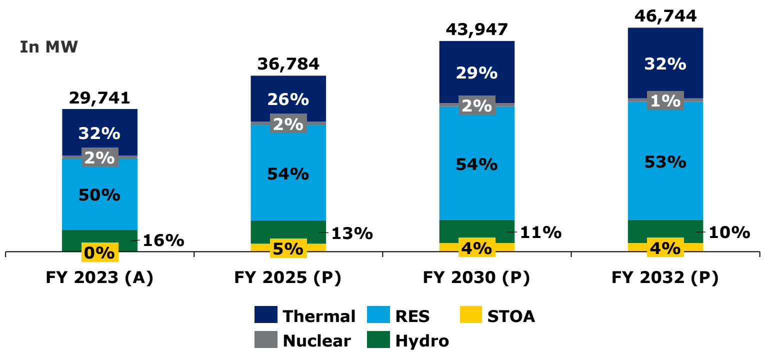
Table **54**: Energy Demand of Karnataka

Particulars	Units	2021-22 (Actual)[269]	2024-25 (Actual)[270]	2026-27 (Projected)[271]	2031-32 (Projected)[272]
Energy Requirement	TWh	72	93	88	106
Peak Demand	GW	15	18	18	22

Fuel Mix of Karnataka [273]

Karnataka's contracted power capacity reached ~ 30 GW by the end of March 2023, positioning it as a key contributor to India's power infrastructure. Of this capacity, 32% is derived from fossil fuel-based sources, primarily coal and 68% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2031-32 is ~ 47 GW, out of which 68% through non-fossil fuel and 32% from fossil fuel based. The state's energy capacity is projected to grow at a CAGR of approximately 4%, increasing from 37 GW in FY 2025 to 47 GW by FY 2032. This growth is driven by rising energy demand and the accelerated adoption of clean technologies. Capacity wise generation mix from FY 2023-32 has been illustrated below:



*Type	Source
Thermal	Coal
STOA	Short-term Open Access (STOA)
RES	Bioenergy
RES	Solar
RES	Wind
RES	Storage
Nuclear	Nuclear
Hydro	Hydro

Figure 97: Fuel-wise capacity mix of Karnataka*

Financial Health of Karnataka's Distribution Companies:

The analysis of payable dues is a key indicator of the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments:

[267] All India Installed Capacity, Central Electricity Authority of India, 2025
[268] Resource Adequacy Report, Karnataka
[269] Load Generation Balance Report, 2022
[270] Load Generation Balance Report, 2025
[271] National Electricity Plan 2022-32
[272] National Electricity Plan 2022-32
[273] Resource Adequacy Report, Karnataka

Overdue and Current. Overdue dues, which make up 50% of the total, refer to payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues—accounting for the remaining 50%—are still within their respective due dates but have not yet been settled:[274]

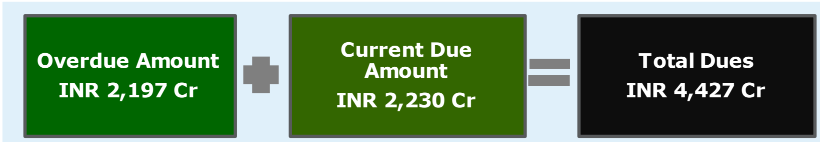


Figure 98: Snapshot of PRAAPTI portal – Karnataka

Utility Name 1: Electricity Supply Companies (ESCOMs)

Karnataka's electricity distribution is managed by five regional Electricity Supply Companies (ESCOMs):

- Bangalore Electricity Supply Company Limited (BESCOM)
- Mangalore Electricity Supply Company (MESCOM)
- Hubli Electricity Supply Company (HESCOM)
- Gulbarga Electricity Supply Company (GESCOM)
- Chamundeshwari Electricity Supply Company (CHESCOM)

distinct geographic regions and consumer segments across the state. Together, these DISCOMs form the backbone of Karnataka's power distribution network, playing a critical role in ensuring reliable electricity supply, minimizing losses, and advancing the state's clean energy transition. Collectively, they serve approximately 30.9 million consumers[275], with the majority being industrial, followed by agricultural, domestic, commercial, and other categories.

Performance History:

Between FY 2021 and FY 2024, energy input in the state grew at a compound annual growth rate (CAGR) of approximately 12%, reflecting a steady rise in power demand. During the same period, the ACS-ARR gap narrowed by only 13%, indicating modest improvements in revenue realization and billing and collection efficiency. However, the Average Cost of Supply (ACS) has continued to increase since FY 2021, largely due to rising power purchase costs. While payable days have been declining over the years, they still remain above the national average (excluding private DISCOMs) of 130 days recorded in FY 2023–24 [276]

Table 55: Performance Snapshot Report

Key Parameters	Units	FY 2020-21[277]	FY 2021-22	FY 2022-23	FY 2022-23
Gross Energy Sold	GWh	54,784	58,462	62,270	76,911
Gross Input Energy	GWh	65,116	69,139	73,493	91,324
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.49	8.58	9.55	9.97
Average Revenue Realized (ARR)	INR/kWh	7.67	8.19	9.18	9.24
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.82)	(0.39)	(0.37)	(0.72)
PAT (Profit/(Loss))	INR in Crore	(4,175)	(4,592)	(3,146)	(8,550)

[274] Dashboard for Karnataka, PRAAPTI, as assessed on 27th May 2025
[275] India Climate & Energy Dashboard
[276] Report on Performance of Power Utilities, Power Finance Corporation, 2023-24
[277] Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Payable Days	Days	201	186	169	177

Capacity Addition Plans of Karnataka:

From FY 2023 to FY 2032, Karnataka's projected energy mix indicates that thermal power will consistently hold a dominant share above 50%, while Renewable Energy Sources (RES) are expected to contribute between 63% and 67% from FY 2024 to FY 2032. This trend reflects a measured energy transition, shaped by the need to balance energy security, cost-effectiveness, and infrastructure readiness. The sustained reliance on thermal power highlights its role in providing stable base-load capacity, even as the state gradually integrates cleaner and more flexible energy alternatives. The required capacity additions to support this transformation are illustrated below[278]:

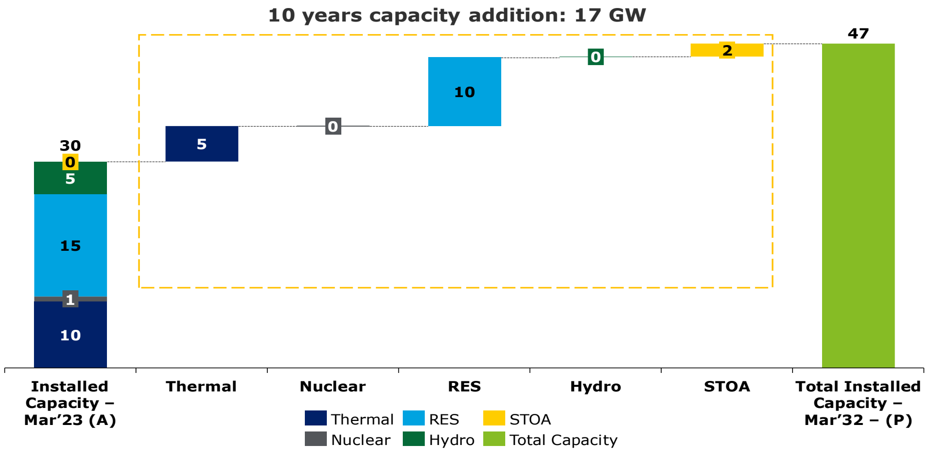
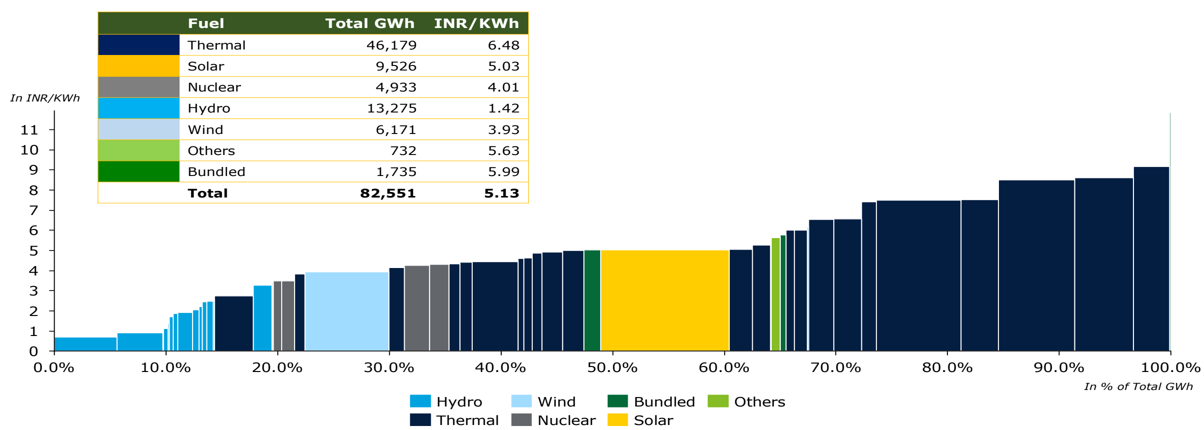


Figure 99: Capacity Addition Plan for the next 8 years of Karnataka

Merit Order Analysis[279]:-

Karnataka's power generation is mainly dependent on thermal energy, which is also among the costliest sources. As per the revised APR order for FY 2024–25 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio is primarily composed of thermal and bundled solar, with an average generation cost of ₹8.86 per kWh during FY23-24. Notably, generation from Bundled Power Solar report significantly higher power purchase costs of ₹11.9/kWh, due to the feed in tariffs for old solar plants locked by CERC for 25 years.



Fuel	Total GWh	INR/KWh
Thermal	46,179	6.48
Solar	9,526	5.03
Nuclear	4,933	4.01
Hydro	13,275	1.42
Wind	6,171	3.93
Others	732	5.63
Bundled	1,735	5.99
Total	**82,551**	**5.13**

Figure 100: Merit Order Analysis of Karnataka

278 Report on Resource Adequacy of Karnataka, Central Electricity Authority, GoI
279 Annual Performance Review for FY 23 and Approved Revised ARR for FY 25, Karnataka Electricity Regulatory Commission, 2024

7.1.5. Key Takeaways from DISCOM analysis

The Offtake Analysis conducted above provides valuable insights on the potential for select State and Private Discoms to absorb electricity generated from DHPP. A snapshot of the analysis is provided below:

Table 56: Summary of Discom Offtake Analysis

State	Forward Year Projections for Capacity Available till	Total Capacity Addition Requirement (from current levels) (in GW)	Total Hydro Capacity Addition Required (in GW)	Year of Power Purchase Cost	Weighted Average Power Purchase Cost (INR/kWh)[280]	Highest Power Purchase Cost		Lowest Power Purchase Cost		Average Cost of Supply - FY 2023-24 (INR/kWh)	Average Revenue Realized - FY 2023-24 (INR/kWh)
						Fuel Type	INR/kWh	Fuel Type	INR/kWh		
Western Region											
Maharashtra	2034	64.89	1.06	FY 25-26	4.73	Gas	7.65	Hydro	2.32	8.96	8.67
Gujarat	2035	86.07	1.10	FY 24-25	4.01	Waste to Energy	6.31	Hydro	1.73	7.12	7.78
Madhya Pradesh	2035	57.54	0.34	Fy 25-26	3.35	Biomass/Biogas	7.65	Hydro	3.05	8.22	7.54
Chhattisgarh	2035	26.03	0.57	FY 24-25	3.74	RE	5.08	Others	3.16	6.13	6.14
Eastern Region											
West Bengal	2035	44.52	0.09	FY 24-25	3.8	Solar	3.94	Wind-Solar Hybrid	2.48	7.86	7.65
Jharkhand	2034	10.33	0.02	FY 24-25	4.33	Thermal	4.55	Hydro and Wind/ Solar	2.8	11.27	8.90
Bihar	2034	31.35	1.00	FY 24-25	3.01	Biomass	5.62	Wind	2.63	8.42	8.79
Odisha	2034	18.09	1.10	FY 24-25	3.37	Biomass Energy	7.6	Hydro	1.56	6.47	6.34
North-Eastern Region											
Assam	2035	9.34	0.33	FY 24-25	4.95	Thermal & Hydro	6.01	Wind	3.08	9.51	9.77
Northern Region											
Uttar Pradesh	2034	116.15	1.81	FY 24-25	4.76	Others	7.09	Wind	2.98	9.20	8.57
Rajasthan	2032	39.17	1.34	FY 24-25	4.22	Thermal	4.48	Solar	2.92	8.81	8.86

[280] 'Weighted Average PP Cost (INR/kWh)' represents the mean cost of power procurement for each state over the specified fiscal year, calculated by weighting the cost of different fuel types based on their share in the total power purchased. This average provides a comprehensive view of the cost burden associated with power procurement.

State	Forward Year Projections for Capacity Available till	Total Capacity Addition Requirement (from current levels) (in GW)	Total Hydro Capacity Addition Required (in GW)	Year of Power Purchase Cost	Weighted Average Power Purchase Cost (INR/kWh)[280]	Highest Power Purchase Cost		Lowest Power Purchase Cost		Average Cost of Supply - FY 2023-24 (INR/kWh)	Average Revenue Realized - FY 2023-24 (INR/kWh)
						Fuel Type	INR/kWh	Fuel Type	INR/kWh		
Punjab	2030	24.48	0.83	FY 23-24	4.36	Non Solar	7.33	Wind	2.57	7.06	7.19
Haryana	2035	35.19	1.35	FY 25-26	4.56	Thermal	5.03	Solar	2.90	7.16	7.21
Delhi - BRPL	2034	7.41	0.20	FY 22-23	5.31	Gas	13.73	Solar	2.66		
Delhi - BYPL	2034	2.11	0.03	FY 22-23	4.40	Gas	13.68	Solar	2.50	8.33	8.99
Delhi - TPDDL	2034	2.02	0.27	FY 24-25	4.58	Gas	5.72	Hydro	3.30		
Southern Region											
Tamil Nadu	2035	77.93	0.02	FY 23-24	5.78	Others	6.61	Nuclear	4.05	8.96	10.66
Telengana	2035	54.24	0.00	FY 25-26	5.12	Others	7.35	Hydro	1.96	8.74	7.85
Andhra Pradesh	2035	38.28	1.19	FY 23-24	4.71	Others	5.84	Hydro	1.75	9.24	8.86
Karnataka	2032	17.00	0.14	FY 24-25	5.13	Thermal	6.48	Hydro	1.42	9.97	9.24

States such as Maharashtra, Gujarat, Bihar, Odisha, Uttar Pradesh, Rajasthan, Haryana and Andhra Pradesh are projected to need more than 1 GW of hydropower capacity to meet their Resource Adequacy needs over the next decade. Further States of Tamil Nadu, Telengana, Karnataka and BRPL in Delhi have an average power procurement cost of over Rs. 5 /kWh providing an opportunity to rationalize their power procurement costs through replacement of expensive fuel based power in their over power portfolio. **With DHPP envisaged to have a tariff range between Rs. 5-6 / kWh in FY2031, it is well suited to be positioned as a cheaper alternative for power procurement by many States as showcased above**.

Further hydropower also offers significant advantages as a base load plant. It provides a long term, clean and low-cost energy source, as evidenced by hydro being the lowest cost fuel type in several states such as Gujarat (1.73 INR/kWh), Odisha (1.56 INR/kWh), and Karnataka (1.42 INR/kWh). This not only helps in reducing the overall cost of supply but also enhances grid reliability. **Hydropower projects such as DHPP also offers a reliable supply source during the wet seasons in India when generation from thermal (coal/lignite) and renewable energy (solar in particular) are usually under stress due to coal supply chain and weather related challenges respectively.**

7.2. Offtake Analysis in Short/Medium Term Power Market through Indian Power Exchanges and Bilateral trade

The short and medium term power market in India has been evolving rapidly since the last few years due to changing customer preferences, favourable regulations and higher margin potential. Some of the key drivers are enumerated below:

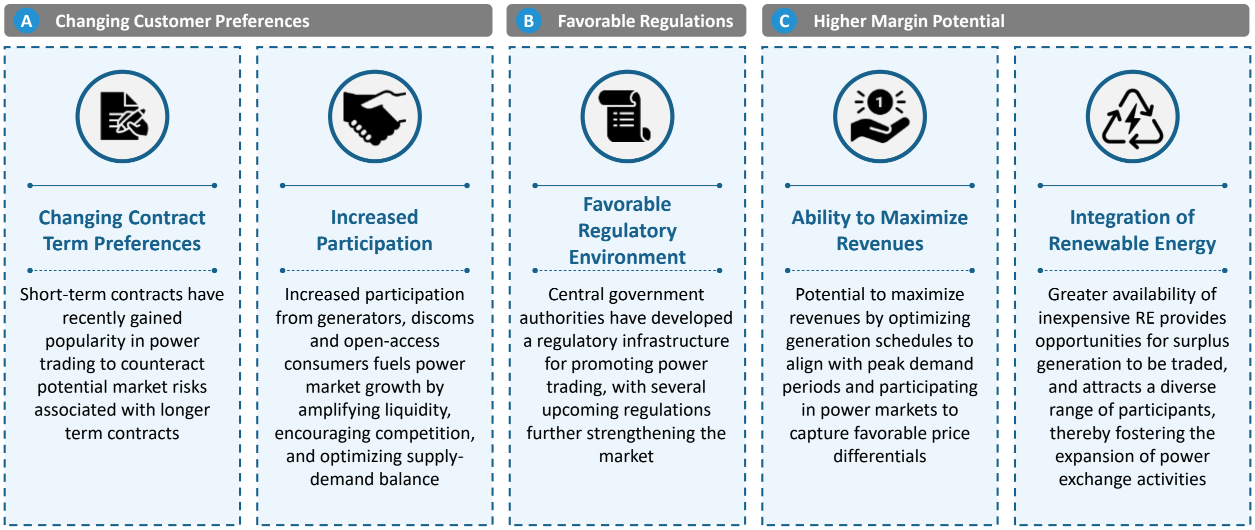


Figure 101: Key Drivers impacting Short/Medium Term Power Markets

A snapshot of market segments in the short/medium term market is shown:

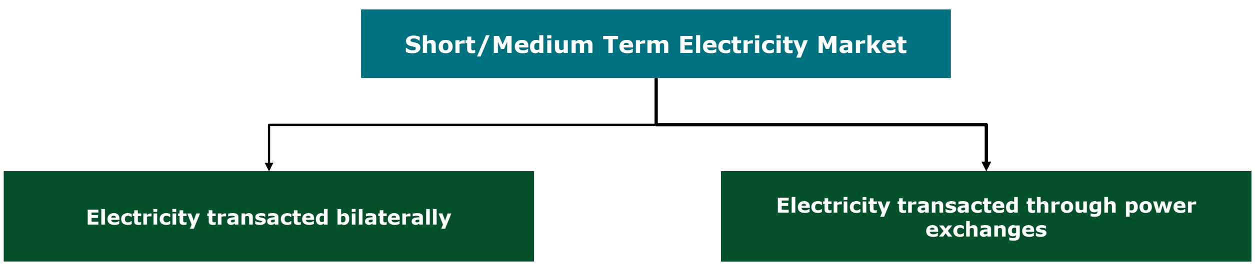


Figure 102: Market Segments in the Short/Medium -Term Power Market

In 2023-24, the short-term market constituted about 12% of the total electricity generation in India.[281] Of the total short-term transactions in 2023-24, the volume of electricity transacted through power exchanges was maximum (63%), followed by bilateral transactions through traders (21%), and direct bilateral transactions between entities (15%).

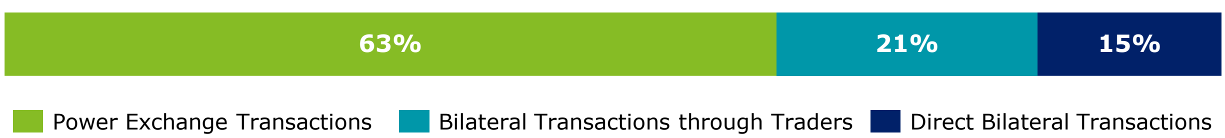


Figure 103: Share of different market segments in Short-Term Power Transactions in 2023-24

Electricity traders act as intermediaries between power generators and distribution companies. In addition to connecting clients, they can also trade on power exchanges on their behalf.

The volume of electricity transacted through power exchanges increased at a CAGR of 22.4%, and the volume of electricity transacted through traders bilaterally increased at a CAGR of 3.1% in the last 15 years.

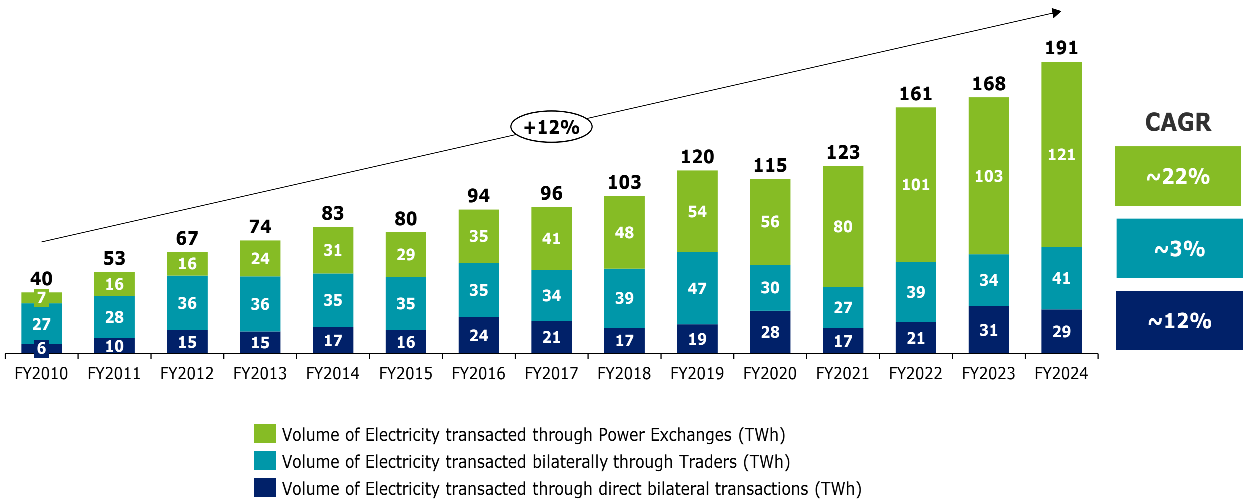


Figure 104: Trend of volume of transaction through Traders and Power Exchanges[281]

Electricity trading from 2009 to 2019 was dominated by bilateral transactions. Since 2020, power exchanges have taken the lead, with transaction volumes exceeding those of the bilateral transactions.

However, the weighted average price of transaction through traders bilaterally was higher than the transactions recorded in the power exchange in 2023-2024 and this has broadly followed a similar trend in the last 15 years.

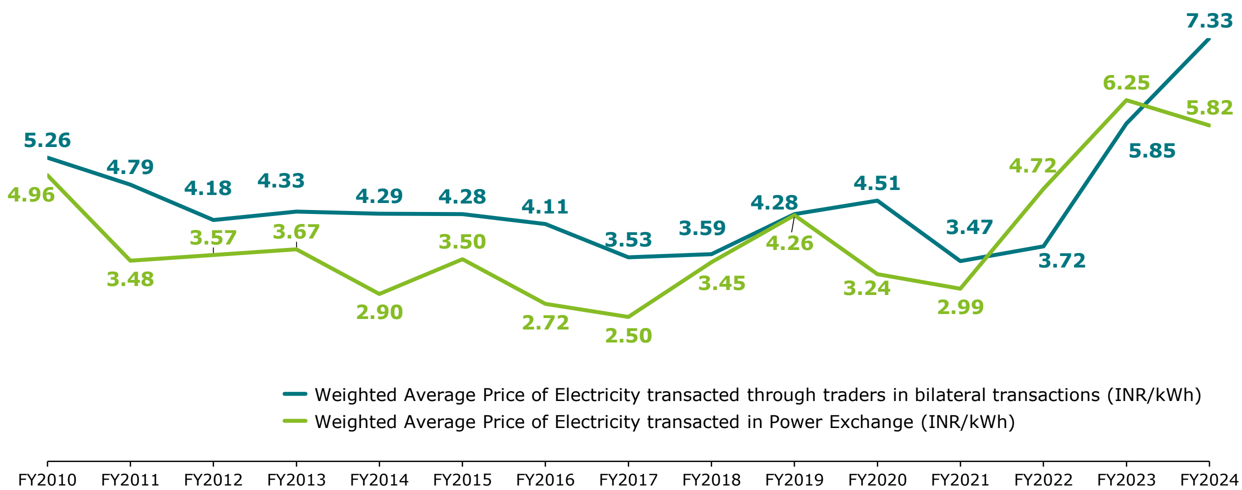


Figure 105: Trend of weighted average price of transaction through Traders and Power Exchanges[281] [281]

7.2.1. Offtake through Power Exchanges

Power exchanges in India are electronic platforms that facilitate the trading of electricity. They operate under the regulatory oversight of the Central Electricity Regulatory Commission (CERC) and play a crucial role in the short-term power market. These exchanges provide a transparent, efficient, and competitive environment for buyers and sellers to trade electricity through various market segments.

[281] Report on Short Term Power Market in India 2023-2024, Central Electricity Regulatory Commission

Power exchanges in India offer several market segments to cater to diverse trading needs.

However, for Cross border Power Market Trading, only trading in Day-Ahead Market (DAM) and Real Time Market (RTM) is allowed by the regulations[282]. Any Indian trader, after obtaining approval from Designated Authority, can trade in Indian Power Exchanges on behalf of any entity of neighboring country complying with CERC regulations. There are three major exchanges in India for Power Trading – Indian Energy Exchange (IEX), Power Exchange of India Ltd. (PXIL) and Hindustan Power Exchange Limited (HPX).

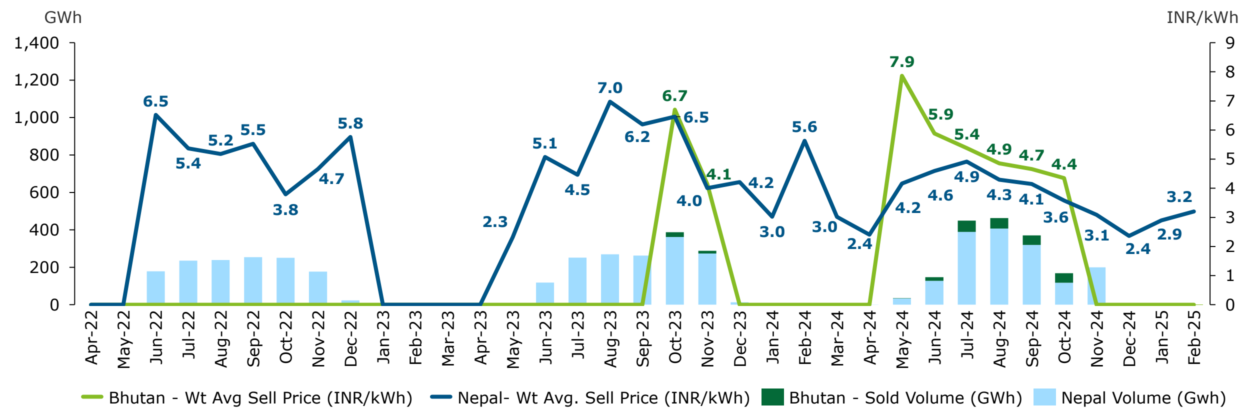
The Cross-Border Trade of Electricity was commenced in the Day Ahead Market of IEX in 2021-22. The trade with Nepal commenced on 17.04.2021, whereas the trade with Bhutan commenced on 01.01.2022. Cross-Border Trade in Real Time Market commenced from October 2023 in Nepal and in December 2023 in Bhutan. As per CERC market monitoring reports, no cross-border transactions were recorded in PXIL and HPX.

The below table shows the transactions by Bhutan in the IEX in the DAM and RTM segments in the last 4 years :

Table 57: Power Exchange Transactions of Bhutan recorded in IEX (DAM and RTM)[283]

Financial Year	Sell		Buy	
	Volume Sold (GWh)	Weighted Average Price (INR/kWh)	Volume Purchased (GWh)	Weighted Average Price (INR/kWh)
2021-22	-	-	240	2.89
2022-23	-	-	319	4.39
2023-24	40	5.78	1299	3.74
2024-25 (till Feb 2025)	240	4.94	902	3.02
Total	**280**		**2,760**	

Bhutan started selling power in the Indian power exchange from 2023-24. Previously, Bhutan used to only purchase power from the exchange. The below illustration shows the power sold in IEX by Bhutan and Nepal in the last 3 years :



Note : The months showing a zero price (0 INR/kWh) in the graph indicate that no power was sold by the country during those periods

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[282] Amendment-1 to the Procedure for approval and facilitating Import/Export (Cross Border) of electricity by the Designated Authority
[283] CERC Monthly and Annual Market Monitoring Reports

Figure 106: Power sold in IEX (DAM and RTM) by Nepal and Bhutan

Bhutan has been selling power in the exchange only during the wet season which is from May to November. Although, the volume sold by Bhutan has been lower in the exchanges but the price at which the electricity sold is higher when compared with Nepal.

7.2.1.1. Price and Volume Trends in Indian Energy Exchange (IEX)

The below illustrations showcase the Market Clearing Price (MCP) and the Final Scheduled volume (FSV) in the DAM and the RTM segments of IEX in the last 3 years to visualize the trends during the months allotted for export of power from DHPP which is from May to November.

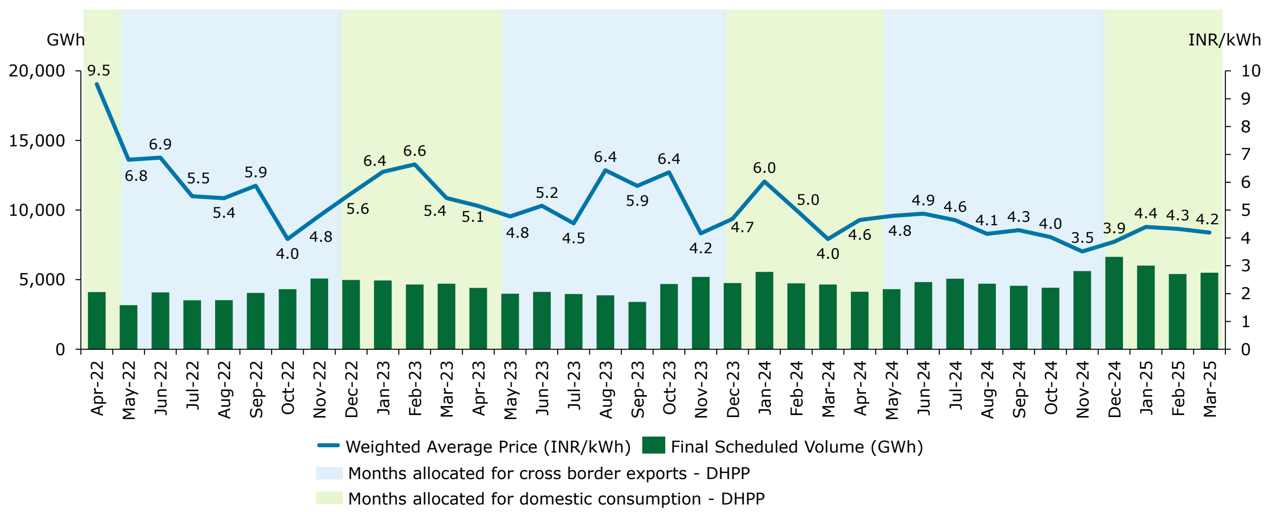

Figure 107: Monthly FSV and MCP in DAM of IEX during last 3 financial years

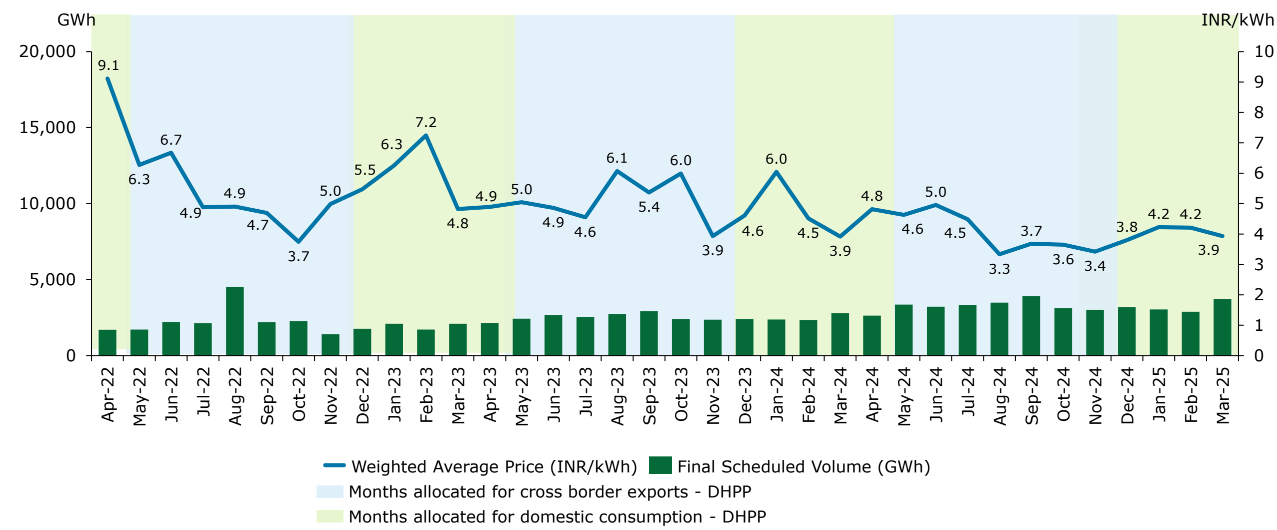

Figure 108: Monthly FSV and MCP in RTM of IEX during last 3 financial years

The MCP shows significant volatility throughout the period, indicating a dynamic interplay of supply and demand. However, there's a noticeable pattern in MCP during the DHPP export months (May to November) :

Pre-Monsoon/Summer Peaks : MCP rises during the pre-monsoon and summer months (May-June), likely due to the higher demand for cooling requirements during summer in India.

Monsoon/Post-Monsoon Dip: MCP is lower during the monsoon months (July-August) likely due to increased domestic hydropower generation.

The MCP in both DAM and RTM has decreased over the years. The transactions and price in the DAM has been higher when compared with RTM.

The hourly trend during the May-November period is shown below for the last 3 financial years for DAM and RTM :

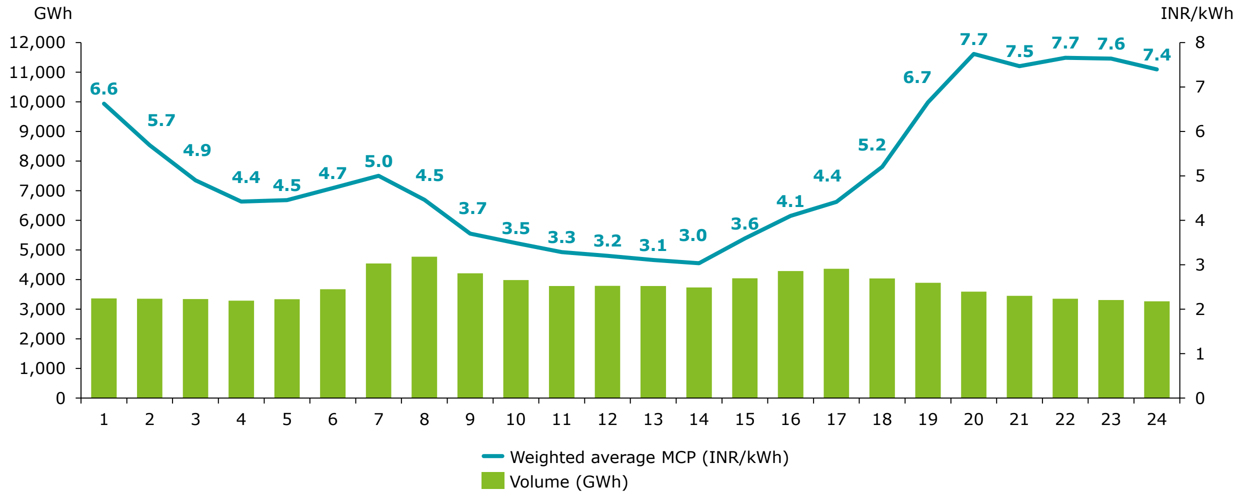

Figure 109: Hourly FSV and MCP in DAM of IEX during last 3 financial years from May to November

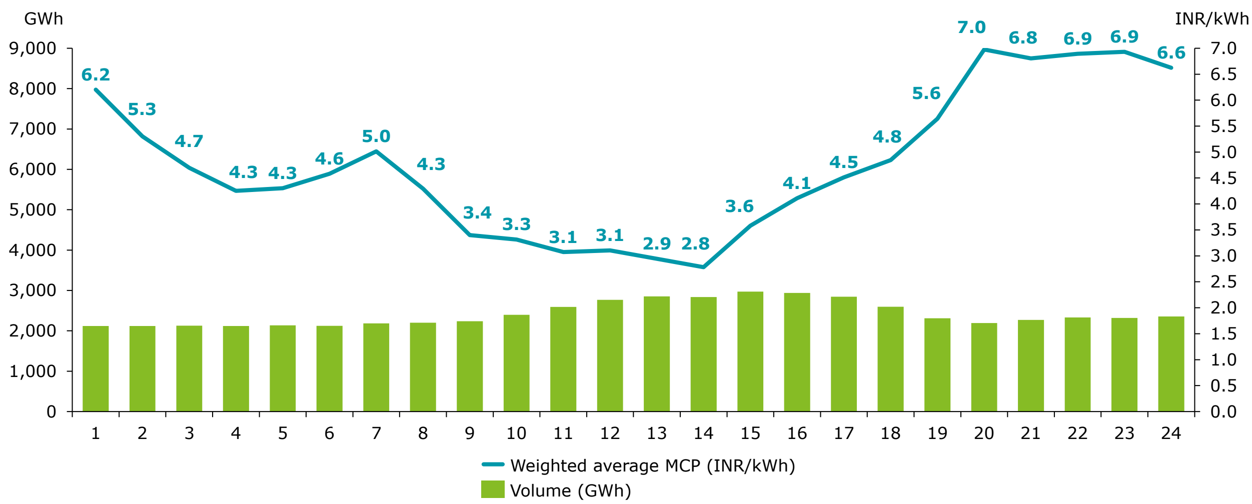

Figure 110: Hourly FSV and MCP in RTM of IEX during last 3 financial years from May to November

The hourly trend shows MCP peaking in the evening after 8 pm for both DAM and RTM segment and lower during the solar hours. This is primarily because during daylight, abundant solar power generation significantly increases electricity supply and due to its low cost, the overall MCP is decreased. As Solar energy decreases, post sunset, the grid relies more on expensive conventional power sources to meet the increased demand resulting from residential sector.

Recent trends in the Day-Ahead Market (DAM) and Real-time Market (RTM) on the Indian Energy Exchange (IEX) from FY 23 to FY 25 show a consistent decline in hourly electricity prices across most time blocks during the year. The heatmap in the Figure below highlights this shift, with red zones (high prices) gradually transitioning to green zones (low prices), reflecting increased renewable energy penetration, improved grid efficiency, and changing demand patterns. Prices remain lowest during solar hours due to high solar generation and peak during evening hours when demand rises and solar output drops.

		FY 2022-23												FY 2023-24												FY 2024-25											
Hour		Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22	Nov-22	Dec-22	Jan-23	Feb-23	###	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23	Jan-24	Feb-24	Mar-24	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25
1		11.82	9.13	10.51	7.04	5.26	6.03	3.42	3.56	2.86	2.96	5.16	5.77	7.58	7.59	8.90	6.55	8.73	7.38	7.04	3.35	2.92	3.03	3.25	4.29	8.69	9.66	9.74	8.50	5.78	4.98	4.91	2.80	2.37	2.54	3.10	4.44
2		11.52	8.06	8.90	5.56	4.43	5.11	3.22	3.36	2.68	2.77	4.42	4.93	7.15	6.36	8.05	4.94	8.16	6.48	6.67	3.16	2.79	2.79	3.08	3.70	7.26	8.12	8.30	6.80	4.49	3.96	3.78	2.65	2.29	2.38	3.09	4.26
3	Night Hours	11.22	7.29	7.61	4.57	3.91	4.58	3.02	3.23	2.55	2.65	4.07	4.36	6.72	5.07	6.67	3.99	7.38	5.70	6.28	3.03	2.69	2.61	2.95	3.46	6.00	5.37	6.63	4.98	3.70	3.55	3.48	2.62	2.20	2.29	3.00	3.85
4		10.93	6.77	6.68	4.11	3.69	4.18	2.87	3.14	2.54	2.62	3.89	4.21	6.33	4.34	5.27	3.60	6.57	5.30	5.87	2.92	2.63	2.56	2.86	3.36	5.33	4.31	5.34	4.66	3.59	3.34	3.26	2.58	2.20	2.27	2.97	3.69
5		10.74	6.66	7.17	3.99	3.87	4.32	3.02	3.32	2.78	2.78	4.49	4.78	6.64	4.14	4.82	3.42	6.74	5.13	5.84	3.11	2.77	2.71	3.03	3.44	4.88	4.30	5.38	4.51	3.67	3.35	3.16	2.60	2.29	2.38	3.04	3.89
6		11.11	6.72	6.78	4.76	5.40	5.43	3.51	4.02	3.58	3.64	6.53	5.77	7.30	4.38	4.85	3.50	7.55	5.74	6.82	3.71	3.16	3.34	3.46	3.76	4.96	4.31	4.95	3.78	3.45	3.33	3.34	2.85	2.81	2.93	3.47	4.52
7	Morning Peak	11.26	6.39	6.03	5.46	6.14	7.05	4.11	4.79	5.52	6.35	9.59	6.85	6.03	3.88	3.94	4.06	8.39	6.89	7.15	3.88	4.28	6.05	5.97	4.32	5.33	4.25	4.98	4.70	4.18	3.78	3.49	3.36	3.73	4.11	5.95	6.15
8		10.05	5.15	4.23	5.14	5.28	5.52	3.99	5.61	8.55	9.35	10.39	7.10	4.12	3.23	3.16	3.21	7.38	6.21	6.48	4.07	7.05	9.34	8.78	5.15	4.33	3.44	3.64	3.70	3.67	3.25	3.40	3.86	6.19	8.63	9.39	5.99
9		8.59	4.17	3.19	3.79	4.00	4.00	3.35	5.22	8.09	9.97	9.86	5.65	3.70	2.92	2.67	2.79	5.22	4.80	5.46	3.92	7.58	9.97	8.55	4.44	3.53	2.93	2.81	3.04	2.85	2.55	2.95	3.45	6.93	9.26	7.52	3.54
10		8.35	4.14	2.93	3.34	3.62	3.51	3.07	5.01	7.71	10.28	9.31	5.56	3.52	2.91	2.77	2.86	4.83	4.72	5.13	3.87	6.75	9.90	7.16	3.96	3.32	2.84	2.59	2.68	2.51	2.29	2.60	3.05	5.88	7.77	5.78	3.33
11		8.45	4.44	3.27	3.14	3.34	3.34	2.94	4.66	6.97	9.39	7.93	4.80	3.25	2.93	2.89	2.75	4.35	4.40	4.72	3.52	5.32	9.67	5.68	3.48	3.00	2.79	2.55	2.38	2.28	2.14	2.21	2.43	3.86	4.38	3.64	2.90
12		8.20	4.86	3.59	3.01	3.29	3.36	2.79	4.43	5.81	7.46	6.34	4.26	3.09	3.00	3.07	2.75	4.09	4.27	4.58	3.34	4.21	7.21	4.16	3.26	2.81	2.67	2.43	2.23	2.04	2.09	2.06	2.15	3.12	3.40	3.35	2.74
13	Solar Hours	7.46	4.84	3.84	2.93	3.05	3.37	2.63	4.02	4.82	6.11	5.08	3.58	2.95	2.98	3.16	2.73	4.07	4.16	4.50	3.14	3.44	4.71	3.73	3.01	2.61	2.71	2.50	2.03	1.86	1.87	1.98	1.91	2.75	3.06	3.12	2.56
14		6.83	5.05	4.01	2.79	2.82	3.13	2.52	3.81	4.01	4.58	4.00	3.19	2.77	2.86	3.17	2.70	3.94	4.17	4.32	2.99	3.14	3.57	3.35	2.72	2.52	2.68	2.53	1.96	1.77	1.84	1.98	1.81	2.48	2.68	2.78	2.37
15		8.14	6.17	5.09	3.06	3.22	3.71	2.73	4.03	4.30	4.53	4.04	3.51	3.11	3.28	3.77	2.90	4.90	4.82	5.16	3.23	3.28	3.74	3.41	3.04	2.92	3.46	3.39	2.52	2.17	2.38	2.50	2.29	2.80	2.97	2.84	2.57
16		9.31	7.14	6.18	3.69	3.81	4.54	3.03	4.37	4.96	5.69	4.89	4.21	3.42	3.76	4.32	3.10	5.58	5.51	6.12	3.57	3.53	4.40	3.69	3.29	3.37	3.84	3.74	3.01	2.68	2.91	2.96	2.83	3.19	3.21	3.13	2.92
17		9.99	6.95	5.46	3.90	4.34	5.01	3.43	4.79	6.78	7.24	6.88	5.29	3.79	3.85	4.29	3.12	6.20	5.80	7.31	4.73	5.60	7.43	4.75	3.69	3.87	4.20	3.87	3.21	3.13	3.25	3.41	3.57	4.18	4.04	3.64	3.47
18		10.30	6.68	4.77	4.08	4.38	5.78	4.55	6.58	8.01	8.43	8.00	5.07	4.10	3.88	3.97	3.15	6.49	6.03	8.55	8.03	8.85	9.63	6.49	3.71	4.16	4.00	3.86	3.29	3.20	3.92	4.88	7.22	8.31	6.94	4.76	3.69
19	Evening Peak	10.86	6.88	5.31	6.52	6.62	9.00	8.71	8.70	9.28	10.40	10.56	6.90	6.32	4.52	4.52	4.10	7.92	7.83	9.00	8.19	8.86	9.80	7.70	4.70	5.02	4.64	4.22	4.12	4.15	5.81	8.06	8.15	9.02	9.65	8.53	6.93
20		11.27	8.26	9.29	9.88	9.63	11.23	8.45	6.04	7.82	10.38	10.93	8.72	7.68	7.24	7.82	7.62	9.28	9.31	8.37	5.12	6.34	9.44	8.23	5.69	7.01	7.79	7.41	8.45	8.14	9.10	7.08	4.18	4.88	8.01	7.79	9.00
21		10.90	8.23	9.34	10.18	9.39	9.33	5.00	4.85	5.10	7.52	7.32	7.03	7.33	6.94	8.48	8.44	9.32	8.93	7.48	4.20	4.36	6.97	6.12	4.49	6.43	8.92	9.13	9.46	8.69	8.41	5.88	3.56	3.50	4.38	4.14	7.75
22		11.22	9.15	10.42	10.12	9.12	8.49	4.18	4.49	4.30	5.74	5.96	6.33	7.57	7.64	9.25	8.99	9.58	8.85	7.46	3.78	3.49	4.38	4.57	4.45	7.76	9.63	9.83	9.82	8.85	8.29	5.82	3.30	3.13	3.50	3.74	5.67
23	Night	11.25	9.55	10.72	9.85	8.43	7.91	3.83	4.50	3.57	4.13	5.50	6.31	7.91	8.00	9.61	9.14	9.54	8.67	7.27	3.52	3.08	3.44	3.73	4.23	8.47	9.85	9.98	9.90	8.41	7.30	5.55	3.02	2.72	3.00	3.34	5.88
24		11.78	9.49	10.52	8.91	6.96	7.22	3.49	3.82	3.12	3.33	5.20	6.05	7.63	7.99	9.46	8.73	9.27	8.48	7.28	3.32	3.00	3.25	3.49	4.22	9.11	9.89	10.00	9.86	7.61	6.77	5.43	2.86	2.61	2.78	3.03	5.10
		Dry Season	Wet Season							Dry Season				Dry Season	Wet Season							Dry Season				Dry Season	Wet Season							Dry Season			

Figure 111: Average monthly day ahead short-term power prices on exchanges in DAM market across 24 hours for FY 23, 24,25 [INR/kWh]

Hour		FY 2022-23											
		Apr-22	May-22	Jun-22	Jul-22	Aug-22	Sep-22	Oct-22	Nov-22	Dec-22	Jan-23	Feb-23	Mar-23
1		11.47	9.49	10.02	6.03	5.98	5.80	3.37	3.93	3.07	3.13	5.90	5.13
2		11.08	8.79	8.83	4.64	4.81	4.76	3.22	3.66	2.77	2.73	4.83	4.62
3	Night Hours	10.69	7.94	8.01	4.22	4.17	3.98	3.01	3.55	2.56	2.56	4.54	4.12
4		10.28	6.63	6.81	3.92	3.90	3.79	2.85	3.40	2.44	2.58	4.30	3.86
5		10.50	6.19	6.86	3.89	4.01	4.04	3.14	3.71	3.27	3.15	4.99	4.39
6		10.67	6.19	6.45	4.39	4.93	4.18	3.78	4.40	3.96	3.76	7.09	5.63
7	Morning Peak	10.82	5.81	6.48	4.96	5.55	5.57	4.63	4.72	4.91	5.70	9.23	6.43
8		10.07	4.76	4.79	4.64	4.88	4.59	4.29	5.30	6.83	7.11	10.22	6.48
9		7.53	3.10	3.55	3.73	3.81	3.60	3.45	5.19	7.09	8.48	9.09	4.90
10		6.99	2.89	3.44	3.56	3.53	3.46	3.12	5.17	7.20	9.34	9.25	5.15
11		7.08	3.66	3.67	3.23	3.39	3.16	3.10	4.98	6.81	8.34	7.98	4.31
12		7.65	3.51	4.03	3.30	3.34	3.38	2.83	4.63	6.31	6.88	7.12	3.97
13	Solar Hours	7.18	3.83	4.05	2.89	3.07	3.09	2.50	4.28	5.43	6.27	5.76	3.66
14		6.58	3.87	4.05	2.58	2.68	2.64	2.30	3.94	4.47	4.94	4.38	3.23
15		7.78	4.43	5.16	3.29	3.51	3.37	2.97	4.63	5.07	5.27	4.84	3.76
16		8.40	5.44	6.16	3.51	3.88	4.00	3.42	4.99	5.61	6.06	5.72	4.34
17		9.59	6.05	5.95	3.60	4.18	4.28	3.99	5.67	6.87	6.98	6.82	5.39
18		9.80	5.01	5.11	3.42	4.19	4.78	5.49	7.48	7.94	6.86	6.38	4.74
19	Evening Peak	10.27	5.29	5.37	4.84	5.73	7.68	7.93	7.97	8.45	9.00	9.56	5.39
20		10.80	7.02	8.75	8.13	8.74	9.44	7.01	5.26	7.24	9.37	10.66	6.92
21		10.57	8.17	9.50	8.13	8.72	7.48	4.06	4.60	4.82	6.98	8.03	5.73
22		10.83	8.78	9.85	8.16	8.09	6.68	3.78	4.32	3.92	4.85	6.38	5.33
23	Night	11.07	8.92	10.31	7.66	7.72	6.50	3.74	4.48	3.86	4.17	6.47	5.95
24		11.33	9.31	10.24	6.82	6.76	5.48	3.24	3.93	3.34	3.41	6.31	5.47
		Dry Season	Wet Season							Dry Season			

Hour		FY 2023-24											
		Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23	Nov-23	Dec-23	Jan-24	Feb-24	Mar-24
1		6.38	7.06	7.68	5.85	7.96	6.97	6.36	3.21	2.96	3.09	3.04	4.14
2		5.98	6.02	6.34	4.35	6.90	5.62	5.73	2.89	2.54	2.74	2.88	3.48
3	Night Hours	5.53	5.12	5.29	4.22	6.13	5.12	5.57	2.82	2.37	2.65	2.70	3.28
4		4.95	4.49	4.79	3.74	5.43	4.73	5.31	2.86	2.40	2.56	2.59	3.18
5		5.34	4.37	4.61	3.96	5.72	4.76	5.46	3.24	2.90	2.94	2.90	3.36
6		5.88	4.42	4.58	4.14	6.71	5.34	5.90	3.63	3.39	3.38	3.48	3.86
7	Morning Peak	5.33	4.05	4.31	4.92	7.98	6.41	6.59	3.94	4.15	5.77	5.52	4.65
8		4.11	3.07	3.37	3.93	6.70	5.41	6.14	4.38	6.81	8.59	7.67	5.36
9		3.41	2.63	2.77	3.27	5.01	4.06	5.17	4.17	7.52	9.68	7.50	4.50
10		3.58	3.05	2.81	3.45	5.06	4.16	4.98	4.20	7.01	9.85	6.75	3.99
11		3.12	2.96	2.80	2.95	4.65	3.67	4.82	3.83	5.79	9.23	5.18	3.43
12		3.44	3.32	3.26	2.95	4.86	3.83	4.74	3.57	4.85	7.48	4.34	3.22
13	Solar Hours	3.17	3.36	3.14	2.72	4.59	3.50	4.35	3.25	4.13	5.83	3.78	3.02
14		2.81	3.20	3.06	2.41	4.44	3.46	4.23	2.95	3.10	4.18	2.98	2.42
15		3.57	4.09	4.00	3.25	5.54	4.49	5.41	3.64	3.83	4.37	3.28	2.97
16		4.12	4.69	4.83	3.55	5.98	4.78	6.57	4.27	4.45	5.46	3.63	3.35
17		4.47	5.02	4.65	3.92	5.91	4.80	7.78	5.33	6.64	7.30	4.52	3.96
18		4.37	4.66	3.84	3.99	6.26	5.58	8.69	7.61	8.54	9.21	5.34	3.52
19	Evening Peak	5.48	4.41	3.57	4.53	6.95	7.42	8.86	7.23	8.23	9.17	7.15	4.90
20		7.37	7.45	6.30	7.70	8.95	8.72	7.66	4.56	6.70	8.80	7.50	6.43
21		6.49	6.77	6.51	7.74	8.83	8.06	6.74	3.94	4.64	7.32	5.14	4.70
22		6.94	7.41	7.83	7.87	8.67	7.36	6.57	3.49	3.49	5.07	4.07	4.18
23	Night	7.03	7.52	8.59	7.72	8.71	7.30	6.60	3.57	3.36	3.89	3.49	4.45
24		6.51	7.83	8.42	6.92	8.27	6.93	6.19	3.16	2.95	3.25	3.22	4.09
		Dry Season	Wet Season							Dry Season			

Hour		FY 2024-25											
		Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25
1		7.88	7.78	8.33	7.18	4.51	4.56	4.11	2.70	2.31	2.67	3.34	4.43
2		6.52	5.76	7.04	6.23	3.75	3.93	3.58	2.50	2.12	2.50	3.19	3.95
3	Night Hours	5.24	4.68	5.83	5.30	3.18	3.58	3.42	2.49	2.03	2.40	3.02	3.66
4		4.53	3.99	4.79	5.07	3.06	3.43	3.26	2.41	2.00	2.38	2.98	3.50
5		4.67	4.05	4.78	4.79	2.88	3.40	3.34	2.79	2.24	2.65	3.18	3.91
6		5.40	4.20	4.79	4.66	2.95	3.45	3.52	3.28	3.13	3.33	3.86	4.64
7	Morning Peak	6.29	4.08	4.73	4.95	3.69	4.29	3.72	3.79	3.86	4.58	5.93	5.80
8		3.87	2.77	3.30	3.84	2.96	3.39	3.68	4.27	5.63	8.06	8.97	5.26
9		3.29	2.55	2.66	2.97	2.37	2.73	2.90	3.81	6.17	8.80	7.14	3.57
10		3.19	2.69	2.66	2.67	2.31	2.70	2.84	3.44	5.40	7.28	5.71	3.51
11		2.88	2.71	2.52	2.39	2.04	2.34	2.52	2.83	4.15	4.57	3.94	2.96
12		2.87	2.99	2.76	2.29	1.96	2.19	2.53	2.70	3.55	3.55	3.41	2.68
13	Solar Hours	2.56	3.03	2.96	1.98	1.80	2.15	2.49	2.54	3.07	3.11	3.08	2.53
14		2.14	2.98	2.95	2.07	1.55	2.02	2.29	2.05	2.59	2.40	2.43	2.00
15		3.02	3.87	3.81	2.76	2.20	2.81	3.04	2.97	3.05	2.84	2.73	2.42
16		3.72	4.48	4.40	3.27	2.57	3.15	3.44	3.43	3.64	3.29	3.23	2.93
17		4.27	4.81	4.25	3.45	2.84	3.86	3.85	4.70	5.01	4.43	4.08	3.64
18		4.38	4.01	3.70	3.64	2.98	4.55	4.93	6.90	7.25	6.33	4.76	3.75
19	Evening Peak	4.54	3.88	3.65	5.76	3.98	5.88	6.88	7.17	8.09	8.97	8.00	6.08
20		6.58	6.57	6.03	8.29	6.92	7.90	6.46	3.70	5.00	7.85	7.74	8.43
21		6.22	7.13	7.84	8.83	7.06	6.41	4.93	3.43	3.69	4.50	4.48	5.78
22		7.11	7.81	8.93	9.01	6.84	6.08	4.56	3.25	3.23	3.63	3.90	4.99
23	Night	7.92	8.70	9.32	9.06	6.41	5.69	4.25	3.11	2.81	3.26	3.59	5.21
24		8.15	8.83	9.26	8.66	5.51	5.16	4.02	2.67	2.39	2.79	3.22	4.03
		Dry Season	Wet Season							Dry Season			

Figure 112: Average monthly short-term power prices on exchanges in RTM market across 24 hours for FY 23, 24,25 [INR/kWh]

During the wet season, Bhutan's rivers experience abundant flow, enabling increased hydropower generation. By strategically aligning electricity exports with high-price periods—particularly during morning and evening peaks—Bhutan can optimize revenue. Conversely, the consistently low prices during solar hours suggest that midday exports may be less economically viable unless driven by specific demand. However, the observed general downward trend in prices from FY 23 to FY 25 indicates that current prices may not hold in the future. Further with energy storage projects (specifically BESS projects) tariffs falling to Rs ~3.3 / kWh and below, the morning / evening peak prices are expected to taper further in the future. This has direct implications for the Dorjilung Hydropower Project (DHPP), as market conditions at the time of its commissioning may differ significantly, potentially resulting in lower price realization. Therefore, while current price patterns seem attractive, future market uncertainties necessitate cautious offtake arrangement for DHPP for the short term market.

7.2.1.2. Offtake through Bilateral Transactions

The Bilateral transactions happen through the traders or directly between the utilities. To increase transparency, the Government of India has introduced DEEP (Discovery of Efficient Electricity Price) e-Bidding & e-Reverse Auction portal for procurement of short and medium-term power[284]. The portal can be accessed through the MSTC website.

Majority of the utilities prefer the DEEP portal for procurement of short and medium-term power. The below table showcases the power procured for short and medium term from the DEEP portal :

Table 58: Bilateral transactions in short and medium term[285]

States	Total Volume Procurred (MW)	Short Term*		Medium Term**											
		(0-1) years		(1-2) years		(2-3) years		(3-4) years		(4-5) years		(5-6) years		(6-7) years	
		Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)
Andhra Pradesh	29,103	100%	9.30	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Himachal Pradesh	27,661	100%	8.16	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
West Bengal	26,263	98%	7.64	1%	5.88	0%	0.00	0%	0.00	1%	3.88	0%	2.24	0%	0.00
Tamil Nadu	9,206	88%	9.02	0%	0.00	0%	0.00	0%	0.00	12%	4.95	0%	0.00	0%	0.00
Kerala	10,806	90%	8.00	0%	0.00	2%	7.48	0%	0.00	7%	6.47	0%	0.00	0%	0.00
Delhi	16,243	99%	7.18	0%	0.00	0%	0.00	0%	0.00	1%	5.99	0%	0.00	0%	0.00
Rajasthan	10,951	99%	7.16	0%	0.00	0%	0.00	0%	0.00	1%	5.58	0%	0.00	0%	0.00
Haryana	12,247	82%	6.90	0%	0.00	1%	5.61	0%	0.00	8%	5.65	8%	6.03	0%	0.00
Maharashtra	11,928	74%	6.00	0%	0.00	7%	5.71	0%	4.95	16%	4.83	0%	0.00	3%	6.95
Gujarat	9,900	90%	6.15	10%	3.68	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Uttar Pradesh	7,245	90%	4.99	1%	4.25	6%	5.13	0%	0.00	3%	5.40	0%	0.00	0%	0.00
Bihar	1,805	100%	9.99	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Chattisgarh	1,493	100%	9.85	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Odisha	3,349	85%	9.01	6%	5.49	0%	0.00	9%	6.23	0%	0.00	0%	0.00	0%	0.00
Punjab	370	100%	8.41	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Madhya Pradesh	400	100%	6.69	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Assam	4,184	98%	6.63	2%	7.98	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Uttarakhand	3,682	91%	5.96	1%	6.51	0%	0.00	8%	7.77	0%	0.00	0%	0.00	0%	0.00

The volume and price of bilateral transactions in the short term (upto 1 year) is higher when compared with the medium term.

The Top 10 States with greater requirement of power and higher price in the short term are: Andhra Pradesh, Himachal Pradesh, West Bengal, Tamil Nadu, Kerala, Delhi, Rajasthan, Haryana, Maharashtra, and Gujarat.

Among these 10 States, 80% of them have price higher than INR 7/kWh.

Among the medium-term contracts, 4-5 years is the most preferred duration for procurement among the States. The Top 3 States with greater requirement of power and higher price in the 4-5 years duration are Maharashtra, Tamil Nadu and Kerala.

Since the volume of bilateral transactions in the short term market is higher, the power procurement pattern of the entities in the short term is analyzed. The following table captures the pattern of power procurement by the states in the short term:

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[284] Guidelines for short term procurement of power by Distribution Licensees through tariff based bidding process

[285] The Price mentioned is in INR/kWh; The Short term Data is taken from bids floated in FY 2025 and the Medium term Data is taken from bids floated 2017-2025

Table 59: Power procurement pattern in the Short term bilateral transactions[286]

States	Short Term (0-1 years)														
	Short Term Volume (MW)	Price (INR/kWh)	Total No of Bids	Total No of Bids											
				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Andhra Pradesh	29,103	9.30	44	5	4	3	3	3	3	1	4	4	5	4	5
Himachal Pradesh	27,661	8.16	187	36	36	34							13	33	35
West Bengal	25,823	7.64	85	2	1	1	12	11	11	9	8	8	20	1	1
Tamil Nadu	8,089	9.02	21		2	4	7	8							
Kerala	9,746	8.00	33	3	3	3	5	7	1	2	1	1	2	2	3
Delhi	16,146	7.18	71				5	14	10	13	14	15			
Rajasthan	10,791	7.16	14	1	1			1	2	1	1	2	2	2	1
Haryana	10,094	6.90	15					1	3	3	3	3	2		
Maharashtra	8,831	6.00	97	9	6	7	12	6	10	13	6	6	9	6	7
Gujarat	8,900	6.15	24	1	1	2	2	3	3	1	1	3	3	2	2
Uttar Pradesh	6,485	4.99	98	6	6	6	6	9	9	10	11	11	10	7	7
Bihar	1,805	9.99	18				2	2	2	4	4	4			
Chattisgarh	1,493	9.85	6			2	2	2							
Odisha	2,854	9.01	16				4	6	6						
Punjab	370	8.41	5						1	1	1	1	1		
Madhya Pradesh	400	6.69	1										1		
Assam	4,084	6.63	27				6	5	4	4	4	4			
Uttarakhand	3,350	5.96	21	2	2	2	1	1	1	1	1	1	3	3	3
Sum			783	65	62	64	67	79	66	63	59	63	71	60	64

The power procurement pattern clearly shows that highest number of bids is floated in May, followed by October. This is likely due to the increasing demand for cooling requirements in the Summer (April-June) and increasing demand during the festival season (September – November).

Further during the export months of DHPP (May – November), the top 3 states with maximum number of bids and price greater than INR 6/kWh are West Bengal, Delhi and Maharashtra.

The Short Term Bilateral Market offers higher volumes compared to Medium Term market. The number of bids for Short Term Market also showcases higher demand during the wet season in India which align with DHPP export potential.

[286] The Price mentioned is in INR/kWh; The Short term Data is taken from bids floated in FY 2025

7.3. Offtake Analysis through C&I

The Commercial and Industrial (C&I) sector in India plays a vital role not only in driving the country's economic growth but also in supporting its broader energy transition. Over the past eight years, ie., from FY 2016 to FY 2024, the sector has maintained a steady growth rate of approximately 5%. It currently accounts for around 40% of the total electricity consumption in India and is also responsible for ~25%[287] of the total CO2 emissions. The details of the same is provided in the figure below:-

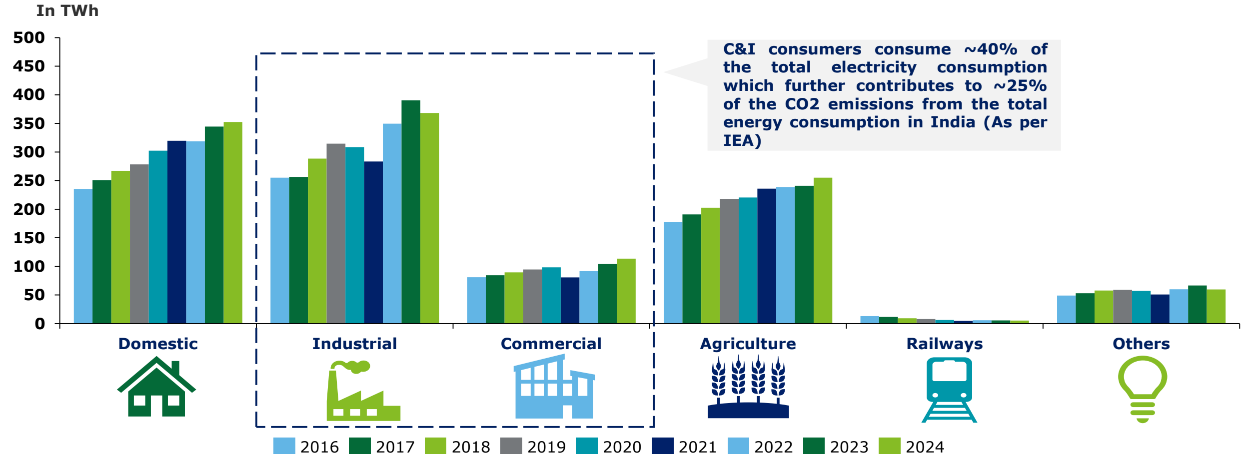

Figure 113: Category wise Electricity Consumption in TWh in India[288]

At COP26, India announced its ambitious target to achieve net-zero emissions by 2070—a critical milestone in the global fight against climate change, especially given India's position as one of the world's fastest-growing economies. The Commercial and Industrial (C&I) segment is expected to play a pivotal role in this decarbonization journey. Many major C&I players have already begun aligning with net-zero pathways, incorporating a defined share of renewable energy into their overall energy consumption mix. The table below highlights the current share of renewables in their energy portfolios, along with their respective net-zero target timelines.

Table 60: Share of Renewable Energy of major C&I consumers in India

S. N	Company Name	Share of Renewable Eenergy		Net-Zero Emission target	Source
		Non Renewable Sources	Renewable Sources		
1	Reliance Industries Ltd	99%	1%	2035	BRSR Report 2023-24
2	Vedanta	79%	21%	2050	BRSR Report 2023-24
3	Aditya Birla Group	12%	88%	2050	BRSR Report 2023-24

[287] Source: IEA, 2022
[288] Source: India's Climate and Energy Dashboard

S. N	Company Name	Share of Renewable Eenergy: Non Renewable Sources	Share of Renewable Eenergy: Renewable Sources	Net-Zero Emission target	Source
4	ITC limited (in TJ)	48%	52%	2040	BRSR Report 2023-24
5	Tata Motors	67%	33%	2030	BRSR Report 2023-24
6	Sunpharma	72%	28%	2050	BRSR Report 2023-24
7	Ambuja Cement	94%	6%	2050	BRSR Report 2023-24
8	Adani Enterprise	72%	28%	2050	BRSR Report 2023-24
9	Infosys	67%	33%	2040	Annual Report 2023-24
10	Asian Paints	45%	55%	2070	BRSR Report 2023-24

The Commercial and Industrial (C&I) segment is one of the largest open access consumer bases in India, allowing these consumers to procure power directly from generators, bypassing traditional distribution companies (DISCOMs). In many cases, open access consumers in India are subject to Renewable Purchase Obligation (RPO) targets, which require them to source a specified percentage of their electricity from renewable energy. These targets are established by the Central and State Electricity Regulatory Commissions (CERC and SERC).

With India's ambitious net-zero goals, RPO targets are expected to become more stringent, driving increased renewable energy procurement within the C&I segment. Procuring renewable energy is particularly advantageous for this segment, as DISCOM tariffs for C&I consumers are typically higher compared to other consumer categories.

We have identified the top ten states where average billing rates for C&I consumers are highest which provides opportunity for C&I consumers located in these states to look for alternative source of supply.

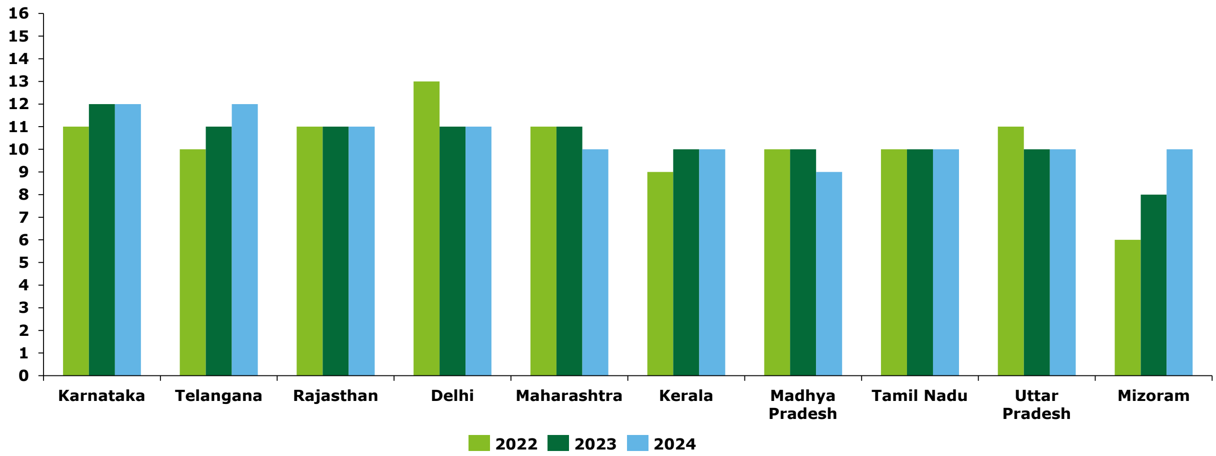

Figure 114: ABR (in INR/kWh) of top 10 States for Commercial Consumers[289]

As can be seen that for the top ten states based on average billing rate of FY 2023-24, the tariffs charged by the state DISCOMs is in the range of INR 10-12 in FY 2023-24.

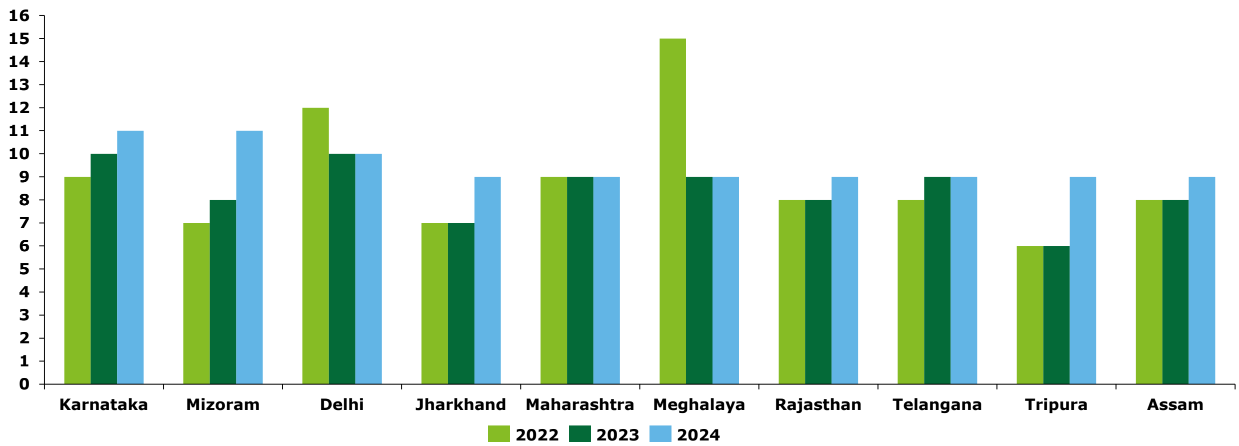

Figure 115: ABR (in INR/kWh) of top 10 States for Industrial Consumers[290]

For Industrial consumers, the average billing rate (ABR) charged by the state DISCOMs is in the range of INR 9-11 per unit in FY 203-24.

It is therefore crucial for major corporate houses to procure cheaper sources of energy, such as renewables including hydropower, to reduce their electricity costs. This not only positively impacts profitability but also enhances process efficiency through the use of cleaner fuels. However, power procured via open access is subject to additional charges, including cross-subsidy surcharge (CSS), additional surcharge, SLDC charges, distribution and wheeling charges, reactive energy charges, and Green Cess.

In the context of DHPP, it is important to engage with major corporate houses to better understand their specific requirements for hydropower integration.

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289 India Climate and Energy Dashboard
290 India Climate and Energy Dashboard

7.4. Benchmarking of Offtake Strategy for Hydropower Projects

The offtake arrangements of major hydro projects across India have been analyzed to evaluate how offtake structuring varies based on project ownership—ranging from fully government-owned to fully private entities. Based on this structuring, the associated risk of participating in the short-term power market has been assessed. This analysis can help inform DHPP's approach to balancing long-term PPAs with short-term market participation.

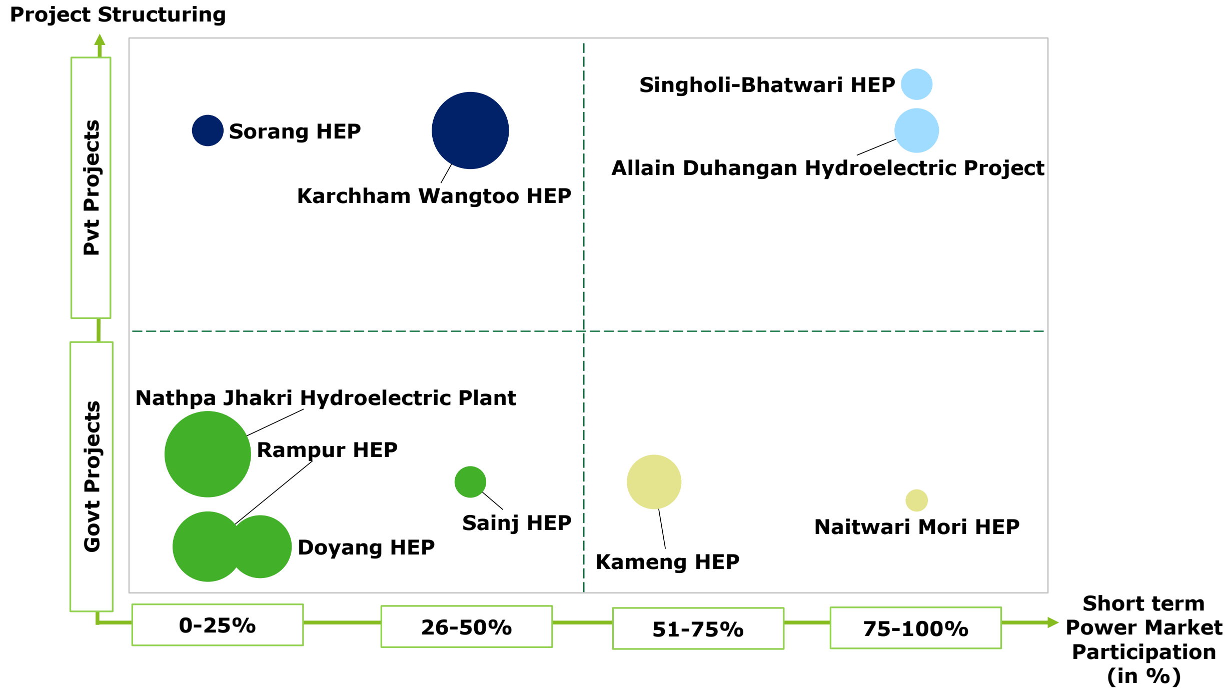


Figure 116: List of Offtake Arrangement of Hydro Projects

The Study of various offtake models across different project structures reveals that most hydro projects are fully committed (100%) under long-term Power Purchase Agreements (PPAs) with State DISCOMs or through central government allocations However, a few projects, such as Himachal Pradesh's hydroelectric plants, participate in short-term power markets due to the absence of long-term PPAs[291]. For the remaining projects, short-term market participation is generally limited to around 30%. In contrast, private hydro projects which is driven by a higher risk appetite exhibits significantly greater exposure, with some participating up to 100% in the merchant power market. These trends are illustrated in the graph below

Table 61: The detailed list of key Hydro project is provided in the table below

S.N	Project	Capacity (MW)	Location	Ownership	Mode of Sale	Offtake
1	Naitwar Mori HEP	60	Uttarakhand	Government – SJVN Ltd.	Merchant (short-term sale)	Sold via exchanges (no long PPA)
2	Singholi-Bhatwari HEP	99	Uttarakhand	Private – ReNew Power Ltd.	Merchant (short-term sale)	No long PPA; entire generation sold on exchange.

[291] Petition for approval of capital cost for Sainj HEP, HPPCL

S.N	Project	Capacity (MW)	Location	Ownership	Mode of Sale	Offtake
3	Sainj HEP[292]	100	Himachal Pradesh	Government-HPPCL	Long term and Merchant (short-term sale)	50% Sold via exchanges (Remaining Long term PPA)
4	Kameng HEP	600	Arunachal Pradesh	Government – NEEPCO Ltd.	Merchant (short-term sale)	345 MW untied (out of 600 MW) sold on exchange
5	Doyang HEP	75	Nagaland	Government – NTPC NEEPCO Ltd.	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
6	Nathpa Jhakri Hydroelectric Plant	1500	Himachal Pradesh	Government-SJVNL	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
7	Karchham Wangtoo HEP[293]	1000	Himachal Pradesh	Private-JSW Energy	Long term and Merchant (short-term sale)	70.4% with PTC under Long term PPA, remaining-Short term
8	SORANG HEP	100	Himachal Pradesh	Private-HSPL (subsidiary of Greenko East Coast Power Projects Pvt Ltd.)	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
9	RAMPUR HEP	412	Himachal Pradesh	Government-SJVNL	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
10	Allain Duhangan Hydroelectric Project[294]	192	Himachal Pradesh	Private-Aditya Birla Group	Merchant (short-term sale)	Short term PPAs (1-3 years) and Power Exchange

Based on the analysis of offtake structures for hydro projects in India, it is observed that most of these projects, whether privately or government-owned, are tied up in long-term PPAs or allocated through central government mechanisms. However, participation in the short-term power market varies depending on the risk appetite. Government-owned projects typically participate in the short-term market up to approximately 30% in addition to their long-term commitments, whereas private projects, driven by higher risk tolerance, can have up to 100% exposure in the merchant power market.

[292] As per petition for determination of tariff from COD to FY 2023-24
[293] As per True Up Order 2014-19 of KW plant
[294] As per IFC disclosure

8. Bangladesh Power Sector Landscape

Purpose of the Section
This section provides an overview of Bangladesh's power sector, focusing on historical generation patterns, current energy security challenges, and strategic responses through regional cooperation and power imports.

Key Conclusion and takeaways
Natural gas has historically accounted for over 50% of Bangladesh's power generation, but declining reserves now present a significant energy security challenge. To address this, the government is increasingly turning to electricity imports from neighboring countries as a strategic measure to diversify and stabilize its energy supply. Bangladesh is rapidly expanding its clean energy capacity, targeting 40% of its total supply by 2041 and 66% by 2050. With approximately 20% of this clean energy expected to come from imports, Bhutanese hydropower is well-positioned to play a crucial role in helping Bangladesh achieve its ambitious sustainability goals.

Bangladesh is one of Asia's fastest growing economies and envisages to be in the 'high-income' country by 2041[295].The electricity sector in Bangladesh has made significant progress over the last decade in terms of generation capacity addition, transmission and distribution network strengthening and enhancing access to electricity as outcomes of various reform initiatives since the 1990s.

A snapshot of Bangladesh's power sector as of June 2024 is set out as follows:

Figure 117 : Power Sector Snapshot of Bangladesh[296]

Generation	• **28,098 MW** installed capacity • **Gas: 44%**, **Furnace Oil: 23%,** Coal: 18%, Import: 9%, Solar: 2%, Diesel: 2%, Hydro: 0.8% • **Public Share** : 49% • **95,996 GWh** Energy Generation
Transmission	• **15,624 Ckt. Km transmission lines** • **66,869 MVA Grid capacity** • **~3%** Transmission Loss
Distribution	• Key Distribution Entities: **DPDC, DESCO, WZPDCL, REB, NESCO, BPDB** • **56% Domestic** Consumption, 27% Industrial consumption • **~8% Distribution Loss**
Consumers	• **640 kWh per capita generation (including captive and off-grid renewable energy)** • **~4 million consumers** • **100% access to electricity**

295 Making Vision 2041 a Reality PERSPECTIVE PLAN OF BANGLADESH 2021-2041
296 Annual Report 2023-2024, Bangladesh Power Development Board (BPDB)

Bangladesh's electricity sector is navigating a period of rapid transformation, characterized by increasing demand. The electricity peak demand has grown at a CAGR of ~7% between FY15 and FY24, backed by high economic development and the Government of Bangladesh's ambitious Power for All program. Although, the demand shortfall has significantly declined over the years since FY15 but from FY20 onwards, the shortfall has been following an increasing trend.

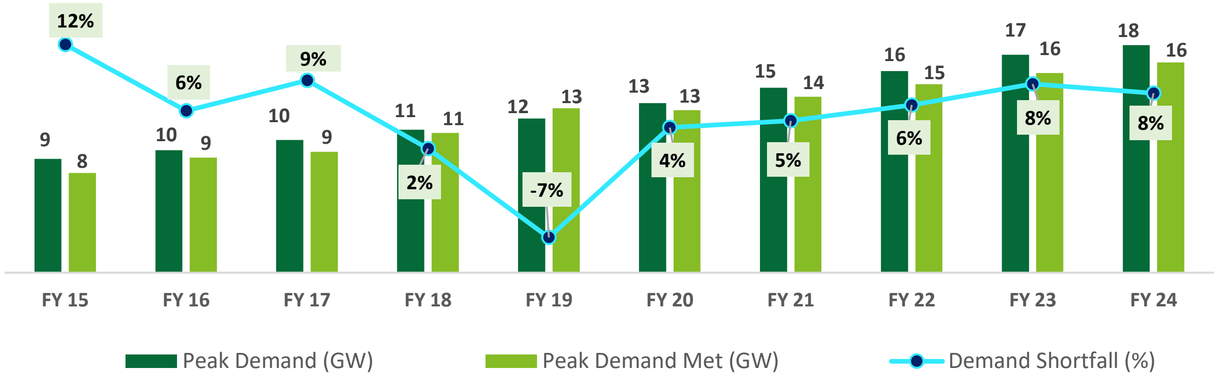


Figure 118 : Electricity Peak Demand and Peak Supply (GW)[296]

To meet the burgeoning electricity demand, Bangladesh has also been increasing its power supply. The country has historically been reliant on gas which contributes over ~50% of its total generation capacity. Considering the depleting natural gas reserves and to ensure energy security for the country the Government of Bangladesh has also been increasing power imports from neighbouring countries over the years. Bangladesh has also made a commitment to decrease GHG emissions by almost 22% below the "business as usual" level by 2030 (89.47 MT CO_{2e}). Therefore, over the years, the country has also scaled up the domestic renewable energy capacity.

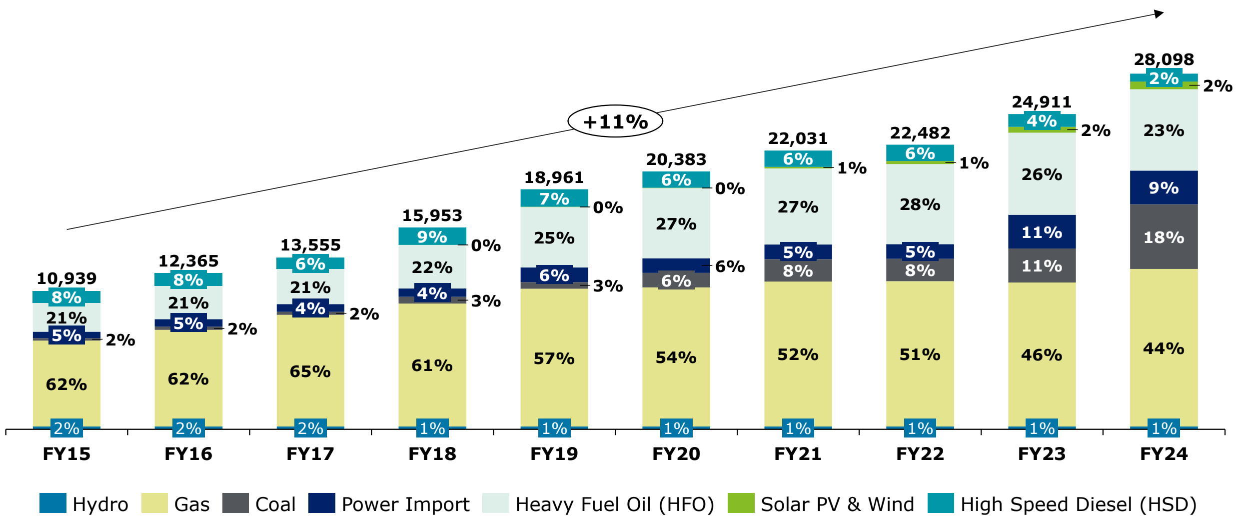


Figure 119 : Installed Capacity trend of Bangladesh by fuel type (MW)

In FY24, Bangladesh imported 2,656 MW (~9% of total installed capacity) of power. All the imported power was from India. The below table shows further details on the imported power :

Table 62: Import of power by Bangladesh in FY2024[297]

Particulars	Capacity	Units	Price

[297] Annual Report 2023-2024, Bangladesh Power Development Board (BPDB)

	(MW)	(GWh)	(INR/kWh)
NVVN Ltd. - India	250	1880	2.93
NVVN Ltd. - India (Tripura)	160	972	6.42
NVVN Bharamara Ph. II - India	300	2278	5.66
PTC India Ltd.	200	1672	6.50
Sembcrop Energy India Ltd.	250	1845	7.30
Adani Power (Jharkhand) Limited, India	1,496	8167	10.41
Total	**2,656**	**16,814**	

Further, a landmark tripartite agreement was signed in October 2024 between Nepal, India, and Bangladesh, enabling Nepal to export 40 MW of electricity to Bangladesh via India's transmission grid. The agreement is set for a five-year term, with the power to be traded at approximately 8.17 Bangladeshi Taka (about INR 9.30 or $0.11) per unit[298] The first transaction occurred on November 15, 2024, with Nepal exporting 40 MW to Bangladesh via India's grid[299]. This successful transaction, which took place in November 2024, sets a clear precedent for similar trilateral transactions involving other countries in the region.

The Bangladesh Power Development Board (BPDB) serves as the single buyer in the power market of Bangladesh. Besides its own generation, BPDB also procures electricity from private companies, commonly referred to as Independent Power Producers (IPPs), rental power plants and public power plants to meet the rising demand and sells bulk electricity to all the distribution utilities. Most of the IPPs (>70%) are Gas and HFO based with just ~5% of renewable energy capacity. All the Rental plants are HFO (65%) and gas based (35%). Majority (~70%) of the public sector plants (BPDB and others) are gas based with just ~1% of renewable energy capacity.

The per unit generation cost of BPDB's own plants and electricity purchases from external sources for the last 3 financial years are shown below[300]:

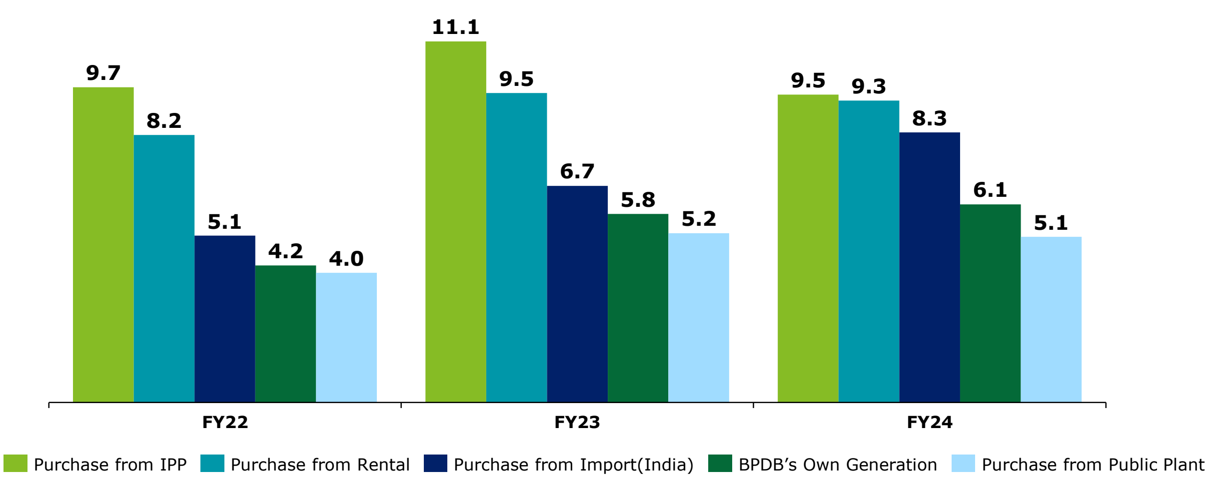


Figure 120: Per Unit cost of electricity generation and purchase by plant type in last 3 years (INR/kWh)

The fuel wise per unit generation cost for BPDB's own generation is given below by comparing with the import prices:

[298] South Asia Subregional Economic Cooperation (SASEC)
[299] Ministry of External Affairs, Government of India
[300] BPDB Annual reports. Average Exchange rate from BDT to INR for each year is taken from https://www.exchangerates.org.uk/BDT-INR-spot-exchange-rates-history-2024.html

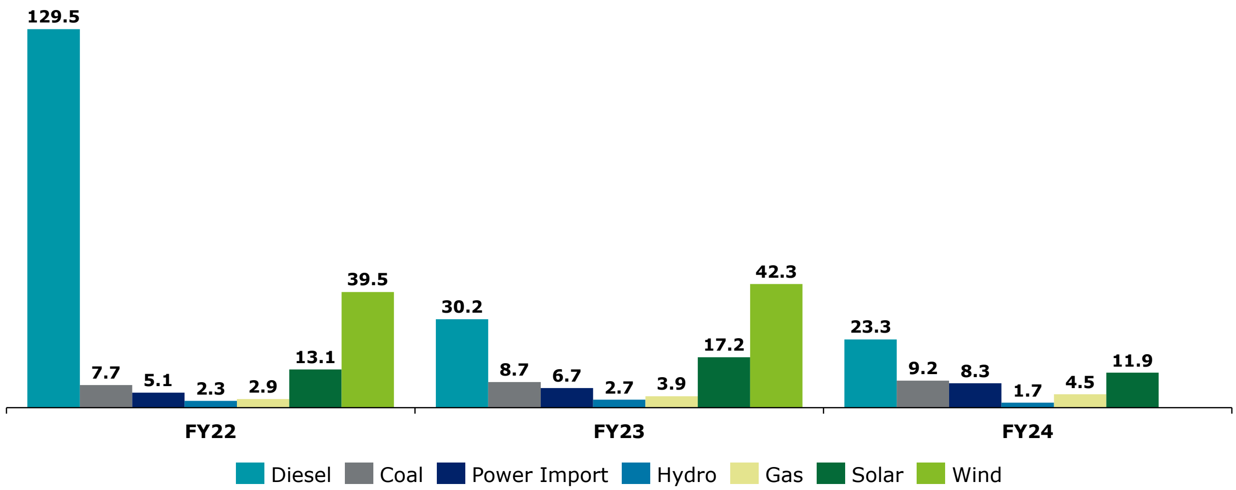


Figure 121 : Per Unit cost of fuel for BPDB's electricity generation in last 3 years (INR/kWh)

The cost of renewable energy generation (solar and wind) by BPDB is very high. Moreover, there is almost negligible hydro capacity in Bangladesh (1% of total installed capacity). Therefore, to increase renewable capacity, in addition to scaling up domestic renewables, Bangladesh needs to leverage regional cooperation to import clean electricity.

Further, as per Integrated Energy and Power Master Plan (IEPMP) 2023, overall electricity demand and supply projections under In-Between scenario (between BAU and Optimistic scenario) from 2030 to 2050 have been mapped (considering 1 Mtoe = 11.63 TWh for demand).

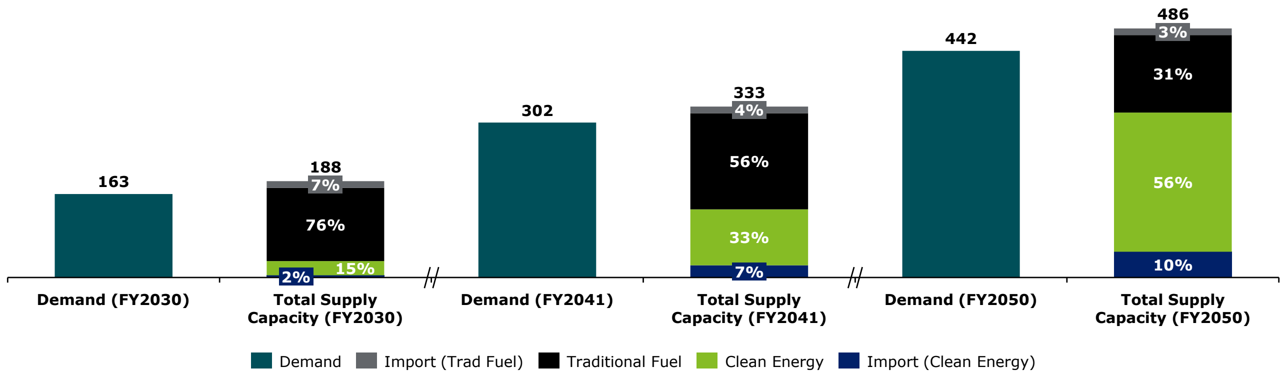


Figure 122 : Electricity Demand and Supply forecast (in TWh)

Bangladesh is significantly expanding its clean energy capacity, targeting 40% of total supply by 2041 and 66% by 2050. With imported sources expected to contribute roughly 20% of this clean energy, Bhutanese hydropower can be a key partner in achieving these ambitious governmental goals.

9. Regulatory Landscape in India and Bangladesh

Purpose of the Section

In this chapter, we examine the regulatory landscape across the South Asian region, with a particular focus on cross-border electricity trade (CBET) regulations and the tariff determination processes. The Indian regulatory framework has been analyzed in detail, including case studies highlighting instances where regulators have disallowed certain cost components—providing important insights into the potential regulatory risks. Additionally, the tariff determination process in Bangladesh has also been studied to identify regulatory considerations. These insights help outline the key regulatory cautions that DHPP should be aware of when planning power exports to these markets.

Key Conclusion and takeaways

The determination of tariffs for hydropower generation plants in India is governed by a well-established regulatory framework administered by the Central Electricity Regulatory Commission (CERC) and the respective State Electricity Regulatory Commissions (SERCs), based on a cost-plus methodology. Recent tariffs for projects commissioned over the past 4–5 years have largely ranged between INR 4 to 5 per unit. Common reasons for tariff disallowances in some hydro projects include cost overruns due to project delays and under-approval of annual fixed charges. While Bangladesh has a regulatory framework in place, including institutions such as the Bangladesh Energy Regulatory Commission (BERC), its regulatory maturity and market mechanisms are still evolving and are not yet comparable to the more advanced regimes found in developed power markets.

9.1. Cross Border Electricity Trade Regulations

9.1.1. Background

Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. In 2021, the Central Electricity Authority (CEA) issued the "Procedure for Approval and Facilitating Import/Export (Cross Border) of Electricity by the Designated Authority," enabling power trading through the Indian Energy Exchange (IEX). By October 2021, Nepal began participating in the day-ahead market (DAM) on the IEX, initially trading around 39 MW (24 MW from the Trishuli Hydropower Plant and 15 MW from the Devighat Powerhouse). Subsequently Bhutan also began trading electricity in the Indian Energy Exchange (IEX) on January 1, 2022 onwards.

Over the past three years, cross border trade volume in the DAM and RTM segments has increased. The Nepal Electricity Authority (NEA) has since secured medium-term power sales agreements with distribution companies in Haryana and Bihar. Additionally, India now recognizes hydropower imports from other countries as part of the Hydropower Purchase Obligation (HPO), encouraging Indian buyers to source power from Bhutan and Nepal.

In a landmark development, a trilateral agreement was also signed between Nepal, India, and Bangladesh for export of 40 MW of electricity from Nepal to Bangladesh via India’s grid. The Cabinet Committee on Public Purchase (CCPP) in Bangladesh has also approved importing approximately 500 MW from Nepal’s proposed 900 MW Upper Karnali Hydroelectric Project, developed by GMR, which will flow to Bangladesh via NTPC Vidyut Vyapar Nigam Limited through Indian transmission corridors. These developments further strengthen the creation of an integrated South Asian power market

Some of the key framework agreement for Cross Border Electricity Trade in the region have been enumerated below:

BIMSTEC MoU for Establishment of BIMSTEC Grid Interconnection

The Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) came into being on 6 June 1997 through the Bangkok Declaration.[301] The BIMSTEC Energy Centre (BEC) was established on 22 January 2011.

A Memorandum of Understanding (MoU) for Establishment of the BIMSTEC Grid Interconnection was signed on 31 August 2018 at the Fourth BIMSTEC Summit held in Kathmandu, Nepal. The MoU came in to force on 07 April 2019. The MoU broadly provides, the framework for parties to cooperate towards the implementation of the grid interconnections for the trade in electricity with a view to promoting rational and optimal power transmission in the BIMSTEC Region.

Table 63: BIMSTEC MoU Agreements

Article	Details
Article 3 (Institutional Arrangements)	For successful implementation of grid interconnections and trade in electricity, the BGICC may engage BIMSTEC Energy Sector Committee of Experts/Officials, Task Force for BIMSTEC Trans-Power Exchange and Development Projects, BIMSTEC Energy Center and other institutions to provide technical support. Furthermore, the BIMSTEC Grid Interconnection Coordination Committee (BGICC) shall determine modalities to implement BIMSTEC Grid Interconnections Master Plan and regional trade arrangements.
Article 4 (Settlement of Disputes)	Member parties implementing the MoU will seek amicable resolution through negotiations/consultations, which would be further referred to BGICC. If BGICC is unable to resolve the differences, that issue shall be referred to BIMSTEC Senior Officials' Meeting on Energy Resolutions.
Article 8 (Relation to Other Agreements)	Member States are not restricted to enter into agreements with a non-BIMSTEC countries on bilateral/multilateral and would not impair a BIMSTEC Member State who is Party to such agreements from fulfilling its obligations under the MoU.

9.1.2. Salient Features of Indian Cross Border Electricity Trade Regulations

Cross-border electricity trade (CBET) in India is primarily managed by the Ministry of Power in coordination with key power sector entities, including the Central Electricity Regulatory Commission (CERC) and the Central Electricity Authority (CEA). The Ministry of Power oversees policy-level interventions, while CERC and CEA handle regulatory and procedural aspects as the central regulating and planning bodies, respectively. The major policies and guidelines for CBET in India include the following:

- Cross Border Trade of Electricity Regulations, 2019
- Guidelines for Import/Export (Cross Border) of Electricity – 2018 and its amendment in 2023
- Procedure for Approval and Facilitating Import/Export (Cross Border) of Electricity by the Designated Authority, 2021 and 2023

Additionally, power exchange operations and transactions are regulated by CERC through specific regulations, with the National Load Dispatch Centre (NLDC) ensuring grid stability during power flows on the exchange.

Under the "Procedure for approval and facilitating Import/Export (Cross Border) of Electricity) by the Designated Authority, 2021", Indian entities are not able to trade with any generating company that is, "owned, directly or indirectly by any natural/ legal personality(ies) whose effective control or source of funds or residence of beneficial owner, is situated in/ citizen of a third country with whom India shares land border and that third country does not have a bilateral agreement on power sector cooperation

[301] BIMSTEC member states are Bangladesh, Bhutan, India, Myanmar, Nepal, Sri Lanka, and Thailand.

with India." For any relaxation in this provision, the Designated Authority must consult with the Indian Ministry of Power and Ministry of External Affairs.

The Government of India has introduced facilitating measures as well, including an amendment by the Central Electricity Authority (CEA) on 31st July 2023 to the procedure for cross-border electricity import/export approval, wherein Indian power traders can trade on Indian power exchanges (Day Ahead Market, Real-Time Market, or both) on behalf of entities from neighbouring countries, after obtaining approval from the designated authority. This is permitted up to a specified quantum and duration, provided the neighbouring country has a power sector cooperation agreement with India, and the generating asset involved is owned or controlled by that country. Both Bhutan and Nepal are trading in Indian Power Exchanges through this route.

However, on 12th August 2024, the Ministry of Power announced revisions to the "Guidelines for Import/Export (Cross Border) of Electricity - 2018" to regulate and streamline electricity exports from coal and gas-based plants in India. Under the updated guidelines, coal-based plants can only export electricity if the coal is sourced through imports, commercial mining, or spot e-auctions. Similarly, gas-based plants must use gas that is imported or sourced from approved suppliers. The guidelines also include provisions for Indian generating stations that supply electricity exclusively to neighbouring countries. These stations may build dedicated transmission lines to connect with the neighbouring country's transmission system, considering technical and strategic factors, with approval from the designated authority as per the Electricity Act, 2003. The costs for such construction will be outlined in the contract agreement between the Indian and neighbouring entities. Additionally, the guidelines allow these stations to connect to the Indian grid (interstate or intra-state) to sell power domestically if there is a sustained non-scheduling of full or partial capacity, or if a default notice is issued by the generator for reasons such as delayed payments under the power purchase agreement (PPA).

Settlement Nodal Agency (SNA)

NTPC Vidyut Vyapar Nigam Ltd (NVVN) has been designated as the SNA[302] for Bangladesh, Bhutan, Nepal and Myanmar, under the Guidelines for Import /Export (Cross Border) of Electricity 2018, from India's side. It is therefore responsible for settlement of grid operation related charges like operating charges, charges for deviation. NVVN's key functions include coordination with Load Dispatch Centers on behalf of Cross Border Entity, provide information on Open Access, undertake weekly meter reading among others.

Other Cost Components for Cross Border Power Trading

The other billing and payment related activities include raising bill to Cross Border Entity for deviation in scheduling, regional transmission deviation charges, reactive energy charge & congestion charge, system operation fees & charges, Settlement Nodal Agency charges / fees, other statutory and necessary charges and late payment surcharge (if applicable).

9.1.3. Bangladesh Rules, Regulation and Guidelines

Bangladesh's policies and regulatory framework to promote CBET are distributed across several instruments including:

- Bangladesh Electricity Act 2018
- Bangladesh Electricity Regulatory Commission Electricity Grid Code Regulations, 2018
- Power Purchase Agreements
- Joint Steering Committee / Working Groups

Some of the salient coverage are summarized below:

[302] Source: http://nvvn.co.in/wp-content/uploads/2020/11/Nomination-of-NVVN-as-SNA.pdf

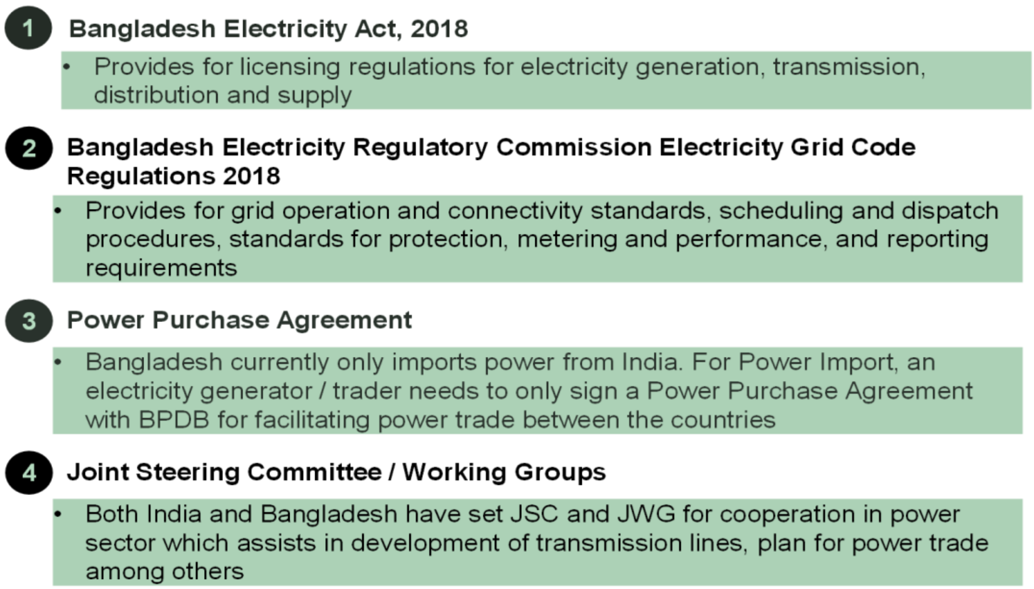


Figure 123: Key Instruments guiding Bangladesh Cross Border Power Trade

Some of the other regulations which may impact cross border trade are:

- Provisions related to CBET in the Quick Enhancement of Electricity and Energy Supply (Special Provisions) Act, 2010;
- Setting of fair rules and procedures for non-discriminatory open access in: Policy Guidelines for Enhancement of Private Participation, Electricity Act 1910, and BERC Power Transmission Tariff Regulations, 2016;
- Transmission pricing mechanisms in the Policy Guidelines for enhancement of Private Participation in the Power Sector; and
- Dispute resolution processes in BERC dispute settlement regulation 2014.

However, there is a lack of detailed and explicit regulations and institutions dedicated towards governing bilateral and multilateral CBET, arrangements for transmission planning and pricing, clarity of roles and responsibilities of institutions for CBET.

9.2. Tariff Regulations - India

The determination of tariff for hydropower generation plants in India is governed by a well-established regulatory framework administered by the Central Electricity Regulatory Commission (CERC) and the respective State Electricity Regulatory Commissions (SERCs). The objective of this framework is to ensure a fair, transparent, and balanced pricing mechanism that protects consumer interests while ensuring financial viability and reasonable returns for generating companies.

The jurisdiction for tariff determination is clearly defined. The CERC is responsible for determining tariffs for inter-state generating stations (ISGS), which supply electricity to more than one state. On the other hand, SERCs are responsible for determining tariffs for intra-state generating stations that operate within a single state, as well as for state-owned generating companies and distribution licensees within their respective states. While SERCs have the autonomy to frame their own regulations, they typically align their methodologies and principles with those established by the CERC, ensuring consistency and uniformity across the national and state regulatory frameworks.

As per the CERC (Terms and Conditions of Tariff) Regulations, 2024, the approach adopted by both CERC and generally followed by SERCs for tariff determination is based on a cost-plus methodology, regulated under a Multi-Year Tariff (MYT) framework. This ensures recovery of prudent costs and allows a fair return on investment for generating companies, including hydropower plants. As per CERC Tariff Regulations (2024-29)[303], following are the Annual Revenue Requirement (ARR) components as determined for Hydro projects across India:

Table 64: Normative Parameters as per CERC Tariff Regulation, 2024-29

Component	Description
Return on Equity (RoE)	15.50% for run-of-river hydro generating station and 17% for storage type hydro generating stations (pre-tax) allowed return, subject to normative parameters (30% normative equity)
Interest on Loan Capital	On normative or actual debt as approved by the regulator (70% normative debt). If actual loan details are unavailable, a weighted average interest rate (capped at 14%) is used; if no loans exist, State Bank of India's 1-year MCLR as of April 1 of the relevant year is applied
Depreciation	Based on straight-line method at rates in Appendix-II; After 15 years, the remaining value is spread over the balance useful life; for new hydro stations, a lower rate may be used with beneficiary consent.
O&M Expenses	For hydro stations commissioned on or after 1.4.2024, O&M expenses for the first year are set at 3.5% (capacity >200 MW) or 5.0% (≤200 MW) of project cost (excluding R&R, IDC, IEDC), with 5.47% annual escalation thereafter
Interest on Working Capital	Rate of interest on working capital shall be on a normative basis, based on the Reference Rate as of April 1 of each year during 2024–29. working capital includes 45 days of fixed cost, 15% of O&M (including security) as spares, and one month of O&M expenses
Additional Capitalization	Post-COD capex allowed for justified reasons
Debt-Equity Ratio	The normative debt-equity ratio is 70:30.

9.2.1. Details of Key Hydro projects in India-

The following table represents some hydropower plants for which tariff was determined by CERC for the control period 2014-19 and 2019-24. Although the CERC has issued the latest regulation, Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2024, the prevailing control period is 2024–29; however, only a limited number of large-scale hydro projects (with capacities

[303] Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2024, issued by Central Electricity regulatory Commission, Dated 15th March, 2024

greater than 400 MW) have been commissioned over the past decade. Several of these projects were commissioned under the 2014 and 2019 tariff regulations, resulting in tariff determination across multiple control periods—namely, 2014–19, 2019–24, and continuing into 2024–29. The table below presents the tariffs as determined by the Central Electricity Regulatory Commission (CERC) under the respective control periods.-

Table 65: Key Parameters of similar Hydro Projects in India

Particulars	Kameng	Teesta III	Koldam	Parbati III	Rampur
Project Description					
Location	Arunachal Pradesh	Sikkim	Himachal Pradesh	Himachal Pradesh	Himachal Pradesh
Owner	NEEPCO	Sikkim Urja Limited	NTPC	NHPC	SJVNL
Type	ROR with Pondage	ROR with Pondage	Storage	ROR with Pondage	ROR with Pondage
COD	2021	2017	2015	2014	2014
Project Capacity (in MW)	600	1200	800	520	412
Tariff rate as per Tariff order 2014-19					
Claimed by Petitioner Tariff for FY 2018-19 (in INR/kWh)	-	5.09[304]	4.68[305]	2.71[306]	4.15[307]
Approved by CERC (2018-19), INR/kWh	-	5.06	4.29	2.65	3.71
Tariff rate as per True-up order 2014-19					
Claimed (2018-19), INR/ kWh	-	5.11[308]	4.47[309]	2.60[310]	4.00[311]
Approved (2018-19), INR/ kWh	-	4.85	4.41	2.57	3.79
Tariff rate as per Tariff order 2019-24					
Claimed (2023-24), INR/ kWh	4[312]	4.75[313]	3.88[314]	2.35[315]	4.02[316]
Approved (2023-24), INR/ kWh	4	4.36	3.83	2.34	3.59
Core Financial Parameters as per Tariff order 2019-24					
Date of CERC Tariff Order	8th Aug 2023	15th July 2024	14th Jan 2024	31st March 2024	24th Jan 2022
Capital Cost Disallowed, in Crore (2019-24)		2,324	-	79	492
Return on Equity(Base Rate),%	-	16.5	16.5	16.5	16.5

[304] Tariff Order 2014-19, CERC, 9th Jan 2020
[305] Tariff Order 2014-19, CERC, 5th April, 2018
[306] Tariff Order 2014-19, CERC, 5th April, 2019
[307] Tariff Order 2014-19, CERC, 15th Feb, 2017
[308] True-up Order 2014-19, CERC, 15th July, 2024
[309] True-up Order 2014-19, CERC, 5th Oct, 2023
[310] True-up Order 2014-19, CERC, 23rd Apr, 2019
[311] True-up Order 2014-19, CERC, 4th June, 2021
[312] Tariff Order 2019-24, CERC, 8th Aug, 2023
[313] Tariff Order 2019-24, CERC, 15th July, 2024
[314] Tariff Order 2019-24, CERC, 14th Jan, 2024
[315] Tariff Order 2019-24, CERC, 31st Mar, 2024
[316] Tariff Order 2019-24, CERC, 24th Jan, 2022

Particulars	Kameng	Teesta III	Koldam	Parbati III	Rampur
Debt- Equity Ratio	-	80:20	70:30	70:30	70:30

*The tariff rate for the Kameng project has been fixed at a provisional rate of Rs. 4 per kWh as per Tariff Order No. 51/GT/2021 dated 8th August 2023.

The insights reflected in the table indicate a consistent trend of tariff rationalization across various control periods. In most instances, the approved tariff rates were lower than the corresponding claimed rates, both at the initial tariff determination stage and during the subsequent true-up process. The assessment indicates significant changes in the approved tariff rates between the tariff orders and subsequent true-up orders, highlighting the regulator's intent to align tariffs with verified costs and audited financial statements while ensuring controllable costs are kept within normative parameters. These consistent reductions in approved tariffs during true-up stages underline the role of regulatory oversight in ensuring that only justified and prudent expenditures are allowed to be recovered through tariffs, ultimately benefiting consumers through lower energy costs.

An in-depth analysis of the regulatory and financial frameworks governing tariff determinations for the FY 2014–19 and FY 2019–24 control periods is undertaken to underscore the challenges encountered and the key observations arising from tariff petitions and the regulatory approval process. The following case studies provide a detailed examination of these elements and their impact:

Case Study 1: Teesta Stage III Hydroelectric Project (1200 MW)

Project COD	28/02/2017
Design Energy (GWh)	Approved by Commission in Tariff Order for FY 2019-24: 5,214 GWh Claimed by Petitioner in Tariff Order for FY 2019-24: 5,214 GWh
Average Tariff (INR/kWh) (2023-24)	Approved by Commission: INR 4.36/kWh Claimed by Petitioner: INR 4.75/kWh
Major Disallowances	• Capital Cost Disallowance: A capital cost of Rs. INR 12,967 Crore was claimed in the tariff petition. However, the Central Electricity Regulatory Commission (CERC), in its final true-up order, approved INR 12,749 Crore, leading to a disallowance of INR 218 Crore. • Additional Capitalization: An additional capital expenditure of INR 159 Crore was proposed for approval. However, the Commission approved only INR 94 Crore, resulting in a disallowance of approximately 41% of the proposed amount.
Reason for Disallowance	• Procurement of spares beyond the cut-off date, capital spares for hydro generating stations shall be allowed separately as additional O&M expenses. • It has been observed that the submitted documentation does not include specific details regarding the plant's availability or an assessment of the improvement in Plant Availability Factor (PAF) following the capitalization of the claimed asset/work. • It has been observed that no information has been provided regarding the claim.
Impact of Disallowances	A significant portion of the Annual Fixed Charges (AFC) initially claimed was not approved by the Commission. While a total AFC of INR 12,623 Crore was proposed, the Commission approved only INR 11,826 Crore, resulting in a disallowance of INR. 796 Crore, or approximately 6% of the total claimed amount. This disallowance affected key financial components including Interest on Loan, Depreciation, Return on Equity, Interest on Working Capital, and Operation & Maintenance (O&M) expenses—all of which were proportionately

	reduced. As a result, the recoverable amount through tariffs was also correspondingly lowered.

Case Study 2: Rampur Hydropower Station (412 MW)

Project COD	13/05/2014
Design Energy (GWh)	Approved by Commission in Tariff Order for FY 2019-24: 1878 GWh Claimed by Petitioner in True Up Order for FY 2019-24: 1878 GWh
Average Tariff (INR/kWh) (2023-24)	Approved by Commission: INR 3.59/kWh Claimed by Petitioner: INR 4.02/kWh
Major Disallowances	• Capital Cost Disallowance: A capital cost of INR 4,149 Crore was claimed in the tariff petition. However, the Central Electricity Regulatory Commission (CERC), in its final true-up order, approved only Rs. INR 4,083 Crore, resulting in a disallowance of INR 65 Crore. • Additional Capitalization: An additional capital expenditure of INR 52 Crore was proposed for approval. However, the Commission approved only INR 15 Crore, resulting in a disallowance of approximately 70% of the proposed amount.
Reason for Disallowance	• Procurement of spares beyond the cut-off date, capital spares for hydro generating stations shall be allowed separately as additional O&M expenses. • No justification has been provided to demonstrate that the claim was made in response to any advice, direction, or mandate issued by the Appropriate Government Instrumentality or a statutory authority. • No justification has been submitted in support of the claim. • It has been observed that no information has been provided regarding the claim.
Impact of Disallowances	A significant portion of the Annual Fixed Charges (AFC) initially claimed was not approved by the Commission. While a total AFC of INR 3,672 Crore was proposed, the Commission approved only INR 3,366 Crore, resulting in a disallowance of INR. 305 Crore, or approximately 8% of the claimed amount. This disallowance affected key financial components such as Interest on Loan, Depreciation, Return on Equity, Interest on Working Capital, and Operation & Maintenance (O&M) expenses—which were all proportionately reduced. As a result, the overall recoverable amount through tariffs was also correspondingly lowered.

Case Study 3: Tuirial Hydroelectric Power Plant (60 MW)

Project COD	30/10/2017
Design Energy (GWh)	Approved by Commission in True-up Order for FY 2014-19: 251 GWh Claimed by Petitioner in True-up Order for FY 2014-19: 251 GWh
Average Tariff (INR/kWh) (2018-19)	Approved by Commission: INR 5.21/kWh Claimed by Petitioner: 4.26/kWh
Major Disallowances	• Time and cost overrun: A total project delay of 1,380 days was claimed. The Commission condoned 490 days of delay, attributing them to factors beyond the control of the project developer, such as poor working conditions, local disruptions, and logistical challenges. However, the remaining 890 days

	linked to issues like slope failure, road restrictions, and increased scope of work were not accepted due to the absence of adequate supporting evidence. • Capital Cost ceiling: An additional capital expenditure of INR 18 Crore was claimed for the period up to March 31, 2019. However, the Commission did not approve this expenditure, as the ceiling capital cost of INR 1,117 Crore agreed upon earlier—was considered the maximum permissible cost for the original scope of the project. Since the proposed expenditure fell within this original scope and exceeded the approved ceiling, no further capital expenditure was allowed under this head.
Reason for Disallowance	• The disallowance of the delay is attributable to the lack of sufficient supporting evidence to substantiate the claims of delay. • The proposed expenditure remained within the original scope but exceeded the approved ceiling, and therefore, no additional capital expenditure was permitted under this category
Impact of Disallowances	These disallowances may affect the project's financial planning, timelines, and overall execution strategy

9.2.2. Key Challenges observed in determining Tariff for above Hydro projects:-

1. **Project Delays and Cost Overruns-** The regulator assesses each project delay and resultant cost overrun on the basis of their nature controllable and non-controllable. Controllable costs such as construction delays, availability fo manpower, material management related costs are allowed as per norms and prudence check by regulator. Any deviation in such costs are allowed in case-to-case basis subject to the satisfaction of the regulator on the reasonsings provided by developers. Natural disasters, slope failures, and geological risks are some of the non-contollable costs which may be allowed by the regulator subject to appropriate technical studies and planned mitigation activities undertaken by the developers.

2. **Capital Cost -** The disallowance majorly occurs due to reduction in the approved equity base on which regulated returns are calculated, inadequate justification for the additional capital cost provided by the petitioner and procurement of capital spares beyond the cut-off date. The disallowance has a direct financial impact as it reduces the approved capital cost used to calculate tariff components such as depreciation, interest on loan, and—most importantly—Return on Equity (RoE), lowering actual returns.

3. **Return on Equity (RoE) -** Regulators allow RoE only on the approved equity portion of the capital cost (typically 30% of approved capital), since RoE is allowed only on the equity portion of the approved capital cost, any disallowed amount leads to a lower return, even though the actual equity investment remains unchanged. This results in reduced earnings from the project, lowers overall profitability, and increases the opportunity cost of capital for investors.

4. **Fixed Return on Equity (RoE) Rates Without Risk Adjustment-** As per Tariff Regulation, RoE rates are fixed at 15.5% in India and do not always reflect market risks, project-specific risks, or inflation. Thus, it does not adequately compensate investors for the higher risk profile and long development horizon of hydro projects.

5. **Under-approval of Annual Fixed Charges-** The petitioner's total Annual Fixed Charges (AFC) were not fully accepted by the Commission. The reduction in approved AFC significantly lowered the recoverable tariff and impacts the overall Annual Fixed Charges (AFC), reducing net project cash flows. This further compresses the margin available to meet equity return expectations.

These factors often lead to significant disallowances, which must be carefully reviewed, addressed, and incorporated both by the regulatory body and the project developers. It is crucial that the project team proactively engages with these regulatory expectations to ensure that all relevant aspects are transparently accounted for in the tariff determination process.

9.3. Tariff Regulations - Bangladesh

In recent years, Bangladesh has made significant progress in expanding its power generation capacity and enhancing electricity access across the country. However, when it comes to cross-border electricity trade, the country is still in the early stages of development. Unlike many other nations with established power markets, power trade in Bangladesh is not yet recognized as a distinct or standalone commercial activity. Instead, electricity import and export activities are primarily managed and executed through the Bangladesh Power Development Board (BPDB), which remains the central agency for power procurement and distribution. While Bangladesh has a regulatory framework in place, including institutions such as the Bangladesh Energy Regulatory Commission (BERC), the regulatory maturity and market mechanisms are still evolving and are not yet comparable to the advanced regulatory regimes found in more developed power markets. For example, tariff determination methodologies and market-based trading mechanisms are still limited in scope and application.

As a result, electricity tariffs for cross-border trade are often determined through negotiated agreements, either via direct bilateral (B2B) discussions or through limited competitive bidding processes.

The Bangladesh Energy Regulatory Commission (BERC) introduced the Generation Tariff Regulations in 2008, adopting a "cost-plus" approach for determining tariffs for electricity generated from conventional power projects. The tariff structure is primarily designed to ensure recovery of the plant's fixed (capacity) costs, with BERC prescribing specific norms and parameters for these components. These guidelines are particularly relevant for hydroelectric projects. A detailed breakdown of the allowable fixed cost elements is provided below:

1. **Return on Rate Base**: It is computed based on the following formula:

   Tariff Rate of Return = [(Equity Capital × Cost of Equity) + (Debt Capital × Cost of Debt)]/
   [Equity Capital + Debt Capital]\

   The debt component of the capital structure is evaluated using the effective interest rates on the licensee's current borrowings. In contrast, the cost of equity does not follow a standard formula and is determined on a case-by-case basis, subject to regulatory discretion. For state-owned entities, the return on equity is equated to the prevailing government cost of capital, which is typically derived from the latest auction rate of Treasury Bills issued by the Central Bank.

2. **O&M Costs:** The Bangladesh Energy Regulatory Commission (BERC) does not set fixed norms for Operation and Maintenance (O&M) expenses. Instead, these costs are approved based on actual expenditures submitted by the licensee across various categories. Generation expenses include operational costs like supervision, engineering, electricity usage, rents, and maintenance. Customer account expenses cover billing, record-keeping, and bad debt provisions. Sales expenses involve marketing and promotional activities, while Administrative and General (A&G) expenses include salaries, office supplies, rents, insurance, pensions, and regulatory fees. This flexible approach ensures that actual costs are accounted for while maintaining regulatory oversight.

3. **Depreciation Cost:** Straight Line Method is used for computing depreciation costs based on the useful life of the asset.

4. **Income Tax:** When determining the tariff rates for the financial year, the amount of income tax included is the actual income tax paid to the government. Other types of taxes, such as VAT and customs duty, are considered part of the cost of the asset. The proposed increase in tariff, based on

the revenue gap approved by the Commission, is grossed up using the prevailing income tax rate to compensate the licensee for the tax paid.

The policy of Government of Bangladesh for setting the electricity tariff stipulates the achievement of the cost reflective tariff and generates a surplus to expand coverage and supply but maintain least cost to the consumers. The tariff setting policy addresses the social commitments of the Government as well.

The tariff structure for the end customers is based on principles of cross-subsidies from commercial consumers to residential consumers. With the uniform tariffs across the country, some distribution companies achieve higher profits while others are unable to make profit and struggle to achieve their commercial targets. The distribution costs and the average revenues differ significantly between rural and urban areas due to the different load density and different customer mix.

10. Electricity Export Risk Analysis

Purpose of the Section

This section evaluates the risks associated with cross-border electricity exports. It focuses on transmission constraints, market dynamics, financial exposures, and operational challenges that may affect the viability and profitability of export strategies.

Key Conclusion and takeaways

Curtailment risks due to transmission constraints and demand fluctuations can limit effective power evacuation. Foreign exchange mismatches introduce currency risks, impacting financial stability. Additionally, participation in merchant markets exposes projects to price volatility and liquidity challenges, necessitating strong risk mitigation strategies.

Cross Border Electricity Trade is subject to a host of export related risks. Some of the key risks and mitigation measures which may impact DHPP have been enumerated below:

Table 66: Electricity Export Risk Analysis

Risk Category	Types of risk	Risk Description	Risk Mitigation
Curtailment Risk	Grid Infrastructure Limitations	India's transmission network may sometimes lack the capacity to evacuate the full power generated, especially during peak generation periods or in regions with underdeveloped transmission. This leads to forced curtailment of power.	• Discussion on the requirements for transmission system access in bilateral forums such as Joint Working Group • Secure grid connectivity approvals early and factor transmission upgrades into project timelines. • Explore storage projects to shift generation to off-peak hours when grid availability is higher
	Load Demand Variability	Hydropower output may not always align with regional or national demand cycles. During low demand or oversupply situations, grid operators may curtail power to maintain grid stability.	• Integrate advanced forecasting tools for demand and generation alignment. • Negotiate flexible PPA that allow for scheduling adjustments and energy banking • Explore participation in ancillary services markets or demand response programs to utilize surplus generation.
	Prioritization of Domestic Generation	In some cases, domestic power plants or renewable sources like solar and wind might be given priority dispatch over imported or foreign-backed hydropower, increasing curtailment risk for foreign projects.	• Regulatory support for "deemed generation" from DHPP for revenue certainty in case alternate power sources are provided priority in dispatch • Structure back to back, long-term PPA with firm offtake commitments to reduce reliance on merit-order dispatch.
	Regulatory and Contractual Constraints	Power purchase agreements (PPAs) and grid codes may include clauses permitting curtailment under certain conditions, such as grid emergencies, transmission congestion, or force majeure events.	• Ensure inclusion of deemed generation provisions to recover revenue in curtailment scenarios. • Provision of alternate transmission evacuation plan in case of grid emergies, transmission congestion

Risk Category	Types of risk	Risk Description	Risk Mitigation
	Market and Policy Changes	Policy shifts or changes in renewable energy incentives might alter dispatch priorities, affecting foreign hydro projects' ability to supply power continuously.	• Ensure long term offtake arrangement under Take or Pay regime
PPAs with DISCOM	Credit Risk	Many Indian DISCOMs face financial stress and liquidity issues, which can lead to delays in payments or defaults, affecting the cash flow and financial viability of the project.	• Include payment security mechanisms in Power Purchase Agreements (PPAs) such as Letters of Credit (LC), escrow accounts, or bank guarantees to ensure timely payments. • Incorporate penalty clauses for delayed payments to incentivize prompt settlement.
	Regulatory and Policy Risk	Changes in government policies, tariffs, or regulations can alter the terms of PPAs or introduce additional compliance burdens, impacting project returns.	• Diversify project portfolio across different states or segments to reduce dependency on a single regulatory regime. • Balance long-term fixed contracts with shorter-term agreements or merchant market exposure to maintain flexibility.
	Scheduling Risk	DISCOMs may curtail power procurement due to grid constraints or demand fluctuations, limiting the ability to sell the contracted power fully and reducing revenue	• Include deemed generation clauses in PPAs that compensate the generator for curtailed power. • Negotiate minimum off-take guarantees or take-or-pay provisions to secure revenue even during curtailment.
	Force Majeure and Political Risk	Events such as natural disasters, political instability, or changes in cross-border relations (especially for projects sourcing water or power from neighboring countries) can affect project operations and contractual obligations.	• Structure Insurance to cover Force Majeure and Political Risks • Include dispute resolution mechanisms (such as international arbitration) specifically designed for cross-border conflicts.
Merchant market risk	Price Volatility Risk	As observed in last three years, the MCP in exchange during peak hours is very low due to excess supply of solar power. Thus, it is important to align the generation profile of DHPP with non-peak hours for better returns.	• Prefer Long Term Offtake Arrangements to secure revenues against fluctuating prices • Use flexible operation strategies (e.g., pumped storage or reservoir management) to shift generation away from low-price peak solar periods. • Consider hybrid models combining hydro with solar or wind to balance supply and improve revenue stability.
	Market Liquidity Risk	At times, liquidity in the power exchange may be limited, affecting the ability to sell the full volume of generated power at competitive prices.	• Capping participation in merchant power market.

11. Transmission Infrastructure between India, Bhutan and Bangladesh

Purpose of the Section

This section presents a detailed overview of the transmission infrastructure linking India, Bhutan, and Bangladesh. It supports the strategic planning of power evacuation from DHPP by analyzing existing and proposed transmission capabilities.

Key Conclusion and takeaways

Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. Currently, the power system of Bhutan is connected to the Indian grid, through 13 interconnecting lines with eight feeders at 400kV and three feeders at 220kV voltage levels from western Bhutan, and two S/C 132kV lines from eastern Bhutan and around 1,000 MW evacuation capacity is additionally planned between India and Bangaldesh.

The South Asian countries like Bhutan, India, Bangladesh and Nepal have growing energy demands and an increasing emphasis on sustainable energy. Enhanced regional cooperation is critical to boosting energy security and overall climate resilience in the region. Moreover, the seasonal demand peaks of these nations are complementary. The peak demand seasons of India and Bangladesh correspond to generation surpluses in run-of-river hydropower systems in Bhutan and Nepal. Similarly, when Bhutan and Nepal have supply shortages (during their dry season) there is excess capacity in India and Bangladesh.

Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. The table below sets out details of the existing cross-border electricity transactions between Bhutan, India and Bangladesh.

Table 67: Power Trade Agreements between South Asia

Particulars	Source	Agreement Type
Bhutan → India (2380 MW)	1020 MW Tala	Government-to-Government
	336 MW Chukha	Government-to-Government
	60 MW Kurichhu	Government-to-Government
	126 MW Dagachhu	Commercial
	720 MW Mangdechhu	Government-to-Government
	118 MW Nikachhu	Government-to-Government
India → Bangladesh (2856 MW[317])	250 MW NVVN Ph-I	Government-to-Government
	200 MW PTC Ph-II	Commercial
	160 MW Tripura	Government-to-Government
	300 MW NVVN Bharamara Ph-II	Government-to-Government
	250 MW Sembcorp	Commercial
	1496 MW Adani Godda	Commercial
	200 MW PTC India	Commercial
India → Nepal (600-800 MW)	237 MW India	Government-to-Government
	400-500 MW Market	Commercial
Nepal → India (800-1000 MW)	941 MW from 28 hydro projects	Commercial
Nepal → Bangladesh (40 MW)	40 MW (through Indian grid)	Government-to-Government (Tripartite Agreement)

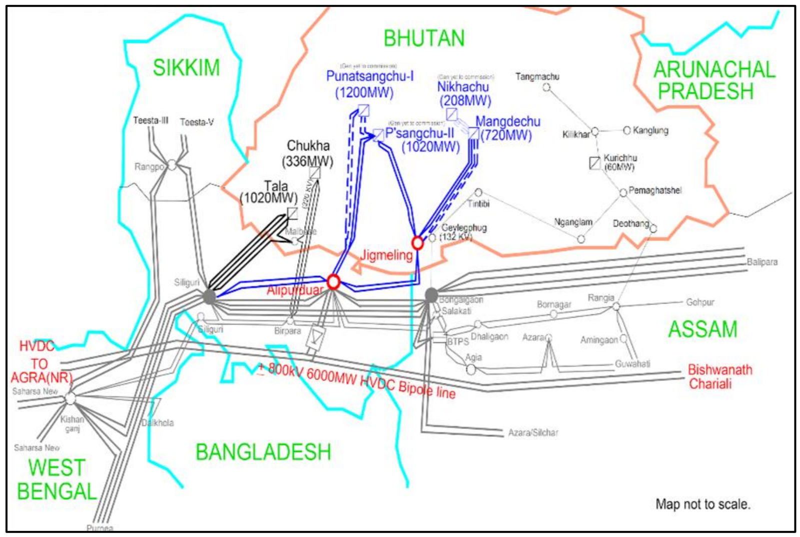
[317] Source: BPDB Annual report 2023-24

To facilitate the trade, at present the interconnection lines amongst South Asian countries is provided in the table below:

Bhutan-India

Currently, the power system of Bhutan is connected to the Indian grid, through 13 interconnecting lines with eight feeders at 400kV and three feeders at 220kV voltage levels from western Bhutan, and two S/C 132kV lines from eastern Bhutan as below:

- 2xD/C, 400 kV Twin Moose from Tala to Silliguri with one circuit LILOed at Malbase Substation;
- 1xD/C, 400 kV Quad Moose from Jigmeling to Alipurduar;
- 1xD/C, 400kV Quad Moose from Lhamozingkha to Alipurduar;
- 1xD/C, 220 kV from Chukha to Birpara;
- 1xS/C, 220kV from Malbase to Birpara;
- 1xS/C, 132 kV from Gelephu to Salakati; and
- 1xS/C, 132 kV from Motanga to Rangia



With additional planned hydropower and solar projects, following new cross-border transmission lines are required to be built for export of surplus power:

- 2xD/C, 400kV Quad Moose Durungri - Rangia/Rowta/Bornagar
- 1xD/C, 400 kV Twin Moose Phuntshothang - Rangia/Rowta/Bornagar

As per the National Transmission Grid Master Plan (NTGMP-2012), the power from Bhutan is to be pooled at Alipurduar and Rangia/Rowta/Bornagar pooling stations which will further be evacuated to the deficit regions of India (NR/WR) via ±800kV, 3,000 MW HVDC sub-station at Alipurduar, and establishment of a ±800kV, HVDC bipole line with converter capacity of 6,000MW at Rangia/Rowta/Bornagar.

India-Bangladesh

Currently, Bangladesh's power system is connected to the Indian grid via two interconnecting lines operating at 400 kV, utilizing both HVDC and HVAC technologies. The proposed Katihar–Porbotipur–Bornagar transmission line is expected to have an evacuation capacity of 1,000 MW. Therefore, DHPP can evacuate power to Bangladesh through this line, as it has sufficient capacity to accommodate the transfer. The details of present status and upcoming line is provided in the table below:-

Table 68: Details of Transmission Infrastructure between India and Bangladesh

Sno.	Interconnection Project	Voltage and Configuration	Intercn. Type	Evac. Capacity (MW)	Intercn. Point in India	Intercn. Point in Bangladesh
Status – Existing Projects						
1.	Baharampur – Bheramara Line	400 kV D/C	HVDC	1000	West Bengal (Baharampur)	Bheramara

2.	Surjyamaninagar – North Comila – South Comila	400 kV *(operated at 132 kV)*	HVAC	160	Tripura (Suraj Maninagar)	North Commila
Status – Proposed Projects						
1.	Katihar – Parbotipur - Bornagar	765 kV D/C	HVDC	1,000	- Bihar (Katihar) - Assam (Bornagar)	Parbotipur

12. Hydropower and Bhutan's Macroeconomic Structure

Purpose of the Section

In this chapter, we assess the macro-fiscal landscape of Bhutan to evaluate the nature and sustainability of public debt, with particular focus on the borrowings undertaken for the DHPP. The analysis examines the size and composition of debt, including loans from multilateral institutions and commercial sources. Special emphasis is placed on assessing whether these borrowings are self-liquidating and align with the sustainability criteria outlined in the Bhutan Debt Management Policy, 2023. The objective is to determine whether the debt generates adequate economic returns to meet repayment obligations or whether it poses long-term fiscal risks. By analyzing Bhutan's debt servicing capacity, revenue generation potential, and overall macroeconomic fundamentals, this chapter aims to provide a clear perspective on the fiscal space available to support the DHPP.

Key Conclusion and takeaways

An analysis of principal repayments, interest outflows, and revenue collections, both domestic and export, indicates that the debt is self-liquidating. Total revenue generated from the project is sufficient to meet debt servicing obligations from the very first year. However, the second condition outlined in the Bhutan Debt Management Policy, which requires that debt servicing remain within 50% of total export revenue, is met only in the later years, around 2045.

While this condition is not fulfilled in the initial years, it is important to recognise that the DHPP project's contribution goes well beyond export earnings. It is expected to act as a catalyst for broader economic transformation by enabling the growth of energy-intensive industries, improving energy security, and supporting infrastructure development. The availability of reliable and surplus electricity can attract manufacturing and service-sector investments, create jobs, and stimulate downstream economic activity. In this way, the project serves as an enabler of long-term economic diversification, reducing Bhutan's dependence on hydropower exports alone and creating the foundations for a more resilient and self-sustaining domestic economy.

12.1. Background

[318]Hydropower has long been the cornerstone of Bhutan's development strategy, serving as both a major source of public revenue and a key driver of foreign exchange earnings. As of FY 2023-24, hydropower accounts for approximately 14% of GDP, over 25% of exports, and contributes nearly 30–35% of total government revenue through royalties, dividends from Druk Green Power Corporation (DGPC), and corporate taxation[319]. Beyond its fiscal role, hydropower is integral to Bhutan's green growth narrative, providing the foundation for carbon-neutral energy exports and regional cooperation in South Asia.

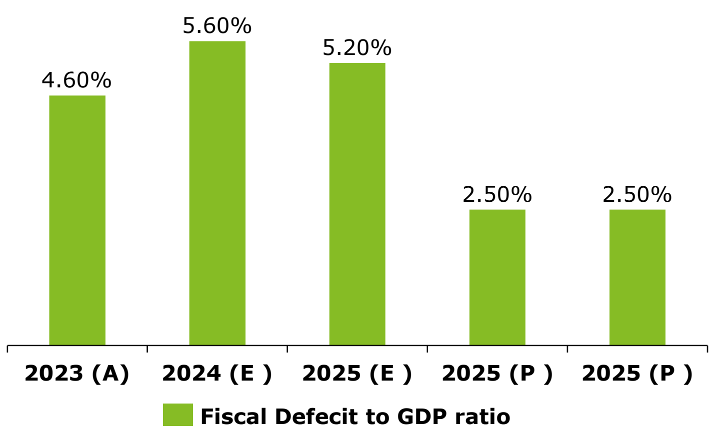


Figure 124: Fiscal Deficit to GDP ratio.

The government has now taken up the mantle of advancing the Dorjilung Hydropower Project (DHPP), a planned 1,125 MW run-of-the-river plant, marking a significant step forward towards the expansion of this strategic sector. Envisioned as a flagship project under Bhutan's long-term hydropower roadmap,

[318] National Budget Financial Year 2024-25, Royal Govt. of Bhutan
[319] https://thewire.in/energy/bhutan-hydropower-electricity-energy-geopolitics

DHPP is expected to enter the construction phase in 2025 and be commissioned by 2032[320]. Once operational, it will rank among Bhutan's largest power plants, at par with Tala (1,020 MW) and Punatsangchhu I (1,200 MW) and is likely to further solidify the country's position as a regional green energy supplier.

DHPP is being developed as a trilateral initiative involving Bhutan, India, and the World Bank. Its financing mix includes grants, concessional loans, and domestic counterpart funding. Its size, complexity, and financing structure necessitate careful fiscal planning, particularly in light of Bhutan's limited revenue base and external debt sensitivities.

12.2. Macro Fiscal Assessment

The total estimated cost of DHPP stands at approximately USD 1.5 billion, equivalent to nearly 50% of Bhutan's nominal GDP as of 2023[321]. Currently, the total public debt comprises external debt of 93 percent of GDP and domestic debt of 14.1 percent of GDP in 2023-24. The major portion of the external debt disbursed was for the development of hydropower projects, as hydropower debt constitutes 57.8 percent and non-hydropower debt constitutes 35.2 percent of total external debt. The Public Debt Management Policy 2023 requires the Central Government's debt stock to be within 55 percent of GDP annually and is, therefore, within the threshold[322,323].

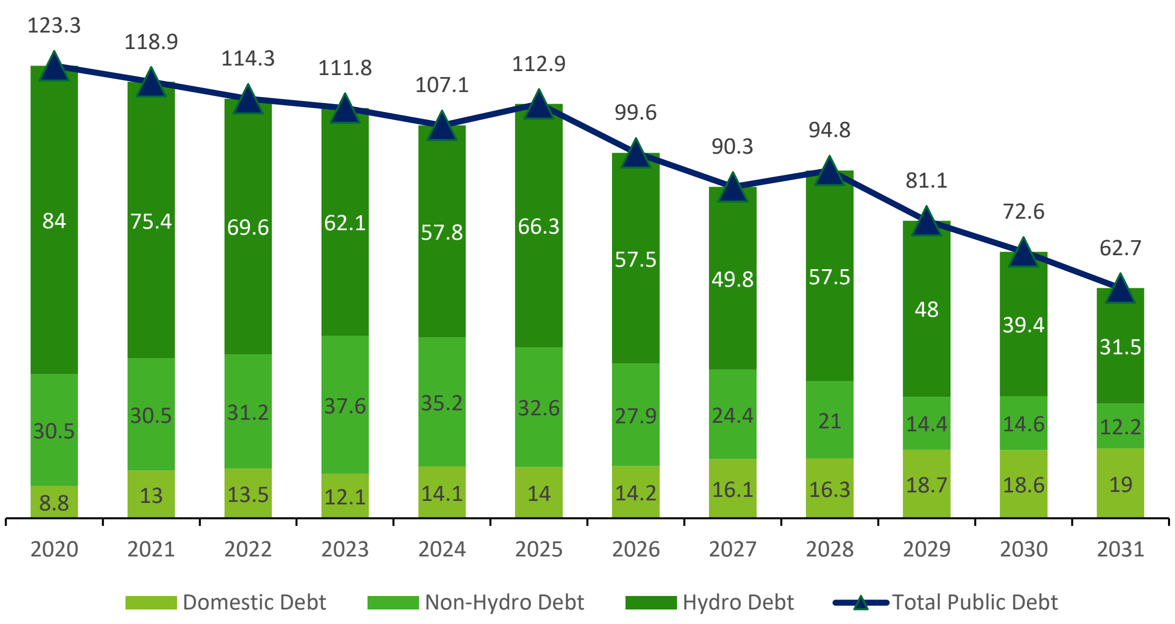


Figure 125: Debt Projections

320 https://thebhutanese.bt/dorjilung-hydropower-project-set-for-completion-by-2032/
321 Haver Analytics
322 Public Debt Management Policy 2023, RGoB
323 *National Budget Financial Year 2024-25, Royal Govt. of Bhutan*

With the DGPP project's initiation, which will be implemented over a 6-year period, the scale of borrowing, even at concessional rates, may have implications for Bhutan's debt trajectory and fiscal space.

However, Bhutan's most recent IMF Article IV Consultation (2023) reaffirmed that while total debt levels are high, the hydropower debt is considered sustainable given that both financial and construction risks of the project are covered[324]. The project has a strong potential revenue-generating capacity as power exports rise. The repayment structure linked to power exports.

From a lender's perspective, it is crucial that DHPP is treated within the framework of "self-liquidating" infrastructure assets, wherein debt incurred is expected to be repaid through the project's own cash flows rather than relying on the general fiscal budget. This approach has precedent in Bhutan's previous large-scale hydro projects and is aligned with the IMF and World Bank recommendations on large infrastructure financing for small economies.

The scale of DHPP, though large, is not unprecedented in Bhutan's hydropower history. However, the evolving fiscal landscape marked by post-pandemic recovery pressures, exchange rate risks, and grant tapering warrants a refreshed macro-fiscal assessment to ensure the project does not compromise Bhutan's development financing envelope or debt sustainability thresholds. Hence, Deloitte has conducted a detailed analysis of long-term fiscal sustainability and economic contribution.

12.3. Debt Structure and Fiscal Implications

Bhutan's Public Debt Policy 2023 emphasises the importance of differentiating hydropower-related debt, which is self-liquidating and linked to revenue-generating assets, from non-hydropower-related debt, which directly affects fiscal space and public service delivery. Based on this policy, the debt incurred for DHPP will be fully classified as hydropower debt.

In order to estimate the fiscal road map post the initiation of the project, we have considered the following:

- The Dorjilung Hydropower Project (DHPP), with a total estimated cost of USD 1.5 billion, represents one of the most ambitious hydropower ventures in Bhutan's recent history.
- The proposed financing mix comprises public equity, multilateral concessional loans, and commercial borrowing, which marks a significant evolution in Bhutan's hydropower financing model, from a historically grant- and bilateral loan-driven approach to a more blended finance architecture. It combines,
    - Approximately USD 1.05 billion in debt from the IFC, IBRD and Indian Commercial Banks. The debt from IBRD will be dollar-denominated, while debt from IFC and Indian Commercial Banks would be rupee-denominated, and
    - USD 450 million in equity,
- The amount of USD 1.05 billion in funding will be infused in equal tranches over six years to support the Project's capital expenditure (financed through a combination of debt and equity as mentioned above).
- Interest accrued on the debt taken (USD 1.05 billion) during the construction phase (up to 2031) will be capitalized into the project cost, while interest during the operations phase will be accounted for in the project's profit and loss statement.
- Debt repayments are scheduled to begin post-2032, following the project commissioning in 2031 and a one-year principal moratorium.
    - The principal must be repaid in equal annual instalments over the next 25 years.
    - The rate of interest is at 9.5% for IFC and Indian Commercial Banks and 5% for the debt from IBRD
- The project will start generating revenues after commissioning, starting from 2031 and should be able to finance the debt service obligations

[324] 2023 IMF Article IV Consultation

- 13% of power generated will be free, 17.6% will be sold domestically, and 69.4% will be exported.
- Tariff for both domestic and export is considered at INR 5.4.
- For the repayment in USD, a historical depreciation rate of 2.89% has been considered.
- The revenue generated in INR will depreciate at the historical trend rate of 3% against the USD.
- The anticipated tariff-based power purchase agreement (PPA) with Indian off-takers will ensure INR-denominated revenue inflows, providing a natural hedge against currency risk and supporting debt servicing.
- Furthermore, the scale of the project will unlock economies of scale, enhancing financial viability, reducing the levelized cost of energy (LCOE), and improving long-term fiscal returns.

The estimation:

Under the Bhutan Public Debt Management Policy 2023, while there is no limit to hydro debt, there are two conditions for hydro debt that needs to be met:

- *The debt service coverage ratio for hydropower debt shall be above 1.2.*
- *The ratio of hydropower debt service to hydropower export revenue shall be maintained within fifty percent (50%).*

The incremental fiscal implications of the project for the next 25 years have been highlighted in the figure below[325]:

Debt Service Coverage Ratio

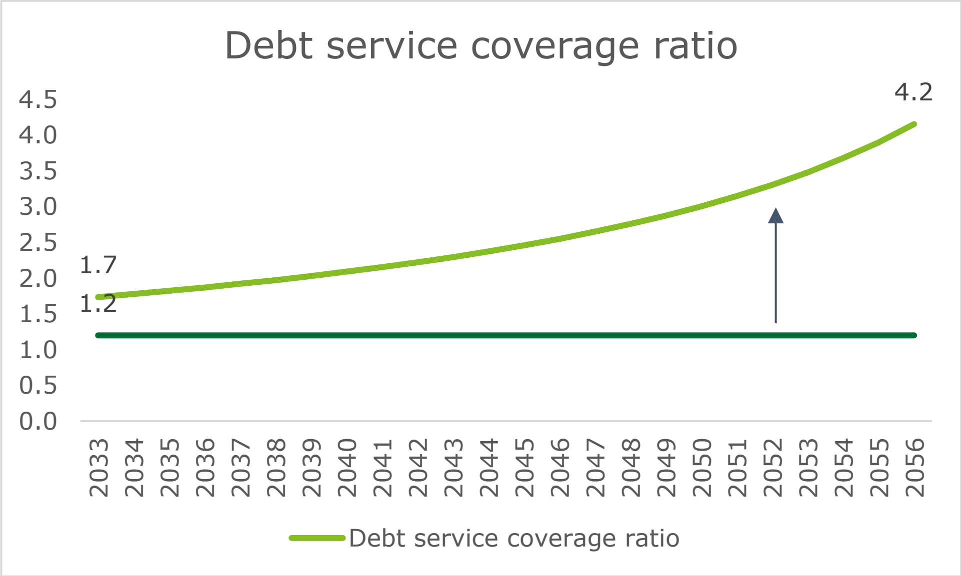


Figure 126: Debt service ratio during period of debt servicing

Debt Service Outgo to Export Revenue

[325] Deloitte Analysis

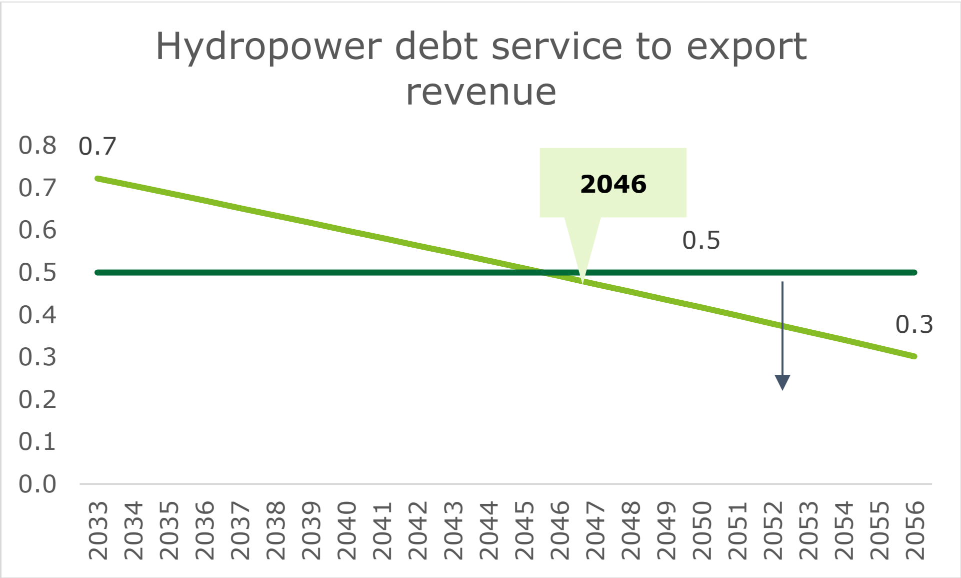


Figure 127: Debt service outgo to export revenue (with currency risk)

Observation:

- The first condition is fulfilled from the very beginning of the repayment period, i.e., 2033. As required, the revenue generated in 2033 exceeds the debt servicing obligations by 73%, with this margin progressively increasing over the years, indicating the debt's sustainability.
- The second condition is not met in the initial years. It is only in the later years when debt servicing obligations decline that they fall to 50% or less of the export revenue. In this case, the condition is met from 2046 onwards.

Therefore, securing appropriate tariff structures and managing interest rates will be critical to ensuring compliance with this requirement.

But it is also important to recognize that the DHPP project's impact extends well beyond export revenues. By providing reliable and large-scale energy supply, it has the potential to power Bhutan's broader economic growth. As a key input across sectors, energy availability will support domestic industrial development, foster new economic opportunities, and contribute to building a more diversified and resilient economy.

12.4. Risk Mitigation

Given the multi-source financing model and Bhutan's exposure to external vulnerabilities, a robust risk mitigation framework is essential to preserve debt sustainability, maintain macroeconomic stability, and ensure long-term fiscal and financial resilience.

Given the multi-source financing model and Bhutan's exposure to external vulnerabilities, a robust risk mitigation framework is essential to preserve debt sustainability, maintain macroeconomic stability, and ensure long-term fiscal and financial resilience.

Key risk areas and corresponding mitigation strategies are outlined below.

12.4.1. Currency Risk Management Strategy

Overview of Exposure:

The DHPP's financing mix introduces multi-currency exposure, with:

- USD-denominated debt from IBRD,
- Rupee denominated debt from IFC and Indian Commercial Banks
- INR-denominated commercial borrowing in the form of equity (~USD 450 million), and
- Revenues (from exports) expected in INR through long-term Power Purchase Agreements (PPAs) with Indian utilities.

Risk Implications:

- Any significant USD/INR appreciation would raise the domestic cost of debt servicing on USD loans.
- Exchange rate volatility may affect imported capital costs, repayment liabilities, and the fiscal buffer.

Proposed Currency Risk Mitigation Tools:

1. Natural Hedging:
    - Structuring all revenue inflows from DHPP in INR, matching commercial loan obligations, which will help limit exposure.
    - Using local procurement and services, where feasible, will reduce losses due to imports made in USD during construction.
2. INR Currency Risk Stress Tests:
    - Integrate exchange rate stress scenarios (e.g., 10–15% INR depreciation) into fiscal forecasts to quantify the impact on debt service ratios and budget balances.
3. Phased Hedging Instruments:
    - Explore the use of cross-currency swaps or forward contracts for a portion of the USD exposure, especially during high-risk repayment years.
    - These could be coordinated with World Bank or IFC risk management products under the umbrella of the broader financing agreement.
4. Build INR Revenue Buffer:
    - Establishing an INR stabilisation account at the central bank or Ministry of Finance (MoF to ring-fence excess revenues during high-earning years, which can be used to smooth repayment obligations during currency stress.

12.4.2. Project Execution Risk

- Mitigation: Rigorous project management through a dedicated project SPV jointly managed by DGPC and the strategic investor to ensure strict adherence to timelines.
- Use of Engineering, Procurement and Construction (EPC) contracts with performance-linked milestones and penalties for delays.

- Independent technical supervision, quarterly progress audits, and proactive dispute resolution frameworks.

12.4.3. Fiscal overstretch & Contingent Liabilities

- Mitigation would require
  - Keep fiscal deficit within the range, in order to ensure strict adherence to Bhutan's non-hydro external debt ceiling (<35% of GDP).
  - Ensure that the equity by DGPC and strategic investors don't need grosss budgetary support or loans from the Government to reduce fiscal strain
  - Cap equity contribution by DGPC to avoid over-leveraging of DGPC and avoid the circumstances mentioned above.
  - Create Fiscal buffers (e.g., Hydropower Stabilisation Fund) to absorb shocks in revenue or cost assumptions.

12.4.4. Governance & Transparency Risk

- Mitigation would require:
  - Full compliance with the Public Financial Management Act, transparent reporting of project debt, and regular disclosure in the Budget and Debt Management Reports.
  - Incorporation of World Bank's environmental and social safeguards, as well as procurement standards in the project regulatory standards

13. Identifying the trends of hydropower financing and investment in Bhutan

Purpose of the Section
The section aims to evaluate the evolving landscape of hydropower financing in Bhutan, particularly focusing on the financing trends for Druk Holding and Investments Power Projects (DHPP). It explores the diversification of funding sources beyond traditional bilateral support, examines legal and technical frameworks such as the MIPA Act, and draws insights from international case studies to identify viable financing structures. Additionally, it seeks to understand the key risks associated with hydropower projects and how these can be effectively mitigated.
Key Conclusion and takeaways
Bhutan has significantly diversified its hydropower financing sources since 2010, incorporating lessons from global best practices. The MIPA Act provides a legal foundation for borrowing from foreign lenders, and financing processes typically take 5–7 months post-PPA finalization. Key risks such as land acquisition challenges, geological uncertainties, hydrology shocks, and cost overruns—have been identified, with mitigation strategies informed by international experiences. These insights collectively strengthen Bhutan's approach to sustainable and resilient hydropower development.

13.1. Considerations for Indian/International Lenders:

a. Key appraisal requirements

Banks / Financiers typically require a thorough and comprehensive appraisal of under-construction hydropower projects to assess their financial viability and risk profile before extending loans or investments.

Multiple teams of lending institutions are generally involved in appraising various categories of risks / project aspects. This appraisal involves evaluating various aspects including but not limited to technical feasibility of the project, environmental impact, hydrological and seismological risks, socio-economic considerations and commercial outcomes (profitability and sustainability dependent on market dynamics).

i. Technical Appraisal of the Project

Lenders' consideration:

- **Hydrological studies**: Assessing water availability, flow rates, and potential for storage or run-of-river projects.
- **Geotechnical investigations**: Evaluating soil and rock conditions at the dam site and surrounding areas to ensure stability and prevent landslides.
- **Design and Engineering:** Reviewing the project design, including dam type, turbine selection, and power plant layout, to ensure it is optimized for power generation and cost-effective.
- **Power generation capacity:** Evaluating the potential power output and assessing the 90% dependable output to ensure reliability and meet demand.

DHPP's preparedness:

- A robust Detailed project report (DPR) has been prepared for DHPP, encompassing a comprehensive study of all critical elements required for technical evaluation of the project. The report includes thorough hydrological studies to validate water availability and dependable power generation estimates. It incorporates extensive geotechnical and geological investigations to confirm site stability and inform sound engineering decisions. The design and layout of project components—such as the dam, turbines, and powerhouse—are optimized for performance and cost-efficiency.

ii. Financial Appraisal

Lenders' consideration:

- **Project cost estimation**: Accurately estimating project costs including construction costs, land acquisition, plant and machinery components, and environmental mitigation, soft costs (viz. preliminary and pre-operative expenses, interest during construction, margin money for working capital requirement etc.) to ensure they are realistic and achievable.
- **Revenue projections**: Forecasting power generation revenue through proposed mix of domestic and export of energy along-with assessment of respective tariff rates, ongoing regulatory scenario (for tariff and/or free power), and delivery point along with appropriate sensitivity analysis to fairly assess project's ability to generate sufficient profit and cash for servicing of debt availed by the project.
- **Operating cost projections:** This analysis includes elements viz. assessment of operational and maintenance cost, periodic repair and maintenance costs, employee costs, insurance cost, administrative overhead allocation and other operating costs along with consideration for suitable escalation rates dependent on contractual and/or inflation related estimations.
- **Financial Modelling**: Based on above analysis, a detailed financial model is prepared to analyze the project's cash flows for the period of asset life thereby assessing its debt servicing capabilities, and financial sustainability.
- **Sensitivity analysis:** Assessing the project's vulnerability to changes in key parameters, such as tariff rates, operating costs, and power demand, to understand the potential range of outcomes.

DHPP'S preparedness:

- Comprehensive financial model shall be prepared to provide a clear picture of the project's financial viability and resilience. Model will include detailed projections of project cost also incorporating sensitivity analysis to assess the potential impact of key risks, including variations in hydrology, construction cost overruns, and project delays. These stress tests will help identify financial vulnerabilities and guide risk mitigation strategies, ensuring that the project remains robust under different scenarios and aligns with lender expectations for financial soundness.

iii. Environmental and social impact:

Lenders' consideration:

- **Environmental impact assessment**: Evaluating the environmental impacts of the project, including those on water resources, ecosystems, and biodiversity, and assessing mitigation measures.
- **Social impact assessment**: Evaluating the social impacts of the project, including resettlement, displacement, and livelihood impacts on local communities, and developing mitigation measures.
- **Resettlement and rehabilitation**: Assessing the need for resettlement and rehabilitation of project-affected populations and ensuring fair and equitable compensation and support.
- **Stakeholder engagement**: Conducting consultations with local communities, affected populations, and other stakeholders to ensure their concerns are addressed.

DHPP's preparedness:

- Key Environmental and social risks have already been identified keeping in view similar projects afflicted with these issues. Accordingly, a comprehensive and detailed risk mitigation plan has been developed by the company and it shall be implemented accordingly. Statutory approvals for such Risk mitigation plans have also been obtained.

iv. Legal and regulatory appraisal

Lenders' consideration:

- **Permits and licenses**: Ensuring the project obtains all necessary permits and licenses from relevant government agencies including but not limited to environment and forest clearance, transmission related approvals, land acquisition and project implementation related approvals.
- **Compliance with regulations**: Ensuring the project complies with all applicable environmental regulations, social impact regulations and other legal requirements.
- **Risk management**: Identifying and managing potential legal and regulatory risks, such as changes in environmental regulations or land acquisition disputes.

DHPP's preparedness:

- Forestry Clearance from Department of Forest and Park Services obtained on 11th and 22nd April 2025.
- No Objection Clearances (NOCs) duly obtained from all the PAHs on 9th August 2024, 18th August 2024, 14th August 2024, 12th August 1st August 2024 from Jarray, Saling, Tsakaling, Tsamang, Tsenkhar gewogs respectively.
- Administrative Clearance from Mongar and Lhuentse Dzongkhags have been obtained on February 25, 2025, and March 27, 2025, respectively.
- Statutory departments like DoST, MoIT, DoE, MoENR have granted the necessary technical and administrative approvals.

v. Commercial Considerations – Off-take related market dynamics

Lenders' consideration:

- **Demand assessment**: Assessing the demand for electricity and the potential market for power generated by the project.
- **Tariff rates**: Evaluating the tariff rates and power purchase agreements (PPAs) to ensure they are favorable and sustainable.
- **Power market analysis:** Analyzing the competitive landscape of the power market and assessing the project's ability to compete effectively.
- **Transmission and distribution:** Assessing the feasibility of connecting the project to the grid and ensuring reliable transmission and distribution of power.

DHPP's preparedness:

- We understand that:
  - DHPP is in advanced stages of discussion for tie-up of long-term power off-take agreement.
  - DHPP will be following international competitive bidding process for selecting vendors on key contracts for civil works and hydro-mechanical components

- Market reports are being prepared with the help of advisors which are aimed at providing suitable analysis for commercial consideration of DHPP.

b. Security enforcement by Indian / International Lenders

We understand that DGPC has obtained a legal opinion dated 14th Oct 2014 for Tangsibji Hydro Energy Limited (THyE) Project from Bhutan Law Services regarding enforceability and validity of the proposed security package in Bhutan.

We have perused this legal opinion to summarize the key observations on enforceability of security by foreign lenders to DHPP. These are detailed out hereunder:

- **Borrowing from Foreign Lenders** - DHPP is within its right to borrow from foreign lenders, provided that DHPP obtains prior written permission of the Royal Government of Bhutan as required under section 13(1) of Moveable and Immovable Property Act (MIPA) of Bhutan for the proposed borrowing before executing any financing agreement with the lenders.
- **Creating Security** - It is permissible under MIPA to create security interest in movable and immovable property. However, for the foreign lenders, the rights and remedies relating the securities are only good when the Borrowing entity has the pre-requisite written permission of the Bhutanese government permitting it to borrow as required under Section 13(1) of MIPA.
    - Land Act of Bhutan doesn't permit lessee of leasehold land in encumbering any charges hereon the leased land other than the use of the leasehold land as per lease terms. This difficulty spills over the charges that the lenders may have on the structures built on the same leasehold land. Thus, it will be critical to obtain written consent from Ministry of Economic Affairs (MoEA) and National Land Commission (NLC) for DHPP to have a right to transfer its remaining lease tenure to the lenders, in the event of its default.

- **Step-in-Right and Security Enforcement** - The Bhutanese counsel has advised, inter alia, that once the right to use the leasehold rights as security for the borrowings is perfected by making a suitable provision in the lease agreement, the step-in-rights of the lenders, in the event of default, are also concurrently perfected. He has also advised that as long as the rights and remedies are expressly provided by the security agreement, the same read with the applicable provisions of MIPA are sufficient to secure the step-in-rights of the lenders.
    - It is possible to create charge/ assignment of the right, title and interest of the borrower company in project contracts, insurance contracts, approvals, etc.
    - A non-Bhutanese person cannot buy immovable property in Bhutan under any circumstances except by being a proprietor of a FDI incorporated company under the Companies Act of Bhutan. Thus the enforcement of securities and disposal of assets by a foreign lender can take place only through recovery of debt from the sale of proceeds of the defaulting company's immovable and movable assets. Hence, although there is no restriction in selling movable assets of a company to any nationals, the immovable properties can only be sold to Bhutanese nationals.
    - In terms of enforcement procedure, the secured party shall have the right to take peaceful possession of the collateral without first obtaining the court order provided the taking of the possession does not involve confrontation or breach of the peace.

c. Financing Timeline

For commercial lenders, **the financing process for DHPP is expected to take approximately 5 to 7 months following the finalization of the Power Purchase Agreement (PPA)**.

This timeline accounts for lender due diligence, credit assessments, internal approvals, and the negotiation of loan terms and security arrangements.

Given the complexity and scale of the project, this duration is considered standard for securing debt financing from commercial institutions.

14. Project Risks - Benchmarking and Mitigations

Purpose of the Section
This section outlines the key material risks typically associated with large hydropower projects, using real-life project examples to underscore the gravity and frequency of such challenges. It further elaborates on specific risk areas considered by lenders and investors and presents a project-specific view of how Dorjilung Hydropower Project (DHPP), has addressed these risks through advance planning, clearances, and proactive stakeholder engagement.

Key Conclusion and takeaways
Hydropower projects are inherently complex and face a multitude of risks including land acquisition challenges, geological surprises, and regulatory delays, which often result in time and cost overruns. In the case of DHPP, these risks have been systematically identified and addressed by DGPC through a well-defined risk mitigation framework. Early-stage regulatory and statutory approvals, detailed geological studies, community consultations, robust vendor selection, and foreign exchange risk mitigation strategies are in place to ensure timely, cost-efficient, and sustainable execution of the project.

14.1. Risk benchmarking and Mitgations

Hydropower projects typically face a range of material risks, including Land acquisition issues, Geological surprises, construction delays leading to time and cost overruns, and regulatory challenges.

For the Dorjilung project, key risks and their mitigation measures have been carefully identified and addressed by DGPC. The following section outlines these risks along with the strategies implemented to manage them effectively.

Land Acquisition and Right of way:

Key factor / risk element considered by Lenders:

- Availability of land for construction of plant.
- In case, private land is to be acquired, whether compensation has been paid and No-Objection-Certificate has been obtained from private stakeholders.
- In case forest land is to be acquired, whether regulatory process and compensatory afforestation measures have been planned for implementation.

Sample project (outside Bhutan) impacted by above mentioned risk:

- **1,500 MW Tipaimukh Hydroelectric project in Manipur, India, initially entrusted to NEEPCO and later switched to NHPC** has been stalled for multiple years, due to issues with land acquisition due to concerns over displacement of indigenous people.
- **World Bank funded project example – 2,160 MW Dasu Hydropower Stage 1 project in Pakistan** has been having land acquisition related delays.

Mitigation in place for DHPP:

> The DHPP project requires 900.08 acres of State Reserved Forest and 18.89 acres of private and institutional land, impacting 58 plots owned by 50 households, one business lessee, and two institutions—totaling 53 affected parties. Only one household will be physically displaced, and the overall resettlement impact is minimal compared to similar projects.
>
> To mitigate these impacts, a comprehensive Land Acquisition and Livelihood Restoration Plan (LALRP) has been finalized. It includes detailed assessments, compensation frameworks, and livelihood support measures developed through stakeholder consultations. **Necessary NOCs and clearances have been obtained, and implementation costs are fully budgeted, with a 10% contingency for unforeseen needs.**

Statutory Approvals:

Key factor / risk element considered by Lenders:

- Whether Cabinet / Government approval, Forest and National Park clearances, Wildlife clearance, Culture clearance, District administrative approval, Environment Clearance, Department of Roads approval, NOC from the affected households losing land to the Project etc. and any other relevant approvals are in place.

Sample project (outside Bhutan) impacted by above mentioned risk:

- **Delay of c. 21 months in construction start of 1,091 MW Karcham Wangtoo Hydroelectric Project** of Jaiprakash Power Ventures Group on account of delay in grant of environment and forest clearance.
- **World Bank funded 444 MW Vishnugad Pipalkoti Hydroelectric Project of THDC India Limited-** Cabinet approval was obtained in 2008, works could not be awarded due to Forest clearance / diversion of forest land. Forest land was acquired in January-14 and subsequently works awarded in January-14. Further, there was disruption of work by local people.

Mitigation in place for DHPP:

DHPP has either already obtained or initiated the process for getting material requisite approvals from the concerned ministries and departments. A list of these approvals has been presented below:

Statutory approvals obtained:

Particulars	Concerned Authority	Approval Date
Approval for Implementation	Department of Energy, Bhutan Ministry of Energy and Natural Resources, Bhutan	August 9, 2024
Approval for Implementation	Ministry of Finance	September 16, 2024
Approval for Implementation	Druk holding and Investments	October 14, 2024
Technical Clearance	Department of Energy, Bhutan Ministry of Energy and Natural Resources, Bhutan	February 12, 2025
Access Road Construction	DoST, MoIT	February 19, 2025
District Administrative Clearance	Mongar and Lhuentse Dzongkhags	February 25, 2025 and March 27, 2025

No Objection Clearances from all Project affected Households	Project affected Households	August 2024
Forestry Clearance	Department of Forest and Park Services	11th and 22nd April 2025

Clearances under process:

Application for Environment Clearance (EC) applied to Department of Environment and Climate Change (DECC) on 5th March 2025 and is at the advanced stage of review process, considering 100 days of turnaround time (TAT). The EC is expected by June 2025 before the start of access road construction works.

Geological uncertainties

Key factor / risk element considered by Lenders:

- Geological uncertainties in hydropower projects such as unexpected rock types, faults, groundwater inflows, and unstable slopes, which can impact design, safety, and cost. These conditions are difficult to predict and can lead to construction delays or failures to implement.

Sample projects impacted by above mentioned risk:

- **2,000 MW under-construction Lower Subansiri Hydroelectric project of NHPC India** has encountered serious geological challenges including weak and ruptured rock strata on dam's right bank. This raised significant safety concerns and lead to multiple re-design, resulting in significant time and cost over-run.
- Work on the **1200 MW Punatsangchhu-I hydroelectric project** commenced in 2008 and was initially expected to commission by 2016 but the project **is running behind schedule**. In July 2013, the project witnessed a geological surprise – the right bank of the project site had sunk, with the loose rock-face gradually moving down to the base of the dam site being excavated. The sinking riverbank necessitated stabilizing the riverbank and design changes in the project. The strengthening work has compelled the project to make technological changes in constructing the dam.

Mitigation in place for DHPP:

Extensive geological and geotechnical investigations were conducted during the 2015 DPR and 2024 DPR Update, including boreholes, seismic and resistivity profiles, exploratory drifts, and expert site inspections. The geology has been thoroughly vetted by consultants, the Panel of Experts, and World Bank specialists.

Given the depth of prior studies, adverse geological scenarios are not anticipated. The headrace tunnel design accounts for worst-case geological conditions,

and a 20% cost contingency has been included. Any unforeseen geological issues during construction will be addressed with site-specific mitigation measures.

Further, sensitivity analysis for cost and time overrun would be carried out to ascertain impact on further fund requirement, debt serviceability and financial stability of the project.

Key factor / risk element considered by Lenders:

- Hydrological shocks from seasonal variations or glacial outbursts etc. can cause water flows to fluctuate, potentially resulting in reduced power generation or flooding. These effects can undermine the project's financial viability and increase the risk for lenders.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- Hydropower plants are prone to hydrology shocks triggered by earthquakes, floods, siltation, droughts etc. Historically, numerous hydropower plants have faced damages or shutdowns due to these hydrology shocks including but not limited to:
  - **6,400 MW Sayano-Shushenskaya Hydroelectric Power Station, Russia** which faced shutdown due to floods in 2009.
  - **2013 floods in Uttarakhand region of India** lead to shutdown/damages in multiple Indian plants including 400 MW Vishnuprayag Hydroelectric projects of Jaiprakash Power Ventures Limited, India and 280 MW Dhauliganga Hydro project of NHPC, India.

Mitigation in place for DHPP:

To address flood risks, future Probable Maximum Flood (PMF) values have been assessed under different climate scenarios through the DPR's Climate Resilience Study. **The project design accounts for projected PMF increases, with significant safety margins for the far future (2060–2079).**

Energy production impacts due to climate variability (2020–2059) have also been analysed, with expected variations ranging from -7% to +18%, and **appropriate design measures have been incorporated to mitigate these risks.**

Environmental and ecological concerns

Key factor / risk element considered by Lenders:

- Impacts on aquatic ecosystems, biodiversity loss, and greenhouse gas emissions from reservoirs.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- **1,070 MW Nam Theun 2 Hydropower Project (Laos)** - Despite being considered a success, has faced criticism for methane emissions and long-term ecological impacts on the Xe Bang Fai River.

Mitigation in place for DHPP:

The DPR identifies potential impacts on aquatic ecosystems from construction activities such as land clearing, earthworks, and river diversion. These are addressed through dedicated plans like the Biodiversity Management Plan (BMP) and Environmental Flow Management Plan (EFMP). These plans outline strategies to protect aquatic life and water quality, supported by regular performance monitoring.

Impacts on plants, birds, and mammals are assessed as **low risk based on ecological data** and are addressed in the BMP. **The DHPP Biodiversity Unit is also working with government agencies to develop Operational Conservation Plans for each identified Critical Habitat.**

Social resettlement and livelihood disruptions

Key factor / risk element considered by Lenders:

- Large hydropower projects often require relocating entire communities, impacting their housing, agriculture, culture, and access to resources. Poorly managed resettlement plan can lead to long-term social unrest, reduced quality of life, and impoverishment of affected people.
- Failure to involve local communities, NGOs, and civil society groups from the early planning stage can lead to misinformation, protests, and even legal challenges.
- Such instances may lead to litigations thereby resulting time and cost overrun.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- 44 Hydropower projects on Siang belt of Sikkim, India.
- 444 MW Vishnugad Pipalkoti Hydroelectric project in Uttarakhand of THDC India Limited.

Mitigation in place for DHPP:

Based on feedback from DHPP, we understand that all affected stakeholders have already been identified through a detailed mapping exercise during the ESIA process**. Consultations were held at local to national levels, and concerns were integrated into mitigation instruments such as the Cultural Heritage Management Plan (CHMP) and Biodiversity Management Plan (BMP)**. Ongoing engagement is ensured through a dedicated Stakeholder Engagement Plan (SEP).

Cost overruns and delays

Key factor / risk element considered by Lenders:

- Cost overruns and delays are the culmination/results of all the risks explained above. Many hydropower projects exceed initial budget and timeline estimates due to unforeseen geological conditions, technical complexities, Environment and social objections and procurement or contractor inefficiencies. This escalates financial risk and affects project returns.

Mitigation in place for DHPP:

In earlier DGPC projects, such as the Dagachhu Hydropower Project, cost overruns were shared between the equity partners—DGPC and TATA. A similar structure is proposed for the DHPP, with any cost overruns to be jointly financed by the equity partners. To further mitigate financial risks, the cost estimates include contingency allowances: 20% for Civil and Hydromechanical Works, and 5% for Electromechanical Work

Further, sensitivity analysis for cost and time overrun would be carried out to ascertain impact on further fund requirement, debt serviceability and financial stability of the project.

Vendor related risk

Key factor / risk element considered by Lenders:

- Financial stability and experience of vendors (Civil contractors, Electromechanical component suppliers and Hydro-mechanical component suppliers) is essential for ensuring quality and timely construction of project.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- Delay in completion of **444 MW Vishnugad Pipalkoti HEP** and **1,000 MW Tehri Pumped Storage Project** of THDC India is also attributable to financial health of main civil contractor. THDC had to approve special arrangement of gap funding to contractor for expediting project progress.

Mitigation in place for DHPP (based on feedback from DGPC):

The implementation of two hydropower projects with **ADB as lead financier**, alongside commercial banks, provided DGPC with critical exposure to **external commercial lending requirements**, compliance standards, and rigorous monitoring/reporting obligations. These experiences enhanced DGPC's capacity to manage financing risks and strengthened internal human resources. **DGPC/DHPP will apply these insights to anticipate and mitigate challenges in future project execution.**

Key Lessons from Dagachhu and Nikachhu Hydropower Projects to be applied on DHPP:

1. **EPC Oversight:** Effective implementation of EPC contracts requires active client oversight; quality control cannot be left solely to the contractor.
2. **Capacity Building:** Deep involvement of DGPC staff during execution led to valuable hands-on experience, strengthening institutional capacity for future projects.
3. **Technical Evaluation:** For complex components (e.g., tunneling, diversions), construction methodology should be a key evaluation criterion during procurement.
4. **Financial Controls:** Use of escrow accounts ensures contractor payments are directly used to pay material and service vendors, improving financial discipline.
5. **Working Capital Management:** Facilitation of working capital advances helps address short-term cash flow mismatches between receipts and payments.
6. **Variation Management:** A structured mechanism is needed for the **early identification and settlement** of variations and claims to prevent delays and cost escalation.

Foreign exchange related risk

Key factor / risk element considered by Lenders:

- Cross-border infrastructure projects with a mismatch between the currency of project revenues and that of liabilities or costs can face asset liability mismatch if foreign exchange rates move unfavorably. Hedging instruments may be a solution to currency risk in these circumstances, but they may not offer a cost-effective solution in certain markets due to the costs involved and the lack of long-term hedging options for local currencies. Another alternate would be to create balance sheet reserve to cater to adverse foreign exchange fluctuation thereby smoothening the impact over time.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- World bank funded **1,070 MW Nam Theun 2 hydropower project in Laos**, faced a similar issue as there was a mismatch between the currency denomination for revenue stream, project cost and debt financing. Currency risk was mitigated in this project by structuring the currency profile of the financing to match that of the project costs (pre-completion of the project) and the revenues (post-completion of the project). This also provided a natural hedge against the tariff structure, which required half of the underlying long-term debt structure to be denominated in Thai baht, and the other half in US dollars.

Mitigation in place for DHPP

In the absence of an approved hedging policy in Bhutan, **the financing agreement may include a provision for establishing a reserve to manage foreign exchange fluctuations**, as has been implemented in previous completed projects.

Same would need to be factored in the financial projections for DHPP.

15. Evaluation of Financing Structure models for DHPP

Purpose of the Section

This section aims to provide a structured analysis of Bhutan's evolving approach to hydropower financing, with a focus on identifying the most suitable financing model for the Dorjilung Hydropower Plant (DHPP). It traces the historical progression from bilateral sovereign-funded models to more diversified structures, including public ownership and Public-Private Partnerships (PPP). By evaluating past domestic projects and international case studies, the section outlines available financing structures and compares their strategic fit for DHPP based on factors such as fiscal impact, risk distribution, technical capability, autonomy, and bankability.

Key Conclusion and takeaways

Each financing model—bilateral sovereign support, full public ownership, and PPP—offers unique advantages but also presents distinct limitations. Bilateral arrangements, while proven, can limit market flexibility and strategic autonomy. Fully public models demand significant fiscal commitment and expose the government to concentrated project risk. In contrast, the PPP model demonstrates a more balanced approach, allowing risk-sharing, access to private capital and technical expertise, and improved bankability. Given these considerations, the PPP structure aligns best with the strategic, financial, and policy objectives of the DHPP and is therefore considered the most appropriate model for its implementation.

15.1.1. Evolution of Hydropower financing in Bhutan

Bhutan's hydropower development journey has undergone significant transformations over the decades, reflecting changes in its funding models and strategic partnerships. This evolution can be divided into two distinct phases:

Phase 1: Bilateral Funding Model with India as Primary Partner (1988-2010s)

Indo-Bhutan hydropower cooperation began in 1961 with the signing of the Jaldhaka agreement. The Jaldhaka project is situated on the Indian side of Indo-Bhutan border in West Bengal. The major part of power produced at Jaldhaka hydropower plant was exported to southern Bhutan.

Bhutan's first mega power project, **Chukha Hydropower Project (CHP) with a capacity of 336 MW, was commissioned in 1988**. This was the first project to be developed on a bilateral funding model with India as the partner. Project was funded by India through a mix of Grant (60%) and Concessional Debt (40%), serving as the model for subsequent collaboration with India in Hydropower sector.

Successful implementation of CHP paved way for other bilaterally funded projects including **Kurichhu Hydropower plant (KHP)** which became **operational in 2002**. KHP has an installed capacity of 60 MW and was funded by grant and concessional debt from India.

Basochhu Hydropower plant (BHP) with a capacity of 64 MW, was the first notable hydropower plant in Bhutan to be developed without India's assistance. Plant became **operational in 2004** and was developed with equity from **Royal Government of Bhutan and support from Austria in form of Grants and Concessional Debts.**

The **Tala Hydropower Plant** with a capacity of 1,020 MW became another milestone in Bhutanese Hydropower landscape. **Commissioned in 2006**, this project followed the successful implementation of the Chukha project and reinforced the Bhutan-India hydropower cooperation model. Tala was funded by a combination of 60% grant and 40% concessional loans from India.

A significant restructuring of the hydropower sector took place in 2007, when the Chhukha, Kurichhu, and Basochhu project companies were consolidated into the newly formed Druk Green Power Corporation (DGPC). This was followed by the integration of the Tala Hydropower Project into DGPC in 2009.

Later in this phase, **commissioned in 2019**, the Mangdechhu Hydroelectric Project (720 MW) was also developed with Indian financial assistance, though with a modified structure of **70% loan and 30% grant.** This shift reflected evolving dynamics in bilateral funding arrangements, as Bhutan took on more financial responsibility while still leveraging Indian backing.

Though there were a few non-Indian-funded exceptions—such as the Basochhu Project (64 MW), supported by Austria—the overwhelming majority of hydropower investment during this phase was India-led, with Bhutan heavily reliant on this funding model for infrastructure expansion.

Phase 2: Diversification (2010s-Now)

By the early 2010s, DGPC actively started diversifying its financing structures to reduce reliance on bilateral sovereign funding. While India remained a vital partner, Bhutan began exploring multilateral institutions, private investors, and export credit agencies to broaden its funding base and reduce dependency.

The most notable milestone in this period was the **Dagachhu Hydroelectric Project (126 MW), commissioned in 2015**. It became Bhutan's first public-private partnership (PPP) in the power sector, co-developed by Druk Green Power Corporation (DGPC), Tata Power (India), and the Asian Development Bank (ADB). Project was developed at a total project cost of $237.6 million funded at a debt-equity ratio of 65:35 involving financing from the Multilateral Organisations, Commercial bank, local public stakeholders and Strategic Private Investors.

Commissioned in 2019, Nikachhu Hydropower Project (NHP) marked another advancement in the evolution of the hydropower development models employed by Bhutan. NHP is Bhutan's **first hydropower project to be funded on a fully public model** with 100% of the equity being held by DGPC. Project was developed at a cost of Nu. 16,007 million with major investments of $120.5 Million from Asian Development Bank comprising of USD 50.5 million loan and grant from the ADB's Asian Development Fund (ADF), and a USD 70 million loan from ADB's ordinary capital resources (OCR) which was extended. directly to the project company Additionally, part debt was funded by a syndicate of Indian Commercial Banks, contributing Nu. 3,274 million to the project. Remaining funds were brought in by DGPC as its equity contribution to the project.

Bhutan has continued diversifying its funding sources for its upcoming projects as well. Similar to Dagachhu Hydropower Plant, **the Khorluchhu Power Plant with envisaged capacity of 600 MW, is being developed under a Public-Private partnership model** with Tata Power Company Limited (TPCL) as the off-taker and strategic investor. Project is envisaged to be developed at a project cost of Nu. 68,992.38 million with Debt-Equity Ratio of 70:30. Equity shareholding will be divided between DGPC and TPCL in the ratio of 60% and 40% respectively. The required debt funding is expected to be sourced from Indian Commercial Banks.

15.1.2. Financing Structure options for development of Dorjilung Hydropower Plant

This section outlines the various financing structure options available to DGPC for the development of the Dorjilung Hydropower Plant (DHPP). Drawing reference from existing DGPC projects and examples from international market, DGPC has three main available structures for development of DHPP (defined below). Each of these models offers distinct strategic advantages while also presenting certain limitations and risks.

The following section presents a comparative analysis of these financing structures, highlighting their key advantages and disadvantages while considering their suitability with respect to development of Dorjilung Hydropower Plant.

- **Bilateral Sovereign Support**

Financing Structure: Financing solely through mix of government loans and grants from bilateral sovereign Partners countries

Precedent Project: Initial hydro projects of DGPC funded by mix of grant and concessionary debt from India and Austria. Notable Projects includes: Chhukha Hydropower Plant, Basochhu Hydropower Plant, Kurichhu Hydropower Plant, Tala Hydropower Plant, Mangdechhu Hydropower Plant.

Key Benefits:

1. **No Drainage of financial resources:** Projects funded primarily through grants or highly concessional loans reduce the fiscal burden on the national budget, enabling Bhutan to pursue capital-intensive projects without upfront financial outflows.
2. **Gaining experience of operating plants without investing:** Early exposure to large-scale hydropower development allowed Bhutan to build technical, managerial, and institutional capabilities without bearing full investment risk.
3. **Access to concessional financial terms:** Bilateral arrangements often include below-market interest rates, longer repayment periods, and grace periods, which make the financing terms more favorable compared to commercial borrowing.
4. **Accelerated Infrastructure Development:** Reliable funding from sovereign partners enabled rapid construction of key hydropower infrastructure in Bhutan which was critical for national revenue generation.

Drawbacks:

1. **Limiting Bhutan's Autonomy in Project development:** Heavy reliance on external sovereign partners can result in reduced control over project design, timelines, and implementation priorities.
2. **Affecting Fiscal Space and Economic Dependence:** Even concessional debt adds to Bhutan's external debt profile, and recurring dependence on a single country may constrain fiscal flexibility and increase economic vulnerability.
3. **Less Advantageous Terms in Power Purchase Agreements (PPAs):** In case of bilateral funding, power is typically exported to the country supporting the project. Such funding linked PPAs can limit Bhutan's negotiation power, often leading to long-term contracts with fixed rates or limited pricing flexibility, especially when the financier is also the buyer.
4. **Restricted Market and Offtake Options:** Tying projects to specific bilateral partners may confine power exports to a single market, limiting Bhutan's ability to diversify energy buyers and optimize revenues.
5. **Limited Private Sector Participation and Innovation:** Sovereign-funded models often exclude private or multilateral stakeholders, reducing opportunities for innovative financing, risk-sharing mechanisms, and technology upgrades.

- **Public ownership model (Nikachhu Model)**

Funding Structure: 100% equity with DGPC with funding support from International Financial Institutions/Multilateral Organizations and Commercial bank.

Precedent Project: Nikachhu Hydropower Plant

Key Benefits:

1. **Ability to control the design and construction of the project:** As the sole owner, DGPC has complete autonomy over design, construction, and operational decisions-enabling optimization of technical solutions and maximization of long-term economic benefits.

2. **Retained Revenues and Strategic Asset Ownership:** 100% public ownership ensures that all project returns attributable to equity stakeholders remain with DGPC while strategically holding the asset within the country, enhancing long-term fiscal returns.
3. **Enhanced Institutional Capacity:** Managing projects in-house builds the project development and financial management capabilities of DGPC, reducing future dependence on external developers.
4. **Potential for Greater Export Value:** With control over power sales, Bhutan can potentially secure more favorable Power Purchase Agreements (PPAs) and explore diversified regional export markets.

Key Drawbacks:

1. **Limited Availability of Public Funds:** Fully publicly owned models require significant upfront capital. This poses challenge for Bhutan as it will have to deplete its own financial resources or depend on multilaterals or commercial bank for debt, both of which can strain fiscal capacity and potentially erode Bhutan's credit profile over the long term. Further, with DGPC or the government bearing the full financial risk, cost overruns, construction delays, or underperformance directly impact national finances.
2. **Reduces the public sector's ability to pursue other opportunities**: Allocating large sums to a single infrastructure project can restrict the government's ability to fund future projects or to support other critical sectors such as health and education.
3. **Lack of Private Sector Participation:** Excluding the private sector limits innovation, risk-sharing, and operational efficiency. It also means Bhutan misses opportunities to attract private capital and expertise.
4. **Requirement for Government Guarantees:** In absence of a reputed strategic investor, commercial financiers may require sovereign guarantee from the Royal Government of Bhutan to back the debt. While this may secure better financing terms, it exposes the government to contingent liabilities, potentially affecting Bhutan's credit profile and limiting its ability to raise debt for its future projects.

- **Public-Private Partnership (PPP) Model (Dagachhu Model)**

Funding Structure: Investment from Strategic partners with funding support from International financial Institutions/Multilateral Organizations and Commercial banks.

Precedent Project: Dagachhu Hydropower Plant

Key Benefits:

1. **Eases Fiscal Burden on the Royal Government of Bhutan (RGoB):** By leveraging private or strategic partner equity, the government can undertake large-scale hydropower development without committing significant public funds or sovereign borrowing.
2. **Access to International Expertise and Best Practices:** Strategic partners often bring global experience in engineering, project management, and sustainability standards, improving the overall efficiency and quality of project execution.
3. **Risk Sharing Between Public and Private Stakeholders:** Project risks—financial, construction, operational—are distributed between DGPC, the strategic partner, and lenders, reducing the government's exposure to potential losses.
4. **Improved Access to Commercial Financing:** The presence of a reputable private or international partner improves project bankability, making it easier to attract commercial loans without sovereign guarantee from RGOB.

Key Drawbacks:

1. **Reduced Control for DGPC and RGoB:** Sharing ownership and decision-making authority with a strategic partner means DGPC and RGoB may have less autonomy in key areas such as project design, procurement, tariff setting, and power sales negotiations.
2. **Potential Strategic Dependency:** Overreliance on external partners for capital and technical leadership could constrain Bhutan's strategic maneuverability in the hydropower sector, especially if partnerships influence project priorities or restrict market access.
3. **Profit-Sharing Dilutes Long-Term Returns:** Since profits are shared among stakeholders, Bhutan may forgo a portion of long-term revenue in exchange for upfront capital and risk-sharing, reducing the national benefit from its hydropower resources create a dependency.

For selection of an appropriate financing structure for Dorjilung Hydropower plant, DGPC must weigh multiple factors - ranging from fiscal impact and risk exposure to technical capability and access to commercial financing. The structure should take into consideration both Bhutan's evolution in Hydropower development and the complexity and scale associated with development of Dorjilung Hydropower plant.

15.1.3. Financing Structures for other International Hydropower Projects

Ruzizi III Hydroelectric PowerStation - Africa

Ruzizi III Hydroelectric power station is a 147 MW hydropower plant being developed via a public private partnership (PPP) model in Africa. Plant will be one of the largest infrastructure projects in the region comprising Burundi, the Democratic Republic of Congo, and Rwanda hereafter referred to as "Contracting States".

Ruzizi III Energy Limited (REL), a special purpose vehicle, has been established for design, construction, funding, operation and maintenance of the project over a 25-year concession period post operationalization of the project, commencing in either 2026 or 2027.

Industrial Promotion Services (IPS), the infrastructure and industrial development arm of Aga Khan Fund for Economic Development (AKFED) and SN Power AS (SNP) wholly owned by Scatec ASA shall hold 70% equity in the project company while remaining 30% is being held by the contracting States.

Stakeholders involved:

1. Public Stakeholders: Government of Burundi, the Democratic Republic of Congo, and Rwanda
2. Multilateral Organization: African Development Bank Group (ADFB), World Bank, European Investment Bank (EIB), KfW Development Bank, European Union
3. Strategic Partners: Industrial Promotion Services and SN Power AS
4. Commercial Banks

Funding Structure:

- Project is envisaged to be funded at gearing ratio of 70% debt and 30% equity.
- Equity for the project shall be contributed by the shareholders in their ownership ratio. Strategic Partners i.e. IPS and SN Power will bring in equity from its own sources while the contracting States may use grants and concessionary debt received from development agencies/multilateral organizations to fulfill their equity commitment.
- Debt component will be in the form of blended finance consisting of grants, commercial debt and concessionary debt on lent by contracting States to the project company.

Envisaged funding structure of the project has been enumerated below:

Table 69: Envisaged Funding Structure of the Ruzizi III Hydroelectric PowerStation Project

S.No	Description	% Capital Class	Amount (Million USD)	% Capital Outlay
1	**Equity**	**100%**	**143**	**24%**
1.1	SN Power/IPS	70%	100	17%
1.2	Three Sovereigns	30%	43	7%
2	**Debt**	**100%**	**461**	**76%**
2.1	Concessional	80%	369	61%
2.2	Commercial	20%	92	15%
	Total	**100%**	**604**	**100%**

<u>Funding commitment by Development Agencies/Multilateral Organizations</u>

About 80% of Ruzizi III's debt requirements (i.e. USD 369 million) as well as the three Countries' equity contributions (i.e. USD 43 million) will be co-financed by six development partners who have committed concessional loans and grants to the Project. Their commitments, which will be subject to the completion of due diligence process, are summarised in the table below:

Table 70: Funding Commitments by Development Agencies/ Multilateral Organizations

Organisation	Committed Funds (Million USD)	Funding Type
African Development Bank	138	Concessional Loans and Grants
World bank	150	Concessional Loans
European Investment Bank	108	Concessional Loans and Grants
KfW Development Bank	31	Grants
European Union	35	Grants
African Development Bank	15	Concessional Loans
Total	**477**	

The envisaged financing structure of the project is outlined in the chart below:

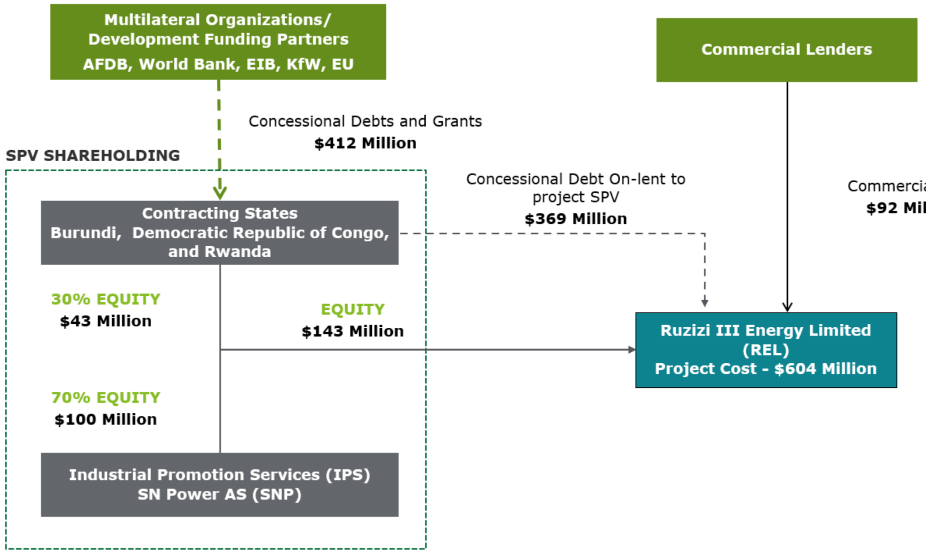


Figure 128: Funding structure of Ruzizi III Hydroelectric PowerStation - Africa

Nam Ngiep 1 Hydropower Project – Laos

Nam Ngiep 1 is a 290-megawatt hydropower project under operation in Bolikhamxay and Xaysomboun provinces of the Lao PDR. Power station generates around 272 MW of electricity for export to Thailand and releases water to a regulating pond where a second dam and power station generates around 18 MW of electric

ity for local use.

The project was implemented at a total cost of $980 million, with a gearing ratio of 66:34. It was executed under a Public-Private Partnership (PPP) model, supported by strategic equity investments from corporations in Japan, Thailand, and Laos. The debt component was financed through a combination of funding from International Financial Institutions (IFIs) and commercial banks based in Japan and Thailand.

Stakeholders involved:

- **Public Stakeholders**: Lao Holding State Enterprise (a State corporation of Laos)
- **Strategic Investors**: EGAT International (Thailand) and KPIC Netherlands (Japan)
- **Multilateral Organization / IFIs**: Asian Development bank (ADB), Japan Bank for International Cooperation (JBIC)
- Commercial Banks from Thailand and Japan

Table 71: Funding structure of Nam Ngiep 1 Hydropower Project– Laos

S.No	Description	% Capital Class	Amount (Mn USD)	% Capital Outlay
1	**Equity**	**100%**	**336**	**34%**
1.1	Lao Holding State Enterprise	25%	84	9%
1.2	KPIC Netherlands (Japan)	45%	151.2	15%
1.3	EGAT International (Thailand)	30%	100.8	10%
2	**Debt**	**100%**	**644**	**66%**
2.1	ADB	23%	145	15%
2.2	JBIC	31%	200	20%
2.3	Thai Commercial Banks	35%	227	23%
2.4	Japanese Commercial Bank	11%	72	7%
	Total Capital Outlay	**100%**	**980**	**100%**

Further, financing structure of the project is outlined in the chart below:

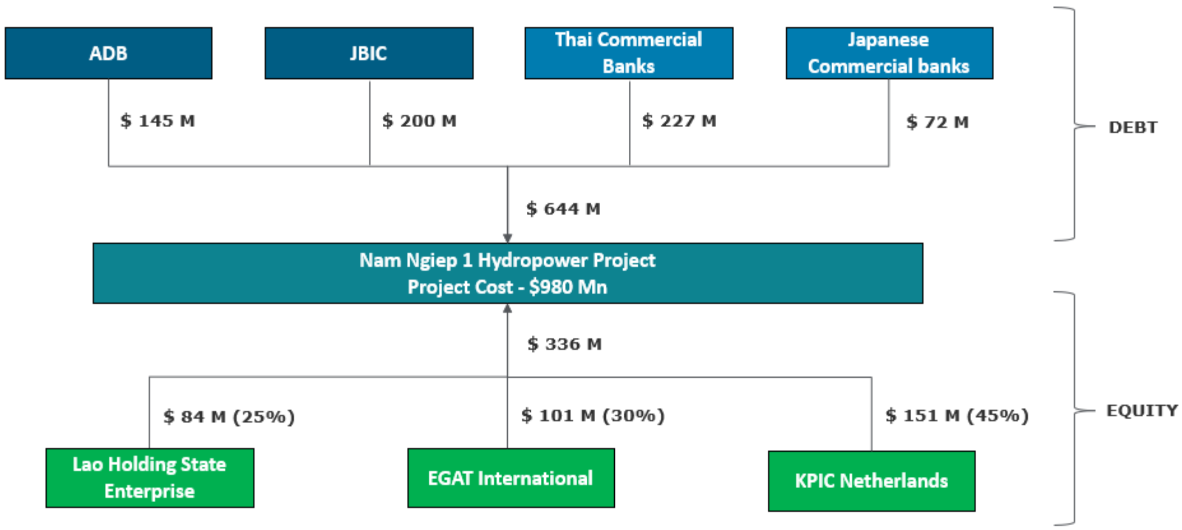


Figure 129: Funding structure of Nam Ngiep 1 Hydropower Project – Laos

Xayaburi Dam Project - Laos

The **Xayaburi Dam** is a run-of-river hydroelectric dam located on the Lower Mekong River in northern Laos. The power plant houses seven 175 MW Kaplan turbine-generators and one 60 MW Kaplan turbine generator. The total installed capacity is 1,285 megawatts commissioned in Oct 2019, with a total annual energy production of 7,406 GWh. Around 95% of the power is supplied to Electricity Generating Authority of Thailand ("EGAT") pursuant to the Power Purchase Agreement for a period of 25 years from the Commercial Operation Date.

Project was implemented on a PPP basis by Xayaburi Power Company Limited (XPCL) a JV between CK Power Public Company Limited (a Thai Construction Company), Natee Synergy Company Limited, Electricity Generating Public Company Limited (EGCO), PT Sole Company Limited and EDL Generation Public Company (Lao Government Entity). The debt component was financed through a combination of funding from Thai Commercial Banks.

Plant was developed at a total project cost of $3.8 billion at a gearing ratio of 84:16 with equity infusion from the JV partners of XPCL and debt participation from a consortium of Thai Commercial banks including Bangkok Bank, Kasikorn Bank, Krung Thai Bank, Siam Commercial Bank and other small Thai banks.

<u>Stakeholders involved at time of project implementation:</u>

- **Public Stakeholders**: EDL - Generation Public Company (a State corporation of Laos)
- **Strategic Investors**: CK Power Public Company Limited, Electricity Generating Public Company Limited, PT Sole Company Limited and Natee Synergy Company Limited
- **Commercial Banks** from Thailand

Table 72: Funding structure of Xayaburi Dam Project – Laos

S.No	Description	% Capital Class	Amount (Mn USD)	% Capital Outlay
1	**Equity**	**100%**	**600**	**16%**
1.1	CK Power Public Company Limited	37.5%	225	6%
1.2	Natee Synergy Company Limited	25%	150	4%

S.No	Description	% Capital Class	Amount (Mn USD)	% Capital Outlay
1.3	EDL - Generation Public Company	20%	120	3%
1.4	Electricity Generating Public Company Limited	12.5%	75	2%
1.5	PT Sole Company Limited	5%	30	1%
2	**Debt**	**100%**	**3,200**	**84%**
2.1	Syndicate of Thai Commercial banks	100%	3,200	84%
	Total Capital Outlay	**100%**	**3,800**	**100%**

Further, financing structure of the project is outlined in the chart below:

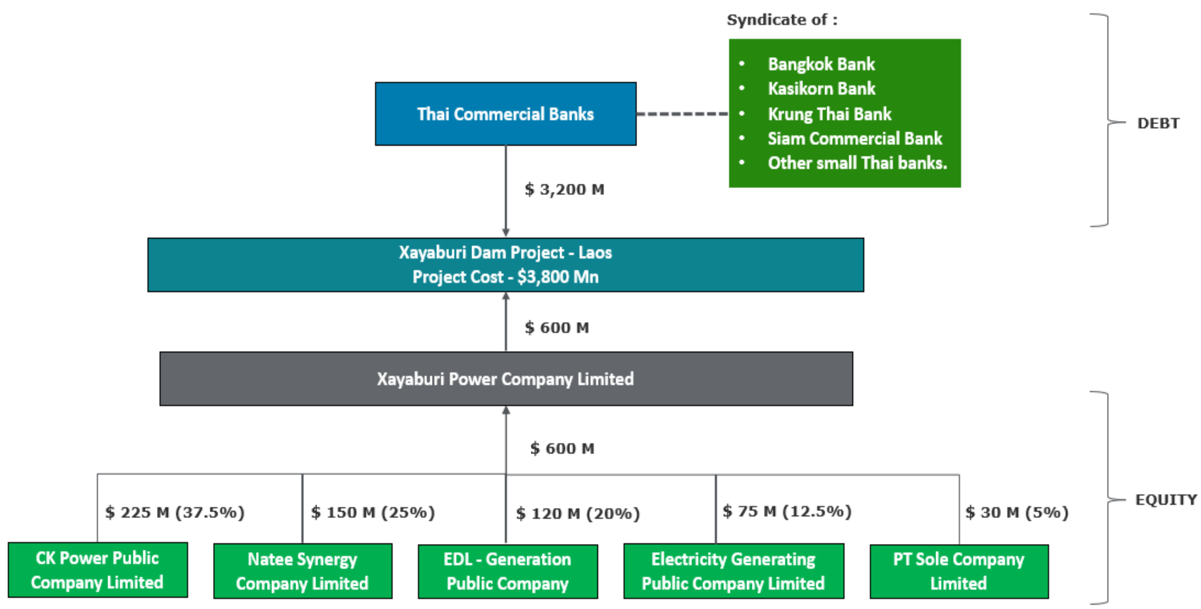


Figure 130: Funding structure of Xayaburi Dam Project – Laos

15.2. Key factors that should be considered while selecting the optimal structure:

1. **Fiscal Impact and Financing Flexibility:** Selected structure should minimize strain on Bhutan's public finance and reduce the need for large sovereign borrowing or guarantees.
2. **Risk Distribution:** Given the scale DHPP, financing structure shall allow construction, financing and operational risks to be distributed among multiple stakeholders which can help safeguard DGPC and the government from concentrated exposure. Further, hydropower projects are prone to cost and time overruns and it would be essential to have a reputed strategic partner / stakeholder that has capacity to bring in additional proportionate capital as the need arises in future.

3. **Access to Technical Expertise:** Bringing in experienced partners can enhance project design, implementation, and management. This becomes increasingly important for large scale projects like DHPP.
4. **Autonomy and Long-term value:** Funding models such as PPP may involve revenue-sharing or shared governance. In such cases, it is important to maintain strategic oversight to safeguard and enhance the national interest. Bhutan's Sustainable Hydropower Development Policy 2020 requires any RGOB entity i.e. DGPC in this case, to hold minimum of 51% stake in a PPP structure. **This stipulation makes sure that national interests remain central in decision-making,** even when external partners are involved.
5. **Bankability of the project:** Selected funding structure should be likely to improve investor confidence which would help in accelerating the financial closure without need of RGOB guarantee.

Suitability of different financing structures for DHPP's development considering these:

Table 73: Suitability of Different Financing Structures for DHPP's Development

Financial Objectives	Suitability for DHPP (High/Low/Neutral)		
	Bilateral Sovereign Support	Public Ownership	PPP
Fiscal Impact	High	Low	High
Risk Distribution	High	Low	High
Access to Technical Expertise	Low	Low	High
Autonomy and Long-term value	Low	High	Neutral
Bankability of the project	Not applicable	Neutral	High
Recommendations	**Not Suitable**	**Not Suitable**	**Suitable**

Conclusion:

After evaluating the different funding structures, each model has its strengths and limitations. **Bilateral Sovereign Support**, while offering lower immediate fiscal impact due to grants and concessional loans, can lead to long-term dependency on a single partner and limits strategic flexibility. **Public Ownership** provides greater autonomy and control over the project, but it requires substantial government resources and may lack access to the technical expertise needed for optimal project execution. In contrast, a **Public-Private Partnership (PPP)** offers a balanced approach, facilitating risk-sharing, improving project bankability, and providing access to technical expertise and private sector efficiency. Therefore, a PPP model may be considered as a suitable option for the Dorjilung Hydropower Plant, as it enables Bhutan to leverage external investment while retaining oversight and aligning with long-term development goals.

16. Next Steps

Purpose of the Section

This section outlines the upcoming deliverables following stakeholder consultations, with a focus on aligning with project objectives, integrating feedback, and ensuring high standards of quality and timeliness. It aims to address critical aspects such as financial analysis and modeling, identification of suitable financing sources, evaluation of offtake arrangements and their impact on project returns and tariffs, and the development of implementation and risk mitigation strategies.

Key Conclusion and takeaways

The key deliverables will include a comprehensive Financial Analysis Report featuring a detailed financial model, scenario analysis for offtake structures, and assessments of royalties, taxes, and returns. It will also evaluate potential public, donor, and commercial financing sources, propose risk mitigation strategies, and outline an implementation roadmap for DGPC financing. Additionally, an Information Memorandum and Request for Proposal (RFP) will be developed, including indicative financing terms and formal documentation to support stakeholder and investor engagement.

The team has initiated the development of the next set of deliverables after conducting stakeholder consultation and incorporating necessary feedback from them. These deliverables have been clearly defined to ensure alignment with project goals and stakeholder expectations, with a continued emphasis on quality, precision, and adherence to established timelines. The deliverables have been enlisted below:

Financial Analysis Report

- Constructing a financial model for the project based on proposed financing plan
- Identifying suitable financing plan
- Prepare Scenario analysis for different offtake arrangement
- Assessing level of royalties and tax that may accrue to Bhutan, as well as overall financing return to project participant.
- Assessing the availability of public, donor financing commercial debt financing
- Examining risk mitigation tools for obtaining commercial funding
- Proposing an implementation structure for the project
- Analyzing the interplay between financing arrangement, offtake arrangement and tariffs
- Assessing the viability and preparing holistic roadmap for DGPC financing
- Propose a work program to conduct a preliminary market soundness assessment

Information memorandum & RFP

- Determining indicative terms to ensure optimal financing.
- Preparation of Information Memorandum document

A. Annexure

Bhutan Power Corporation Limited (BPCL)

Bhutan Power Corporation Limited (BPCL) was established on 1st July 2002 as a public utility company, following its separation from the former Department of Power under the Ministry of Trade and Industry. The corporatization aimed to enhance efficiency and improve electricity service delivery in the power sector. In 2007, ownership of BPCL was transferred to Druk Holding and Investment Limited (DHI), the commercial arm of the Royal Government of Bhutan, created under a Royal Charter to manage the government's investments for the long-term benefit of the Bhutanese people.

Bhutan Power Corporation Limited (BPCL) envisions being an innovative and efficient power utility driving Bhutan's socio-economic transformation. Its mission is to provide affordable, adequate, reliable, and quality electricity services to customers. The organization upholds a culture rooted in integrity, mutual respect, professionalism, accountability, care, and the Bhutanese value of "Tha Dhamtse."

Bhutan Power Corporation Limited has demonstrated strong operational and financial progress from 2020 to 2024. The data reflects rising energy demand, expanding customer base, and improved efficiency across key metrics. Notable gains include reduced system losses, increased energy sales per employee, and stronger financial ratios, indicating enhanced productivity and service delivery in the power sector.

Further, BPCL showcases **strengthening financial position over the years**. The Fixed Assets Turnover Ratio improved from 0.33 in 2020 to 0.69 in 2024, reflecting more efficient utilization of fixed assets as revenue generation ramps up over time. Liquidity ratios such as the Current Ratio and Quick Ratio, though slightly below 1 in later years, are indicative of a tightly managed working capital cycle, **BPCL's debt servicing capacity have also showcased marked improvement** over the last 5 years with both ICR and DSCR improving to 3.92x and 3.28x in 2025 from 2.15x and 1.56x respectively in 2020. Leverage levels remain within a manageable range. The Gearing Ratio and Debt-Equity Ratio rise moderately, reaching 0.39 and 1.28 respectively. Profitability indicators also show a gradual upward trajectory. Return on Assets improves from 2% to 5%, and Return on Equity from 4% to 10%.

Table 74: Key Performance Indicators of BPCL[326]

S No	Indicators	2020	2021	2022	2023	2024
	KPIs					
1	Energy Requirement (MU)	2,154	2,671	3,663	5,922	7,549
2	Purchase from TD by DCSD	2,003	2,518	3,536	5,744	7,193
3	Energy Wheeled	9,157	8,074	7,240	5,135	5,456
4	Energy Sales in MU	1,961	2,474	3,465	5,690	7,125
5	Number of Customers (Categorywise)	2,13,629	2,22,058	2,32,465	2,43,285	2,55,517
6	Number of Customers / Employee	89.8	99.58	104.76	109.64	114.99
7	Energy Sale (MU) / Employee	0.82	1.11	1.55	2.56	3.21
8	Revenue Earned (Million Nu.) /Employee	3.8	4.88	6.65	9.11	11.3
9	Revenue Earned (Million Nu.) /Total Customer	0.04	0.05	0.06	0.08	0.1

[326] Bhutan Power Databook 2024

S No	Indicators	2020	2021	2022	2023	2024
10	LV line length / MV line length	1.47	1.45	1.5	1.37	1.34
	Losses %					
11	Global System losses (including Wheeling) (%)	1.70%	1.83%	1.81%	2.14%	2.15%
12	Global System losses (Excluding Wheeling) (%)	8.95%	7.36%	5.40%	4.12%	3.87%
13	Transmission Loss including wheeling (%)	1.08%	1.11%	1.23%	1.49%	1.49%
14	Distribution Losses including HV Industries (%)	2.82%	3.34%	2.30%	1.35%	0.16%
	Ratios					
15	Fixed Assets Turnover Ratio (Sales / Average Net Block)	0.33	0.39	0.48	0.58	0.69
16	Current Ratio (Current Assets/Current Liabilities)	1.24	1.16	0.83	0.98	0.79
17	Quick Ratio (Current Asset-Inventory / Current Liabilities)	1.11	1.01	0.68	0.79	0.65
18	Interest Coverage Ratio (Profit before interest & tax/Interest)	2.15	2.79	2.45	2.79	3.92
19	Debt Service Coverage Ratio (PAT+Depreciation+Interest / (Interest +Principal Repayment)	1.56	2.36	2.54	2.83	3.28
20	Gearing Ratio (Debt/Net Fixed Asset)	0.35	0.33	0.33	0.37	0.39
21	Debt Equity Ratio (Debt /average Equity)	0.94	0.89	1.12	1.17	1.28
22	Return on Asset (Profit after tax/average net total assets)	0.02	0.04	0.03	0.03	0.05
23	Return on Equity (return available/ Average Equity)	0.04	0.07	0.05	0.07	0.10

Deloitte.